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Chapter

1

Matter in Our Surroundings

INTRODUCTION

Chemistry can be defined as the study of matter and the changes it undergoes.

Matter is defined as anything that occupies space and has mass. It is the stuff our universe is made of, and all the chemicals that make up tangible things, from rocks to pizza to people, are examples of matter.

The early scientists thought that there were only four kinds of matter, or elements - Earth, Air, Fire and Water.

All of the different kinds of matter we see around us were believed to come from mixtures of these four. While this was a good start, four elements alone did not seem to provide enough diversity to account for all matter.

The forms of matter that we see on the earth are often changing. Trees grow, die, and decay. Rocks undergoes weathering thus crumble, and form the fertile soil of the plains or deposits in the oceans. The changes that matter undergoes are another concern of chemistry. But there is even more. In addition to matter, the universe is also composed of energy. When a log burns in the fireplace, it is obvious that a change in matter has occurred. The log seems to have disappeared, leaving a small pile of ash. But the burning of the log warms us as it has given off heat. The heat and light that were liberated by the burning process are forms of energy. Energy is a more abstract concept than matter.

GENERAL CHARACTERISTICS OF MATTER

There are various forms of matter in the nature. The various forms of matter have some common characteristics. The general characteristics have been discussed below :

(i) The Matter has Mass

If we take two pieces of bricks of different size, we will find that one is lighter while the other is heavy. The small sized brick is lighter because it contains less matter, while big sized brick is heavier because it contains more matter. Hence, the amount of matter present in anything is called its mass.

The things may be invisible even then they will have some mass e.g., the air. Although air is invisible even then it has some mass.

(ii) The Matter Occupies Space

You must have experienced this phenomenon during your everyday life. If one stool is placed in the corner of your drawing room, you know another stool cannot be placed in the same corner. If the space on your bookshelf is completely filled with the books, you cannot keep more books on it. The two trucks coming from the opposite directions in a narrow lane cannot cross each other as there is space only for one. If they will try to cross each other they will collide. Therefore, it is clear that matter occupies space.

(iii) The Matter has Inertia

If any thing is lying on any place, it will remain there until and unless some external force disturbs its position. Similarly if anything is in motion, it will remain in motion till some external force stops it. This is known as property of inertia. The matter itself cannot change its position. Suppose if a foot ball is placed in the centre of a playground, it will remain there till it is pushed by anybody. Similarly if the football is pushed by a player it will move on till it is stopped by other players or by the friction of the play-ground.

(iv) The Matter is Effected by the Gravity

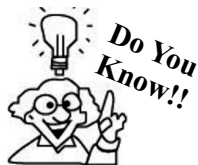
If anything is thrown upward with a certain force it will automatically come to the ground because of force of attraction exerted by the earth. This attraction is called gravity.

When you throw a ball upward with a certain force, it will automatically come down on the earth because of earth's gravity. The fruits always fall downwards from the trees because of this gravity. Similarly water always flows from higher level to lower level.

(v) The Matter cannot be Destroyed

In all physical and chemical changes, total mass of the matter before and after the change remains the same. This can be proved by a simple experiment performed by Lavoisier.

Lavoisier's experiment – Lavoisier took tin metal in an airtight flask. The flask (with tin) was weighed. It was then heated. Tin metal reacted with the oxygen present in the air of the flask and was converted into tin oxide. The flask was cooled and again weighed. It was found that the weight practically remains unchanged.



Law of conservation of mass – This law was given by Lavoisier after carrying out experiment. According to this law “ In any chemical reaction mass of reactants always remains equal to the mass of products. i.e. during any chemical reaction mass remain conserved.

CHECK Point

- ☛ Take one football bladder filled with air and hang it on one side of the beam of a physical balance and put weight on the pan hanging on other side untill balance becomes horizontal. Now remove the air from the bladder. What will you find and what does it indicates?

SOLUTION

You will find that the end of beam (on which the bladder has been hanged) moves upward. This indicates that air has a mass. Therefore, all the forms of matter have mass.



idea box

Perform this experiment in your school chemistry lab. Take few ml of aqueous solution of NaCl in a test tube (20ml) tied with a thread. Also take few ml of CuSO_4 aqueous solution in 250 ml conical flask. Now put test tube containing NaCl solution in conical flask containing CuSO_4 solution. One end of the thread is tied with test tube while other end is kept out of conical flask and flask is closed with rubber cork and then weigh the flask. Now pull the free end of thread to tilt the test tube so that content of test tube and flask get mixed (chemical change). Now again weigh the flask and record your observations what you conclude from this activity?

PHYSICAL NATURE OF MATTER

Every substance has a unique set of properties that allow us to recognize it and to distinguish it from other substances. The properties of matter can be categorized as physical or chemical. Physical properties can be measured without changing the identity and composition of the substance. These properties include color, odour, density, melting point, boiling point, and hardness. Chemical properties describe the way a substance may change or react to form other substances. A common chemical property is flammability, the ability of a substance to burn in the presence of oxygen.

PARTICLE NATURE OF THE MATTER

The concept about the nature of matter is very old. According to the ancient Hindu and Greek philosophers the matter is composed of very small particles which cannot be further sub-divided. John Dalton was the first person who gave scientific explanation about the nature and the composition of the matter.

Evidences for Particles in Matter

Most of the evidence for the existence of particles in matter and their motion comes from the experiments on *diffusion* and *Brownian motion*.

(a) Dissolving a solid in a liquid

Potassium permanganate is a purple coloured solid substance and water is a liquid.

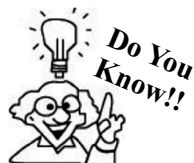
Take 2-3 crystals of potassium permanganate and dissolve them in 100 ml of water.

Now take out 10 ml of this solution & put into another 90 ml of clear water.

Keep diluting the solution like this 5 to 8 times. Everytime solution will become purple colour.

This experiment shows that just a few crystals of potassium permanganate can colour a large volume of water.

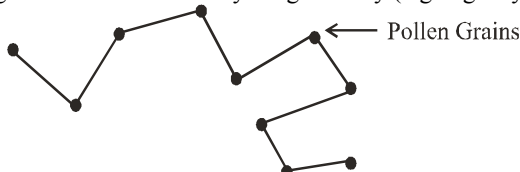
It means a crystal of KMnO_4 is made up of millions of tiny particles. They divide themselves into smaller & smaller particles. So we concluded that matter is made up of small (tiny) particles. Particles of KMnO_4 and particles of water spread into each other, it means they are moving. This self movement of different particles among each other, so they become mixed uniformly, is called *diffusion*.



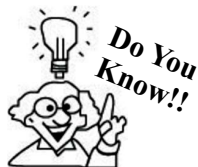
Reason for diffusion of particles of matter is that they are continuously in random motion as they possess kinetic energy. As kinetic energy depends upon temperature. Thus rate of diffusion becomes faster with increase in temperature.

(b) Movement of pollen grains in water

Pollen grains move rapidly throughout the water in a very irregular way (zig-zag way).



The pollen grains move on the surface of water because they are constantly being hit by fast moving particles of water. This type of zig-zag movement of the small particles suspended in a liquid (or gas) is called *Brownian motion*.



Brownian motion increases on increasing the temperature because of increase in kinetic energy of particles of matter.

The existence of Brownian motion gives two conclusions.

(i) Matter is made up of tiny particles. (ii) Particles of matter are constantly moving.

CHECK Point

What is nature of matter is it continuous or particulate?

SOLUTION

Particulate

Characteristics of Particles of Matter

(i) **The particles of matter are very, very small**

The matter is composed of small particles : To understand particle nature of the matter, let us perform one experiment. Take a beaker full of water. Now place a small crystal of blue vitriol (copper sulphate) in the water. You will observe that the water begins to become blue coloured and slowly the size of the crystal becomes smaller and smaller. The crystal has divided itself into number of smaller particles and ultimately it dissolves in water. This experiment suggests that matter is composed of small particles.

(ii) **The particles of matter have spaces between them**

Is there vacant space between the particles of matter ? Let us perform the following experiments –

Experiment (1) – Take a piece of chalk (used for writing on black board) and dip it into water. Some water is absorbed by the chalk. There are vacant spaces or pores in the chalk which are occupied by the water.

Experiment (2) – Take a wide mouthed test tube, almost half filled with water. Now put a sugar cube in the test tube and mark the water level with a glass marking pencil. The sugar cube will dissolve in water and the level of water will go down to a small extent. Why it so happened ? There are vacant spaces between the particles of water. This vacant space is occupied by the particles of the sugar and water level goes down.



idea box

Perform this experiment in your school chemistry lab. Take a big sized test tube and fill it with water. Add a few drops of Phenolphthalein solution. Now tie a cellophane paper on the mouth of the test tube. Invert the test tube on the open mouth of a bottle containing ammonium hydroxide. Record your observation and discuss your observation with your friends and teacher.

(iii) **The particles of matter are constantly moving**

This property can be explained by Brownian motion and diffusion.

Diffusion : It is the phenomenon in which the movement of molecules or particles occur from their higher concentration towards the lower concentration. “Intermixing of particles of two different types of matter on their own is called diffusion”.

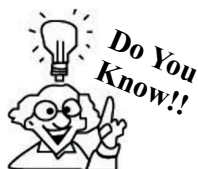
For eg : When a perfume bottle is open in one corner of a room, its fragrance spreads in the whole room quickly. The particles of perfume move rapidly in all directions and mix with the moving particles of air in the room.

CHECK Point

- ☛ Why liquids diffuse slowly as compared to gases?

SOLUTION

In liquids, the molecules are less free than in a gas, i.e., the intermolecular forces (van der Waals forces) in a liquid are greater than in a gas.



We have seen that the particles of matter in liquid and gaseous state are mobile. But it is difficult to visualise that the particles of solid matter are also mobile. We do not find any movement of the particles in case of solid matter like wood, iron, gold, copper etc. even if these substances are placed in water or air. If you dip a zinc rod in mercury, you will find that after sometime the particles of mercury enters into the zinc metal. Similarly if a gold rod is placed in mercury, the mercury particles slowly enters into gold. This process is very slow.

Both heat as well as temperature are responsible for the motion. When the temperature is increased there will be increase in the motion also. We may conclude that particles of matter are mobile, whether the matter is in solid, liquid or gaseous state. The motion of the particles is fast in case of gaseous state, slow in liquid state and very slow in solid state.

**idea box**

Take two 250 ml beakers. Now fill one beaker with cold water and the other with hot water. Now with the help of dropper add 1-2 drops of ink in both the beakers. Record your observations and draw a conclusion on the basis of your observations.

(iv) **Particles of matter attract each other**

There are some forces of attraction between the particles of matter which bind them together.

Cohesive force : The force of attraction between the particles of same substances is called Cohesive Force.

Adhesive force : The force of attraction between the particles of different substances is called Adhesive Force.

For eg : If we take a piece of chalk, a cube of ice and an iron nail and beat them with a hammer.

Observation : Chalk will easily break into smaller pieces.

More force is required to break a cube of ice. Iron nail does not break.

Conclusion : Force of attraction is quite weak in between the chalk particles.

Force of attraction in between the particles of ice cube is a bit stronger.

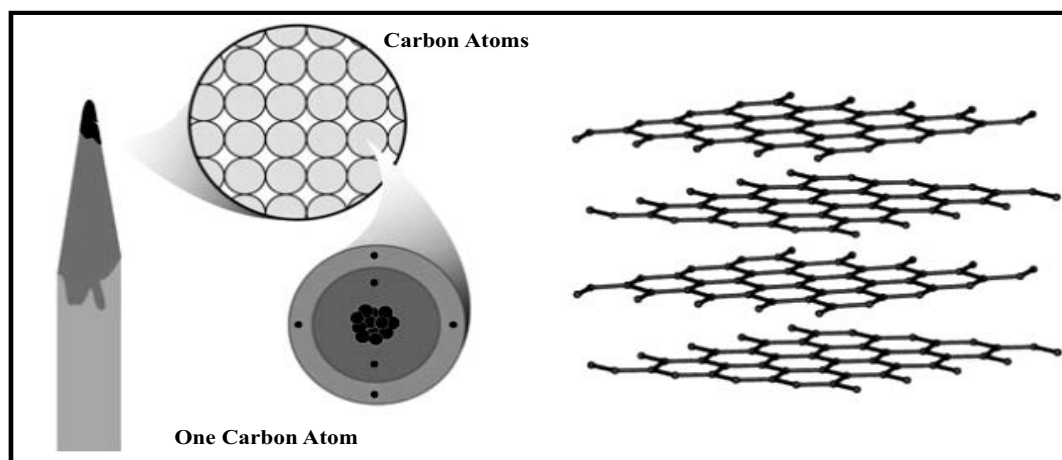
Force of attraction in between the particles of iron is very, very strong.

CONNECTING TOPICS**THEORIES RELATED TO MATTER****(1) Atomic Theory of Matter**

The earliest ideas about matter and atoms were developed by Greek philosophers between 380 and 450 B.C. At that time, the question under discussion was whether matter had the property of being continuous or discontinuous. These concepts can be visualized if you took the “lead” or graphite of a broken pencil point and divided it in half, then divide that piece in half and again half of that piece. This process could be continued as long as possible. If matter was continuous, the process could be continued indefinitely without ever “running out” of graphite. If matter was discontinuous, then at some point, the dividing process would end when the last particle which could still be called graphite was all that remained. A further division would result in destroying the matter called graphite.

Democritus used the word “*atomos*” meaning “*indivisible*” or “*uncuttable*” to describe the ultimate building blocks of matter. Later word atoms is used for atomos. Atoms result from the repeated division of matter to very very tiny, permanent, indivisible, invisible bits of matter with definite size and shape. This Atom Theory was applied in support of the *Discontinuous Theory of Matter*. The Continuous Theory of Matter received wide spread support until the 1800’s when *John Dalton* revived the atom concept to explain certain aspects of chemical reactions.

The final division of matter such as graphite results in atoms of the element carbon.



(2) Kinetic Theory of Matter :

The Kinetic Theory explains the differences between the three states of matter. It states that all matter is made up of moving particles which are molecules or atoms. In solids, the particles are so tightly bound to each other that they can only vibrate but not move to another location.

In liquids, the particles have enough free space to move about, but they still attract one another. In gases, the particles are far apart and can move about freely since there is much free space. Solids change into liquids, and liquids into gases, when the particles gain more kinetic energy, like when being heated and are able to move apart from one another. When the molecules vibrate more quickly upon heating, some of it escapes from the matter.

The main postulates of kinetic Theory of matter are following :

- Matter consists of molecules. These are the smallest particles, which are capable of free existence and retain all the chemical properties of the parent substance.
- The molecules are always in a state of random continuous motion.
- The molecules exert forces on one another. These forces depend upon intermolecular distance
- All collision between the particles of matter are perfectly elastic.

ILLUSTRATION : 1

What would have happened to the gas if the molecular collisions were not elastic?

SOLUTION :

On every collision, there would have been loss of energy. As a result, the molecules would have slowed down and ultimately settle down in the vessel. Moreover, the pressure would have gradually reduced to zero.

MOLECULAR NATURE OF THE MATTER

Matter is composed of the molecules formed by combination of atoms. As already stated, the molecules are very minute in size and mass. These molecules cannot be seen even with the most powerful microscope of your school. A substance appears to be composite, but there are wide gaps (interstices) between molecules which may expand or contract, the expansion or contraction of the bodies on heating or cooling is due to the change in the distances between the molecules.

The molecules of the substances are always in the state of constant motion. The molecules of the different substances move with the different speed while molecules of the same substance move with the same speed. The lighter molecules move with greater speed as compared to the heavier molecules. Further with the increase of temperature motion of the molecules also increases.

At high temperature the molecules move faster and push one another with greater force and the distance between the molecules increase. This explains why the bodies expand on heating. When a solid substance is heated upto its melting point, it changes to liquid state because the force of attraction between the molecules decreases and the distance between the molecules increase. The molecules of a liquid have a random motion, i.e., they can move from one place to another or in other words liquid can flow easily. If a liquid substance is heated upto its boiling point it changes to gaseous state. In this state the force of attraction becomes negligible. The distances between the molecules increase to a very great extent. Therefore, the molecules in gaseous state have free movement.

INTER-MOLECULAR FORCES

The particles of the matter are bonded together with a force called Inter-molecular force. In solid, the particles are bonded together with strong inter-molecular forces. When energy (say in the form of heat) is given, the bonds break, the arrangement of the molecules is disturbed and the solid is converted into liquid. If more energy is given, the inter-molecular forces are further decreased, the arrangement of the molecules is further disturbed and the liquid is converted into gas. Different solids possess inter-molecular force of different magnitude. If this force is more in any solid, more energy will be required to break the bonds between the molecules, i.e., more energy will be required to convert it into liquid.

As stated above there are forces of attraction between the molecules and these forces continuously scatter them because of their velocity. However, these forces are not uniform in the molecules of all the substances. In solids the force of attraction between the molecules is maximum and in gases it is minimum or negligible. In liquids the molecules are held together less firmly as compared to the solids. Therefore, liquid molecules get separated from one another easily. This explains that why some liquids evaporates even at low temperature.

CHECK Point

1. Arrange the solids, liquids and gases in the increasing order of inter molecular forces of attraction.
2. Arrange solid, liquid and gas in order of energy, giving reasons.

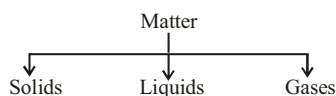
SOLUTION

1. Gases < Liquids < Solids.
2. Solid < Liquid < Gas. This is because a solid absorbs energy to change into a liquid which further absorbs energy to change into a gas.

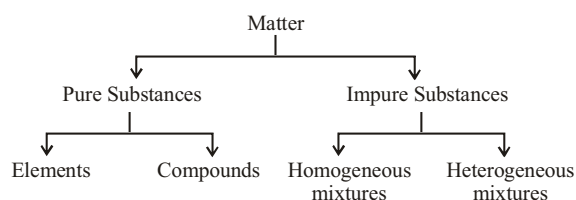
CLASSIFICATION OF MATTER

Matter can be classified into different categories depending upon their physical or chemical nature.

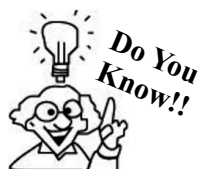
- (i) Based on state (Physical Classification)



- (ii) Based on purity/composition (Chemical Classification)

**STATES OF MATTER**

Matter is everything around us. Matter is anything made of atoms and molecules. Matter is anything that has mass and occupy space. Matter is also related to light and electromagnetic radiation. Up to 1995 scientists have identified five states of matter. These five main states of matter are solids, liquids, gases, plasmas and Bose-Einstein condensates. These all are different states of matter.



Each of these states is also known as a phase. Elements and compounds can move from one phase to another phase when special physical forces are present. One example of those forces is temperature. The phase or state of matter can change when the temperature changes. Generally, as the temperature rises, matter moves to a more active state. Phase describes a physical state of matter. The keyword to notice is physical. Things only move from one phase to another by physical means.

In the solid state, substances are rigid and have definite shapes. Volumes of solids do not vary much with changes in temperature and pressure. In the liquid state, the individual particles are confined to a given volume but they do not have definite shape. A liquid flows and assumes the shape of its container up to the volume of the liquid. Volumes of liquids changes to appreciable extent with change in temperature and pressure. Gases neither have definite shape nor volume. They occupy all parts of any vessel in which they are confined.

SOLID

The solids are characterised by incompressibility, rigidity and mechanical strength. It indicates that the molecules, atoms or ions that make up a solid are closely packed or in other words they are held together by strong forces and cannot move about. Due to the strong intermolecular attractive forces between the constituent particles, solids are rigid and possess a definite shape and volume. These have definite melting point, high density and low compressibility.

PROPERTIES OF SOLIDS

Important properties of solids can be described as follows:

- (1) **Shape and volume** : A solid has a definite volume and definite shape. The molecules in a solid are in fixed positions and are close together because the intermolecular force of attraction between the particles is very strong. The molecules are strongly held and arranged in order. Although the molecules can still vibrate, they cannot move from one part of the solid to another part. As a result, a solid does not easily change its shape or its volume. If you force the molecules apart, you can change the shape and the volume of a solid by breaking it into pieces. However, each of those pieces will still be a solid and have its own particular shape and volume.
- (2) **Compressibility** : A-solid is rigid and cannot be compressed easily. The molecules in a solid are closely packed and the intermolecular distance is very small which cannot be reduced further.
- (3) **Diffusion** : A solid does not diffuse into another solid easily because intermolecular force of attraction in solid is very strong.
- (4) A solid does not flow and can be stored in a container because intermolecular force of attraction is so strong that molecules do not flow and leave the surface of the solid.
- (5) **Melting** : A solid on heating usually changes into its liquid state. This is because heating provides sufficient energy to molecules to overcome the intermolecular force of attraction. Intermolecular separation increases, changing the solid into liquid.

CONNECTING TOPIC

CLASSIFICATION OF SOLIDS

Solids are classified into two groups based on the arrangement of constituent particles.

- (a) Crystalline solids (b) Amorphous solids

(i) Amorphous Solids

Amorphous solids are those solids in which the constituent particles are arranged in haphazard manner and not in a regular fashion. These resemble liquids as they flow very slow at room temperature and thus termed as **super cooled liquids**.

The properties showing the difference between the crystalline and amorphous solids are listed in table.

These lose their shape on standing and flow under their own weight, e.g., glass, pitch, rubber, plastics, starch and proteins.

(ii) A Crystalline Solid

Crystalline solid consists of a large number of small units, called crystals, each of which possesses a definite geometric shape bound by plane surfaces called faces. A crystal is thus a solid figure with a definite geometrical shape, flat faces and sharp edges. The angle between two faces (interfacial angle) is a characteristic property of a crystal. A crystal may be cleaved not in all directions but in certain preferred directions to give smaller crystal which has the same characteristic as the parent.

Table : Distinction between Crystalline and Amorphous Solids

	Property	Crystalline solids	Amorphous solids
1.	Shape	They have a definite and regular geometrical form.	They do not have definite geometrical shape.
2.	M.P.	They have a sharp (definite) melting point.	They melt over a wide range of temperatures.
3.	Compressibility	They are rigid and incompressible.	They too are usually rigid and cannot be compressed to any appreciable extent. However, graphite is soft because of its unusual structure.
4.	Cutting with sharp edged tool	They give clean cleavage, i.e., they break into two pieces with plane surfaces.	They give irregular cleavage, i.e., they break into two pieces with irregular surfaces.
5.	Heat and fusion	They have a definite heat of fusion.	They do not have a definite heat of fusion.

6.	Isotropy/Anisotropy	Anisotropic. i.e., their mechanical and electrical properties depend on the direction along which they are measured.	Isotropic i.e., they have similar physical properties in all directions because the constituents are arranged in random manner.
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Alternatively, solids can be classified into four types depending upon the nature of interparticle forces present between the constituent units.

- (a) **Ionic solids** : Here, constituent units are ions of opposite charges and interparticle forces are ionic forces, e.g., NaCl, BaCl₂, KNO₃, etc.
- (b) **Covalent solids** : Here, the constituent particles are atoms which are held together by covalent bonds. These solids are very hard because of strong interparticle covalent bonds. Examples are quartz, SiO₂, diamond, graphite, etc.
- (c) **Metallic solids** : Here, the constituent units are positively charged kernels which are held together by a cloud of loosely bound electrons (metallic bonds). These electrons are free to move throughout the lattice, e.g., Na, Cu, Fe, Al, Ag, etc.
- (d) **Molecular solids** : Here, the constituent units are covalent molecules and the interparticle forces are hydrogen bonds or van der Waal's forces, e.g., solid CO₂, I₂, S₈, urea, sugar etc.

CHECK Point

- ☛ Why urea has a sharp melting point but glass does not?

SOLUTION

Urea is a crystalline solid whereas glass is amorphous. Crystalline solids have sharp melting points whereas amorphous solids do not possess that melting points.

ILLUSTRATION : 2

Classify the following substances into ionic, covalent, molecular or metallic :

MgO, SO₂, I₂, H₂O (ice), SiO₂(quartz), brass.

SOLUTION :

MgO = Ionic, SO₂ = Polar molecular, I₂ = non-polar molecular, H₂O = Hydrogen bonded molecular, SiO₂ = Covalent or network, brass = metallic.

Knowledge ENHANCER

Interesting Property of Some Solids : Elasticity :

When a piece of rubber is stretched its length is increased, but on removing the force, the rubber gains its original shape. Similarly if a weight is hanged with a spring, the length of the spring is increased, but on removing the weight, the spring gets its original length. This property of a solid body by virtue of which it tends to regain its original state, when the deforming forces are removed, is known as elasticity. The bodies in which complete restoration of the initial state takes place, are called perfectly elastic bodies, while those bodies which do not show any such tendency are termed as non-elastic or plastics.

The property of elasticity can be explained on the basis of the molecular force between the molecules. The molecular force works between the molecules in such a way as if they are joined together by a spring. In normal conditions the solid retaining its shape and size. The molecules remain balanced unless some external force is applied. When force is applied the molecules begin to move towards the direction of the force. When the distance between the molecules increases they experience force of attraction and when the distance between the molecules decreased, they experience repulsion. When the external force is removed the molecules regain their original position.

Ice a Unique Solid :

Ice is considered as unique solid because it is less dense as a solid than it is as a liquid. Normally, a liquid becomes denser as it freezes and its molecules pack closer together. However, as water freezes, its molecules form a lattice network of hexagons containing empty spaces. These hexagons cause water to expand about 11% in volume when it freezes. This expansion means that ice has fewer molecules per cubic centimeter than water has, so it is less dense than water. To be precise, ice's density is 0.917 g/cm³ and water's maximum density is 1.00 g/cm³ at 3.98° C.

LIQUID

Liquids have definite volume but do not have definite shape. They take the shape of the vessel in which they are kept. Molecules of liquid are more close to each other in comparison to the gas molecule due to strong attractive forces. Liquids have enough vacant space in comparison to solids. Thus liquid is an intermediate state between the complete disorder found in gas and highly ordered molecular state in solid. Example: water and milk.

Properties of Liquids :

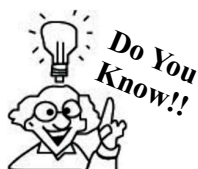
Important properties of liquid can be described as follows :

(1) Shape and volume :

Molecular forces between the liquid molecules are not so strong to keep molecules in a fixed position i.e., intermolecular attraction between molecules is less than that of solids. Therefore, liquids do not have a definite geometric shape. Due to appropriate attractive force they have definite volume in comparison to gas.

(2) Density :

In comparison to the gas, in liquids the molecules are more close to each other. Therefore, density of liquids is higher than that of gaseous state. For example, at one atmospheric pressure and 100°C the density of water is 0.958 gm cm^{-3} whereas under same conditions steam (gaseous water) has a density of $0.00059\text{ gm cm}^{-3}$. That is water is almost 1600 times more dense than vapour.



It can be concluded that density of liquid is more in comparison to the gas due to their more closely packed molecules. There is negligible effect of pressure on the density of liquid. Liquids are incompressible.

$$\text{Density of any substance} = \frac{\text{Mass of that substance}}{\text{Volume of that substance}}$$

(3) Diffusion :

When two soluble liquids are placed close to each other then molecules of one liquid enter in between the molecules of the other and the two liquid mixed with each other. It is called diffusion. Diffusion is seen in liquids like gases.

It is a comparatively slow process, because the velocity of liquid molecules is very less in comparison to the gas molecules. Moreover vacant space is comparatively quite less in liquid than gas.

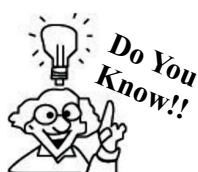
(4) Compressibility :

The vacant space in between the molecules of liquids is very less, so the compressibility of liquids is very less, as compared to gases. For example at 300K when the pressure of an ideal gas is changed from one atmosphere to two atmosphere then its volume reduces by 50% whereas when the same pressure is increased on water (liquid) its volume decreases by only 0.0045%.

(5) Evaporation :

Though the molecules of liquid are bonded to each other by, attractive force still they have a tendency of evaporation.

Therefore evaporation is that process in which liquid changes into vapour at room temperature. The kinetic energy of all liquid molecules is not same. But it is equal to the average value or near about it i.e. some molecules have more or some less. That is known that kinetic energy of molecules of liquid opposes the intermolecular force of attraction which keeps them close. Therefore some molecules evaporate from the surface and go to the vapour state. We will learn more about evaporation later in this chapter.



The pressure exerted by the vapours in equilibrium with its liquid at a given temperature is called vapour pressure of that particular liquid. This is also called saturated vapour pressure. The magnitude of vapour pressure depends upon the following two factors.

(i) Nature of liquid : The vapour pressure of a liquid depends upon the nature of the liquid.

(ii) Temperature of liquid : The vapour pressure of a liquid increases with increase in temperature.

**CHECK** *Point*

- ☛ A liquid is transferred from a smaller vessel to a bigger vessel at the same temperature. What will be the effect on the vapour pressure?

SOLUTION

No effect as it depends only on the nature of the liquid and temperature.

CONNECTING TOPIC**SURFACE TENSION**

It is an important property of liquids, which is directly related to the intermolecular forces between the molecules. Liquid surface feel stretched due to surface tension. As mentioned previously, the molecules in liquids are held closely and hence attract each other. A molecule in the bulk of the liquid is attracted equally on all sides so that the net attractive pull on the molecule is zero. However, a molecule which lies at the surface (known as surface molecule) is subjected only to the attractive forces of the molecules below it. This is because there are no molecules above it. Therefore, surface molecules experience a resultant downward attractive force within the liquid. This creates an imbalance of forces at the surface. In other words, the liquid surface is under tension due to imbalanced forces. This effect is called **surface tension**.

Effect of temperature

Surface tension decreases with rise in temperature. The decrease of surface tension with increase of temperature is because of the fact that the kinetic energy (or speed) of the molecules increases. As a result, the intermolecular forces decrease and therefore, surface tension also decreases. For example, the clothes are washed more efficiently in hot water than in cold water due to decreased surface tension in hot water.

Surface tension and nature of liquid

Since, surface tension of a liquid is due to intermolecular attractive forces, therefore, the magnitude of surface tension is a measure of intermolecular attractive forces. When the attractive forces between the molecules are large, the surface tension is large.

Importance of Surface Tension**(i) Capillary action**

When one end of a capillary tube is put into a liquid that wets glass the liquid rises into the capillary tube to a certain height and then stops. The rise of a liquid in capillary is called **capillary action**. The rise of liquid in a capillary is due to the inward pull of surface tension acting on the surface which pushes the liquid into the capillary tube. This phenomenon is very important. For example, water below the surface of the earth rises to the plants through the roots, oil rises into the wick of an oil lamp, ink rises in a blotting paper, are examples of capillary action.

(ii) Spherical shape of drops

The liquid drops have nearly spherical shapes. We have learnt that surface tension tries to decrease the surface area of a liquid to the minimum. Since the sphere has minimum surface area for the given volume of liquid.

GAS

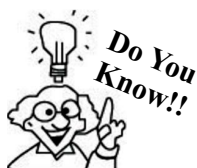
Gas is the third state of matter. A gas differs from a solid and a liquid in a number of ways. A gas fills entire space available to it and therefore it has no definite shape or volume.

The molecules in a gas are in a continuous motion in all possible directions in the complete space available within the container in which it is enclosed. The average velocity of a gas particle is nearly 15000 ms^{-1} at ordinary temperature.

Due to very weak intermolecular forces in gases its molecules are in greater motion and show irregular movement. Therefore gases do not possess definite size, shape and volume. Gases have very low density and high compressibility.

Properties of Gases**(1) Compressibility**

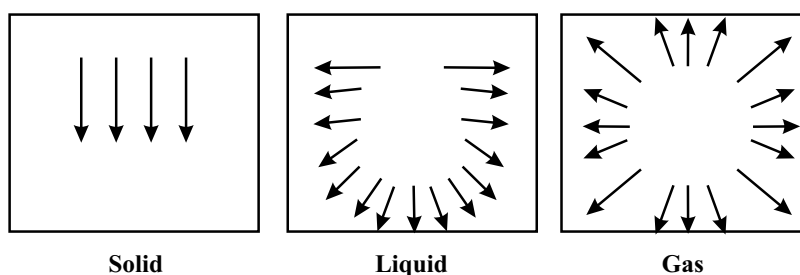
The gases can be compressed on applying external pressure. In case of gases the distances between the molecules are much greater as compared to the solids and liquids. Therefore, on applying external pressure the gases can be compressed easily and under such conditions the gas molecules come closer to each other and they occupy less space than before.



The property of liquids of incompressibility is useful for us in hydraulic machines. A simple system of automobile hydraulic brakes are a good example of this. The brake system cannot work correctly if there is any air (gas) in the system because the gas is compressible.

(2) Exertion of pressure

Solids exert pressure only in downward direction. Liquids exert pressure downward as well as to the sides. But gases exert pressure in all directions (a good example is a balloon) as shown in figure below. This pressure is due to bombardment of the particles against the walls of the container. Thus gases exert pressure equally in all directions.



(3) Homogeneous nature

Gases have similar composition in all parts and are therefore homogeneous in nature.

(4) Liquefaction

Gases can be liquefied by cooling and by applying pressure.

(5) Diffusion

When a bottle of ammonia is opened in one corner of the laboratory, its pungent smell can be experienced all over the laboratory. Similarly, when a bottle of perfume is opened in the drawing room its smell spreads and can be detected all over the room, In fact all gases possess this property of intermixing with one another without any mechanical aid to form a homogeneous mixture called diffusion.

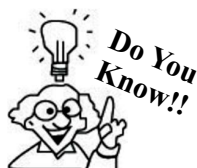
Uses of Gaseous Diffusion

1. Separation of gases :

Gases can be separated from the mixture of gases by the use of process of diffusion. When a mixture of gases is diffused, the lighter gases will diffuse quickly than the heavier gases. In this way by repeating the process of diffusion, gases can be separated from the mixture.

2. Relative densities and molecular weights of the gases :

Graham's Law of diffusion (explained later in this chapter) can be used for the determination of relative densities and molecular weights of the gases.



Air, becomes impure because of the process of diffusion. Poisonous and foul smelling gases diffuses in the air and because of this dispersion, the air becomes foul smelling.

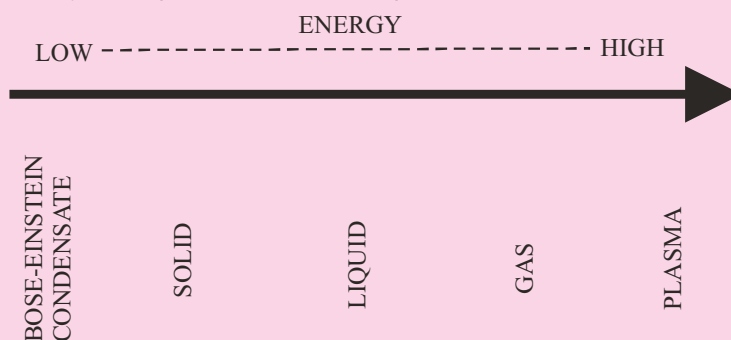
Knowledge ENHANCER

Plasma : Plasmas are a lot like gases, but the atoms are different because they are made up of free electrons and ions of the element. You don't find plasmas too often when you walk around. They aren't things that happen regularly on Earth. If you have ever heard of the Northern Lights or ball lightning, you might know that those are types of plasmas. It takes a very special environment to keep plasmas going. They are different and unique from the other states of matter. You won't find plasmas just anywhere. However, there may be some in front of you. Think about a fluorescent light bulb. They are not like regular light bulbs. Inside the long tube is a gas. Electricity flows through the tube when the light is turned on. The electricity acts as that special energy and charges up the gas. This charging and exciting of the atoms creates glowing plasma inside the bulb.

Another example of plasma is a neon sign. Just like a fluorescent light, neon signs are glass tubes filled with gas. When the light is turned on, the electricity flows through the tube. The electricity charges the gas, possibly neon, and creates plasma inside of the tube. The plasma glows a special color depending on what kind of gas is present inside the plasma. You also see plasma when you look at stars. Stars are big balls of gases at really high temperature. The high temperatures charge up the atoms and create plasma. Fluorescent lights are cold compared to really hot stars. They are still both forms of plasma, even with different physical characteristics.

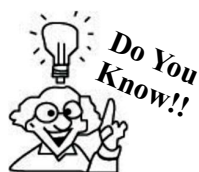
BOSE-EINSTEIN CONDENSATES : Condensation happens when several gas molecules come together and form a liquid. It all happens because of a loss of energy. Gases are really excited atoms. When they lose energy, they slow down and begin to collect. They can collect into one drop. Water condenses on the lid of your pot when you boil water. It cools on the metal and becomes a liquid again. You would then have a condensate.

In 1995, two scientists, Cornell and Weiman, finally created this new state of matter. Two other scientists, Satyendra Bose and Albert Einstein, had predicted it in the 1920. They didn't have the equipment and facilities to make it happen in the 20s. Now, we do. If plasmas are super hot and super excited atoms, the atoms in a Bose-Einstein condensate (BEC) are total opposites. They are super-unexcited and super-cold atoms.

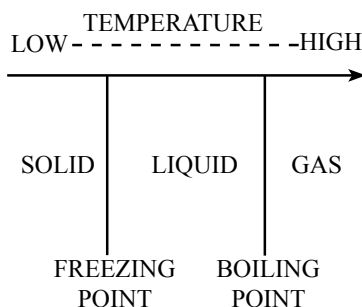


INTERCONVERSION OF MATTER INTO DIFFERENT STATE

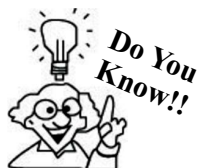
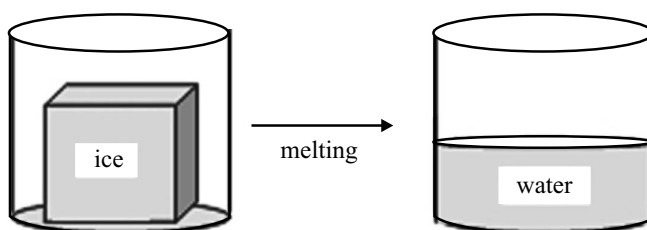
The phenomenon of change of matter from one state to another state and back to original state, by altering the condition of temperature and pressure, is called interconversion of matter. All matter can move from one state to another. It may require very low temperatures or very lower or high pressures, but it can be done. Phase changes happen when certain points are reached. when a liquid turns in to solid. Scientists use parameter called a freezing point to measure when that liquid turns into a solid.



There are physical effects that can change the freezing point. Pressure is one of those effects. When the pressure surrounding a substance goes up, the freezing point also goes up. That means it's easier to freeze the substance at higher pressures. When it gets colder, most solids shrink in size. There are a few which expand but most shrink.

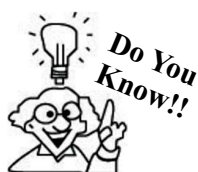


If you have a solid (cube of ice) and want to make it liquid water. You need some energy. Atoms in a liquid state have more energy than the atoms in a solid state. The easiest energy around is probably heat. There is a particular temperature for every solid substance called the melting point. When a solid reaches the temperature of its melting point it melts and get converted in to a liquid. For water the temperature has to be a little over zero degrees Celsius.



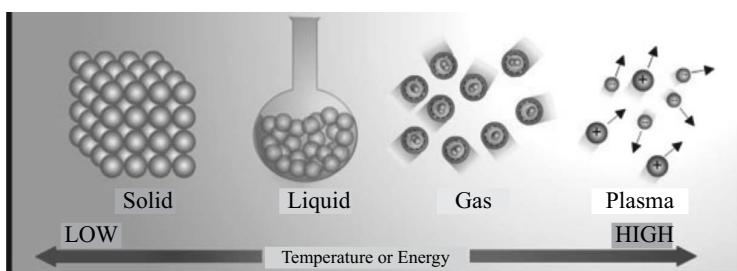
Melting point of a solid also depends upon the surrounding pressure. More will be the pressure more lower will be the melting point. Thus solid melts easily at higher pressure.

The reverse is true if you have a gas. You need to lose some energy from very excited gas atoms. The easy answer is to lower the surrounding temperature. When the temperature drops, energy will be sucked out of gas atoms. When the temperature of the condensation point is achieved it become a liquid. If you have the steam of a boiling pot of water and it hit the wall, the wall would be so cool that steam would quickly become a liquid.



Condensation point of gas also depends upon the surrounding pressure. More will be the pressure more closer will be molecules and thus greater will be the intermolecular forces of attraction. Thus more will be the surrounding pressure more easily gas gets condensed in to liquid.

A plasma can be made from a gas if a lot of energy is pushed inside. All of this extra energy makes the neutral atoms break apart into positively and negatively charged ions and free electrons. They wind up in a big gaseous ball.



The various state of matter can be interchanged into one another by altering the conditions of (1) Temperature (2) Pressure.

Matter in Our Surroundings

1. Altering the Temperature of Matter

(a) Interconversion of solid into liquid & vice versa:

The solids can be converted into liquids by heating them. Similarly, liquids can be cooled to form solids.

For e.g. : Ice at 0°C changes into water at 0°C , when heat energy is supplied to it.

The water at 0°C changes into ice at 0°C on freezing.

(i) Melting or fusion :

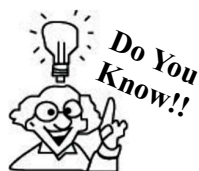
The process due to which a solid changes into liquid state by absorbing heat energy is called melting or fusion. The constant temperature at which a solid changes into liquid state at atmospheric pressure by absorbing heat energy is called melting point.

(ii) Freezing or solidification :

The process due to which a liquid changes into solid state by giving out heat energy is called freezing or solidification. The constant temperature at which liquid changes into a solid state by giving out heat energy is called freezing point.

Explanation : On increasing the temperature of solids, the kinetic energy (K.E.) of particles increases. Due to increase in K.E., the particles start vibrating with greater speed. The energy supplied by heat overcomes the force of attraction between the particles. Then particles leave their fixed positions & start moving freely and thus solid melts.

Latent Heat of Fusion : The amount of heat energy that is required to change 1 kg of solid into liquid at atmospheric pressure at its melting point is known as the latent heat of fusion. Latent heat of fusion of ice = $3.34 \times 10^5 \text{ J/kg}$.



The numerical value of freezing point and melting point is same

• Melting point of ice is 0°C (273.16K)

• Freezing point of water is 0°C

$$0^{\circ}\text{C} = 273.16 \text{ K}$$

Particles of water at 0°C (273 K) have more energy as compared to particles in ice at the same temperature.

CHECK Point

☛ Why during melting of solid temperature remains constant?

SOLUTION

When melting point of solid is reached. Energy supplied further in the form of heat is utilised in phase transition from solid to liquid till solid gets converted into liquid.

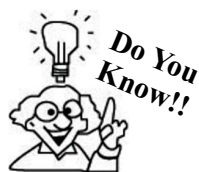
(b) Interconversion of liquid into gaseous state & vice versa:

Liquids can be converted into gases by heating them. Similarly, gases can be converted into liquids by cooling them.

(i) Boiling or vaporisation:

The process due to which a liquid changes into gaseous state by absorbing heat energy is called boiling.

Boiling point: The constant temperature at which a liquid rapidly changes into gaseous state by absorbing heat energy at atmospheric pressure is called boiling point.



For water, the vapour pressure reaches the standard sea level atmospheric pressure of 760 mmHg at 100°C . Since the vapour pressure increases with temperature, it follows that for pressure greater than 760 mmHg (e.g., in a pressure cooker), the boiling point is above 100°C and for pressure less than 760 mmHg (e.g., at altitudes above sea level), the boiling point will be lower than 100°C . As long as a vessel of water is boiling at 760 mmHg, it will remain at 100°C until the phase change is complete. Rapidly boiling water is not at a higher temperature than slowly boiling water. The stability of the boiling point makes it a convenient calibration temperature for temperature scales.

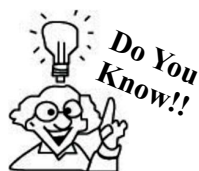
(ii) Condensation or liquefaction

The process due to which a gas changes into liquid state by giving out heat energy is called condensation.

Condensation Point: The constant temperature at which a gas changes into liquid state by giving out heat energy at atmospheric pressure.

Explanation : When heat is supplied to water, particles start moving faster. At a certain temperature, a point is reached when the particles have enough energy to break the forces of attraction between the particles. At this temperature the liquid starts changing into gas. Boiling is a bulk phenomena. Condensation is the opposite of evaporation. It takes place when water vapour in the air condenses from a gas, back into a liquid form, and leaves the atmosphere, returning to the surface of the Earth. You must have seen water droplets on glass containing ice cubes, it is due to condensation.

Latent heat of vaporisation : The amount of heat which is required to convert 1 kg of the liquid (at its boiling point) to vapour or gas without any change in temperature. Latent heat of vaporisation of water = 22.5×10^5 J/kg.



The numerical value of condensation point and boiling point is same.

Water changes into steam at 100°C

(273 + 100) = 373 K

Steam changes into water at 100°C

Particles in steam, that is water vapour at 373 K, have more energy than water at the same temperature.

CHECK Point

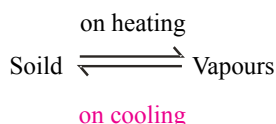
- ☛ Why steam cause more severe burns than boiling water (100°C)?

SOLUTION

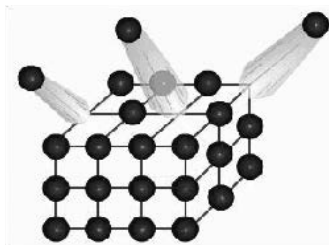
As steam have absorbed extra energy in the form of latent heat of vaporisation. Steam at 100°C will be more dangerous than water at 100°C.

(c) Direct Interconversion of solid into gaseous state & vice versa : Sublimation

Certain substances when heated, pass directly from the solid state to vapour state without being converted into liquid. The vapours when cooled give back the solid substance. This phenomenon is known as sublimation and the substance is called sublimate.



The examples of substances which can easily sublime are camphor, iodine, naphthalene, ammonium chloride, etc. The force of attraction is not uniform in the molecules of all the solids. In some solids, the molecules are bonded together with stronger inter-molecular forces and in the other, the molecules are bonded together with comparatively weak inter-molecular forces.



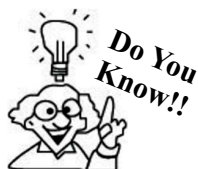
Sublimation depicted at submicroscopic level

The solids, having weak inter-molecular forces, when heated are directly converted into vapours (gaseous state) without being converted into liquids. Small amount of energy is sufficient to make the inter-molecular force of attraction negligible. This increases the inter molecular distance to a very great extent. Therefore, the solid is directly converted into vapours.



idea box

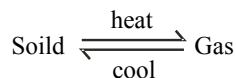
Take appropriate quantity of ammonium chloride in a china dish. Cover the dish with a perforated filter paper. Now place an inverted funnel over the dish. Plug the upper open end of the funnel with cotton. Now keep the entire apparatus as such for 3 days and after that check inverted funnel and record your observation. Now repeat the same after regular interval of 3 days for 15 days. Now repeat this experiment with camphor, iodine and naphthalene also. Record your observations and find out the role of perforated filter paper in this experiment. Also draw a conclusion on the basis of your experiment.



The process of sublimation is very helpful in separating a mixture of solids and also in the purification of the substances.

CHECK Point

- ☛ Which phenomena is represented by below equation?



SOLUTION

Sublimation

2. By Altering Pressure

The difference in various states of matter is due to the difference in intermolecular spaces between their particles. So when a gas is compressed the intermolecular space between its particles get decreases & ultimately it will convert into liquid. So high pressure & low temperature can liquify gases.

For eg : Carbon dioxide (CO_2) is gas under normal conditions of temperature and pressure. It can be liquefied by compressing it to a pressure 70 times more than atmospheric pressure.

EVAPORATION

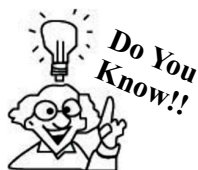
Perhaps one of the most accepted facts of life is that wet things eventually become dry. The liquid water has changed to the gaseous state in a process known as *vaporization*. When this process occurs below the boiling point, it is known as *evaporation*. In order for a molecule of a liquid to escape to the vapour state, however, it must overcome the intermolecular forces attracting it by its neighbours in the liquid. Two conditions allow a molecule in a liquid to escape the liquid state to the gaseous state. First, it must be at or near the surface of the liquid. Second, it must have at least the minimum amount of kinetic energy to overcome the intermolecular forces.

CHECK Point

- ☛ Why do we see water droplets on the outer surface of a glass containing Ice-cold water ?

SOLUTION

The water vapour present in air on coming in contact with cold glass of water, loses energy. So water vapour gets converted to liquid state, which we see as water droplets



Evaporation is important to life on Earth. The heat of the sun evaporates water from earth's surface. The evaporated water goes high into the air. Then it cools down, forms clouds, and falls from the sky as rain or snow.

Evaporation is important for people too. When we sweat, water on our skin evaporates. The evaporation makes the skin feel cooler.

Why Evaporation Occurs :

Molecules in the liquid state are constantly moving, but at different speeds. Faster moving molecules at the surface of a liquid break away from the attraction of the other molecules, and escape into the air. Furthermore, heating makes liquids evaporate faster because there are more fast moving molecules and, therefore, more molecules can escape.

CHECK Point

☛ Why should we wear cotton clothes in summer ?

SOLUTION

During summer, we perspire more because of the mechanism of our body which keeps us cool. During evaporation, the particles at the surface of liquid gain energy from the surroundings or body surface. The heat energy equal to latent heat of vaporisation is absorbed from the body leaving the body cool. Cotton, being a good absorber of water helps in absorbing the sweat.

Factors Affecting Evaporation :

- (i) Temperature (ii) Surface area (iii) Humidity (iv) Wind speed (v) Pressure

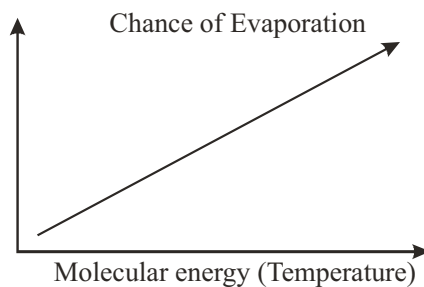
(i) Temperature:

With the increase in temperature the rate of evaporation increases.

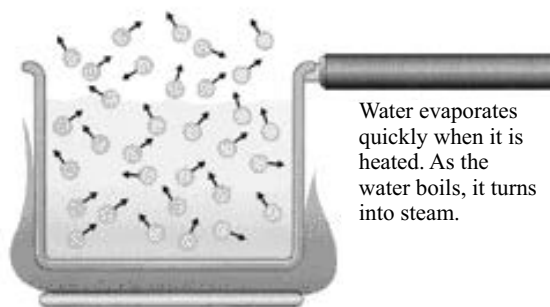
$$\text{Rate of evaporation} \propto T.$$

Reason:

On increasing temperature more number of particles get enough K.E. to go, into the vapour state.



An increase in heat makes things evaporate faster. For example, if a person put the pan of water on the stove and then turned on the heat, the water would soon begin to boil. The person might see steam rise off on top of the water. Steam is water turning into vapour.



Water evaporates quickly when it is heated. As the water boils, it turns into steam.

Matter in Our Surroundings

(ii) Surface area :

Rate of evaporation $\propto$ surface area

Since evaporation is a surface phenomena. If the surface area is increased, the rate of evaporation increases. So, while putting clothes for drying up we spread them out.

CHECK Point

- ☛ Two vessels one is of cylindrical and another is of cuboidal shape have same volumes. Both vessels are filled with equal amount of water. In which case rate of evaporation will be more.

SOLUTION

Evaporation will be more in case of cuboid vessel as it has greater surface area.

(iii) Humidity of air :

Rate of evaporation $\propto \frac{1}{\text{Humidity}}$

Humidity is the amount of water-vapour present in air.

When humidity of air is low, the rate of evaporation is high and water evaporates more readily.

When humidity of air is high, the rate of evaporation is low and water evaporate very slowly.

(iv) Wind speed :

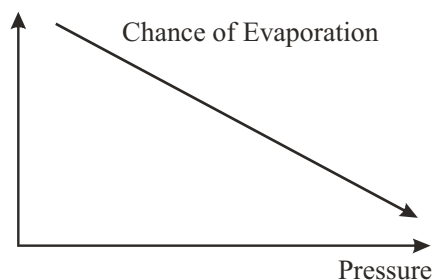
Rate of evaporation $\propto$ wind speed.

With the increase in wind speed, the particles of water-vapour move away with the wind.

So the amount of water-vapour decreases in the surrounding.

(v) Pressure :

The rate of evaporation can also increase with a decrease in the gas pressure around a liquid. Molecules like to move from areas of higher pressure to lower pressure. The molecules are basically sucked into the surrounding area to even out the pressure. Once the vapour pressure of the area increases to a specific level, the rate of evaporation will slow down.

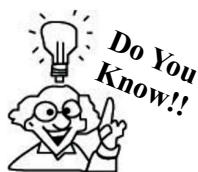


Some Common Student's Misconceptions About Evaporation :

- (a) When liquid water evaporates, it changes from liquid to air.
- (b) When liquid water evaporates, water molecules change into air molecules.
- (c) When liquid water evaporates, it disappears and is gone.
- (d) When liquid water evaporates, it goes to clouds or forms clouds (but not necessarily stays in the air).

Cooling Caused by Evaporation

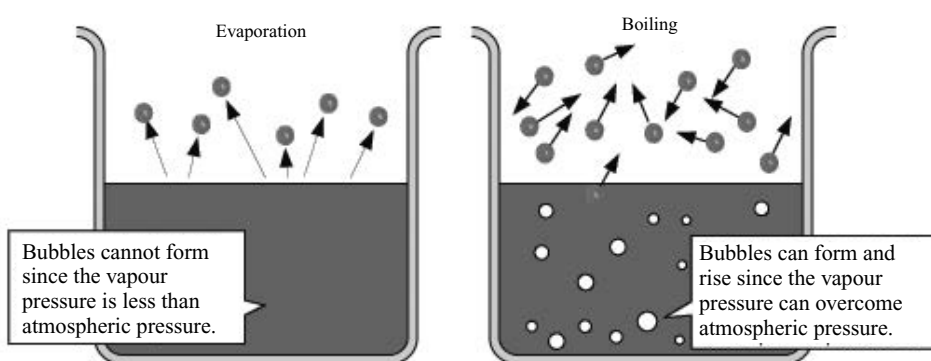
The process of evaporation is always accompanied by a cooling effect. For example, when a liquid evaporates from the skin, a cooling sensation results. The reason for this is that only the most energetic molecules of liquid are lost by evaporation, so that the average energy of the remaining molecules decreases. The surface temperature, which is a measure of this average energy also decreases. Many refrigeration processes are based on this principle.



The cooling effect of evaporating water is important to health maintenance in warm climates. Perspiration covers our bodies with a layer of water when it is warm. The evaporation of this water cools our bodies and us along with it. The cool feeling after a hot shower is not just a feeling but a reality. Evaporation cools the liquid but heats the air. If water is allowed to evaporate under a vacuum, the evaporation process occurs faster. Infact, the water cools enough to freeze.

Difference between Boiling and Evaporation :

- (1) Boiling takes place only at a particular temperature for a liquid. Whereas evaporation occurs at all temperatures.
- (2) Boiling is Bulk phenomena i.e., the bubble formation occurs even below the surface. Whereas evaporation is surface phenomena, i.e., bubble formation occurs only on the surface of liquid.



idea box

Take two plastic containers and fill them with equal amount of water. Now tightly cap one of the containers and wrap the cap's diameter with plastic tape. Such that the container becomes air tight. Keep the other container as such open. Now place both the containers together at side of your room. After 2–3 days check level of water in both containers. Record your observations and on the basis of that draw a conclusion.

CONNECTING TOPIC

GAS LAWS

By changing the pressure or temperature of any gas its volume also changes. The nature of the change is almost same for all gases. Therefore many scientists studied them in detail and gave some laws which are known as Gas Laws.

Boyle's Law :

Boyle's Law is useful when we compare two conditions of the same gas with no change in temperature. (Remember, "Always Boyle's at the same temperature!")

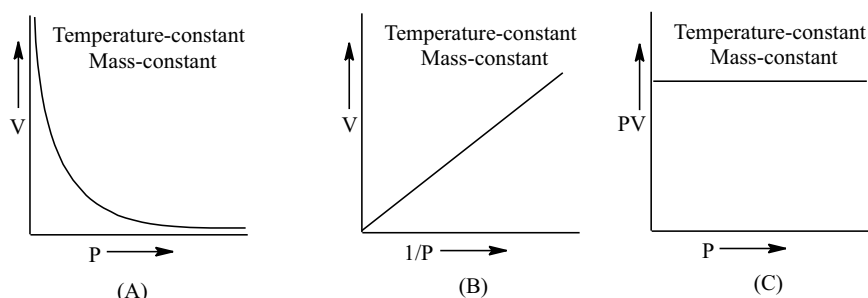
Boyle's law states that, "at constant temperature, the volume of a sample of a gas varies inversely with the pressure".

$$\therefore P \propto \frac{1}{V} \text{ (when temperature and number of moles are kept constant)}$$

The proportionality can be changed into an equality by introducing a constant k, i.e. ,

$$P = \frac{k}{V} \text{ or } PV = k$$

Boyle's law can be verified by any one of the following three ways graphically.



Alternatively, Boyle's law can also be stated as follows :

“Temperature remaining constant, the product of pressure and volume of a given mass of a gas is constant”.

The value of the constant depends upon the amount of a gas and the temperature.

Mathematically, it can be written as, $P_1 V_1 = P_2 V_2 = P_3 V_3 = \dots\dots\dots$

It may be noted that Boyle's law is obeyed only by a hypothetical gas called ideal gas. However, gases like nitrogen, oxygen, carbon dioxide, etc. obey Boyle's law at low pressures and high temperatures and behave as ideal gases. At higher pressures and low temperatures, these gases do not obey Boyle's law.



idea box

Take a small baloon it should be so small that you can push all sides of it together between your hands without any of the baloon pouching out at any point. Now increase the pressure on air inside the baloon by squeezing one end of baloon. Record your observation and draw a conclusion on the basis of that.

ILLUSTRATION : 3

A given mass of a gas occupies 240 ml at 800 mm of Hg. What volume will the gas occupy if the pressure is increased to 1200 mm of Hg, temperature remaining constant?

SOLUTION :

$$P_1 = 800 \text{ mm } P_2 = 1200 \text{ mm, } V_1 = 240 \text{ ml, } V_2 = ?$$

$$\text{Applying Boyle's Law } P_1 V_1 = P_2 V_2$$

Substituting the values

$$800 \times 240 = 1200 \times V_2$$

$$V_2 = \frac{800 \times 240}{1200}$$

Hence the new volume = 160 ml.

ILLUSTRATION : 4

A sample of gas occupies 10 litre under a pressure of 1 atmosphere. What will be its volume if the pressure is increased to 2 atmospheres? Assume that the temperature of the gas sample does not change.

SOLUTION :

Mass and temperature is constant hence we can use Boyle's law

$$P_1 V_1 = P_2 V_2$$

$$\text{Given } P_1 = 1 \text{ atm, } V_1 = 10 \text{ litre, } P_2 = 2 \text{ atm, } V_2 = ?$$

$$1 \times 10 = 2 \times V_2 \Rightarrow V_2 = 5 \text{ litre}$$

Charles' Law

It relates the volume and temperature of a given mass of a gas at constant pressure.

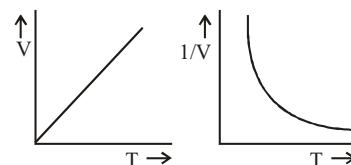
Charles' law can be stated as follows :

"The volume of a given amount of a gas at constant pressure varies directly as its absolute temperature".

$$V \propto T \text{ (if pressure is kept constant)}$$

$$\text{or } \frac{V_t}{T} = \frac{V_0}{T_0}$$

The graphs between V and T are shown.



idea box

Take a baloon such that at point of time between the begining of filling of a ballon and the maximum streching of a baloon, the change in internal pressure of a baloon is negligible as the baloon increases in size. Now fill that baloon partially with air and first place it in a car on a hot day in summer. Now take that ballon out of car and put in to a refrigerator for some period of time. Record your observations and draw a conclusion on the basis of that.

Gay-Lussac Law

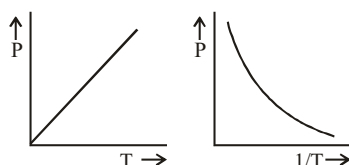
It relates the pressure and absolute temperature of a given mass of a gas at constant volume.

$$P \propto T \text{ (if volume and number of moles are kept constant)}$$

At constant volume, the pressure of a given amount of a gas is directly proportional to its absolute temperature.

$$\text{At constant volume } \frac{P}{T} = \text{constant or } \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

The graphs between P and T are shown :



Avogadro's Law

(i) For solid, liquid and gas :

1 mole of any substance contains Avogadro's number (N_A) of Particles (molecules/atoms/ions) etc.

$$N_A = 6.023 \times 10^{23}$$

(ii) For gases:

In 1812, Amadeo Avogadro stated that samples of different gases which contain the same number of molecules (any complexity, size, shape) occupy the same volume at the same temperature and pressure. It follows from Avogadro's hypothesis that $V \propto n$ (T and P are constant).

$$V \propto n \text{ (} T, P \text{ constant)} \Rightarrow \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Graham's Law of Diffusion

Diffusion is the tendency of any substance to spread throughout the space available to it. Diffusion will take place in all directions and even against gravity. The streaming of gas molecules through a small hole is called effusion.

According to Graham, the rate of diffusion (or effusion) of a gas at constant pressure and temperature is inversely proportional to the square root of its molecular mass.

$$r \propto \sqrt{\frac{1}{M}}, \text{ at constant P and T} \quad \therefore \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}, \text{ at constant P and T}$$

Since, molecular mass of gas = $2 \times$ vapour density, $\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}}$, at constant P and T

ILLUSTRATION : 5

N_2O and CO_2 have the same rate of diffusion under same conditions of temperature and pressure. Why?

SOLUTION :

Both have same molar mass ($=44 \text{ g mol}^{-1}$). According to Graham's law of diffusion, rates of diffusion of different gases are inversely proportional to the square root of their molar masses under same conditions of temperature and pressure.

Dalton's Law of Partial Pressures

The total pressures of a mixture of gases, which do not react with each other, filled in a vessel is equal to the sum of partial pressures of the mixed gases.

$$P = P_1 + P_2 + \dots + P_n$$

THE COMBINED GAS LAWS

Step I. Let us change the pressure of the gas from P_1 to P_2 at a constant temperature T_1 . The volume will change from V_1 to new volume say V_x . Since temperature remains constant, therefore, by applying Boyle's law,

$$P_1 V_1 = P_2 V_x \quad \dots(i)$$

or
$$V_x = \frac{P_1 V_1}{P_2}$$

Step II. Let the temperature of the gas be now changed to T_2 at constant pressure P_2 so that volume may change from V_x to V_2 . Thus, according to Charles' law,

$$\frac{V_x}{T_1} = \frac{V_2}{T_2} \quad \text{or} \quad V_2 = \frac{V_x T_2}{T_1} \quad (ii)$$

Substituting the value of V_x from Eq. (i), we get

$$V_2 = \frac{P_1 V_1}{P_2} \times \frac{T_2}{T_1}$$

Rearranging, we get

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Thus,
$$\frac{PV}{T} = \text{Constant (K)}$$

ILLUSTRATION : 6

At 25° C and 760 mm of Hg pressure, a gas occupies 600 mL volume. What will be its pressure at a height where temperature is 10°C and volume of the gas is 640 mL.

SOLUTION :

$$P_1 = 760 \text{ mm Hg}$$

$$P_2 = ?$$

$$V_1 = 600 \text{ mL}$$

$$V_2 = 640 \text{ mL}$$

$$T_1 = 273 + 25 = 298 \text{ K} \quad T_2 = 273 + 10 = 283 \text{ K}$$

According to combined gas law equation,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\text{or} \quad P_2 = \frac{P_1 V_1 T_2}{V_2 T_1} = \frac{(760 \text{ mm Hg}) \times (600 \text{ mL}) \times (283 \text{ K})}{(640 \text{ mL})(298 \text{ K})} = 676.6 \text{ mm Hg}$$

IDEAL GAS EQUATION

According to Boyle's laws, for a fixed mass of a gas at constant temperature

$$V \propto \frac{1}{P} \quad (\text{Constant } T \text{ and } n) \quad \dots(i)$$

According to Charles' law

$$V \propto T \quad (\text{Constant } P \text{ and } n) \quad \dots(ii)$$

According to Avogadro's law

$$V \propto n \quad (\text{Constant } P \text{ and } T) \quad \dots(iii)$$

Combining equations (i), (ii) and (iii), the combined gas law can be written as:

$$V \propto \frac{nT}{P}$$

$$\text{or} \quad PV \propto nT$$

$$\text{or} \quad PV = nRT$$

where R is a constant of proportionality and is known as molar gas constant. The value of R is same for all gases. Therefore, it is also called universal gas constant. The above equation is known as ideal gas equation.

Numerical Value of R

$$(1) \quad 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

$$(2) \quad 82.1 \text{ atm cm}^3 \text{ K}^{-1} \text{ mol}^{-1}$$

$$(3) \quad 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

ILLUSTRATION : 7

5g of ethane is confined in a bulb of one litre capacity. The bulb is so weak that it will burst if the pressure exceeds 10 atm. At what temperature will the pressure of gas reach the bursting value?

SOLUTION :

$$PV = nRT \Rightarrow 10 \times 1 = \frac{5}{30} \times 0.082 \times T$$

$$T = \frac{60}{0.082} = 731.70 \text{ K} = 458.70^\circ\text{C}$$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- Matter is made up of small
- The forces of attraction between the particles are in solids, in liquids and in gases.
- is the change of gaseous state directly to solid state without going through liquid state, and vice-versa.
- Evaporation causes
- Latent heat of fusion is the amount of heat energy required to change 1 kg of solid into liquid at its
- Solid, liquid and gas are called the three of matter.
- The smell of perfume gradually spreads across a room due to
- Rapid evaporation depends on the area exposed to atmosphere.
- As the temperature of a system increases, the pressure of the gases.....
- As the volume of a specific amount of gas decreases, its pressure.....
- As the temperature of a gas decreases, its volume.....
- Gas molecules at higher temperatures have more than at cooler temperatures.
- Usually the total charge of a plasma is
- The pressure inside a sealed tube, if you raise the temperature go.....
- Forces of attraction in liquids are ___ than in solids.
- Liquids that move quickly downhill are described as having.....

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- Boiling is a bulk phenomenon.
- Evaporation is a surface phenomenon.
- The rate of evaporation depends only on the surface area exposed to the atmosphere.
- Latent heat of vaporisation is the heat energy required to change 1 kg of a liquid to gas at atmospheric pressure at its melting point.
- Water at room temperature is a liquid.
- Atoms in a liquid are farther apart than the atoms in a gas.
- The molecules in a gas are in constant motion.
- Gases present in air have the same pressure throughout the entire atmosphere.

- All materials move from solid to liquid to gas as the temperature increases.
- Because electrons have been stripped away from atoms in plasma, plasmas have a negative charge.
- It is just as easy to compress a liquid, as it is to compress a gas.
- Evaporation and boiling are the same processes because molecules move from a liquid to gaseous state.
- If we pour liquid nitrogen (N_2) into a glass, it will change its state to a solid.
- You may find plasma in a star.
- A system that changes from a solid state to a liquid state gains energy.
- Plasmas are all made of the same ions. They have different colours due to different amounts of electricity.

Match the Following :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in Column I have to be matched with statements (p, q, r, s) in Column II.

- Column II gives how rate of evaporation changes for factors given in column I match them correctly.

Column I

- | | |
|------------------------------|------------------------------|
| (A) Increase in surface area | (p) Increases |
| (B) Decrease in temperature | (q) Decreases |
| (C) Decrease in humidity | (r) Unchange |
| (D) Increase in wind speed | (s) May increase or decrease |

Column II

- Column II give properties for matter mentioned in column I match them correctly.

Column I

- | | |
|------------------------------|----------------------|
| (A) Liquid | (p) Definite shape |
| (B) Solid | (q) Definite volume |
| (C) Plasma | (r) Super low energy |
| (D) Bose-Einstein condensate | (s) Super energetic |

Column II

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

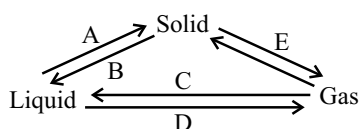
- Name the three states of matter. Give one example of each.
- What are the two ways in which the physical state of matter can be changed ?
- Explain how gases can be liquified ?
- What is sublimation ? Give examples.
- Define latent heat of fusion.
- Define latent heat of vaporization.

7. What produces more severe burns, boiling water or steam?
8. How can the boiling point of a liquid be raised, without adding any impurity?
9. In how many forms did the earlier scientists classify matter?
10. Why does a summer rainstorm lower the temperature?
11. A beaker of a liquid with a vapour pressure of 350 torr at 25°C is set alongside a beaker of water (Vapour pressure of 23.76 torr) and both are allowed to evaporate. In which liquid does the temperature change at a faster rate? Why?
12. At a given temperature, one liquid has a vapour pressure of 240 torr and another measures 420 torr. Which liquid probably has the lower boiling point? Which probably has the lower heat of vaporization?
13. A drop of dettol got evenly distributed in water. How?
14. Liquid nitrogen is used as a commercial refrigerant to flash freeze foods. Nitrogen boils at -196°C. What is this temperature on the Kelvin temperature scale?
15. What property or properties of gases can you point to support the assumption that most of the volume in a gas is empty space?
5. Explain the diffusion of copper sulphate into water.
6. Which state of matter is compressible? Why?
7. Why do the gases exert more pressure on the walls of the container than the solids?
8. The process in which a solid is converted directly into a gas is called sublimation. Iodine is an element that sublimates. A sample of solid iodine in a stoppered flask was allowed to stand undisturbed for several days. Crystals of solid iodine grew on the sides of the flask. Explain at the molecular level what happened?
9. Describe why a drop of food coloring in a glass of water slowly becomes evenly distributed without the need for stirring?
10. Liquid mix more slowly than gases. Why?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. What is condensation? How is the condensation of a gas carried out?
2. Why do solids not diffuse?
3. The following diagram shows the three states of matter and how they can be interchanged. Name the changes A to E.



4. Convert the following Kelvin temperatures to degrees Celsius.

(i) 175 K	(ii) 295 K
(iii) 300 K	(iv) 225 K

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Explain why?
 - (i) A gas fill a vessel completely.
 - (ii) Camphor disappears without leaving any residue.
 - (iii) The temperature does not rise during the process of melting and boiling, though heat energy is constantly supplied.
 - (iv) Water stored in an earthen vessel becomes cool.
 - (v) An iron almirah is a solid at room temperature.
2. Which phenomenon occurs during the following changes?
 - (i) Size of naphthalene balls decrease.
 - (ii) Wax melts in the sun.
 - (iii) Drying of wet clothes.
 - (iv) Formation of clouds.
 - (v) Density of liquids is more than gases.
3. Define :

(i) Melting point	(ii) Boiling point
(iii) Vapourisation	(iv) Freezing
(v) Brownian motion	

2 EXERCISE

Text - Book Questions :

- Which of the following are matter ?
Chair, air, love, smell, hate, almonds, thought, cold-drink, smell of perfume.
- Give reason for the following observations.
The smell of hot sizzling food reaches you several metres away, but to get smell from cold food you have to go close.
- A diver is able to cut through water in a swimming pool. Which property of matter does this observation show?
- What are the characteristics of particles of matter?
- The mass per unit volume of a substance is called density.

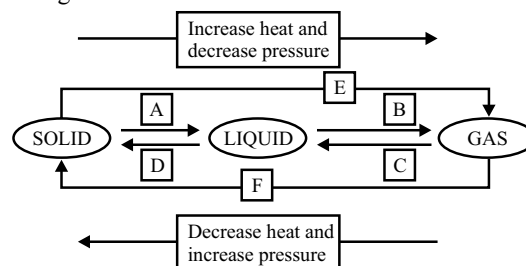
$$\left(\text{Density} = \frac{\text{Mass}}{\text{Volume}} \right)$$

Arrange the following in order of increasing density, air, exhaust from chimneys, honey, water, chalk, cotton, iron.
- Tabulate the differences in the characteristics of states of matter.
 - Comment upon the following :
rigidity, compressibility, fluidity, filling a gas in a container, shape, kinetic energy and density.
- Give reasons:
 - A gas fills completely the vessel in which it is kept.
 - A gas exerts pressure on the walls of the container.
 - A wooden table should be called a solid.
 - We can easily move our hand in air but to do the same through a solid block of wood we need a karate expert.
- Liquids generally have lower density as compared to solids. But you must have observed that ice floats on water. Find out why?
- Convert the following temperature to celcius scale.
 - 300 K
 - 573 K
- What is the physical state of water at
 - 250°C
 - 100°C
- For any substance why does the temperature remain constant during the change of state?
- Suggest a method to liquify atmospheric gases.
- Why does a cooler cool better on a hot dry day?

- How does the water kept in an earthen pot (matka) become cool during summer?
- Why does our palm feel cold when we put some acetone or petrol or perfume on it?
- We can sip hot tea from a saucer faster than from a cup?
- What type of clothes should we wear in summer?

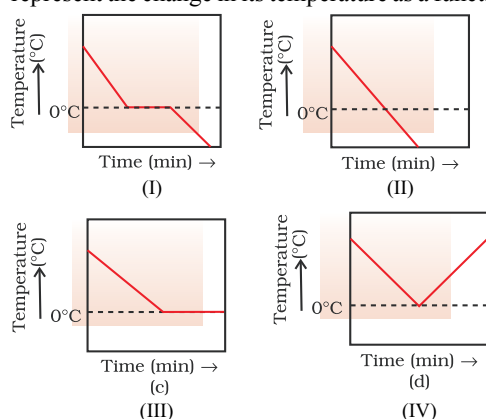
Text - Book Exercise :

- Convert the following temperature to Celcius scale.
 - 293 K
 - 470 K
- Convert the following temperature to Kelvin scale.
 - 25°C
 - 373°C
- Give reasons for the following observations :
 - Naphthalene balls disappear with time without leaving any solid.
 - We can get the smell of perfume sitting several metres away.
- Arrange the following substances in the increasing order of forces of attraction between the particles – water, sugar, oxygen.
- What is the physical state of water at
 - 25°C
 - 0°C
 - 100°C
- Give two reasons to justify :
 - water at room temperature is a liquid.
 - an iron almirah is solid at room temperature.
- Why is ice at 273 K more effective in cooling than water at the same temperature?
- Why does steam produce more severe burns on the skin as compared to boiling water?
- Name A, B, C, D, E and F in the following diagram showing change of state.



Exemplar Questions :

- Alka was making tea in a kettle. Suddenly she felt intense heat from the puff of steam gushing out of the spout of the kettle. She wondered whether the temperature of the steam was higher than that of the water boiling in the kettle. Comment.
- A glass tumbler containing hot water is kept in the freezer compartment of a refrigerator (temperature $< 0^{\circ}\text{C}$). If you could measure the temperature of the content of the tumbler, which of the following graphs (fig.) would correctly represent the change in its temperature as a function of time.



- You want to wear your favourite shirt to a party, but the problem is that it is still wet after a wash. What steps would you take to dry it faster?
- It is a hot summer day, Priyanshi and Ali are wearing cotton and nylon clothes respectively. Who do you think would be more comfortable and why?
- Why does the temperature of a substance remain constant during its melting point or boiling point?

HOTS Questions :

- When a jar of coffee is opened, people in all parts of the room soon notice the smell. Explain why this happens?
- If water is boiling and the flame supplying the heat is turned up, does the water become hotter? What happens?
- Why we close the bottle of nail polish remover immediately after using it?
- Why does a desert cooler cools better on hot dry day?
- Which diffuses faster, the bad smell from a cat-pan due to ammonia or an expensive French perfume with an average molecular weight of 170 g/mol ? How much faster does the faster one diffuse?
- Ethyl chloride boils at 13°C . When it is sprayed on the skin, it freezes a small part of the skin and thus serves as a local anaesthetic. Explain how it cools the skin?
- Why is ice at 273 K more effective in cooling than water at the same temperature?
- A graduated cylinder was filled with water to the 15.0 mL mark and weighed on a balance. Its mass was 27.35 g . An object made of silver was placed in the cylinder and completely submerged in the water. The water level rose to 18.3 mL . When reweighed, the cylinder, water, and silver object had a total mass of 62.00 g . Calculate the density of silver.
- What volume in millilitres does a sample of nitrogen with a mass of 0.245 g occupy at 21°C and 750 torr ?
- The label on a cylinder of a noble gas became unreadable, a student allowed some of the gas to flow into a 300 mL gas bulb until the pressure was 685 torr . The sample now weighed 1.45 g , its temperature was 27.0°C . What is the molecular mass of this gas?
- What change or changes in the state of a gas bring about each of the following effects? (a) The number of impacts per unit time on a given container wall increases. (b) The average energy of impact of molecules with the wall of the container decreases. (c) The average distance between gas molecules increases. (d) The average speed of molecules in the gas mixture is increased.
- 400 mL of a gas at 227°C is to be reduced to a volume of 300 mL . By what degrees Celsius, must the temperature be altered, keeping pressure constant.
- 200 mL of a gas at 27°C and 1140 mm pressure is transferred to a vessel of 450 mL capacity. What temperature is to be applied, if the pressure applied is 1 atmosphere ?
- What are super cooled liquids?
- How water below the surface of earth rises to plants through the roots?
- What is implied by the word “equilibrium” in equilibrium vapour pressure?
- Why do the doctors advise to put strips of wet cloth on the forehead of a person having high temperature?
- When we smell the odour of a rose, our olfactory nerves are sensing molecules of the scent. Explain how smelling a rose demonstrates that molecules are always moving?
- The volume of a given mass of gas, at 150°C is 400 mL . At what temperature, will it occupy a volume of 600 mL at the same pressure?
- A gas occupies 500 mL at 40°C and 800 mm pressure. What volume will it occupy at 353°C and 600 mm pressure?

3

EXERCISE

Multiple Choice Questions :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The quantity of matter present in an object is called its
(a) weight (b) gram
(c) mass (d) density
- At higher altitudes
(a) boiling point of a liquid increases
(b) boiling point of a liquid decreases
(c) no change in boiling point
(d) melting point of solid increases
- The boiling point of alcohol is 78°C . What is this temperature in Kelvin scale
(a) 373 K (b) 351 K
(c) 375 K (d) 78 K
- In which phenomena water changes into water vapour below its B.P. ?
(a) Evaporation (b) Condensation
(c) Boiling (d) No such phenomena exist
- The boiling point of water on celsius and Kelvin scale respectively is :
(a) 373, 273 (b) 0, 273
(c) 273, 373 (d) 100, 373
- The liquid which has the highest rate of evaporation is
(a) petrol (b) nail-polish remover
(c) water (d) alcohol
- When we put some crystals of potassium permanganate in a beaker containing water, we observe that after sometime whole water has turned pink. This is due to
(a) boiling
(b) melting of potassium permanganate crystals
(c) sublimation of crystals
(d) diffusion
- The state of matter which consists of super energetic particles in the form of ionized gases is called
(a) gaseous state
(b) liquid state
(c) Bose-Einstein condensate
(d) plasma state
- The force that binds the particles of matter together is known as
(a) intermolecular space (b) bond
(c) intermolecular force (d) nuclear force
- The change of a liquid into vapour is called
(a) vapourization (b) solidification
(c) sublimation (d) None of these
- Which of the following describes the liquid phase ?
(a) It has a definite shape and a definite volume
(b) It has a definite shape but not a definite volume
(c) It has a definite volume but not a definite shape
(d) It has neither a definite shape nor a definite volume
- When a teaspoon of solid sugar is dissolved in a glass of liquid water, what phase or phases are present after mixing
(a) liquid only (b) still solid and liquid
(c) solid only (d) None of these
- Non-reacting gases have a tendency to mix with each other. This phenomenon is known as
(a) chemical reaction (b) diffusion
(c) effusion (d) explosion
- A gas can be compressed to a fraction of its volume. The same volume of a gas can be spread all over a room. The reason for this is that
(a) the volume occupied by molecules of a gas is negligible as compared to the total volume of the gas.
(b) gases consists of molecules which are in a state of random motion
(c) gases consist of molecules having very large inter-molecular space which can be reduced or increased under ordinary conditions
(d) None of these
- What is the term used to describe the phase change of a liquid to a gas?
(a) Boiling (b) Condensation
(c) Melting (d) None of the above
- What term is used to describe the phase change of a solid to a liquid?
(a) Freezing (b) Melting
(c) Boiling (d) None of the above
- What is the term used to describe the phase change as a liquid becomes a solid?
(a) Evaporation (b) Condensation
(c) Freezing (d) None of the above
- Which has the least energetic molecules?
(a) Solids (b) Liquids
(c) Gases (d) Plasmas
- In which phase of matter would you expect alcohol exists at room temperature?
(a) Solid (b) Liquid
(c) Gas (d) Plasma
- Which of these choices will not change the state of matter?
(a) Temperature (b) Crushing a crystal
(c) Pressure (d) Heat
- If you leave water in a glass and some molecules turn into a gas, it is called...
(a) Egasoration (b) Evaporation
(c) Extinction (d) Solidification

22. As of the 1990s, scientists have proved the existence of how many states of matter?
 (a) Two (b) Three
 (c) Four (d) Five
23. Out of the following which is the densest state of matter?
 (a) Solids (b) Liquids
 (c) Gases (d) Plasmas
24. Select the one that is not a matter.
 (a) Feeling of hot (b) Smoke
 (c) Humidity (d) Water
25. Which is incorrect statement?
 (a) Matter is continuous in nature.
 (b) Of the three state of matter, the one that is most compact is solid state.
 (c) In solid state interparticles space (i.e., empty space) is minimum.
 (d) The density of solid is generally more than that of a liquid.
26. Select the one that when used would be considered as best condition for liquification of a gas.
 (a) Increasing the temperature.
 (b) Decreasing the pressure.
 (c) Increasing the pressure and decreasing the temperature.
 (d) Decreasing the pressure and increasing the temperature.
27. Select the correct order of evaporation for water, alcohol, petrol and kerosene oil
 (a) water > alcohol > kerosene oil > petrol
 (b) alcohol > petrol > water > kerosene oil
 (c) petrol > alcohol > water > kerosene oil
 (d) petrol > alcohol > kerosene oil > water
28. Which one is a sublime substance?
 (a) Table salt (b) Sugar
 (c) Iodine (d) Potassium Iodide
29. S.I. unit of temperature is
 (a) Celcius (b) Fahrenheit
 (c) Kelvin (d) None of these
30. 10 K is equal to
 (a) 283°C (b) -263°C
 (c) 263°C (d) -283°C
31. The process of change of liquid state into gaseous state at constant temperature is known as
 (a) boiling (b) melting
 (c) fusion (d) evaporation
32. Dry ice is
 (a) ice having no water of crystallisation
 (b) ice that has been dried
 (c) solid carbon dioxide
 (d) None of these
33. Which one is a surface phenomenon?
 (a) Evaporation (b) Boiling
 (c) Both (a) and (b) (d) None of these
34. Rate of diffusion depends upon
 (a) temperature
 (b) surface area
 (c) both temperature and surface area
 (d) None of these
35. Which of the following processes is known as fusion?
 (a) Change of liquid to solid
 (b) Change of solid to liquid
 (c) Change of liquid to vapour
 (d) Change of gaseous state to solid state
36. The evaporation of a liquid can best be carried out in a
 (a) flask (b) china dish
 (c) test tube (d) beaker
37. The one, in which interparticle forces are strongest, is
 (a) sodium chloride (b) hydrogen
 (c) ether (d) carbon dioxide
38. The melting point temperature of the solid state of a substance is 40°C. The freezing point temperature of the liquid state of the same substance will be
 (a) 35°C (b) 40°C
 (c) 45°C (d) can't predict.
39. Which one will help to accelerate the process of evaporation of a liquid kept in an open china dish?
 (a) Keeping dish in open
 (b) Blowing air into the liquid
 (c) Keeping the dish under a running fan
 (d) All the above
40. When a gas is compressed keeping temperature constant, it results in
 (a) increase in speed of gaseous molecules
 (b) increase in collision among gaseous molecules
 (c) decrease in speed of gaseous molecules
 (d) decrease in collision among gaseous molecules.
41. Pressure of air at sea level is
 (a) one atmosphere (b) 76 cm of Hg
 (c) 760 mm of Hg (d) All are correct.

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

1. Which of the following substance(s) can sublime ?
 (a) Camphor
 (b) Solid carbon dioxide
 (c) Ammonium chloride
 (d) Sodium bicarbonate
2. The correct statement(s) amongst the following is/are
 (a) gases diffuse at different rates
 (b) diffusion also takes place in liquids
 (c) diffusion of liquid and a gas is known as intimate mixing
 (d) some liquids diffuses at rate equal to gases
3. Which of the following statement(s) about evaporation is/are correct?
 (a) It is a bulk phenomena
 (b) It causes cooling
 (c) Results into increase in temperature
 (d) It is a surface phenomena

Matter in Our Surroundings

4. Which of the following properties of liquid increases with increase of temperature?
 - (a) Vapour pressure (b) Viscosity
 - (c) Surface tension (d) Evaporation
5. Which of the following properties is different for solids liquids and gases?
 - (a) Movement of molecules
 - (b) Particle size of the substance
 - (c) Mass of the substance
 - (d) Density
6. Gases can be liquified either by lowering the temperature, applying pressure. This shows that
 - (a) molecules of a gas repel each other
 - (b) there exists a kind of intermolecular attraction between molecules of a gas
 - (c) molecules of a gas are in a state of random motion
 - (d) intermolecular forces between gas molecules increases when distance between molecules decreases.
7. Which of these statement(s) is/are true?
 - (a) Gases have high density.
 - (b) Gases can be compressed more than solids.
 - (c) Gases have very specific shapes.
 - (d) Gases undergoes diffusion fastest
8. Which of these choices is defined as "Standard Pressure?"
 - (a) 14.7 psi (b) 1 atm
 - (c) 760 torr (d) 1 pascal
9. All liquids have different
 - (a) density (b) viscosity
 - (c) solubility (d) rate of evaporation
10. Which of the following statement(s) is/are applicable for amorphous solids?
 - (a) Are more flexible at higher temperatures.
 - (b) Include glasses.
 - (c) Do not have specific melting points.
 - (d) Have sharp melting points
11. Which of the following statement(s) about solids is/are incorrect ?
 - (a) Expand largely when the temperature rises.
 - (b) Change shape easily.
 - (c) Have a low density.
 - (d) Generally solids have high density
12. Which of these choices is / are example(s) of a plasma?
 - (a) Aurora Borealis (b) Fluorescent Light Bulb
 - (c) Neon Sign (d) Incandescent Light Bulb
13. Which of the following statement(s) is/are true for gases?
 - (a) Gases do not have a definite shape and volume
 - (b) Volume of the gas is almost equal to the volume of the container confining the gas
 - (c) Confined gas exerts uniform pressure on the walls of its container in all directions
 - (d) Mass of the gas cannot be determined by weighing a container in which it is enclosed

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

During evaporation, the heat needed is absorbed from the surroundings which is therefore cooled according to the kinetic theory, particles in a liquid are in continuous random motion at different speeds. However, the average kinetic energy of the particles remains constant provided the temperature of the liquid does not change collisions between particles produce some fast-moving particles these particles, especially those near the surface, can escape from the intermolecular force of attraction of their neighbouring particles and jump out of the liquid. This, the liquid loses its more energetic particles while the less energetic ones are left behind the average kinetic energy of the remaining particles is therefore reduced. This results in a fall in temperature or a cooling effect.

1. During evaporation particles of a liquid change into vapours.
 - (a) from the surface (b) from the bulk
 - (c) from the bottom (d) from all over the liquid.
2. Which of the following liquid on evaporation causes maximum cooling?
 - (a) Acetone (b) Water
 - (c) Ethanol (d) Vinegar
3. The process of evaporation causes
 - (a) Cooling (b) Dryness
 - (c) Heating (d) All of the above

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 - (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
1. **Assertion :** It is easier to cook food at hill.
Reason : The boiling point of water increases at hill.
 2. **Assertion :** During evaporation of liquid the temperature of the liquid remains unaffected.
Reason : Kinetic energy of the molecules is directly proportional to absolute temperature.
 3. **Assertion :** At room temperature, the evaporation of a liquid takes place at constant rate.
Reason : During evaporation of a liquid, the temperature of the liquid remains unaffected.

Multiple Matching Questions :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

- | 1. | Column I | | Column II | |
|-----|---------------------|---|-------------|---------|
| | (A) Diffusion | | (p) Solids | |
| | (B) Definite shape | | (q) Liquids | |
| | (C) Definite volume | | (r) Gases | |
| | (D) Compressibility | | (s) Plasma | |
| | A | B | C | D |
| (a) | q, r | p | p, q | r |
| (b) | q, r | r | p | s |
| (c) | p, q | r | s, p | p, q, s |
| (d) | q | r | p, q | s |

Integer Type Questions :

DIRECTIONS : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

- From the given factors, how many affect Evaporation?
Temperature, Pressure, Surface area, Volume of liquid, Density of liquid, Humidity, Wind speed.
- In how many states does the matter exists?

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1 to 18) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Volume of a gas at a particular temperature and on atmospheric pressure is 200 ml. Keeping the temperature constant if pressure is increased to 5 atmosphere, then volume of the gas will be
(a) 100 ml. (b) 40 ml.
(c) 200 ml. (d) 205 ml.
- Boyle's law states that the
(a) pressure of a gas is directly proportional to the temperature at constant volume
(b) pressure of a gas is inversely proportional the volume at constant temperature
(c) volume is directly proportional to the temperature at constant pressure
(d) None of the above
- All gases will occupy zero volume when the temperature is reduced to
(a) 273°C (b) 273°A
(c) -273°C (d) 0°C
- A gas which obeys the gas laws is known as
(a) an ideal gas (b) a heavier gas
(c) a lighter gas (d) a real gas
- Which of the following expression at constant pressure represents Charle's law?
(a) $V \propto \frac{1}{T}$ (b) $V \propto \frac{1}{T^2}$
(c) $V \propto T$ (d) $V \propto d$
- For an ideal gas number of moles per litre in terms of its pressure P, gas constant R and temperature T is
(a) PT/R (b) PRT
(c) P/RT (d) RT/P
- Rate of diffusion of a gas is
(a) directly proportional to its density
(b) directly proportional to its molecular mass
(c) inversely proportional to the square root of its density
(d) inversely proportional to the square root of its molecular mass
- Boyle's law states that the
(a) pressure of a gas is directly proportional to the temperature at constant volume
(b) pressure of a gas is inversely proportional the volume at constant temperature
(c) volume is directly proportional to the temperature at constant pressure
(d) None of the above
- Equal volumes of all gases under similar conditions of temperature and pressure contain equal numbers of molecules. This statement was made by
(a) Gay-Lussac (b) Avogadro
(c) Berzilius (d) John Dalton
- Select one correct statement. In the gas equation, $PV = nRT$
(a) n is the number of molecules of a gas
(b) V denotes volume of one mole of the gas
(c) n moles of the gas have a volume V
(d) P is the pressure of the gas when only one mole of gas is present.
- Correct gas equation is :
(a) $\frac{V_1 T_2}{P_1} = \frac{V_2 T_1}{P_2}$ (b) $\frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$
(c) $\frac{P_1 T_2}{V_1} = \frac{P_2 V_2}{T_2}$ (d) $\frac{V_1 V_2}{T_1 T_2} = P_1 P_2$
- The total pressure exerted by a number of non-reacting gases is equal to the sum of the partial pressures of the gases under the same conditions is known as
(a) Boyle's law (b) Charle's law
(c) Avogadro's law (d) Dalton's law
- Kinetic theory of gases proves
(a) only Boyle's law (b) only Charles' law
(c) only Avogadro's law (d) All of these
- Generally , liquid drops assume spherical shape because:
(a) a sphere has maximum surface area
(b) a sphere has minimum surface area
(c) sphere is symmetrical in shape
(d) None of these
- The total pressure of a mixture of two gases is :
(a) The sum of the partial pressures
(b) The difference between the partial pressures
(c) The product of the partial pressures
(d) The ratio of the partial pressures.
- Graph between P and V at constant temperature is
(a) straight
(b) curved increasing
(c) straight line with slope
(d) none of these
- The surface tension of which of the following liquid is maximum?
(a) $\text{C}_2\text{H}_5\text{OH}$ (b) CH_3OH
(c) H_2O (d) C_6H_6

18. If temperature increases, the surface tension of a liquid
- (a) increases (b) decreases
- (c) remains the same (d) increases then decreases

DIRECTIONS (Qs. 19 to 21) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

19. Which of the following is/are value(s) of gas constant R ?
- (a) $0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$
- (b) $8.21 \text{ L torr K}^{-1} \text{ mol}^{-1}$
- (c) $82.1 \text{ atm mL K}^{-1} \text{ mol}^{-1}$
- (d) $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
20. At constant temperature, in a given mass of an ideal gas
- (a) the ratio of volume and temperature always remains constant
- (b) volume always remains constant
- (c) pressure always remains constant
- (d) the product of pressure and volume always remains constant
21. If a gas is expanded at constant temperature :
- (a) the pressure decreases
- (b) the kinetic energy of the molecules remains the same
- (c) the kinetic energy of the molecules decreases
- (d) the number of molecules of the gas increases

DIRECTIONS (Qs. 22 to 24) : Study the given paragraph(s) and answer the following questions.

Chlorine is widely used to purify municipal water supplies and to treat swimming pool waters suppose that the volume of a particular sample of Cl_2 gas is 9.22 L at 1124 torr and 24°C .

22. How many grams of Cl_2 are in the sample?
- (a) 39.7 g (b) 37.9 g
- (c) 40.2 g (d) 39.5 g
23. What volume will the Cl_2 occupy at STP?
- (a) 13.5 L (b) 8.5 L
- (c) 12.5 L (d) 8.43 L
24. At what temperature will the volume be 15.00 L if the pressure is 8.76×10^2 torr?
- (a) 400 K (b) 384.5 K
- (c) 373 K (d) 377 K

DIRECTIONS (Qs. 25 and 26) : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
- (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
25. **Assertion :** Ideal gas equation is valid at low pressure and high temperature.
Reason : Molecular interactions are negligible under this condition.
26. **Assertion :** Plot of P Vs. $1/V$ (volume) is a straight line.
Reason : Pressure is directly proportional to volume.

DIRECTIONS (Q. 27) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

27.	Column I	Column II
(A)	Boyle's law	(p) Temperature
(B)	Charle's law	(q) Pressure
(C)	Gay-Lussac law	(r) Volume
(D)	Ideal gas equation	(s) Number of moles of a gas
	A B C D	
(a)	p,q,r q,r p,q r	
(b)	q,r p,r p,q p,q,r,s	
(c)	p,q r s s	
(d)	q r p,q s	

DIRECTIONS (Qs. 28 and 29) : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

28. A gas was allowed to expand from a volume of 400 ml to 2000 ml at a constant temperature. The initial pressure of the gas was 3 atmospheres. If the final pressure of the gas in terms of x is $x/10$. Find the value of x.
29. A gas occupies 3.25 litres at 0°C . What volume will it occupy at -20°C , pressure remaining constant?

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- particles
- maximum, intermediate, minimum
- Sublimation
- cooling
- melting point
- states
- diffusion
- surface
- increases
- increases
- decreases
- kinetic energy
- zero
- up
- weaker
- low viscosity

True / False :

- | | | |
|-----------|-----------|-----------|
| 1. True | 2. True | 3. False |
| 4. False | 5. True | 6. False |
| 7. True | 8. False | 9. True |
| 10. False | 11. False | 12. False |
| 13. False | 14. True | 15. True |
| 16. False | | |

Match the Following :

- (A) → (p); (B) → (q); (C) → (p); (D) → (p)
- (A) → (q); (B) → (p, q); (C) → (s); (D) → (r);

Very Short Answer Questions :

- Solids, liquids and gases are three states of matter and their examples are ice, water and steam respectively.
- Melting and boiling are two ways in which the physical state of matter can be changed.
- Gases can be liquefied by increasing pressure or by lowering temperature.
- Sublimation is a phenomena of interconversion of solid into gases and vice-versa.



e.g., solid CO_2 gets directly converted into gaseous form.

- Latent heat of fusion : The amount of heat energy that is required to change 1 kg of solid into liquid at atmospheric pressure at its melting point is known as the latent heat of fusion.
- Latent heat of vaporisation : The amount of heat which is required to convert 1 kg of the liquid (at its boiling point) to vapour or gas without any change in temperature.
- Steam produces more severe burn as it is in a gaseous phase thus have absorbed extra energy in the form of latent heat of vaporisation.

- By increasing pressure on liquid.
- They classified the matter in four forms: air, earth, fire and water.
- Water evaporates quickly on a hot day and thus lowers the air temperature.
- The liquid other than water. The higher the vapour pressure, faster the liquid evaporates more quickly it cools and lower will be its temperature.
- The liquid with the higher vapour pressure (420 torr) has the lower boiling point. It also probably has the lower heat of vaporization.
- This is because there is enough space between the particles of water and dettol particles gets into the spaces between the particles of other matter (water).
- $K = ^\circ\text{C} + 273 = -196 + 273 = 77\text{ K}$
- The fact that gases are readily compressible supports the assumption that most of the volume of a gas sample is empty space.

Short Answer Questions :

- The process due to which a gas changes into liquid state by giving out heat energy is called condensation. When a gas is compressed it loses its heat energy there by consequently decreasing intermolecular distance between molecules and increasing intermolecular forces of attraction between molecules. Thus, resulting into liquefaction of gas.
- This is because the forces of attraction between the particles of a solids are very strong and there is very little space for the particles to move around.
- A-freezing, B-melting, C-condensation
D-evaporation, E-sublimation
- (i) -98°C (ii) 22°C
(iii) 27°C (iv) -48°C
- Copper sulphate crystals are blue in colour. When we put some crystals of copper sulphate at the bottom of the beaker containing water, the water slowly turns blue. This is because the particles of copper sulphate mix with the particles of water. This mixing of particles will continue till the whole solution turns blue. This phenomenon is called diffusion.
- The gaseous state is compressible because the distance between the molecules are greater and on applying external pressure, the gas molecule come closer to each other and occupies less space than before. Thus gas gets compressed.
- In gases, the particles move randomly at high speed and they collide with each other and with walls of the container. Due to this collision with walls of the container, the gases exert more pressure than solids.

8. Iodine molecules sublime from the crystals at the bottom of the flask into the gas phase, where their presence imparts a pale violet color to the gas. Some of these molecules then condense on the surface (sides) of the flask, forming crystals.
9. Both the liquid molecules and food coloring molecules are in motion. Through constant motion and collisions the food colouring molecules eventually become dispersed.
10. Since gas molecules are far apart, therefore intermolecular forces of attraction between a particular gas molecules are less in comparison to liquids. Thus gas molecules of one type mix more easily with gas molecules of another type in comparison to liquids.

Long Answer Questions :

1. (i) This is because intermolecular forces of attraction between gas molecules is almost negligible. Thus gases occupy the volume of container in which they filled in.
(ii) This is because camphor is a sublime substance. Thus it gets converted into gaseous form directly from solid state without first being converted into liquid.
(iii) This is because during melting and boiling the heat energy supplied is utilised in a phase transition like solid to liquid in case of melting and liquid to gas in case of boiling. Therefore, temperature remains constant.
(iv) As the earthen pot is made up of mud and having many pores in it. These pores serves as air cavities. The water evaporates through the air cavities present in earthen pot which results in cooling of remaining water. This is somewhat similar to the process of sweating in our body which keeps our body cool during summers.
(v) Iron atoms have very less interatomic spaces between them and are thus firmly packed. That's why iron is rigid solid at room temperature and have high melting and boiling point.
2. (i) Napthalene is a sublime substance it gets converted into gaseous form directly without being first converted into liquid state. Thus the size of napthalene balls decreases automatically due to sublimation.
(ii) Wax is a solid it absorbs heat energy from sun and starts melting.
(iii) Wet clothes absorb heat energy from sun and water molecules evaporates off to atmosphere.
(iv) Water vapours formed from ocean and other water bodies of earth when reaches up to atmosphere, there they got cooled up and release their energy and undergoes condensation to form clouds.
- (v) In comparison to gases, in liquids the molecules are close to each other. Therefore, density of liquids is higher than that of gaseous state.
3. (i) The constant temperature at which a solid changes into liquid state at atmospheric pressure by absorbing heat energy is called melting point.
(ii) The constant temperature at which a liquid rapidly changes into gaseous state by absorbing heat energy at atmospheric pressure is called boiling point.
(iii) The process due to which a liquid changes into gaseous state by absorbing heat energy is called vapourisation.
(iv) The process due to which liquid changes into solid state by giving heat energy is called freezing or solidification.
(v) The zig-zag movement of small solid particles suspended in a liquid or gas is called Brownian motion.

2 EXERCISE

Text - Book Questions :

1. The following are matter.
Chair, air, almonds, cold-drink, smell of perfume.
[Note: Love, hate, smell and thought are not matter as they neither have mass, nor occupy space.]
2. The particles of aroma in hot sizzling food mix with the particles of air around us. Due to this spreading of particles, we are able to get the smell even at a distance (i.e., several metres away). Thus, it is the result of diffusion which increases with increase of temperature. We know that the rate of diffusion increases on heating so the smell of hot sizzling food reaches us several metres away but to get smell from cold food we have to go close.
3. It shows the property of compressibility in case of liquid state (water is a liquid). Due to large interparticle space in liquids (water) diver is able to cut through water.
4. The characteristics of particles of matter are :
(i) They have space between them.
(ii) They are continuously moving.
(iii) They attract each other.
5. In increasing order of density, they can be arranged as follows :
air < exhaust from chimneys < water < honey < cotton < iron.

6. (a) Differences in the characteristics of matter :

	Property	Solids	Liquids	Gases
1.	Shape	They have a definite shape.	They take the shape of container and have no definite shape.	They take the shape of the container and have no definite shape.
2.	Volume	They have a fixed volume.	They have a fixed volume.	The volume is not fixed. They completely occupy the available volume. Their volume is equal to the volume of container.
3.	Density	The density of solids is higher than other states of matter.	The densities of liquids are more as compared to gases.	Their densities are least.
4.	Intermolecular forces	The intermolecular forces of attraction are maximum.	The intermolecular forces of attraction are stronger than those of gaseous particles.	The intermolecular forces of attraction are minimum.
5.	Intermolecular space	The intermolecular space is minimum	The intermolecular space is more than those of solids but less than those of gases.	The intermolecular space is maximum.
6.	Compressibility	Compressibility is least	Compressibility more than that of solids but less than that of gases.	Compressibility is maximum.
7.	K.E.	Kinetic energy possessed by particles is minimum.	Kinetic energy possessed by particles is more than those of solids but less than those of gases.	Kinetic energy is maximum
8.	Diffusion	Show negligible diffusion	In them the process of diffusion is slow.	In them process of diffusion is very fast.

- (b) (i) **Rigidity:** It refers to the property shown by solids. Because of the absence of property of flow in solids they are considered as rigid. It is due to low value of K.E. in solids.
- (ii) **Compressibility:** It is the property which depends upon the availability of interparticle space. Because in solids the intermolecular space is quite less. So they are incompressible, whereas gases show the property of compressibility to maximum extent.
- (iii) **Fluidity:** It is the property to flow. It is shown by liquids and gases which are collectively known as fluids. It is maximum for gases.
- (iv) **Filling of gas in cylinder:** The gases have maximum compressibility and due to this a large volume of gas can be compressed under pressure to a very small volume. The LPG and CNG cylinders contain gases compressed under pressure.
- (v) **Shape:** Solids have a definite shape, liquids and gases do not have a definite shape and they take the shape of the container.
- (vi) **Kinetic energy:** It is the energy possessed by molecules by virtue of their motion. The kinetic energy is maximum in case of particles of gases and least in case of particles of solids.
- (vii) **Density:** It is the ratio of mass to volume. The density of solids is more than the density of liquids and gases. The density of gases is lowest due to large intermolecular space in them.
7. (a) Since, the interparticle force of attraction between gaseous particles is very small. So they neither have a definite shape nor a definite volume. The gas completely fills the vessel in which it is kept and its volume is the same as that of the vessel. It is due to very high rate of diffusion in case of gases.
[Note: The actual volume occupied by gaseous molecules is negligible as compared to the volume of the gas].
- (b) Gaseous particles are in a state of random motion at high speed. Due to random motion particles hit each other and also the walls of the container. The pressure exerted by the gas is because of the force exerted by gas particles per unit area on the walls of the container.
- (c) Since, the wooden table has a definite shape and definite volume. So, it should be classified as a solid. The other properties associated with solids are that they have distinct boundaries, rigidity and incompressibility. These are the properties of wooden table also. So, it is a solid.
- (d) Air is a gas. In gaseous state there is maximum interparticle distance and so they are highly compressible. Due to this property of compressibility we can easily move our hand in air but solids are incompressible and in them the constituent particles are quite close. So we need a large force to move our hands through a block of solid. This can be done only by a Karate expert.

8. Ice (a solid at room temperature) is lighter and floats on water (a liquid at room temperature) because ice has a cage-like structure in which there are large vacant space between various water molecules which are linked to form ice. In water molecule, such spaces are negligible. Thus, we can say that ice floats on water because the structure of ice is more porous than that of water. Due to this the density of water is more than that of ice and so ice floats over water.
9. To change temperature on Kelvin scale (K) to celcius scale ($^{\circ}\text{C}$) you have to subtract 273.16 from the given temperature on Kelvin scale.
 - (a) $300\text{ K} = (300 - 273.16)^{\circ}\text{C} = 26.84^{\circ}\text{C}$ (or 27°C)
 - (b) $573\text{ K} = (573 - 273.16)^{\circ}\text{C} = 299.84^{\circ}\text{C}$ (or 300°C)
 [Note: In ordinary calculations we use 273.0 instead of 273.16].
10.
 - (a) The boiling point of water is 100°C and above this temperature it is in gaseous state. At 250°C , water is present in gaseous state only.
 - (b) At 100°C , there is equilibrium between gaseous state and liquid state i.e., both states liquid and gaseous are present.
11. The temperature remains constant during the change of state because the heat supplied is used up in changing the state by overcoming the intermolecular forces of attraction. This heat energy is known as latent heat.
12. The atmospheric gases can be liquified by applying pressure and reducing temperature.
13. On a hot dry day the amount of water vapour present in air is less. (i.e., humidity is low) and so, the rate of evaporation increases. Due to this a cooler cool better on a hot dry day.
14. The earthen pot has a large number of tiny pores which provide a large surface area for evaporation of water kept in it. Evaporation cause cooling. The rate of evaporation increases with rise of temperature. Since, the temperature is high during summer so rate of evaporation increases and the water kept in an earthen pot become cool.
15. Acetone or petrol or perfume particles gain energy from our palm for evaporation and this gives a feeling of coolness and we feel a cooling sensation.
16. Surface area of saucer is more as compared to that of a cup. Due to increase in surface area, the rate of evaporation increases and thus the temperature of the hot tea in saucer gets lowered and we can sip it faster than from a cup.
17. We should wear light colour and cotton clothes in summer. Light colour because it reflects heat. Cotton clothes because it has pores in it. It absorbs sweat and allows the sweat to evaporate faster thereby gives cooling effect.
- (b) The particles of perfume which is a mixture of number of pleasant smelling vapour mix with the particles of air around us and spread out. Due to this spreading we are able to get the smell even sitting several metres away.
4. We know that the forces of attraction between particles are maximum in case of solids and minimum in case of gases. Thus the correct order is
oxygen < water < sugar
(gas) (liquid) (solid)
5.
 - (a) Freezing point of water is 0°C and its boiling point is 100°C between these two temperatures it is in liquid state i.e. at 25°C water is in liquid state.
 - (b) It represent freezing point of water and both solid and liquid states are in equilibrium.
 - (c) It is boiling point of water and at this temperature both liquid and vapour (gaseous) state are in equilibrium.
6.
 - (a) At room temperature water is liquid because it has no definite shape. It takes the shape of the container. It has a definite volume. Water shows the property of fluidity.
 - (b) An iron almirah has a definite shape and also a definite volume. So, it is a solid at room temperature.
7. When ice (a solid) melts at 273 K (m.p. of ice) it absorbs energy from its surroundings in order to overcome the forces of attraction present in solid particles. Due to this the temperature of surroundings gets lowered and cooling is caused. In case of liquid, water at 0°C there is no such tendency of changing to vapour state and so it has almost negligible tendency to absorb any energy from surroundings. Therefore, it will cause comparatively less cooling.
8. When water is boiled at 100°C (373 K) it absorbs certain amount of heat energy (latent energy) to get converted into steam. It results in more energy in steam than in water at 100°C . Due to this (i.e., more energy) steam produces more severe burns on the skin as compared to boiling water.
9. A = Fusion (or melting)
B = Vapourisation
C = Condensation (or liquidification)
D = Solidification (or freezing)
E = Sublimation
F = Sublimation

Exemplar Questions :

Text - Book Exercise :

1.
 - (a) $293\text{ K} = (293 - 273.16)^{\circ}\text{C}$
 $= 19.84^{\circ}\text{C}$ [or 20°C]
 - (b) $470\text{ K} = (470 - 273.16)^{\circ}\text{C}$
 $= 196.84^{\circ}\text{C}$ [or 197°C]
2.
 - (a) $25^{\circ}\text{C} = (25 + 273.16)\text{ K}$
 $= 298.16\text{ K}$ [or 298.0 K]
 - (b) $373^{\circ}\text{C} = (373 + 273.16)\text{ K}$
 $= 646.16\text{ K}$ [or 646.0 K]
3. (a) Naphthalene is a sublime substance. It changes from solid state to gaseous state without passing through the liquid state. When kept in open the naphthalene balls get sublimed and the vapours go to atmosphere and no solid is left and the balls disappear.
4.
 - (i) an increase of surface area by spreading the shirt
 - (ii) an increase in temperature by putting the shirt under the sun
 - (iii) increase the wind speed by spreading it under the fan
 Cotton being a better absorber of the water than nylon helps in absorption of sweat followed by evaporation which leads to cooling. So Priyanshi is more comfortable, whereas Ali is not so comfortable.

Matter in Our Surroundings

5. The temperature of a substance remains constant at its melting and boiling points until all the substance melts or boils because, the heat supplied is continuously used up in changing the state of the substance by overcoming the forces of attraction between the particles. This heat energy absorbed without showing any rise in temperature is given the name latent heat of fusion/latent heat of vapourisation.

HOTS Questions :

- This is due to diffusion. Some particles with high kinetic energy leave the coffee jar and spread out through the air in the room in a haphazard and random way. That's why we notice the smell of coffee in a room.
- The water does not become hotter, but it will boil faster.
- Nail polish remover contains acetone which has a high rate of evaporation. To stop the evaporation of acetone, we have to close the bottle immediately after using it.
- Cooling by a desert cooler is caused by the evaporation of water. A desert cooler cools better on a hot and dry day because the higher temperature increases the rate of evaporation of water and dryness of air (less humidity in air) also increases the rate of evaporation. Evaporation removes all the particles of high kinetic energy and the particles left behind have low kinetic energy and this causes cooling.
- As the molar mass of ammonia is only 17 g/mol which is less than molar mass of expensive French perfume (170g/mol)
According to Graham's law of diffusion rate of diffusion

$$\propto \sqrt{\frac{1}{\text{Molar mass of gas}}}$$

Molar mass of perfume is 10 times more than NH_3 . Thus bad smell of cat-pan diffuses.
10 times faster than French perfume.
- Ethyl chloride is a highly volatile and inflammable gas. It condenses to liquid at 13°C , and is kept in sealed tubes under pressure. On striking the warm skin, it vaporizes with such a rapidity that it freezes the tissues. This makes it a local anaesthesia of a moments duration, during which a small cut, as of an abscess or infected finger may be made without pain.
- Cooling takes place when we remove heat from the system. At 273 K, the ice will take away the latent heat from the medium and convert itself into water. Thus, in case of ice at 273 K there will be a change in physical state whereas in case of water at 273 K, there will be no change in state. Hence, in case of water, lesser energy will be taken away from the medium.
- Mass of silver = $62.00\text{ g} - 27.35\text{ g} = 34.65\text{ g}$
Volume of silver = $18.3\text{ mL} - 15\text{ mL} = 3.3\text{ mL}$
Density of silver = (mass of silver) / (volume of silver)
= $(34.65\text{ g}) / (3.3\text{ mL}) = 11\text{ g/mL}$

9. We will use the ideal gas law. First we need to know the number of moles of nitrogen :

No of moles of N_2

$$= (0.245\text{ g } \text{N}_2) \left(\frac{1\text{ mol } \text{N}_2}{28.02\text{ g } \text{N}_2} \right) = 8.74 \times 10^{-3}\text{ mol } \text{N}_2$$

$$V = \frac{nRT}{P} = \frac{(8.74 \times 10^{-3}\text{ mol}) (0.0821\text{ L atm/mol K}) (294\text{ K})}{(750\text{ torr}) \left(\frac{1\text{ atm}}{760\text{ torr}} \right)}$$

$$= 0.214\text{ L} = 214\text{ mL}$$

10. Since, $PV = nRT$, then $n = PV/RT$

$$n = \frac{PV}{RT} = \frac{(685\text{ torr}) \left(\frac{1\text{ atm}}{760\text{ torr}} \right) (0.300\text{ L})}{\left(0.0821 \frac{\text{L atm}}{\text{mol K}} \right) (300.2\text{ K})}$$

$$\text{molar mass} = \frac{1.45\text{ g}}{0.0110\text{ mol}} = 132\text{ g mol}^{-1}$$

11. (a) Increase in temperature at constant volume or decrease in volume or increase in pressure (b) decrease in temperature (c) increase in volume, decrease in pressure (d) increase in temperature

12. $V_1 = 400\text{ mL}$, $V_2 = 300\text{ mL}$

$$T_1 = (227^\circ\text{C} + 273) = 500\text{ K } T_2 = ?$$

$$\text{Applying Charles's law, } \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\text{Hence, } T_2 = \frac{T_1 \times V_2}{V_1}$$

$$\text{Substituting the values, } T_2 = \frac{500 \times 300}{400} \text{ or } T_2 = 375\text{ K}$$

$$T_2 \text{ in degree Celsius} = 375 - 273 = 102^\circ\text{C}$$

$$\text{Alteration of temperature} = 227^\circ\text{C} - 102^\circ\text{C} = 125^\circ\text{C}$$

The temperature should be reduced by 125°C .

13. $P_1 = 1140\text{ mm}$, $P_2 = 760\text{ mm}$, $V_1 = 200\text{ mL}$,

$$V_2 = 450\text{ mL}$$

$$T_1 = 27^\circ\text{C} + 273 = 300\text{ K}, T_2 = ?$$

$$\text{Applying gas equation, } \frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$$

$$\text{Hence, } T_2 = \frac{P_2 V_2 T_1}{P_1 V_1} = \frac{760 \times 450 \times 300}{1140 \times 200} = 450\text{ K}$$

$$\text{Temperature to be applied} = 450\text{ K} - 273\text{ K} = 177^\circ\text{K}.$$

14. Amorphous solids in which constituent particles are arranged in haphazard manner. They resemble liquids as these flow very slowly at room temperature and thus termed as super cooled liquids.

15. It is due to capillary action. The rise of liquid in a capillary is due to the inward pull of surface tension acting on the surface which pushes the liquid into the capillary tube.

16. Equilibrium refers to a state where opposing forces are balanced in the case of a liquid in equilibrium with its vapour, it means that a molecule escaping to the vapour is replaced by one condensing to the liquid.
17. We put strips of wet cloth on the forehead of a person having high temperature. This is because when the water evaporates, it takes away the necessary heat from the body of the patient and thus lowers the body temperature.
18. The molecules that give roses their aroma evaporate from the surface of the flower. Once in the gas phase, they collide countless times with other gas molecules, moving slowly away from the rose, when they reach a nose, they are sensed by the olfactory sensors.
19. $V_1 = 400 \text{ ml}$ $V_2 = 600 \text{ ml}$
 $T_1 = 15^\circ\text{C} + 273 = 288 \text{ K}$, $T_2 = ?$
- Applying Charles' law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
- Hence, $T_2 = \frac{T_1 \times V_2}{V_1}$
- Substituting the values
- $$T_2 = \frac{288 \times 600}{400}$$
- $$T_2 = 432 \text{ K}$$
- T_2 in degree Celsius = $432 - 273 = 159^\circ\text{C}$
 The new temperature = 159°C .
20. $P_1 = 800 \text{ mm of Hg}$, $P_2 = 600 \text{ mm of Hg}$
 $V_1 = 500 \text{ ml}$, $V_2 = ?$
 $T_1 = 40^\circ\text{C} + 273 = 313 \text{ K}$, $T_2 = 353^\circ\text{C} + 273 = 626 \text{ K}$
- Applying gas equation, $\frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$
- Hence, $V_2 = \frac{P_1 V_1 T_2}{T_1 P_2}$
- $$V_2 = \frac{800 \times 500 \times 626}{313 \times 600} = 1333.33 \text{ ml}$$
- New volume = 1333.33 ml .
4. (a) Boiling point of water is 100°C whereas evaporation of water into water vapours occurs at room temperature.
5. (d) Boiling point of water = $100^\circ\text{C} = 100 + 273 = 373 \text{ K}$
6. (a) As intermolecular forces are least in case of petrol. Thus, it has highest rate of evaporation.
7. (d) This is due to diffusion of particles of potassium permanganate into intermolecular spaces between particles of water.
8. (d) 9. (c)
10. (a) During vapourization energy is supplied to liquid in the form of heat. Liquid reaches its boiling point and get converted into gas (vapours).
11. (c)
12. (a) Sugar molecules occupy the intermolecular spaces present in water molecules. Thus after dissolution only liquid is left.
13. (c) Effusion is the process in which individual molecules flow through a hole without collisions between molecules.
14. (c) 15. (a) 16. (b) 17. (c)
18. (a) As order of energy for different states of matter is following
 Bose-Einstein < Solids < Liquids < Gases < plasmas condensate.
19. (b) Alcohol, because of presence of intermediate magnitude of inter molecular forces is liquid at room temperature.
20. (b) Crystal is a solid and even on crushing it you will get smaller crystals (solid).
21. (b)
22. (d) After 1990s five states of matter are known solid, liquid, gas, plasma and Bose Einstein condensate.
23. (a) 24. (a) 25. (a) 26. (c)
 27. (d) 28. (c) 29. (c) 30. (b)
 31. (a) 32. (c) 33. (a) 34. (c)
 35. (b) 36. (b) 37. (a) 38. (b)
 39. (d) 40. (d) 41. (d)

More than One Option Correct :

1. (c) The mass of an object is a fundamental property of the object, a numerical measure of inertia, a fundamental measure of the amount of matter in the object.
2. (b) At higher altitudes atmospheric pressure decrease. Therefore at lower temperature vapour pressure becomes equal to atmospheric pressure. Thus boiling point of a liquid decreases.
3. (b) $\text{K} = 273 + ^\circ\text{C}$
 $\text{K} = 237 + 78 = 351 \text{ K}$
4. (a, d) More will be temperature more number of molecules possess enough kinetic energy to escape liquid surface thereby increasing vapour pressure. It also increase rate of evaporation.
5. (a, d) 6. (b, d)

3 EXERCISE

Multiple Choice Questions :

Matter in Our Surroundings

7. (b, d) Gases have very low intermolecular force between their molecules thus they can be compressed and undergoes rapid diffusion.
8. (a, b, c)
9. (a, b, c, d) Different liquids have different density, viscosity, solubility and rate of evaporation.
10. (a, b, c) Amorphous solids like glass and plastic do not have a sharp melting point.
11. (a, b, c)
12. (a, b, c) Plasma is a superenergetic state of matter consisting superexcited atoms.
13. (a, b, c) The mass of gas can be determined by weighing the container, filled with gas and again weighing this container after removing the gas. The difference between the two weights gives the mass of the gas.
4. (a) An ideal gas is one which obeys gas laws under all conditions of temperature, pressure and volume.
5. (c) According to Charle's law $V \propto T$ or $\frac{V}{T} = k$.
6. (c) $PV = nRT$ $\therefore \frac{n}{V} = \frac{P}{RT}$
7. (d) According to Graham's law of diffusion,

$$D \propto \frac{1}{\sqrt{\rho}} \text{ or } \rho = \frac{M}{V} \quad (M = \text{molar mass})$$
8. (b) Boyle's law

$$P \propto \frac{1}{V}$$

$$P = \frac{K}{V}$$

$$PV = K$$

Passage Based Questions :

1. (d) 2. (a) 3. (a)

Assertion & Reason :

1. (c) The higher the altitude, the lower the atmospheric pressure. Lower pressure in turn causes water to evaporate more quickly, and water boils at a lower temperature.
2. (d) During evaporation molecules at surface possess more kinetic energy and escapes to atmosphere. Therefore, resulting liquid undergoes cooling.
3. (c) During evaporation of a liquid, the temperature of the liquid falls.
9. (b)
10. (c) In the equation $PV = nRT$, n moles of the gas have volume V .
11. (b) $\frac{PV}{T} = \text{constant}$ $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\Rightarrow \frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$$
12. (d) The statement of Dalton's law is $P = P_A^\circ + P_B^\circ + \dots P_n^\circ$ Where P is the total pressure $P_A^\circ, P_B^\circ, \dots$ are the partial pressure of the components of gaseous mixture.
13. (d) Kinetic theory of gases proves all the given gas laws.
14. (b) Liquid drops assume spherical shape because a sphere has minimum surface area.
15. (a) By Dalton's law of partial pressures, the total pressure of a mixture of two gases is the sum of the partial pressures.
16. (d) $P \propto \frac{1}{V}$ (at constant T)

$$PV = \text{constant}$$
17. (c) Due to intermolecular H-bonding the surface tension of H_2O is more than other liquid. One H_2O molecule is joined with 4 another H_2O molecule through H-bond. Hydrogen bonding is in order $H_2O > C_2H_5OH > CH_3OH$.

Multiple Matching Questions :

1. (a) A-(q, r); B-(p); C-(p, q); D-(r)

Integer Type Questions :

1. 5
Evaporation does not depends upon volume and density of liquid.
2. 5
Solid, Liquid, Gas, Plasmas and Bose-Einstein condensates.

4 EXERCISE

1. (b) $P_1 = 1 \text{ atm}, V_1 = 200 \text{ ml}$
 $P_2 = 5 \text{ atm } V_2 =$
 According to Boyle's law
 $P_1 V_1 = P_2 V_2$
 $V_2 = \frac{P_1 V_1}{P_2} = \frac{1 \times 200 \text{ ml}}{5} = 40 \text{ ml}$
2. (b) Boyle's law
 $P \propto \frac{1}{V}$
 $P = \frac{K}{V}$
 $PV = K$
3. (d)
4. (a) An ideal gas is one which obeys gas laws under all conditions of temperature, pressure and volume.
5. (c) According to Charle's law $V \propto T$ or $\frac{V}{T} = k$.
6. (c) $PV = nRT$ $\therefore \frac{n}{V} = \frac{P}{RT}$
7. (d) According to Graham's law of diffusion,

$$D \propto \frac{1}{\sqrt{\rho}} \text{ or } \rho = \frac{M}{V} \quad (M = \text{molar mass})$$
8. (b) Boyle's law

$$P \propto \frac{1}{V}$$

$$P = \frac{K}{V}$$

$$PV = K$$
9. (b)
10. (c) In the equation $PV = nRT$, n moles of the gas have volume V .
11. (b) $\frac{PV}{T} = \text{constant}$ $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\Rightarrow \frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$$
12. (d) The statement of Dalton's law is $P = P_A^\circ + P_B^\circ + \dots P_n^\circ$ Where P is the total pressure $P_A^\circ, P_B^\circ, \dots$ are the partial pressure of the components of gaseous mixture.
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18. (b)
19. (a, c, d)
20. (a, d) According to Boyle's law, $V \propto \frac{1}{P}$

$$PV = \text{constant}$$
 According to Charle's law at constant pressure

$$\frac{V}{T} = \text{constant}$$
21. (a, b) At constant temp., when gas expands the K.E. of the molecules remains the same, but the pressure decreases.

$$\left(\because P \propto \frac{1}{V} \right)$$

(Sol. 22-24)

$$V = 9.22 \text{ L} \quad P = 1124 \text{ torr} = \frac{1124}{760} \text{ atm} = 1.47 \text{ atm}$$

$$T = 24^\circ\text{C} = 24 + 273 = 297 \text{ K}$$

From ideal gas equation

$$PV = nRT$$

$$n = \frac{PV}{RT}$$

$$= \frac{1.47 \text{ atm} \times 9.22 \text{ L}}{0.0821 \text{ L atm mol}^{-1} \text{K}^{-1} \times 297 \text{ K}} = 0.56 \text{ mol}$$

$$22. \text{ (a) No. of moles of Cl}_2 = \frac{\text{Given gram of Cl}_2 \text{ in sample}}{\text{Molecular mass of Cl}_2}$$

$$0.56 = \frac{\text{Given grams of Cl}_2 \text{ in sample}}{70.906}$$

$$\text{Given grams of Cl}_2 \text{ in sample} = 39.7 \text{ g}$$

$$23. \text{ (c) At STP } P_2 = 1 \text{ atm}$$

$$T_2 = 273 \text{ K}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1.47 \text{ atm} \times 9.22 \text{ L}}{297 \text{ K}} = \frac{1 \text{ atm} \times V_2}{273 \text{ K}}$$

$$V_2 = \frac{1.47 \text{ atm} \times 9.22 \text{ L} \times 273 \text{ K}}{297 \text{ K}}$$

$$V_2 = 12.458 \text{ L} = 12.5 \text{ L}$$

$$24. \text{ (d) } V_2 = 15 \text{ L} \quad P_2 = 876 \text{ torr}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1124 \text{ torr} \times 9.22 \text{ L}}{297 \text{ K}} = \frac{876 \text{ torr} \times 15 \text{ L}}{T_2}$$

$$T_2 = \frac{876 \text{ torr} \times 15 \text{ L} \times 297 \text{ K}}{1124 \text{ torr} \times 9.22 \text{ L}}$$

$$T_2 = \frac{3902580}{10363.28} \text{ K}$$

$$T_2 = 376.577 \text{ K}$$

$$T_2 = 377 \text{ K}$$

25. (a) With the increase in temperature when pressure is low the distance between molecules increases and intermolecular forces becomes almost negligible. Thus under those conditions a gas behaves like ideal gas.

26. (c) Assertion is true but reason is false.

Pressure is inversely proportional to volume (Boyle's law). $P \propto \frac{1}{V}$ (n, T constant)

27. (b) A-(q, r); B-(p, r); C-(p, q); D-(p, q, r, s)

28. 6

$$P_1 = 3 \text{ atm} \quad P_2 = \frac{x}{10}$$

$$V_1 = 400 \text{ ml} \quad V_2 = 2000 \text{ ml}$$

According to Boyle's law

$$P_1 V_1 = P_2 V_2$$

$$3 \times 400 = \frac{x}{10} \times 2000$$

$$x = \frac{3 \times 400 \times 10}{2000} = 6$$

29. 3

$$V_1 = 3.25 \text{ L} \quad V_2 = ?$$

$$T_1 = 0^\circ\text{C} = 0 + 273 = 273 \text{ K}$$

$$T_2 = -20^\circ\text{C} = -20 + 273 = 253 \text{ K}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \text{ (P is constant)}$$

$$\frac{3.25}{273} = \frac{V_2}{253}$$

$$V_2 = 3 \text{ L}$$

Is Matter Around us Pure

INTRODUCTION

All matter can be broadly divided into two major groups “Pure” and “Impure”. The term ‘purity’ has quite a different meaning in chemistry than in our day-to-day life. Normally when we refer to pure water, pure milk, etc., what is implied is that the water, milk etc., are free from harmful substances such as bacteria, fungi, viruses, etc. ‘Purity’ in chemistry is entirely of a different nature. When we say a substance is pure, it means that the substance is made of only one type of constituent particles.

All matter can be classified into two categories: pure substances and mixtures. A pure substance consists of a single element or compound. Iron is formed only of iron (Fe) atoms; table salt is formed only by sodium chloride (NaCl) molecules. A mixture, however, is made up of different compounds or elements. When salt is added to water to make saltwater, it becomes a mixture. The salt and water molecules do not combine to form new molecules, but only “mix” together while still retaining their identities. Air is also a mixture, containing just the right amounts of nitrogen, oxygen, and other gases for life on Earth.

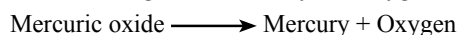
Very little of our surroundings can actually be classified as “pure”. Most of what we see is a mixture of the pure substances known as elements and compounds. Sometimes pure substance mix uniformly with no boundaries between components, but sometimes the pure substances form more obvious mixtures with boundaries. Even when the presence of a mixture is not obvious, however, it has properties that are different than the pure component substance. Physical properties distinguish between a mixture and a pure substance.

According to the chemical classification of matter there are two main categories:

- (i) Pure substances
- (ii) Mixtures

PURE SUBSTANCE

Lavoisier, a French chemist on the basis of quantitative studies classified pure substances into elements and compounds. He showed that when we heat mercuric oxide it changes into mercury and oxygen.



Mercuric oxide is a compound because it decomposes into simpler substances, whereas mercury and oxygen cannot be further decomposed into anything simpler as they are elements.

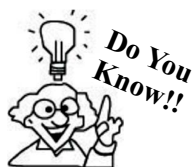
A pure substance has a constant composition. Water, for example, is always 88.81% oxygen by weight. Pure substances also have constant chemical and physical properties. Water freezes at exactly 0°C and boils at exactly 100°C at atmospheric pressure. The composition and properties of a pure substance are the same regardless of where it is found. When pure, the salt found in a salt shaker on the dining room table has the same composition as the salt dug from mines deep beneath the earth or the salt obtained by evaporating seawater. No matter where salt comes from, it always contains 1.54 times as much chlorine by weight as sodium. Since it always has the same composition, salt has the same chemical and physical properties.

CHECK Point

- ☛ Two samples of ammonia are collected one from manufacturing plant and other from urine deposits of animals. Both samples have same composition of nitrogen and hydrogen? if yes then give reason.

SOLUTION

Yes, both samples will have same amount of nitrogen and hydrogen as both samples are pure substance, not mixture.

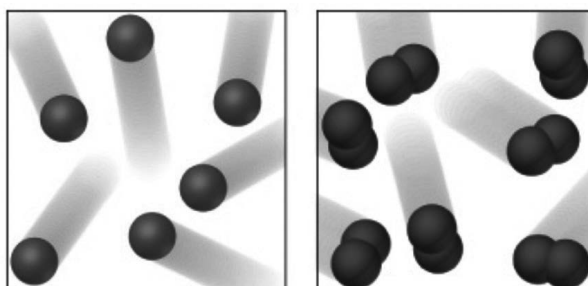


A solution of salt in water or sugar solution being homogeneous appears to consist of one type of particles. But it is made up of more than one kind of particles. Hence, it is not a pure substance. It is a mixture. A material is said to be homogeneous if it has uniform composition and identical properties throughout.

ELEMENT

An element is defined as a pure substance that contains only one kind of particles. These particles may be atoms or molecules. Elements are made up of particles/atoms of only one kind. However, in nuclear reactions, elements having high atomic mass can be split into lighter elements.

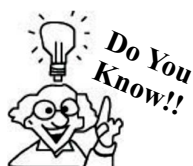
Carbon, sulphur, iron, lead, gold, mercury, hydrogen, oxygen and nitrogen are some examples of familiar elements. Iron, gold, copper etc. contain single atoms as constituent particles held together, whereas in some other cases, the constituent particles are molecules which contain two or more atoms combined together. For example, hydrogen, oxygen, nitrogen etc. consist of molecules, comprising of two atoms combined together.



(a) Atoms of an element

(b) Molecules of an element

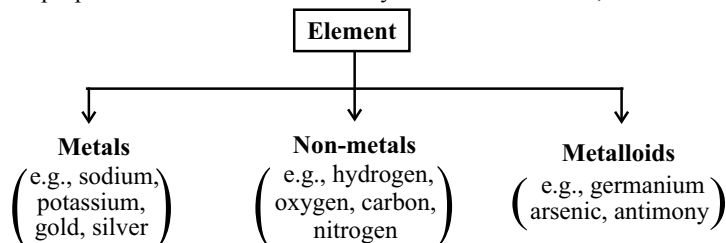
Figure : Elements might consist of individual atoms, as in (a), or molecules, as in (b).



Most of the elements are solids, while eleven of them are gases and only two are liquids. Of the two liquids, mercury, is a metal and bromine is a non-metal. However, two other metals can also exist in the liquid state at around 30°C . These two metals are gallium and caesium (gallium melting point = 29.9°C and caesium melting point = 28.6°C).

Classification of Elements

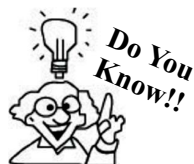
On the basis of variation in their properties elements can be broadly classified as **metals**, **non-metals** and **metalloids**.



Metals

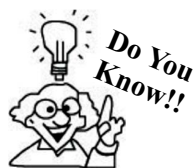
Metals are one of the most important elements that we use in our every day lives, but what we need to understand why it is important. Physical properties of metals which are as follows.

- (i) **Physical state**- Metals are solids at room temperature with the exception of mercury and gallium, which are liquids at room temperature.
- (ii) **Lustre**- Metals have the quality of reflecting light from its surface and can be polished e.g., gold, silver and copper.
- (iii) **Malleability**- Metals have the ability to withstand hammering and can be made into thin sheets known as foils.
- (iv) **Ductility**- Metals can be drawn into wires. 100 gm of silver can be drawn into a thin wire of about 200 meters long.
- (v) **Hardness**- All metals are hard except sodium and potassium, which are soft and can be cut with a knife.
- (vi) **Valency**- Metals have 1 to 3 electrons in the outermost shell of their atoms.
- (vii) **Conduction**- Metals are good conductors because they have free electrons. Silver and copper are the two best conductors of heat and electricity. Lead is the poorest conductor of heat. Bismuth, mercury and iron are also poor conductors.
- (viii) **Density**- Metals have high density and are very heavy.



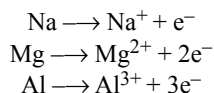
Iridium and osmium have the highest densities whereas lithium has the lowest density.

- (ix) **Melting and boiling point**-Metals have high melting and boiling point.



Tungsten has the highest melting point whereas sodium and potassium have low melting points.

- (x) **Electropositive character**-Metals are elements that have a tendency to lose electrons and form cations. They normally do not accept electrons.



CHECK Point

- ☛ Choose one word or phrase from each pair that describes what most metals are like.

- (a) soft/hard (b) weak/strong (c) shiny/dull (d) tough/brittle (e) heavy or dense/light
(f) conduct heat/do not conduct heat (g) conduct electricity/do not conduct electricity (h) liquid/solid

SOLUTION

- (a) hard (b) strong (c) shiny (d) tough (e) dense (f) conduct heat (g) conduct electricity (h) solid

Knowledge ENHANCER

“Base Metals and Noble Metals”

In chemistry the term base metal is used informally to refer to a metal that oxidizes or corrodes relatively easily, and reacts variably with dilute hydrochloric acid (HCl) to form hydrogen. Examples include iron, nickel, and zinc. Copper is considered as base metal as it oxidizes relatively easily, although it does not react with HCl. It is commonly used in opposition to noble metal.

“Noble metals are metals that are resistant to corrosion or oxidation, unlike most base metals. They tend to be precious metals, often due to perceived rarity. Examples include tantalum, gold, platinum, silver and rhodium.”

Non-Metals

Non-metals are generally poor conductors of heat and electricity. Solid non-metals are generally brittle, with little or non-metallic luster. Most non-metals have the ability to gain electrons easily. Non-metals display a wide range of chemical properties and reactivities.

Physical properties of non-metals are :

- (i) **Physical state**-Most of the non-metals exist in two of the three states of matter at room temperature: gases (oxygen) and solids (carbon). These have no metallic lustre, and do not reflect light.
- (ii) **Nature**-Non-metals are very brittle, and cannot be rolled into wires or pounded into sheets.
- (iii) **Conduction**- They are poor conductors of heat and electricity.
- (iv) **Reactivity**-They generally form acidic or neutral oxides with oxygen.

Comparison Between Metals and Non-Metals

Property	Metals	Non-metals
State of matter	These are usually solid, except mercury, which is a liquid at room temperature. Note: Gallium and Caesium melt below 30°C. So, if room temperature is around 30°C, they may also be in liquid state.	These exist in all the three states. Bromine is the only liquid.
Density	They usually have high density	Their densities are usually low.
Melting point	They usually have a high melting point except mercury, cesium, gallium, tin, lead.	Their melting points are low.
Boiling point	Their boiling points are usually high.	Their boiling points are low.
Hardness	They are usually hard, except mercury, sodium, calcium, potassium, lead, etc.	They are usually not hard except diamond, which is the hardest substance known.
Malleability	They can be beaten into thin sheets.	They are generally brittle.
Ductility	They can be drawn into thin wires, except sodium, potassium, calcium, etc.	They cannot be drawn into thin wires.
Conduction of heat	They are good conductors of heat.	They are poor conductors of heat.
Conduction of electricity	They are good conductors of electricity.	They are non-conductors, except for carbon in the form of graphite and the gas carbon.
Lustre	Newly cut metals have high lustre. Some get tarnished immediately.	Usually not lustrous, except iodine and diamond which is the most lustrous of all the substances.
Alloy formation	They form alloys. However carbon, phosphorus, sulphur, etc., can be present in some alloys.	Generally, they do not form alloys.
Tenacity	These usually have high tensile strength except sodium, potassium, calcium, lead, etc.	These have low tensile strength.
Brittleness	They are hard but not brittle, except zinc at room temperature.	They are generally brittle.



idea box

Make arrangements as shown in figure below



Now here A is made up of aluminium. Record your observation that whether the bulb will light up or not?. Now place bar made up of iron, copper and silver one by one at place of A. On the basis of your observations predict order of electrical conductivity of iron, copper, silver and aluminium. Can you suggest few non-metals if they are placed at position of A, bulb will light up.

Metalloids :

The properties of the metalloids are between those of the metals and non-metals, so the metalloids exhibit characteristics of both classes. The most common metalloid is silicon (Si). It combines with oxygen to form SiO_2 which is used in number of familiar substances, including sand, glass and cement.

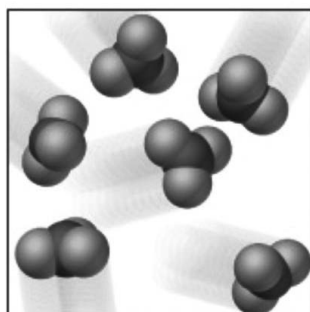
Physical properties of metalloids are

- (i) **Nature :** Metalloids can be shiny like metals or dull like non-metals. They are ductile in nature and can be drawn in shapes of pipes.
- (ii) **Lustre :** They have metallic lustre.
- (iii) **Elasticity :** They are usually brittle in nature.
- (iv) **Conductivity :** Many of the metalloids are semi conductors. Semi conductors are insulators at lower temperature but become conductor at higher temperature.

COMPOUND

It is a pure substance that can be decomposed into simpler substances by some suitable chemical technique. A compound is formed by combination of two or more elements in a definite proportion.

Example, water is a compound of hydrogen and oxygen elements present in the ratio of 1: 8.



Molecules of a compound

Properties of Compounds

1. A compound cannot be separated into its constituents by mechanical or physical means.
Example, if we bring a magnet near a sample of iron sulphide, the iron present in the iron sulphide cannot be separated.
2. Properties of a compound differ entirely from those of its constituent elements.
Example, water is made up of hydrogen and oxygen. However, the properties of hydrogen and oxygen (both gases) are different from water (liquid). Hydrogen is combustible, oxygen is a supporter of combustion whereas water (made up of both hydrogen and oxygen) puts out a flame.
The sodium chloride NaCl is very different from the elemental sodium and the elemental chlorine used in its formation. Elemental sodium, (Na) consists of nothing but sodium atoms, which form a soft, silvery metal that can cut easily with a knife. Its melting point is 97.5°C and it reacts violently with water. Elemental chlorine, Cl_2 consists of chlorine molecules. This material, a yellow green gas at room temperature is very toxic, and it was used as a chemical warfare agent during World War I. Its boiling point is -34°C . The compound NaCl is a translucent, brittle, colorless crystal having a melting point of 800°C . It does not react with water as sodium does and not toxic to humans which chlorine is, but it is an essential component of all living organism.

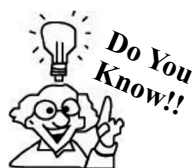
CHECK Point

- ☛ Are the physical and chemical properties of a compound necessarily similar to those of the elements from which it was composed?

SOLUTION

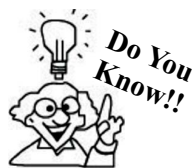
No, it is not necessary that physical and chemical properties of compounds are similar to those of the elements of which they composed.

3. Energy changes are involved in the formation of a compound. For example, iron and sulphur reacts only when heat is supplied.



1. A candle burns to form carbon dioxide and water vapour and liberates heat and light energy. [Exothermic reaction]
2. During the process of photosynthesis, carbon dioxide and water combine to form glucose by absorbing light energy. [Endothermic]

4. The constituent elements in a compound are in a fixed proportion by weight. In water, hydrogen and oxygen are present in a fixed ratio of 1:8 by weight.
5. A compound is a homogenous substance. That is it is same throughout in properties and composition. A compound has a fixed melting point and boiling point. For example, ice melts at 0°C .



Non-stoichiometric compounds : Molecules (constituting units of compounds) have definite atomic composition and are referred as stoichiometric compounds. e.g., CH_4 , H_2O , NH_3 etc.
Certain materials in which atomic composition is variable are called non-stoichiometric compounds. e.g., the composition of cuprous sulphide may vary from $\text{Cu}_{1.7}\text{S}$ to Cu_2S .
Non-stoichiometry is common feature of sulphides and oxides of transition metals.

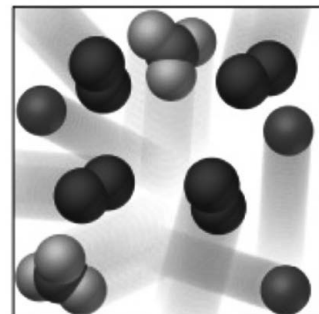


idea box

Make a project report by considering acid rain. Mention the name of elements involved and harmful compounds formed by them. Also explain in detail about the harm of the compounds formed and how their properties are different from constituent elements. You can consult science magazines, science journals, internet and can discuss with your teacher.

MIXTURE

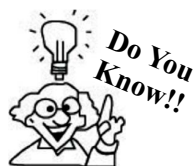
Mixtures are absolutely everywhere you look. Mixtures are the form for most things in nature. Rocks, air, or the ocean, they are just about anything you find. They are substances held together by physical forces, not chemical. That statement means the individual molecules stay near each other, but their fundamental chemical structure does not change as they do not undergoes chemical combination. When you see distilled water, it's a pure substance. That fact means that there are just water molecules in the liquid. A mixture would be water with other things dissolved inside, may be salt. Each of the substances dissolved in water keeps the original chemical properties. So, if you have some dissolved salt, you can boil off the water and still have those dissolved substances left over. Because it takes very high temperatures to boil salt, the salt is left in the container.



Mixture of elements and compound

There are an infinite number of mixtures. Anything you can combine is a mixture. Think of everything you eat. Just think about how many cakes there are. Each of those cakes is made up of a different mixture of ingredients. Even the wood in your pencil is considered a chemical mixture.

The substances that make up a mixture can be separated by physical means because they have different physical properties (such as different melting points) and are not chemically bonded. A mixture can be separated into its parts in a variety of ways, including decantation (letting the sand in a mixture of water and sand settle, and then draining off the water), filtering, and evaporation. You can use a kitchen funnel and coffee filter for filtration, and either use sunlight or low heat for evaporation.



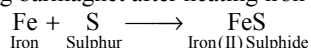
Try out these methods on a salt water and a sand and water mixture to see how they work and compare the results. Evaporation will work for both salt water and sand and water solutions, but filtration will not work for salt water.

ILLUSTRATION : 1

Iron fillings and sulphur are mixed in a watch glass and then heated. Is there any change occurs after heating?

SOLUTION :

Yes, initially before heating we have mixture of iron fillings and sulphur. Mixture retains properties of its components from which iron fillings can be separated by using barmagnet after heating iron and sulphur combines to form iron (II) sulphide.



As iron (II) sulphide is a compound from it iron fillings can not be seperated out.

Differences Between Mixtures and Compounds

Mixture	Compounds
A mixture can be separated into its constituents by physical processes (filtration, evaporation, sublimation, distillation)	A compound cannot be separated into its constituents by physical processes. It can be separated by chemical means
A mixture shows the properties of its constituents	A compound has a new set of properties different from its constituents
Composition of a mixture varies and the constituents are present in any proportion by weight. It does not have definite formula	The composition of a compound is fixed and the constituents are present in fixed proportions by weight. It has a definite formula
The constituents do not react chemically, thus no energy changes take place. A mixture does not have a fixed melting point or boiling point Examples: air, sand and salt	Chemical reactions take place and energy changes in the form of heat and light are involved. A compound has a fixed melting point and boiling point Examples: H ₂ O (water), FeS (iron sulphide)

Types of Mixtures

Depending upon the nature of the components that forms the mixture we can have different types of mixtures.

- (i) **Homogeneous mixture** : It is a mixture that has the same composition throughout. e.g., a solution of sugar in water. Such a mixture has two or more components. The composition of such a mixture is variable (e.g., different quantities of sugar are added to the same quantity of water).

Some other examples of homogeneous mixture are common salt in water, copper sulphate solution, etc.

- (ii) **Heterogeneous mixture** : In such a mixture the particles of each component of the mixture remain separate and can be observed as individual grains under a microscope. e.g., mixture of grains and sand. This type of mixtures contain physically distinct parts and have a non-uniform composition. Some other examples of heterogeneous mixture are a mixture of sodium chloride and iron fillings, a mixture of salt and sulphur, a mixture of oil and water.

CHECK Point

- ☛ Sample A is a mixture of iron and sulphur and sample B is a black mass obtained by heating the mixture of sample A. When is observed for homogeneity?

SOLUTION

A is heterogeneous B is homogeneous

Classification of Homogeneous Mixtures

Homogeneous mixtures can be further classified as follows:

- (i) **Solid-solid mixture**: Examples of such a mixture are brass, bronze, etc.

A homogeneous mixture of two or more metals (or non-metals) is called an **alloy**. Brass and Bronze are alloys. Alloys find a wide range of applications in construction and various other types of industries.

- (ii) **Solid-liquid mixture**: Examples of such a mixture are

- (a) aqueous solution of salt or sugar
- (b) solution of iodine (I_2) in carbon tetra chloride (CCl_4).

- (iii) **Liquid-liquid mixture**: Examples of this type of mixture are

- (a) rectified spirit (a mixture of gasoline and alcohol)
- (b) a mixture of toluene and benzene.

Knowledge ENHANCER

Many of the metallic products that we use in our daily lives are actually alloys. An alloy is a homogeneous mixture of metallic elements with one solid phase. Although an alloy is considered as a solid solution, it is made by mixing the metals in the molten state and then allowing the liquid solution to cool and resolidify.

The main idea with alloys is that they are better at something than any of the metals would be alone. Metallurgists sometimes add chromium (Cr) and nickel (Ni) to steel. While steel is already an alloy of iron and small amount of carbon that is also very strong one, the addition of small amounts of the other metals help steel resist rusting. Depending on what element is added, you could create Stainless Steel or Galvanized Steel. It's always about improving specific qualities of the original metal.

Amalgams are a special type of alloy, amalgams are alloys that combine mercury and other metals. The most obvious place you may have seen amalgams is in old dental work. The fillings in the mouths of your grand parents may have been amalgams. Mercury is a liquid at room temperature. That physical trait was used when they made fillings. Let's say you have an amalgam of mercury (Hg) and silver (Ag). When it is created, it is very soft. As time passes, the mercury leaves the amalgam and the silver remains. The silver that is left is very hard.

Never, ever, play with mercury (Hg) it is very poisonous. You shouldn't even touch it because it will seep into your skin. Dentists don't usually use amalgams with mercury anymore because it may have slowly poisoned people and gotten them sick.

Pure gold (24 K) is a comparatively soft element and is easily deformed. It is made stronger by mixing with other elements. For example, 18 K gold is 75% (by mass) gold with the rest silver or copper. Stainless steel is a mixture of three elements: 80% iron, 12% chromium, and 8% nickel. The percent composition of these alloys can be used to convert between the mass of the alloy and the mass of the component.

Classification of Heterogenous Mixture :

- (i) **Solid-gas mixture:** For example, Hydrogen gas adsorbed on palladium.
- (ii) **Solid-solid mixture:** For example
 - (a) gun powder
 - (b) mixture of sulphur and iron filling
- (iii) **Solid-liquid mixture::** Suspension of sulphur in water.
- (iv) **Liquid-liquid mixture: :** Benzene in water (i.e. immiscible liquids).

CHECK Point

☛ Classify the following substances into elements, compounds and mixtures:

- (i) Milk (ii) 22-carat gold (iii) Iodized table salt (iv) Diamond (v) Smoke (vi) Steel (vii) Brass (viii) Dry ice (ix) Mercury (x) Air (xi) Aerated drinks (xii) Glucose (xiii) Petrol/Diesel/Kerosene oil (xiv) Steam (xv) Cloud.

SOLUTION

Elements — (iv), (ix)

Compounds — (viii), (xii), (xiv), (xv)

Mixtures — (i), (ii), (iii), (v), (vi), (vii), (x), (xi), (xiii).

Points of difference between homogeneous and heterogeneous mixtures :

These are listed in table below :

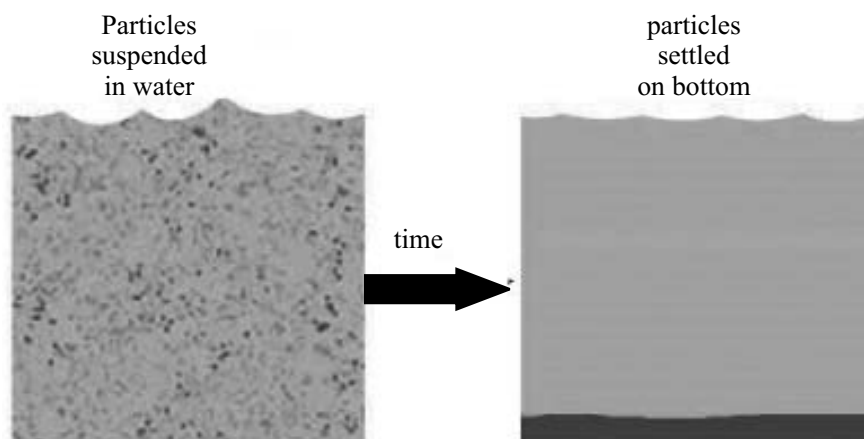
Homogeneous mixtures	Heterogeneous mixtures
(i) Such mixtures have same composition throughout	(i) Such a mixture has different composition in different parts.
(ii) The components of such a mixture cannot be seen even under a microscope.	(ii) The components of such a mixture can be seen even with naked eyes.

METHODS OF SEPARATION OF THE CONSTITUENTS OF MIXTURES

We have already learnt that most of the natural substances are not chemically pure. Different methods of separation are used to get individual components from the mixtures. Heterogeneous mixture can be separated into their respective components by simple physical methods like hand picking, sieving, filtration that we use in our daily life. Sometimes special techniques are used for separation of components of a mixture. Some of these techniques are discussed in details. The constituents of any mixture can be separated by taking advantage of the difference in their physical properties. We have studied that every component retains its own characteristic properties even in the mixtures. Thus, based upon the particular property (e.g., melting point, boiling point, solubility, etc.) in which the constituents of the mixture differ, it can be separated into its constituents. Various techniques employed for the separation of mixtures are discussed below:

1. Decantation or Sedimentation

This separation technique is applicable for a mixture containing one liquid and the other solid component. The solid component should be heavier and insoluble in liquid. Sedimentation is the process by which insoluble heavy particles in a liquid, are allowed to settle down. This is a simple process that most people employ at home. For e.g., suppose you are making some tea. You have boiled the water, and added the tea leaves into the water. Then you realise that you cannot find the strainer. You may look for a clean piece of cloth but do not succeed. What would you do? Keep the tea with the leaves, aside for some time. The tea leaves will settle down. This settling down of the particles in lower part of the container is called sedimentation. Another example of sedimentation is the settling of mud particles in water (Fig.).



Sedimentation

Decantation is the process by which, a clear liquid obtained after sedimentation, is transferred into another container, without disturbing the settled particles. After the tea leaves have settled down, the clear tea (liquor) from the top, can be poured into a cup. This transfer of the clear tea, is called decantation (Fig.).

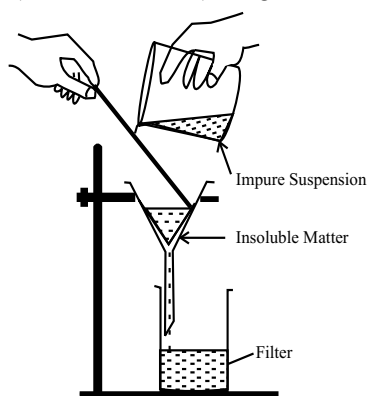


Decantation

2. Filtration

This technique is used when a mixture contains two components, of which only one is soluble in a particular solvent. For separation the mixture is shaken with the solvent and then filtered as shown in figure. The soluble component passes through the filter paper as **filtrate** and the insoluble solid component is retained on the filter paper called **residue**. The soluble component is recovered from filtrate either by evaporating off or distilling off the solvent. For example, the separation of mixture containing.

- (i) Salt (water soluble) and sand (water insoluble) using water as solvent.
- (ii) Sulphur (Soluble in CS_2) and glass powder (insoluble in CS_2) using CS_2 as solvent.
- (iii) Sugar (Soluble in water) and wood dust (insoluble in water) using water as solvent.



Separation by filtration

ILLUSTRATION : 2

When do we use fluted filter paper or hot water funnel for filtration?

SOLUTION :

To avoid crystallization during filtration, fluted filter paper is used when the volume of the solution to be filtered is small and hot water funnel when the volume is large.

3. Centrifugation

This process is used to separate very small solids particles from a liquid mixture. Sometimes the solid particles in a liquid are very small and can pass through a filter paper. For such particles, the filtration technique cannot be used for separation. Such mixtures are separated by centrifugation. So, centrifugation is the process of separation of insoluble materials from a liquid where normal filtration does not work well. The centrifugation is based on the size, shape, and density of the particles. The principle is that the denser particles are forced to the bottom and the lighter particles stay at the top when spun rapidly.

Example – Milk is the mixture of fat, water, and other constituents. Using the method of centrifugation, most of the fat can be separated from milk. In milk, fat is suspended throughout the milk which is separated out using the method of centrifugation.

When milk is churned rapidly, water which is heavier than fat, migrates away from the centre of centrifuge while fat is forced towards the bottom, which is drained out.

4. Evaporation

This technique is applied to separate a non-volatile soluble salt from its mixture in volatile liquid. On heating, the volatile liquid evaporates leaving behind the soluble salt. Many of you might be aware, that for preparing salt, seawater is taken in large pans and water is allowed to evaporate, by the heat of the sun. In the laboratory, salt can be obtained in a similar way, from an aqueous salt solution as under:

Take the solution of salt and water in evaporating dish. Heat the dish carefully till the entire water in the dish gets evaporated. The white crust that remains as residue after evaporation, is salt (Fig.).



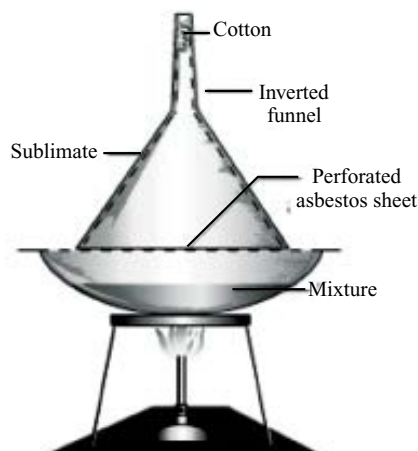
Evaporation

Similarly, sulphur can be separated from a solution of sulphur and carbon disulphide. Keep the solution in a flat dish at room temperature, for some time. The carbon disulphide, being volatile, completely evaporates, leaving behind yellow crystals of sulphur.

5. Sublimation:

This process is used to separate a mixture containing two components, one of which can form a sublimate i.e., direct change of solid to vapour state on heating, while the other do not.

The apparatus, shown in figure, involve a China dish containing mixture covered with perforated filter paper. An inverted funnel whose stem is plugged with cotton is placed on the China dish. The content of the dish is heated gently over sand bath causing the volatile substance to vapourise. These vapours are converted back to solid on cooling on the inner wall of the inverted funnel. The non-volatile substance is left behind in the dish.

**Sublimation**

The sublimates of some mixtures that can be separated by sublimation are given in the table below. Some mixtures that can be separated by sublimation

Mixture	Sublimate
Iodine and sand	Iodine
Ammonium chloride and sodium chloride	Ammonium chloride
Naphthalene and sand	Naphthalene
Camphor and glass powder	Camphor

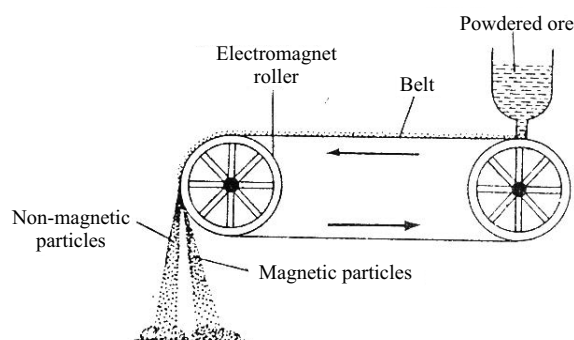
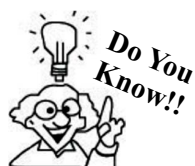
6. Gravity Separation

This technique is used for the separation of a mixture having components differing in densities. The powdered mixture is treated with a stream of running water when the lighter components are washed away leaving the heavier ones e.g., separation of gold particles from rocky substances.

7. Magnetic Separation

This method is applied to separate a mixture containing one magnetic component and the other non-magnetic components. The powdered mixture is allowed to fall on a belt of magnetic separator as shown in figure. When the mixture passes over the rollers, non-magnetic substance falls vertically down whereas the magnetic substance falls a little away. e.g.,

- (i) Separation of iron from sand
- (ii) Iron fillings from sulphur.

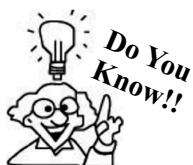
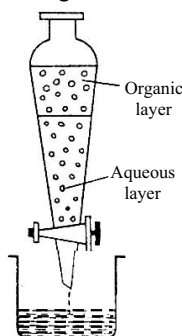
**Magnetic Separation.**

This method is used when one of the components of the mixture is magnetic in nature.

8. Separating Funnel

This method is applicable to recover one of the components from aqueous solution of the mixture by extracting the mixture with a suitable solvent. The solvent should be immiscible with water but should dissolve one of the components of the mixture. The aqueous solution of the mixture is taken in a separating funnel as shown in figure. It is mixed with a small quantity of organic solvent and its contents are shaken well and then the funnel is allowed to stand for sometime. On standing, organic liquid forms a separate layer from aqueous layer. The two layers are collected separately by opening the stop cork. The extraction is further repeated with more quantity of organic solvent. Greater the number of extractions, more will be the amount of substance extracted. For example

- Extraction of nicotine from aqueous solution of tea leaves using ether.
- Extraction of benzoic acid from aqueous solution using benzene.



The separating funnel is used when two immiscible liquids, having different densities, are to be separated.

CHECK Point

- ☛ Suggest a method to purify:
- Camphor containing traces of common salt.
 - Kerosene oil containing water.

SOLUTION

- Sublimation. Camphor sublimes while common salt remains as residue in the china dish.
- Solvent extraction using a separating funnel. The lower water layer is run off when kerosene oil is obtained. It is dried over anhydrous CaCl_2 and then distilled to give pure kerosene oil.

9. Crystallisation or Recrystallisation

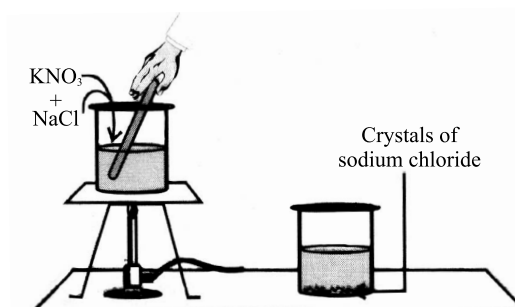
Crystals are the purest form of a substance having definite geometrical shapes. The process by which an impure compound is converted into its crystals is known as crystallisation.

This is one of the most commonly used techniques for purification of solid organic compounds. It is based on the difference in the solubilities of the compound and the impurities in a suitable solvent. The impure compound is dissolved in a suitable solvent in which it is sparingly soluble at room temperature but appreciably soluble at higher temperature. The solution is concentrated to get nearly a saturated solution. When saturated solution is cooled, crystals of pure substance will separate out which are removed by filtration. The filtrate, i.e., mother liquor contains the impurities along with small quantity of the compound.

10. Fractional Crystallisation:

This method of separation is used when two solid components of the mixture have different solubilities in the same solvent. The procedure involves the dissolution of mixture in small quantity of hot suitable solvent. When the concentrated solution is allowed to cool, the less soluble component separates out leaving the more soluble component in the mother liquor. For example, separation of *o*-nitro acetanilide and *p*-nitro acetanilide by crystallization with ethanol.

- In a solution containing potassium nitrate and sodium chloride, sodium chloride is less soluble and crystallizes out first.
- In a solution containing potassium chloride and sodium nitrate, potassium chloride is less soluble and crystallizes out first.



CHECK Point

- A mixture contains two components A and B. The solubilities of A and B in water near their boiling point are 10 grams per 100 ml and 2g per 100 ml respectively. How will you separate A and B from this mixture?

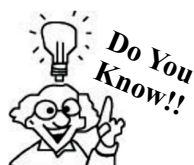
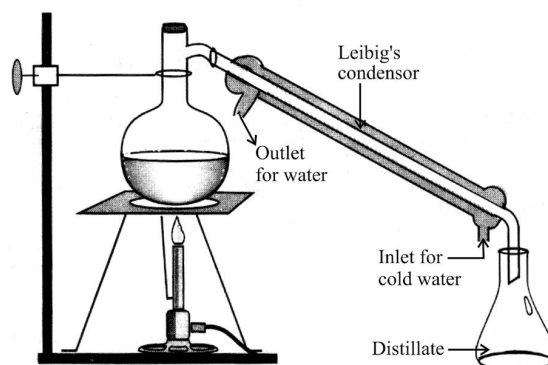
SOLUTION

Fractional crystallization. When the saturated hot solution of this mixture is allowed to cool, the less soluble component B crystallizes out first leaving the more soluble component A in the mother liquor.

11. Distillation

This technique is commonly applied for the separation of a mixture of liquid components having a large difference in boiling point. The lower boiling point component vapourises first and its vapours are condensed by using water condenser and collected as shown in figure 1.5. The higher boiling residue is left behind in the distillation flask. e.g.,

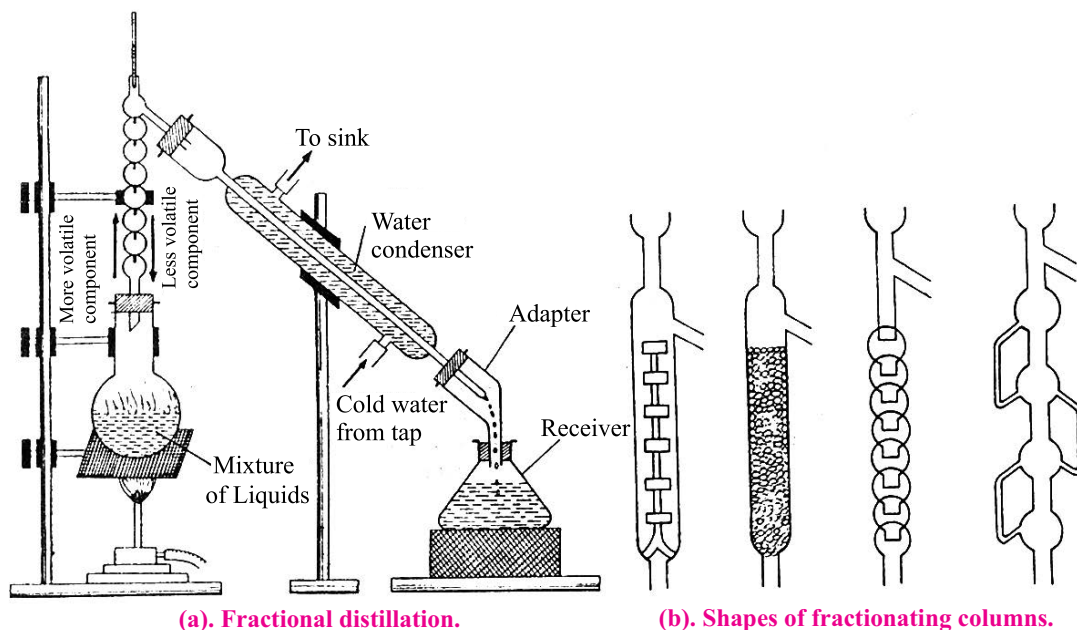
- Separation of benzene (B.P. 353 K) and toluene (B.P. 384 K)
- Separation of chloro benzene and bromobenzene.



- This process allows separation and recovery of both components of a solid-liquid mixture by the use of a distillation apparatus.
- Iodine is the soluble solid which is recovered as residue, while alcohol is the liquid component that is recovered as the distillate from mixture of iodine and alcohol.

12. Fractional Distillation

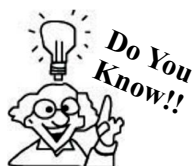
This method of distillation is applied for the separation of the mixture in which the components have a small difference in boiling points. In this method, the distillation flask is fitted with a large fractionating column having a large number of bulbs to increase the surface area as shown in figures (a) and (b). Cold water is circulated in the outer jacket of the condenser. On heating the flask, liquid vapourises and the vapours rise up the column. The vapours of less volatile component condense back in the flask while the vapours of only more volatile component escape from the top of fractionating column, condensed and collected in a receiver. For example,



(a). Fractional distillation.

(b). Shapes of fractionating columns.

- Separation of fractions like gasoline, kerosene, diesel, lubricating oil, etc, from crude petroleum.
- Separation of a mixture of methanol (B.P. 338 K) and acetone (B.P. 329 K).



Fractional distillation method is used when b.p's of the two liquids of the mixture are very close to one another i.e. differ by 10 K.

CHECK Point

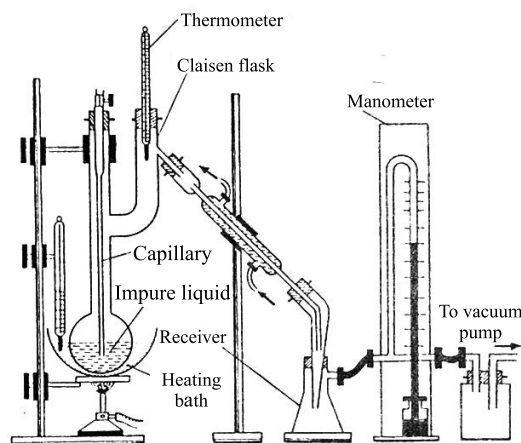
- Suggest methods for the separation of the following mixtures.
- Mixture of liquid A (b.p. 365 K) and liquid B (b.p. 354 K).
 - Mixture of liquid C (b.p. 353 K) and liquid D (b.p. 413 K).

SOLUTION

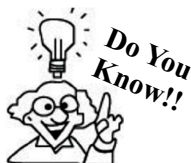
- Fractional distillation because the boiling points of the two liquids differ by just 11K.
- Simple distillation since the boiling points of the two liquids are widely apart.

13. Vacuum Distillation or Distillation Under Reduced Pressure

This technique of distillation is applied to liquids which decompose on heating to their boiling point. We know that the boiling point of a liquid varies with atmospheric pressure i.e., at reduced pressure the boiling point of liquid is also reduced and thus liquid distils at low temperature. The apparatus involves a vacuum pump connected with receiver as shown in figure to carry out distillation under reduced pressure. For example, glycerol is distilled at reduced pressure as it decomposes on heating to its boiling point.



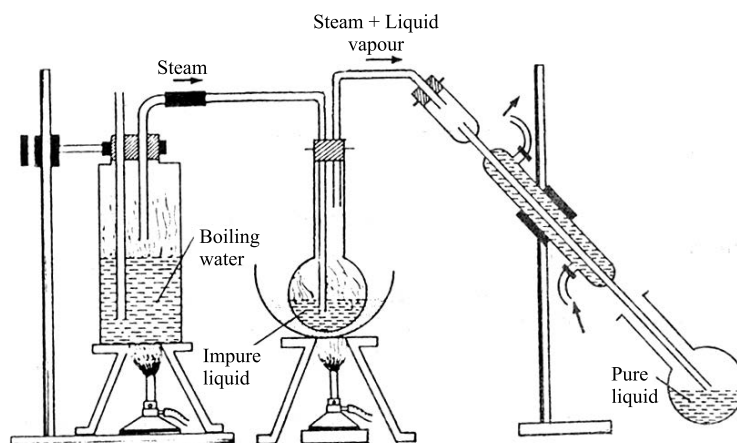
Distillation under reduced pressure.



With the commonly used water suction pumps in the laboratory, a pressure of 10 – 20 mm Hg can be obtained. Under these conditions, the boiling points are reduced by about 100 degrees.

14. Steam Distillation

This is a convenient method for the separation and purification of organic compounds (solid or liquid) from non-volatile organic or inorganic impurities. This method is applicable to only those compounds which are *volatile in steam*, insoluble in water, possess a vapour pressure of about 10 – 15 mm of Hg at 373 K and contain non-volatile impurities. Some of the compounds which can be purified by this process are essential oils, nitrobenzene, chlorobenzene, etc.



Steam distillation apparatus.

The principle of steam distillation is based on Dalton's law of partial pressures. In steam distillation, a mixture of water and an organic liquid is heated by steam as shown in figure. The mixture boils when the combined vapour pressure of water (p_1) and that of the organic liquid (p_2) is equal to the atmospheric pressure (P) i.e.

$$P = p_1 + p_2$$

Evidently, the boiling temperature of the mixture would be lower than the boiling temperature of pure organic liquid and decomposition of organic compound is avoided.

The apparatus involves a steam generator connected to distillation flask. When steam passes through the impure organic compound the mixture starts boiling when the combined vapour pressure becomes equal to atmospheric pressure. At this temperature the vapours of the volatile liquid mixed with steam rise up and condense in condenser. The distillate contains the desired substance and water which can be easily separated with the help of a separating funnel to recover organic substance.

ILLUSTRATION : 3

How will you separate a mixture of *o*-nitrophenol and *p*-nitrophenol?

SOLUTION:

Steam distillation. *o*-nitrophenol being volatile distils over along with water while *p*-nitrophenol being non-volatile remains in the flask.

15. Chromatography

Chromatography is based on the difference in adsorption of different substances on the surface of a solid medium. The technique of separating the components of a mixture in which separation is achieved by the differential movement of individual components through a stationary phase under the influence of a mobile phase. It is a modern technique used for the separation of a mixture of substances. The name chromatography means colour writing, since in its original applications by Michael Tswett in 1906 it involved separation of mixtures containing coloured substances which when separated, formed distinct coloured rings. These zones or rings are named chromatograms. These days, chromatography is applied to coloured as well as colourless substances for analysis.

How it works :

The principle on which paper chromatography works is very simple. The pulp in the filter paper used for chromatography, is composed of thin threads acting as capillaries. When this paper is suspended in a beaker of water, touching the water surface, water ascends through these capillaries. Different soluble dyes move upwards at different speeds and to different levels, by capillary action. In research and industry a special kind of filter paper is used for chromatography. In the laboratory, a fine quality of ordinary filter paper can also be used.

In other type of chromatography, the adsorbent medium used is generally magnesium oxide, alumina or filter paper. The solvent generally used for dissolving a mixture of two or more constituents is water or alcohol.

The different constituents of a mixture get absorbed differently on the same adsorbent material, because they have different rates of movement. The rate of movement of each absorbed material depends upon :

- (i) The relative solubility of the constituents of mixture in a given solvent.
- (ii) The relative affinity of the constituents of mixture for the adsorbent medium.

The stationary phase can be either a *solid* or tightly bound *liquid* on a solid support while the mobile phase can be either a *liquid* or a *gas*. Depending upon the nature of the stationary and the mobile phases, the different types of chromatographic techniques commonly used are

- (1) Paper chromatography (2) Column chromatography (3) Thin layer chromatography (4) Gas chromatography

CHECK Point

- ☛ A liquid (1.0 g) has three components. Which technique will you employ to separate them?

SOLUTION

Column chromatography.



idea box

Take a rectangular strip of chromatography paper. Draw a line with a pencil at about three centimetres height from the base of this paper. Place a drop of black ink on the pencil line. Suspend this chromatography paper in a wide and tall glass cylinder. Gradually, pour water into the cylinder till the lower end of chromatographic paper slightly dips in the water. Cover the cylinder with a glass lid to prevent any evaporation and leave the apparatus undisturbed for an hour. Now record your observations and on the basis of your observation draw a conclusion.

SOLUTION

Solution may be defined as a homogeneous mixture of two or more pure substances, the relative ratio of which can be varied within certain limits. Solution of two substances is the most common type of solution and is termed as **binary solution**. With three or four components it is referred as **ternary** or **quaternary** solution respectively. A solution, in spite of its homogeneous nature, retains the properties of its constituents. The two constituents of solution are

- (i) **Solvent** (ii) **Solute**
- (i) **Solvent** : It is that component of the solution.
 - (a) which is present in relatively large proportion in the solution
 - (b) whose physical state is the same as that of the resulting solution.
- (ii) **Solute** : It is the component of solution which is present in relatively small proportion. For example, a solution of sugar in water (colourless solution) and a solution of copper sulphate in water (blue solution) have small quantities of sugar and copper sulphate respectively as solute dissolved in water as solvent.

Types of Solutions : Solutions can be categorised in two ways.

On the basis of the amount of the solute present in solution, solutions are classified as

- (1) **Dilute solution** : The solution in which quantity of solute in a given mass of solvent is very small is called a dilute solution.
- (2) **Concentrated solution** : The solution in which quantity of solute in a given mass of solvent is very large is called a concentrated solution.
- (3) **Saturated solution** : If a solution contains dissolved solute as much as it is capable to dissolve then it is called saturated solution.
- (4) **Unsaturated solution** : A solution in which amount of solute dissolved is less than its capability at that temperature and more solute can be dissolved in it is called unsaturated solution.
- (5) **Supersaturated solution** : Sometimes it is also possible to dissolve more solute than its capability of dissolving. Such solutions are known as supersaturated solutions.

CHECK Point

- ☛ A solution contains quantity of solute dissolved at temperature T. Now when more quantity of solute is added it remains undissolved, what kind of solution is that?

SOLUTION

Saturated solution.

(ii) On the Basis of Particle Size of Solute :

Based on the size of particles solutions are classified into three main categories **1.** True solution **2.** Colloidal solution **3.** Suspension

1) TRUE SOLUTION

A solution in which the particles of the solute are broken down to such of fine state, that they cannot be seen under powerful microscope is called a true solution.

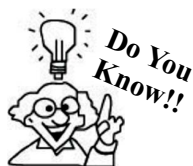
Characteristics of True Solution :

- (i) A true solution is always clear and transparent, i.e., light can easily pass through it without scattering.
- (ii) The particles of solute break down to almost molecular size and their diameter is of the order of 1 nm (10^{-9} metre) or less.
- (iii) A true solution can completely pass through filter paper as particle size of solute is far smaller than the size of pores of filter paper.
- (iv) A true solution is homogeneous in nature.
- (v) In a true solution, the particles of solute do not settle down, provided temperature is constant.
- (vi) From a true solution, the solute can easily be recovered by evaporation or crystallisation.

Types of True Solutions :

Depending upon the physical states of the solute and the solvent, following combinations are possible for a true solution.

Solute	Solvent	Solution	Example
solid	solid	solid	certain alloys
liquid	solid	solid	Hg in Ag
gas	solid	solid	hydrogen gas in palladium metal
solid	liquid	liquid	sugar in water
liquid	liquid	liquid	petrol in kerosene
gas	liquid	liquid	soft drinks
solid	gas	gas	carbon in air (smoke)
liquid	gas	gas	fog
gas	gas	gas	air



Among the given solutions following combinations form the heterogeneous solutions therefore must not be considered as true solutions.

- (i) solid - gas Example ; Storm
- (ii) liquid - gas Example ; Aerosol
- (iii) gas - solid Example ; Styrene foam
- (iv) solid-solid Example ; Minerals

CHECK Point

- ☛ (1) Suggest an alternate name for a homogeneous mixture.
 (2) Does a homogeneous mixture exhibit the characteristics of the constituents present?

SOLUTION

- (1) Homogeneous mixture are also called true solution.
- (2) Yes, for example mixture of salt in water is also salty in taste as common salt.

In most of the cases water is taken as the solvent since it can dissolve a wide range of chemical compounds. Those solutions in which water is taken as the solvent are called aqueous solutions. Besides water other liquids can also be taken as solvents. Such solutions are called non-aqueous solutions.

SOLUBILITY

Solubility is the ability of the solvent (water) to dissolve the solute (sugar). The solubility of a solute is defined as the maximum amount that dissolves in a given amount of solvent at a specified temperature. When a specific amount of solvent contains the maximum amount of dissolved solute, the solution is said to be saturated. If less than the maximum amount is present, the solution is unsaturated. In certain unusual situations, an unstable condition may exist in which there is actually more solute present in solution than its solubility would indicate. Such a solution is said to be supersaturated. Supersaturated solutions often shed the excess solute if a tiny “seed” crystal of solute is added, or if the solution is shaken.

The solubilities of several compounds in water are listed in table.



Note that PbSO_4 , $\text{Mg}(\text{OH})_2$ and AgCl have very low solubilities and thus are considered insoluble.

$$\text{Solubility} = \frac{\text{Weight of solute in saturated solution}}{\text{Weight of solvent in saturated solution}} \times 100$$

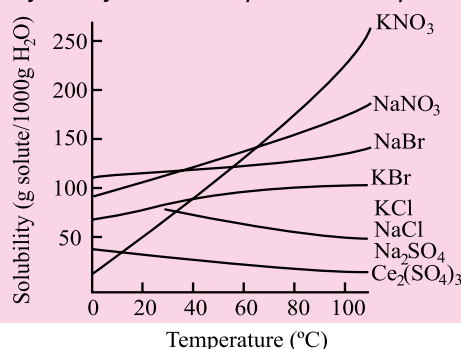
Table : Solubilities of compounds (At 20°C)	
Compound	Solubility (g solute/100g H_2O)
sucrose (table sugar)	205
HCl	63
NaCl	38
KNO_3	34
PbSO_4	0.04
$\text{Mg}(\text{OH})_2$	0.01
AgCl	1.9×10^{-4}

Various Factors that Effect Solubility :

- Effect of temperature :** Most solids and liquids are more soluble in water at higher temperatures. Unlike solids, gases become less soluble as the temperature increases. This can be witnessed by observing water being heated. As the temperature increases, the water tends to fizz somewhat as the dissolved gases are expelled.
- Effect of pressure :** Solubility also depends on pressure. When you increase the surrounding pressure, you can usually dissolve more gases in the liquid. Think about your soda can. They are able to keep the fizz inside because the contents of the can are under higher pressure.
- Nature of solvent :** Solvents with high value of dielectric constant can dissolve polar and ionic compounds to a larger extent than the solvents with low value of dielectric constants.
- Nature of solute :** Ionic compounds (or polar compound) are more soluble in water (i.e., a polar solvent). Non-polar compounds are more soluble in non-polar solvents like benzene.
- Size of solute particles :** Smaller the size of the particles greater is the solubility. Example, it is easier to dissolve powdered sugar than granules of sugar.
- Mechanical stirring :** Mechanical stirring increases solubility. Example, sugar dissolves faster on stirring with a spoon.

Knowledge ENHANCER

Solubility curves : The graphs which show the variation of solubility with temperature are called solubility curves. With the help of solubility curves solubility of any solute at a particular temperature can be found out.



CONCENTRATION OF SOLUTION

The concentration of a solution is defined as the amount of solute present in a given amount of solution or may also be defined as the amount of solute dissolved in a given mass or volume of solvent

$$\text{Concentration of solution} = \frac{\text{Amount of solute}}{\text{Amount of solution}} \text{ or } \frac{\text{Amount of solute}}{\text{Amount of solvent}}$$

Various ways of expressing the concentration of a solution are

(1) Mass Percentage (% by weight).

The mass of the solute in gms per 100 g of the solution is termed as the mass percent.

Mathematically,

$$\% \text{ by weight} = \frac{\text{Weight of the solute}}{\text{Weight of the solution}} \times 100$$

For example, a 15% solution of urea in water (by mass) means that 15 g of urea is present in 100 g of solution. Similarly, a 10% solution of NaCl by weight means that 100 g of solution contains 10 g of NaCl.

ILLUSTRATION : 4

A syrup is prepared by dissolving 250g sucrose in 150 g of water. Calculate the mass percent of sucrose in the solution.

SOLUTION :

We know that

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

However, Mass of solution = Mass of solute + Mass of solvent = 250 + 150 = 400 gm

$$\text{Mass \%} = \frac{250\text{g} \times 100}{400\text{g}} = 62.5\%$$

(2) Volume Percentage (% by volume).

It may be defined as the volume of solute in ml present in 100 ml of the solution.

$$\% \text{ by volume} = \frac{\text{Volume of the solute}}{\text{Volume of the solution}} \times 100$$

For example, a 20% solution of acetic acid (by volume) means 20 ml of acetic acid is present in 100 ml of solution.

CONNECTING TOPIC

Other ways of expressing concentration of solutions are :

(i) Normality (N) :

Normality of a solution may be defined as the number of gram equivalent of solute present per litre of the solution. It is denoted by N.

Mathematically,

$$N = \frac{\text{Gram equivalents of solute}}{\text{Volume of the solution in litres}}$$

$$N = \frac{\text{Weight of solute in gm}}{\text{Eq. wt. of solute} \times \text{volume of solution in litre}}$$

$$N = \frac{\text{Strength of solution in gm/litre}}{\text{Equivalent weight of solute}}$$

The unit of normality is **equivalent lit^{-1}** .

Gram equivalent mass of the solute can be determined by using the formula

$$\text{Gram equivalent mass} = \frac{\text{Molar mass}}{Z}$$

where Z is a whole number and represents

- (i) the number of H^+ ions furnished by an acid molecule in solution i.e., basicity of an acid.
- (ii) the number of OH^- ions furnished by a molecule of base in solution i.e., acidity of base.
- (iii) the number of electrons involved in oxidation of a reducing agent or reduction of an oxidising agent.

ILLUSTRATION : 5

Calculate the normality of NaOH solution containing 50.0 gm NaOH in 5.0 litre solution.

SOLUTION :

$$\begin{aligned}\text{Equivalent weight of NaOH} &= \frac{\text{M.W. of NaOH}}{\text{Acidity of base}} \\ &= \frac{40}{1} = 40\end{aligned}$$

$$\therefore \text{No. of equivalents in 50.0 g NaOH} = \frac{50}{40} = 1.25$$

$$\begin{aligned}\text{Normality of NaOH solution} &= \frac{\text{Equivalents of solute}}{\text{Volume of solution in litres}} \\ &= \frac{1.25}{5} = 0.25N\end{aligned}$$

(ii) Molarity (M) :

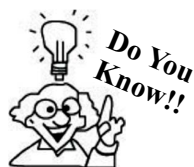
Molarity of a solution is defined as *the number of moles of the solute per litre of solution*. For example, a molar (1 M) solution of sugar means a solution containing 1 mole of sugar (i.e., 342 gm) per litre of the solution.

Thus mathematically,

$$M = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

$$\text{or } M = \frac{\text{Wt. of solute in gms}}{\text{Molecular wt. of solute} \times \text{Volume of solution in litres}}$$

Therefore, unit of molarity is **moles lit.⁻¹**. It is the most common way of representing the concentration of the solution.



Relationship between normality and molarity of a solution can be given by the undermentioned formula

$$\text{Molarity} = \text{Normality} \times \frac{\text{Eq. wt.}}{\text{Mol. wt.}}$$

(iii) Molality (m) :

Molality of a solution may be defined as *the number of moles of the solute dissolved per 1000 g of the solvent*. For example, a 0.2 molal (0.2 m) solution of glucose means a solution obtained by dissolving 0.2 gram mole of glucose in 1000 g (1 kg) of water.

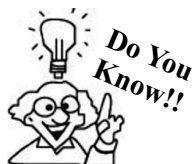
Mathematically,

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent in kilograms}}$$

$$\text{Molality (m)} = \frac{\text{No. of moles of solute}}{\text{Wt. of solvent in gms}} \times 1000$$

$$m = \frac{\text{Mass of solute in grams}}{\text{Molecular mass of solute}} \times \frac{1000}{\text{Wt. of solvent in gms}}$$

It is expressed in units of **moles kg⁻¹**.



The value of normality and molarity varies with temperature as the volume of solution containing equivalents or moles of the solute increases with rise in temperature. However, molality is temperature independent because here volume term is not involved.

Following is the relationship between molarity and molality derived from the formulae of molarity & molality.

$$\frac{\text{Molarity}}{\text{Molality}} = \frac{\text{Mass of solvent in kg}}{\text{Volume of solution in litre}}$$

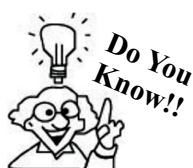
(iv) Mole Fraction (X) :

It is defined as *the ratio of the number of moles of one component (solute or solvent) to the total number of moles of all the components (solute plus solvent) present in the solution. It is denoted by 'X'*

Thus,
$$X_{\text{solute}} = \frac{\text{Moles of solute}}{\text{Moles of solute} + \text{Moles of solvent}}$$

For example, in a solution containing n_1 moles of solute and n_2 moles of solvent, the mole fraction of solute (X_{solute}) and mole fraction of solvent (X_{solvent}) are

$$X_{\text{solute}} = \frac{n_1}{n_1 + n_2}, \quad X_{\text{solvent}} = \frac{n_2}{n_1 + n_2}$$



The sum of the mole fractions of all components present in the solution is always one and has no unit.

$$X_{\text{solute}} + X_{\text{solvent}} = \frac{n_1}{n_1 + n_2} + \frac{n_2}{n_1 + n_2} = 1$$

*Mole fraction of the components is always **temperature independent**.*

Thus, in a binary solution, if we know the mole fraction of one component, the mole fraction of the other component can be calculated.

(v) Mass Fraction :

Similar to mole fraction, mass fraction is defined as *the ratio of the mass of one component (solute or solvent) to the total mass of the solution (both components)*. For a binary solution containing two components A and B, the mass fractions are

$$\text{Mass fraction of A} = \frac{w_A}{w_A + w_B}$$

$$\text{Mass fraction of B} = \frac{w_B}{w_A + w_B}$$

ILLUSTRATION : 6

Calculate mass fraction and mole fraction of ethyl alcohol and water containing 9.2 g alcohol in 18.0 g of water.

SOLUTION :

Mass of ethyl alcohol = 9.2g

Molecular mass of alcohol ($\text{C}_2\text{H}_5\text{OH}$) = 46 g mol^{-1}

Number of moles of alcohol in 9.2 g = $\frac{9.2}{46} = 0.2 \text{ mol}$

Mass of water = 18g

So, number of moles of water = $\frac{18}{18} = 1.0 \text{ mole}$

Mass fraction of alcohol = $\frac{9.2}{18+9.2} = \frac{9.2}{27.2} = 0.338$

$\therefore$ Mass fraction of water = $1 - 0.338 = 0.672$

$X_{\text{ethanol}} = \frac{0.2}{1.0+0.2} = \frac{0.2}{1.2} = 0.167$

$\therefore X_{\text{water}} = 1 - 0.167 = 0.833$

(2) SUSPENSION

If we mix powdered sand with water and shake the mixture constantly, it appears to be somewhat homogeneous, but the sand particles soon settle when we stop shaking it. Nothing like this occurs when we dissolve salt or sugar in water. Small ions and molecules do not settle down in solution. The shaken mixture of sand and water is an example of a suspension. A suspension may be defined as a heterogeneous mixture in which the solid particles are spread throughout the liquid without dissolving in it. They settle as precipitate if the suspension is left undisturbed for sometime. To form suspensions, particles must have dimensions larger than about 1000nm ($1\mu\text{m}$). Suspension that don't settle at once, like powdered sand in water, can nearly always be separated by filtration.

Properties of Suspension

1. A suspension is of heterogeneous nature. There are two phases. The solid particles represent one phase while the liquid in which these are suspended or distributed forms the other phase.
2. The particle size in a suspension is more than 100 nm (or 10^{-7}m).
3. The particles in a suspension can be seen with naked eyes and also under a microscope.
4. The solid particles present in the suspension can be easily separated by ordinary filter papers. No special filter papers are needed for the purpose.
5. The particles in a suspension are unstable. They settle down after sometime when the suspension is kept undisturbed. This is known as precipitate.

Other examples :

- (i) Slaked lime suspension used for white-washing has particles of slaked lime suspended in water.
- (ii) Paints in which the particles of dyes are suspended in turpentine oil.



idea box

You can perform an activity in your school lab or at home to differentiate between solution and suspension. Take two glass beakers of 500 ml and pour 250 mL of water in each of them. Now add 2 grams of common salt to one beaker and 1g of calcium carbonate to another beaker. Then stir the contents in both beakers properly. First observe and list any differences or similarities between the two mixtures. Now darken the room and shine a flash light through each beaker and record your observations.

(3) COLLOIDAL SOLUTION (COLLOIDS) :

When finely divided clay particles are dispersed throughout water, they eventually settle down of the water due to gravity. The dispersed clay particles are much larger than molecules and consist of many thousands or even millions of atoms. These type of mixtures are called suspensions. In contrast, the dispersed particles of a solution are of molecular size. Between these extremes lie dispersed particles that are larger than molecules, but not so large that the components of the mixture separate under the influence of gravity.

These intermediate types of dispersions are called colloidal dispersion, or simply colloids. Colloids form the dividing line between solutions and heterogeneous mixtures. Like solutions, colloids can be gases, liquids, or solids.

CHECK Point

☛ Can air be considered as colloid?

SOLUTION

No, sometimes it acts like a suspension due to smoke particulates, dust and fog. Generally air is considered as homogeneous mixture of gases.

Dispersed phase is the substance whose particles are dispersed in another phase. Actually particles of dispersed phase remain in colloidal range (i.e., 1–200 nm). Whereas dispersion medium provides a medium for dispersion of particles. Dispersion medium consists of continuously inter linked molecules. Substance that tends to keep the colloidal particles far apart and prevents the growth of colloidal particles or coagulation is known as *stabilizing agent*.

Dispersed phase + Dispersion medium $\xrightarrow{\text{stabilizer}}$ Colloidal system

Common Examples of Colloids :

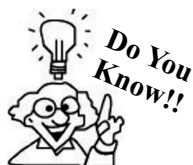
Type	Dispersed phase	Dispersion medium	Common examples
Foam	Gas	Liquid	Soap suds, whipped cream
Solid foam	Gas	Solid	Pumic stone, polyurethane foam
Liquid aerosol	Liquid	Gas	Mist, fog, clouds, liquid air pollutants
Emulsion	Liquid	Liquid	Cream, milk, blood
Solid emulsion (gel)	Liquid	Solid	Butter, cheese, jams, jellies
Smoke	Solid	Gas	Dust, particulates in smog, storm
Sol	Solid	Liquid	Starch in water, proteins, starch, etc
Solid sol	Solid	Solid	Alloys, pearls, opals

CHECK Point

☛ Can we consider an alloy as a colloidal solution?

SOLUTION

Yes, alloys are colloidal solution in which dispersed phase and dispersion medium both are solid.



A substance is said to be in colloidal state when it is dispersed in another medium in the form of very small particles having size in the range 1-200 nm. (1×10^{-7} to 2×10^{-5} cm). The colloidal particle may consist of many atoms, ions, or molecules, or it may even be a single giant molecule. Gas in gas, always form homogeneous mixture and so never form colloids.

CHECK Point

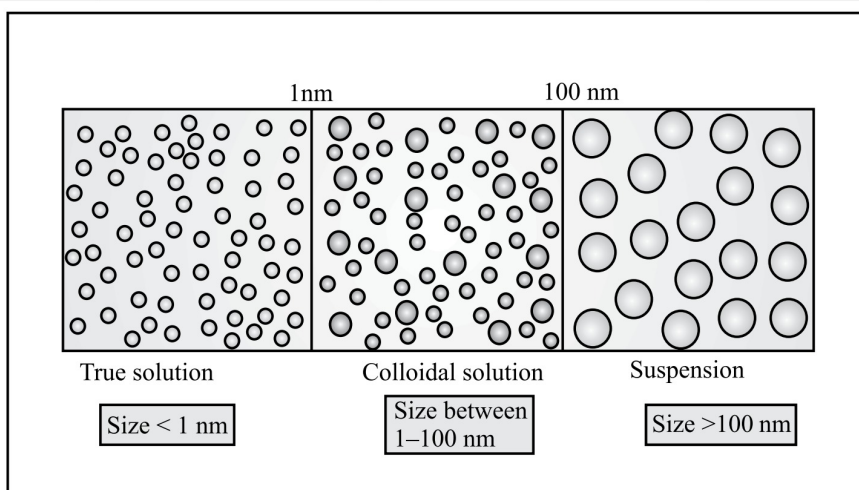
☛ What is common in aqua sols and solid aerosols? How do they differ?

SOLUTION

Aquasol and solid aerosol both have solid as the dispersed phase. They differ in dispersion medium. Aquasols have water as the dispersion medium while aerosols have gas as the dispersion medium.

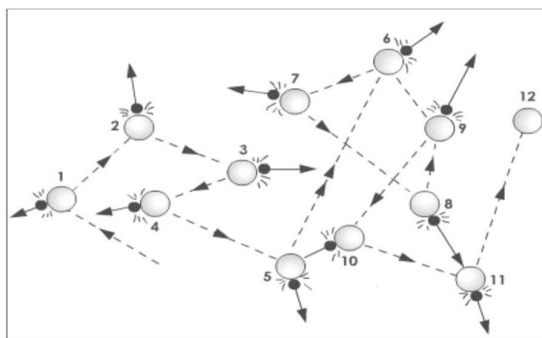
DIFFERENCE BETWEEN THE SOLUTION, COLLOIDAL SOLUTION AND SUSPENSION

Property	True solutions	Colloidal solutions	Suspensions
• Particle size	Less than 10^{-7} cm.	Between 10^{-5} and 10^{-7} cm	Greater than 10^{-5} cm
• Visibility of particles	Invisible to naked eye, not visible under powerful microscope	Invisible to naked eye. Visible under powerful microscope	Easily visible
• Sedimentation of particles	Do not settle down	Settle down under high centrifugation	Settle down due to gravity
• Filtration through filter paper	No residue is formed	No residue is formed	Residue is formed
• Nature	Homogeneous	Heterogeneous	Heterogeneous
• Appearance	Transparent	Turbid	Opaque
• Diffusion through membranes	Readily	Diffuses slowly	Does not diffuse

**Special Properties of Colloids****(1) Brownian movement :**

English Botanist Robert Brown noticed the irregular or chaotic motion of particles suspended in water. This was later on named as Brownian motion.

When a colloidal solution is viewed through a powerful microscope, the colloidal particles can be seen moving in a random zig-zag path. This zig-zag motion of colloidal particles is called Brownian movement. This random motion is due to collisions between the colloidal particles.



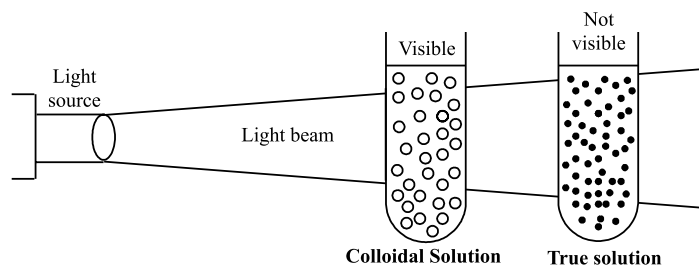
Brownian movement of colloidal particles

Example, when a beam of light passes through a dark room, the dust particles can be seen moving in rapid, random fashion.

(2) Tyndall effect:

All colloidal solutions are capable of scattering light or opalescence.

When a beam of converging rays falls in colloidal solution, scattering of light by sol particles in all the directions, gives rise to a bright glowing cone when looked at it sideways. This is known as Tyndall effect.



Light scattering by colloid dispersions (Tyndall effect)

The solution particles absorb light, become self luminous and then scatter light in all possible directions.

Some examples of Tyndall effect are:

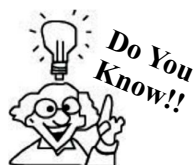
- (i) Blue colour of sky and sea water.
- (ii) Visibility of tails of comets.
- (iii) Twinkling to stars.
- (iv) Visibility of sharp ray of sunlight entering through a slit in dark room.
- (v) Visibility of projector path and circus light.

CONNECTING TOPIC

CLASSIFICATION OF COLLOIDS

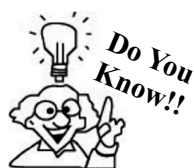
(i) Classification Based on the Nature of Interaction Between Dispersed Phase and Dispersion Medium :

Colloidal systems, depending on the nature of interaction between the dispersed phase and the dispersion medium are classified into lyophobic (solvent hating) and lyophilic (solvent loving).



If water is the dispersion phase, then the colloids are either hydrophilic or hydrophobic.

- (1) **Lyophilic colloids :** In this type of colloids, the dispersed phase has great attraction for the dispersion medium. In such colloids, the dispersed phase does not precipitate easily and the sols are quite stable. If the dispersion medium is separated from the dispersed phase, the sol can be reconstituted by simply remixing with the dispersion medium. Hence, these sols are called reversible sols. Examples of lyophilic sols include sols of gum, gelatin, starch, proteins and certain polymers in organic solvents.
- (2) **Lyophobic colloids :** In this type of colloidal sols, the dispersed phase has little affinity for the dispersion medium. These colloids are easily precipitated on the addition of small amounts of electrolytes, by heating or by shaking and therefore are not stable. Once precipitated, it is not easy to reconstitute the sol by simple mixing with the dispersion medium. Hence, these sols are called irreversible sols. Examples of lyophobic sols include sols of metals and their insoluble compounds like sulphides and hydroxides. Lyophobic sols need stabilizing agents to prevent the dispersed phase from precipitating out.



Hydrophobic sols are often formed when rapid crystallization takes place. With rapid crystallization, many centres of crystallization called nuclei are formed at once. Ions are attracted to these nuclei and very small crystals are formed. These small crystals are prevented from settling out by the random thermal motion of the water molecules.

CHECK Point

☛ Why lyophilic colloidal sols are more stable than lyophobic colloidal sols?

SOLUTION

This is because lyophilic sols are highly hydrated in the solution.

Differences Between Lyophilic and Lyophobic Colloids

	Property	Lyophilic colloids	Lyophobic colloids
1.	Ease of preparation	These are easily formed by direct mixing.	These are formed only by special methods.
2.	Reversible or irreversible nature	These are reversible in nature.	These are irreversible in nature.
3.	Particles nature	These particles of colloids are true molecules and are big in size.	The particles are aggregates of many molecules.
4.	Visibility	The particles are not easily visible even under ultramicroscope.	The particles are easily detected under ultramicroscope.
5.	Stability	These are very stable.	These are unstable and require traces of stabilizers
6.	Action of electrolytes	They are not easily precipitated by small amount of electrolytes. Very large quantities of electrolytes are required to cause coagulation.	They are easily precipitated by the addition of small amount of suitable electrolytes.
7.	Charge on particles	The particles do not carry any charge. The particles may migrate in any direction or even not under the influence of an electric field.	The particles move in a specific direction <i>i.e.</i> , either towards anode or cathode depending upon their charge.

(ii) Classification of Colloids Based on Type of Particles of the Dispersed Phase

(1) Multimolecular Colloids (2) Macromolecular Colloids (3) Associated Colloids.

1. Multimolecular Colloids :

In this type of colloids the colloidal particles are aggregates of atoms or small molecules with molecular size less than one nanometer (1 nm). For e.g., gold sol consists of particles of various sizes which are clusters of several gold atoms. The molecules in the aggregates are held together by Vander Waal forces. They have usually lyophilic character.

2. Macromolecular Colloids :

These are the substances having big size molecules (called macromolecules) which on dissolution form solution in which the dispersed phase particles have size in the colloidal range. Such substances are called **macromolecular colloids**. These macromolecules forming the dispersed phase are generally polymers having very high molecular masses. Naturally occurring macromolecules are starch, proteins and cellulose. Man made macromolecules are polymers such as polyethylene, nylon, plastics and polystyrene. These colloids are quite stable and resemble true solutions in many respects. They have usually lyophobic character.

3. Associated Colloids (Micelles) :

Certain substances behave as strong electrolytes at low concentration but at higher concentrations these substances exhibit colloidal characteristics due to the formation of aggregated particles. These aggregated particles are called micelles. Micelles are called associated colloids. The formation of micelles takes place only above a particular temperature called Kraft Temperature (T_k) and above particular concentration called the critical micelle concentration (CMC). For example, CMC for soaps is about 10^{-4} to 10^{-3} mol L⁻¹. On dilution, these colloids revert back to individual ions. Surface active molecules such as soaps and synthetic detergents form associated colloids in water. Micelles have both a lyophilic and lyophobic parts. Micelles may consists of more than 100 molecules.

CHECK Point

☛ What type of colloidal sols are formed in the following?

- (i) Sulphur vapours are passed through cooled water.
- (ii) White of an egg is mixed with water.
- (ii) Soap solution.

SOLUTION

- (i) Multimolecular because sulphur molecules associate together to form colloidal sol.
- (ii) Macromolecular because protein molecules present in the white of the egg are macromolecules soluble in water.
- (iii) Associated because RCOO^- ions associate together to form micelles.

APPLICATIONS OF COLLOIDS

Colloidal dispersions play a very significant role in natural processes in our daily life and in industries. Some of the important applications of colloids are following.

1. Food:

Most of our food preparations are colloidal in nature. Milk, butter and cream are example of emulsions, jam, jellies, ice-creams, salad dressing, bread etc., are all examples of food colloids.

2. Medicines:

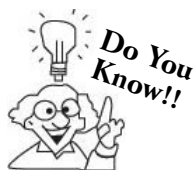
Most drugs (insoluble in water) are administered as colloidal dispersions namely suspensions, emulsions. In the colloidal form the drug is easily absorbed by body tissues and is therefore more effective. Most skin ointments are emulsions.

3. Detergency:

Soaps and detergents by themselves are lyophilic colloids and therefore the entire detergent industry involves colloids.

4. Water Treatment:

Suspended/colloidal particles of undesirable muddy mass need to be removed before making water fit for drinking. The water cleansing process involves the coagulation of the suspended material by alum or high molar mass polyelectrolytes in water treatment plants.



- Rivers carrying muddy water as suspension when come near sea, the high salinity (salt content) of the sea cause suspension to coagulate which ultimately settle down at the point of contact. As a result, the level of the river bed rises. Thus, there is the formation of the delta at the point where the river enters the sea.
- Artificial rain can be made by coagulating the water containing clouds in the sky.
- Colloidal $\text{Fe}(\text{OH})_3$ is given to a person having poisoning due to arsenic because it absorbs arsenic and then can be vomited out.

5. Rubber Industry:

Latex is a colloidal solution of negatively charged colloidal rubber particles. Rubber can be obtained from latex by coagulation.

6. In Metallurgical Operations :

Emulsions play an important role in industry. The metal ores are concentrated by froth-floatation process which involves the treatment of the pulverised ore in emulsion of pine oil.

CONNECTING TOPIC

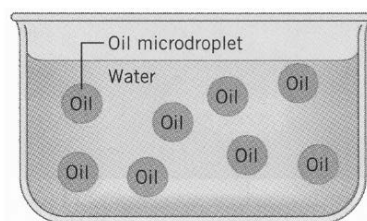
EMULSIONS

Emulsions are colloidal solutions in which both the dispersed phase and dispersion medium are liquids. Emulsions are broadly classified into two types. (i) Oil-in-water emulsion (ii) Water-in-oil emulsion. The substances which are added to stabilize the emulsions are called **emulsifier** or **emulsifying agent**. The substances that are commonly used as emulsifying agents are gum, soap of different forms, gelatin, albumin etc.

(i) Oil in Water Emulsion :

In this type of emulsion oil is the dispersed phase, while water is the dispersion medium. Examples are milk and vanishing cream.

Milk is an emulsion of soluble fats in water and here casein acts as an emulsifier. Vanishing cream is another example of this class. Such emulsions are called **aqueous emulsions**.



An oil-in-water emulsion

CHECK Point

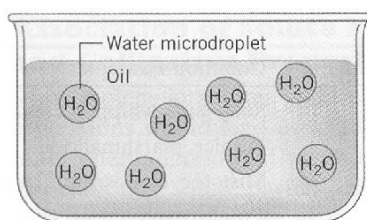
☛ Gelatin is generally added to ice creams. Why?

SOLUTION

Ice cream is an emulsion of milk or cream in water, i.e., oil-in-water type. Gelatin is added to act as an emulsifier, i.e., it helps to stabilize the emulsion.

(ii) Water in Oil Emulsion :

In this type of colloidal system water is the dispersed phase and oil acts as the dispersion medium. Examples of water in oil emulsions are butter, cold cream, cod liver oil and margarine (in which a soyabean product is the emulsifying agent).



A water-in-oil emulsion



It may be noted that the two types of emulsions can be interconverted by simply changing the ratio of the dispersed phase and dispersion medium.

Emulsification is the process of making emulsions. Emulsions are made by shaking the dispersed phase and dispersion medium vigorously and then stabilizing the emulsion with an emulsifier. Most often soaps and detergents are added to stabilize emulsions. Stabilization is obtained by the coating of the drops of an emulsion by the stabilizer. This prevents the drops of the emulsion from combining together and separating out as a separate layer. The separation of an emulsion into its constituent liquids is called demulsification. The various methods used for demulsification are freezing, boiling, centrifugation, electrostatic precipitation or chemical methods which destroy the emulsifying agents.

PHYSICAL AND CHEMICAL CHANGES

To understand the difference between a pure substance and a mixture, let us understand the difference between a physical and a chemical change. As with the properties of a substance, the changes that substances undergo can be classified as either physical or chemical. During physical changes a substance changes its physical appearance, but not its composition. The evaporation of water is a physical change. When water evaporates, it changes from the liquid state to the gas state, but it is still composed of water molecules. All changes of state (for example, from liquid to gas or from liquid to solid) are physical changes.

Example 1 : Mixing iron powder and sulfur powder at room temperature is a physical process. The mixture can be separated into their original constituent without any behavioural change. This can be achieved by using a magnet and the iron will be attracted towards the magnet, leaving the sulfur behind.

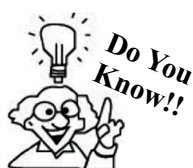
Example 2 : When sugar is dissolved in water, it seems to disappear completely, and not even a very powerful microscope can show us the molecules of sugar in water. However, the solution obtained is evenly sweet. By evaporating the water, the sugar can be recovered in a crystalline form.

Characteristics of Physical Changes

- (i) It is a temporary change.
- (ii) No new substances are formed.
- (iii) No change in mass takes place.
- (iv) Can be reversed by reversing the conditions.
- (v) Change in physical state, size and appearance.

Some Examples Involving Physical Changes

Physical changes	Observation	Change on physical property
1. Switching of an electric bulb.	The bulb glows and gives out heat and light energy.	The physical appearance of the bulb changes.
2. Rubbing a permanent magnet on a steel rod.	The steel rod gets magnetised. If it is brought near iron nails, they get attracted.	The steel rod acquires the property of attracting pieces of iron.
3. Action of heat on iodine	The brownish grey crystals of iodine change to form violet vapours. On cooling the vapours condense to form crystals.	Change in state and colour.
4. Dissolving of common salt in water.	The white crystalline salt disappears in water. However, the water tastes exactly like common salt. Moreover, common salt can be recovered by evaporation.	Change of state.



Some Other Common Examples of Physical Changes :

- (i) Formation of dew
- (ii) Evaporation of water
- (iii) Crystallisation of sugar from its solution.
- (iv) Ringing of an electric bell
- (v) Breaking of a glass pane
- (vi) Making of ice cream.
- (vii) A rock rolling down a hill.
- (viii) Bending of a glass tube by heating
- (ix) Melting of wax.
- (x) Sublimation of camphor

Chemical Change

A chemical change is one in which the identity of the original substance is changed and a new substance or new substances are formed. In a chemical change the properties of the substance before and after the change are entirely different. *e.g. souring of milk, burning of paper, burning of candle, etc.*

Look at the picture of the burning candle. The wax of a candle burns into ash and smoke. The original materials are changing into something different.

Characteristics of a Chemical Change

- (i) A chemical change is permanent change and cannot be reversed to give back the original substance.
- (ii) One or more new substances (called products) are formed.
- (iii) Change in mass of a substance takes place.
- (iv) The composition of the product is different from that of the starting substance.
- (v) A chemical change is always accompanied by the change in energy.



ILLUSTRATION : 7

Out of burning of candle and melting of ice which represents pure physical change?

SOLUTION :

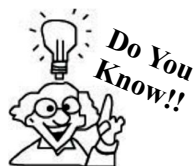
Melting of Ice represents pure physical change as it involves change of state only. In case of burning candle hydrocarbons present in wax is converted into new products on combustion along with simultaneous melting of wax.

Some Examples Involving Chemical Changes :

Chemical change	Observation	Chemical equation
1. Burning of magnesium in air	When a magnesium ribbon is heated in a flame of Bunsen burner, it catches fire and burns with dazzling white flame to form white ash.	Magnesium + Oxygen $\longrightarrow$ Magnesium oxide
2. Rusting of iron	When iron (silver grey) is left exposed to moist air for a few days, reddish brown powdery mass (rust) is found on its surface.	(from air) $\longrightarrow$ Iron + Oxygen + Water vapours $\longrightarrow$ Rust
3. Burning of LPG.	When LPG (liquefied petroleum gas) is burnt, it burns with a pale blue flame and liberates colourless gas carbon dioxide along with steam.	Butane (LPG) + Oxygen $\longrightarrow$ Carbon dioxide + Water

Some Other Common Examples of Chemical Changes :

- | | |
|---|---|
| (i) <i>Burning of wood or charcoal</i> | (ii) <i>Burning of candle</i> |
| (iii) <i>Decomposition of water into hydrogen and oxygen.</i> | |
| (iv) <i>Formation of water from hydrogen and oxygen</i> | |
| (v) <i>Digestion of food</i> | (vi) <i>Formation of biogas (Gobar gas)</i> |
| (vii) <i>Curdling of milk</i> | (viii) <i>Burning of petrol or diesel</i> |
| (ix) <i>Cigarette smoking</i> | (x) <i>Drying of paint</i> |
| (xi) <i>Rusting of iron</i> | (xii) <i>Ripening of fruit</i> |
| (xiii) <i>Blood clotting</i> | (xiv) <i>Fading of the colour of a dyed cloth</i> |

**Comparison Between Physical and Chemical Changes :**

Physical change	Chemical change
(i) It is a temporary change and can be reversed by change of conditions	(a) It is a permanent change and is an irreversible change.
(ii) No new substance is formed and the composition of the substance remains unaltered.	(b) The composition of the substance changes, resulting in the formation of one or more new substances.
(iii) No energy changes occur	(c) It is generally accompanied by energy changes.

CHECK Point

☛ Classify the following as physical and chemical changes.

- (i) Conversion of milk into curd
- (ii) Burning of magnesium ribbon in air
- (iii) Rusting of iron nails
- (iv) Dissolving salt in water
- (v) Burning of coal
- (vi) Electrolysis of sodium chloride solution by passing current
- (vii) Crystallisation of copper sulphate.

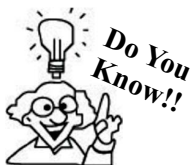
SOLUTION

- (i) Chemical change (ii) Chemical change (iii) Chemical change (iv) Physical change (v) Chemical change (vi) Chemical change (vii) Physical change.



idea box

As you have studied about physical and chemical changes once again go through the concept of physical and chemical changes thoroughly. Now call your friends at your home and also remind them that they should also go through the concept of physical and chemical changes thoroughly. Now play a quiz with your friends in which you become host and make your all friends participants. Given your friends an example and they have to make right guess that whether given example is of physical or chemical change. For right guess 10 points will be awarded and 5 points will be deducted for wrong answer. Let's see who wins.



Changes where physical as well as chemical changes are involved :

- (i) *Burning of a candle (wax)*
 - (ii) *Sublimation of ammonium chloride*
 - (iii) *Action of heat on zinc hydroxide*
 - (iv) *Action of heat on sodium nitrate*
-

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- Common salt is a
- A mixture contains more than substance mixed in proportion.
- Properties of a are different from its constituent elements, whereas a shows the properties of its constituting elements.
- A solution is defined as a mixture that is.....
- We can remove salts from a solution by using the process of.....
- A pure substance has a fixed orat constant temperature.
- An element is made up of only one kind of.....
- Miscible liquids are separated by
- Immiscible liquids are separated by using a
- Filtered tea is amixture
- Alloy is a
- Sublimation of camphor is a change.
- Most common chemical change we observe in our routine life is rusting of
- A solution with solute equal to its solubility is calledsolution.
- Materials that are insoluble in a solvent and have particles that are visible to naked eyes, form a mixture which is a mixture.
- Colloids are mixtures in which the particle size is too to be seen with the naked eye, but is big enough to scatter light.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- Water is homogeneous substance.
- Element is always metal.
- Substance is always homogeneous
- In compound elements combine in definite proportion.
- Iodine can be purified by sublimation.
- Mixtures are always combinations of the same compounds that are at different states.

- We can separate all mixtures by filtration.
- All mixtures are defined as “heterogeneous.
- Only specific compounds can be combined to form mixtures.
- No pure elements are liquids at room temperature.
- Mixtures are combinations of one or more compounds that can be separated with chemical processes.
- Crystals can be made of mixtures.
- Sugar mixed in water is an example of suspension.
- Mechanical stirring increases the solubility.
- In smoke dispersed phase is solid whereas dispersion medium is gas.
- Light beam becomes visible when passed through colloidal solution.
- A suspension is a heterogeneous mixture.
- A colloidal solution is heterogeneous in nature.

Match the Following :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in Column I have to be matched with statements (p, q, r, s) in Column II.

I. Column I	Column II
(A) Steel	(p) Mixture
(B) Air	(q) Compound
(C) Water	(r) Alloy
(D) Diamond	(s) Element
II. Column I	Column II
(A) Miscible liquids	(p) Distillation
(B) Immiscible liquids	(q) Crystallisation
(C) Pure copper sulphate from an impure sample	(r) Sublimation
(D) Salt and ammonium chloride	(s) Separating funnel
III. Column I	Column II
(A) Foam	(p) Whipped cream
(B) Emulsion	(q) Cough syrup
(C) Smoke	(r) Particulates in smog
(D) Suspension	(s) Milk

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- What difference in the property of two miscible liquids enables their separation by fractional distillation ?
- Why is solution not heated to dryness to get crystals ?
- Give one example each of homogeneous and heterogeneous mixture.

4. Name the apparatus by which mixture of oil and water can be separated.
5. How can we check whether the given solid substance is pure or not?
6. A hard substance produces a tinkling sound when beat. Is it metal or a non-metal?
7. Is brass a mixture or a compound?
8. What type of solution is an alloy?
liquid solution or solid solution.
9. A mixture consisting of two miscible liquids 'A' and 'B' whose boiling points differ by 5°C can be separated by which process?
10. Which separation technique is employed to separate complex mixture of organic compounds like carbohydrates, amino acids, vitamins, hormones, etc?
11. In fractional crystallisation two organic compounds get differentiated on the basis of which property.
12. Which method can be used to separate mixture of iron filings and common salt?
13. Give one example of solid-liquid homogeneous mixture.
14. What is an aqua regia?
15. Which method is used to separate two immiscible liquids?
16. Name two elements which are in liquid state at room temperature?
17. What are colloids?
18. Define solubility of solute and how does it vary with temperature?
19. In which range lies the size of colloidal particles?
20. If the path of light becomes visible on being passed through a solution, what type of solution is it?
21. Mass percent is one of the methods of expressing the concentration of a solution. In this method what are the units of mass percent?
22. Name the dispersed phase and dispersion medium in case of dust-storm?
23. Give one example of a solution of gas in a liquid.
24. Do suspensions show the property of Tyndall effect?
25. Can we separate the components of a suspension by process of filtration?
26. Name the type of dispersed phase and dispersion medium in haze.
27. What type of change is burning of candle?
28. Why do colloidal solutions exhibit Tyndall effect?
29. What type of substances form lyophobic sols?
3. What are metals? Give two examples of metals.
4. Which method can be used to separate a mixture of naphthalene and common salt?
5. Define homogeneous and heterogeneous mixtures.
6. Classify the following into elements, compounds and mixtures.
(a) Sodium (b) Soil
(c) Sugar solution (d) Silver
(e) Calcium carbonate (f) Tin
7. Classify each of the following as a homogeneous or heterogeneous mixture.
Sodawater, wood, air, soil, vinegar.
8. Classify the following elements as metal, nonmetal, or metalloid: aluminium, fluorine, gallium, phosphorus, krypton, tellurium, thorium, barium and strontium.
9. Give the names of the elements present in the following compounds:
(a) Quicklime (b) Hydrogen bromide
(c) Baking soda (d) Potassium sulphate
10. Can an element be distinguished from its compound by examination of its physical properties only? Explain.
11. Explain why, water is a compound and not a mixture?
12. How do the properties of a mixture differ from the properties of the components of the mixture?
13. In terms of physical properties, what is there about mixtures that makes it possible to separate out the components?
14. How would you separate the following mixture?
(i) salt and water
(ii) glass powder and iron filings
15. Which of the following mixtures are homogeneous?
(a) Wood (b) Tap water
(c) Soil (d) Cloud
(e) Smoke
16. Why are the metals good conductors of electricity?
17. What are the advantages of chromatography as a method of separation of components of a mixture?
18. Which one shows Tyndall effect and why? True solution or colloidal solution.
19. Will the solubility of common salt in water decrease or increase with increase in temperature? Explain your answer.
20. What will happen if a saturated solution is
(a) heated (b) cooled
21. A solution contains 30 g of sugar dissolved in 370g of water. Calculate the concentration of this solution.
22. A solution is prepared by dissolving 2 g of solute (X) in 100 g H₂O. What is the percentage by mass concentration of solute?
23. A solution contains 40 ml of ethanol mixed with 100 ml of water. Calculate the concentration in terms of volume by volume percentage of the solution.
24. If 110g of salt is present in 550g of solution, calculate the concentration of solution.

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Define element, compound and mixture.
2. What are pure substances? Give two examples of pure substances.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- List some characteristics by which compounds can be distinguished from mixtures.
- What is chromatography ? Explain by giving principle involved. State its important applications.
- Suggest a scheme for the separation of constituents of the following mixture:
Sulphur + sand + sugar + iron filings
- Which method of distillation is employed for the concentration of raw juice in sugar factories. Explain in detail.
- Classify the following as physical change or chemical change.
 - Burning of magnesium ribbon in air
 - Burning of sulphur in air
 - Electrolysis of water
 - Purification of copper sulphate from impure copper sulphate by crystallization.
 - Preparation of sugar solution
 - Conversion of milk into curd
 - Evaporation of alcohol
 - Burning of coal
- Complete the following table based on separation techniques.

Mixture	Types of mixture	Separation technique	Principle
(i) Sulphur + carbon disulphide			
(ii) Iron + sulphur			
(iii) Pigments of flower			
(iv) Petrol + water			
(v) Sand + water			

- How are true solution, colloidal solution and suspension different from each other.?
- Explain in detail about classification of solution on the basis of amount of solute.

2 EXERCISE

Text - Book Questions :

- What is meant by a pure substance ?
- List points of difference between homogeneous and heterogeneous mixtures.
- Differentiate between homogeneous and heterogeneous mixtures with examples.
- How sol, solution and suspension differ from each other?
- To make a saturated solution 36 g of sodium chloride is dissolved in 100 g of water at 293 K. Find its concentration at this temperature.
- How will you separate a mixture containing kerosene and petrol (difference in their boiling point is more than 25°C) which are miscible with each other?
- Name the technique to separate
 - Butter from curd
 - Salt from sea-water
 - Camphor from salt
- What type of mixtures can be separated by technique of crystallization?
- Classify the following as a chemical or physical change :
 - Cutting of trees
 - Melting of butter in a pan
 - Rusting of almirah
 - Boiling of water to form steam.
 - Passing of electric current through water and the water breaking down into hydrogen and oxygen gases.
 - Dissolving common salt in water
 - Making a fruit salad with raw fruits, and
 - Burning of paper and wood
- Try segregating the things around you as pure substances or mixtures.

Text - Book Exercise :

- Which separation techniques will you apply for theseparation of the following?
 - Sodium chloride from its solution in water
 - Ammonium chloride from a mixture containing sodium chloride and ammonium chloride
 - Small pieces of metal in the engine oil of a car
 - Different pigments from an extract of flower petals
 - Butter from curd
 - Oil from water
 - Tea leaves from tea
 - Iron pins from sand
 - Wheat grains from husk.
 - Fine mud particles suspended in water.
- Write the steps that you would use for making tea. Use the words solution, solvent, solute, dissolve, soluble, insoluble, filtrate and residue.

- Pragya tested the solubility of three different substances at different temperatures and collected the data as given below (results are given in the following table, as grams of substance dissolved in 100 grams of water to form a saturated solutions)

Substance dissolved	Temperature in K				
	283	293	313	333	353
	Solubility				
Potassium nitrate	21	32	62	106	167
Sodium chloride	36	36	36	37	37
Potassium chloride	35	35	40	46	54
Ammonium chloride	24	37	41	55	66

- What mass of potassium nitrate would be needed to produce a saturated solution of potassium nitrate in 50 grams of water at 313 K ?
 - Pragya makes a saturated solution of potassium chloride in water at 353 K and leaves the solution to cool at room temperature. What would she observe as the solution cools ? Explain.
 - Find the solubility of each salt at 293 K. Which salt has the maximum solubility at this temperature ?
 - What is the effect of change of temperature on the solubility of a salt?
- Explain the following, giving examples:
 - Saturated solution
 - Pure substance
 - Colloid
 - Suspension.
 - Classify each of the following as a homogeneous or heterogeneous mixture:
 - Soda water
 - Wood
 - Air
 - Soil
 - Vinegar
 - Filtered tea.
 - How would you confirm that a colourless liquid given to you is pure water?
 - Which of the following materials fall in the category of pure substances?
 - Ice
 - Milk
 - Iron
 - Hydrochloric acid
 - Calcium oxide
 - Mercury
 - Brick
 - Wood
 - Air.
 - Identify the solution among the following mixture:
 - Soil
 - Sea water
 - Air
 - Coal
 - Soda water
 - Which of the following will show "Tyndall effect"?
 - Salt solution
 - Milk
 - Copper sulphate solution
 - Starch solution

Is Matter Around us Pure

10. Classify the following into elements, compounds and mixtures:

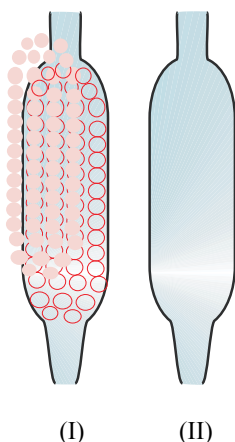
- | | |
|-----------------------|--------------------|
| (a) Sodium | (b) Soil |
| (c) Sugar solution | (d) Silver |
| (e) Calcium carbonate | (f) Zinc |
| (g) Silicon | (h) Coal |
| (i) Air | (j) Soap |
| (k) Methane | (l) Carbon dioxide |
| (m) Blood. | |

11. Which of the following are chemical changes ?

- growth of a plant
- rusting of iron
- mixing of iron filings and sand
- cooking of food
- digestion of food
- freezing of water
- burning of a candle.

Exemplar Questions :

1. Which of the tubes in Fig. (I) and (II) will be more effective as a condenser in the distillation apparatus?



- Sucrose (sugar) crystals obtained from sugarcane and beetroot are mixed together. Will it be a pure substance or a mixture? Give reasons for the same.
- Explain why particles of a colloidal solution do not settle down when left undisturbed, while in the case of a suspension they do.
- During an experiment the students were asked to prepare a 10% (Mass/Mass) solution of sugar in water. Ramesh dissolved 10g of sugar in 100g of water while Sarika prepared it by dissolving 10g of sugar in water to make 100g of the solution.
 - Are the two solutions of the same concentration?
 - Compare the mass % of the two solutions
- While diluting a solution of salt in water, a student by mistake added acetone (boiling point 56°C). What technique can be employed to get back the acetone? Justify your choice.

HOTS Questions :

- How would you confirm that a colourless liquid given to you is pure water ?
- From the following techniques : distillation, filtration, fractional distillation, chromatography, crystallisation, sublimation, evaporation, decantation, sedimentation.
Select the method you will use to separate :
 - the constituents of the colouring matter of ink
 - hydrated CuSO_4 from its aqueous solution
 - sand and water
 - common salt from sea-water
 - petrol from crude oil
- Which of the following materials fall in the category of a "pure substance?"

(a) Ice	(b) milk
(c) Iron	(d) Hydrochloric acid
(e) Calcium oxide	(f) Mercury
(g) Brick	(h) Wood
(i) Air	
- Why does ice float on water ? Is ice water homogeneous or heterogeneous matter ? Pure or a mixture ?
- Classify each of the following as a pure substance or a mixture; if a mixture, indicate whether it is homogeneous or heterogeneous: (a) rice pudding; (b) seawater; (c) magnesium; (d) gasoline.
- Tell whether each of the following properties describes a heterogeneous mixture, a solution (homogeneous mixture), a compound, or an element.
 - a homogeneous liquid that, when boiled away, leaves a solid residue.
 - a cloudy liquid that after a time seems more cloudy toward the bottom.
 - a uniform red solid that has a definite, sharp melting point and cannot be decomposed into simpler substances.
 - a colorless liquid that boils at one unchanging temperature and can be decomposed into simpler substances.
 - a liquid that first boils at one temperature but as the heating continues, boils at slowly increasing temperatures.
- A solid white substance A is heated strongly in the absence of air. It decomposes to form a new white substance B and a gas C. The gas has exactly the same properties as the product obtained when carbon is burned in an excess of oxygen. Based on these observations, can we determine whether solids A and B and the gas C are elements or compounds? Explain your conclusions for each substance.
- How does a gas differ from a liquid with respect to ability to mix with other substances of the same phase to form homogeneous mixtures ?

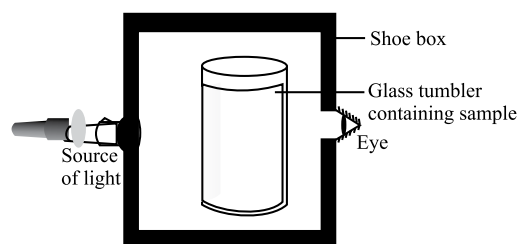
9. Which of the following mixtures is homogeneous and why?
Ethyl alcohol and water

Or

Oil and water

10. Sometimes we refer to alloys as substitutional solids. Why?
11. Discuss the function of fractionating column in fractional distillation.
12. When two substances A and B were made to combine, they formed a third substance C. The following observations were recorded for this :
- During formation of C from combination of A and B a large amount of heat was evolved.
 - The properties of C were different from those of A and B. Predict the nature of C.
13. Is fresh air free of dust particles and impurities of all other kind, a pure substance?
14. Why is a compound considered as pure substance but mixture is not considered as a pure substance?
15. Mixtures can be separated into their components by taking advantage of differences in the chemical properties of the components. Why might this separation method be less convenient than taking advantage of differences in the physical properties of the components?
16. Many dry cereals are fortified with iron, which is added to the cereal in the form of small iron particles. How might these particles be separated from the cereal?
17. How do fog differ from clouds ?
18. Explain why particles of a colloidal solution do not settle down when left undisturbed, while in the case of a suspension they do.
19. A group of students took an old shoe box and covered it with a black paper from all sides. They fixed a source of light (a torch) at one end and made another hole on the other side to view the light. They placed a milk sample contained in a beaker/tumbler in the box as shown in the fig. They were amazed to see that milk taken in the tumbler was illuminated. They tried the same activity by taking a salt solution but found that light simply passed through it?

- Explain why the milk sample was illuminated. Name the phenomenon involved.
- Same results were not observed with a salt solution. Explain.
- Can you suggest two more solutions which would show the same effect as shown by the milk solution?



20. The solubility of sodium chloride in water increases with rise in temperature while that of lithium carbonate decreases. Explain
21. (a) Generally, the solubility of a crystalline substance (solid) increases with rise in temperature. Does it increase uniformly in every case ?
- Name a substance whose solubility
 - increases rapidly with rise in temperature.
 - increases gradually with rise in temperature.
 - increases slightly with rise in temperature.
 - decreases with rise in temperature.
22. A solution contains 50 mL of alcohol mixed with 150 mL of water. Calculate the concentration of this solution.
23. 12 grams of potassium sulphate dissolves in 75 grams of water at 60°C. What is its solubility in water at that temperature ?
24. Explain why filter paper can not be used to separate colloids.
25. Calculate the concentration of a solution in volume percent made when 56g of water is mixed with 0.17 L of ethanol.

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Air is regarded as a mixture because
 - its pressure may vary
 - its temperature may change
 - its volume changes under different conditions
 - its composition may vary
- Which of the following is a compound ?
 - Stainless steel
 - Bronze
 - Graphite
 - Hydrogen sulphide
- The process used to separate oil and water is
 - distillation
 - sublimation
 - separating funnel
 - chromatography
- In which of the following the constituents are present in any ratio?
 - Mixture
 - Compound
 - Solution
 - Colloid
- A mixture of common salt, sulphur, sand and iron filings is shaken with carbon disulphide and filtered through a filter paper. The filtrate is evaporated to dryness in a china dish. What will be left in the dish after evaporation?
 - Sand
 - Sulphur
 - Iron filings
 - Common salt
- Two substances A and B when brought together form a substance C with the evolution of heat. The properties of C are entirely different from those of A and B. The substance C is
 - a compound
 - an element
 - a mixture
 - none of the above.
- Camphor can be purified by
 - distillation
 - filtration
 - sedimentation
 - sublimation
- Which one of the following will result in the formation of a mixture?
 - Crushing of a marble tile into small particles
 - Breaking of ice cubes into small pieces
 - Adding sodium metal to water
 - Adding milk in water
- Purity of a solid substance can be checked by its
 - boiling point
 - melting point
 - solubility in water
 - solubility in alcohol
- A mixture of ethanol and water can be separated by
 - filtration
 - decantation
 - fractional distillation
 - sublimation
- Salt can be obtained from sea water by
 - filtration
 - decantation
 - evaporation
 - sublimation
- A sample contains two substances and has uniform properties. The sample is
 - a compound
 - a heterogeneous mixture
 - an element
 - a homogeneous mixture
- A mixture of ZnCl_2 and PbCl_2 can be separated by
 - distillation
 - crystallization
 - sublimation
 - adding acetic acid
- A mixture of methyl alcohol and acetone can be separated by
 - distillation
 - fractional distillation
 - steam distillation
 - distillation under reduced pressure
- Mixture of sand and sulphur may best be separated by
 - fractional crystallisation from aqueous solution
 - magnetic method
 - fractional distillation
 - dissolving in CS_2 and filtering
- Which component of the mixture ($\text{Fe} + \text{S}$) reacts with dil. HCl and gives hydrogen gas?
 - Sulphur
 - Iron
 - Both
 - None
- Which of the following is considered to be a pure substance?
 - Granite.
 - Sodium chloride.
 - Muddy water
 - Milk of magnesia
- Physical properties of a mixture
 - vary with the amount of substance.
 - depend on the volume of the substance
 - depend on the organization of the substance
 - vary depending upon its components
- Compounds
 - are the same as mixtures
 - can be separated by their physical properties
 - contain only type of element
 - are different kinds of atoms chemically combined with each other.

20. White gold is used in jewelry and contains two elements, gold and palladium. A jeweler has two different samples that are both identical in appearance and have a uniform composition throughout. What can be said about the samples?
- They are homogeneous mixtures and be classified as metallic alloys.
 - The materials are heterogeneous mixtures and can be classified by their components
 - The samples have variable compositions and are classified as metallic solutions.
 - The samples are heterogeneous mixtures that can be separated using magnetic properties.
21. To prepare iron sulphide, by heating a mixture of iron filings and sulphur powder, we should use a
- Copper dish
 - Watch glass
 - China dish
 - Petri dish
22. Which of the following is an example of a heterogeneous substance?
- Bottled water
 - Table salt
 - Pieces of copper
 - Candle
23. Which of the following is an example of a homogeneous substance?
- Granite
 - Copper sulphate
 - M & M candy
 - Muddy water
24. Which flow chart correctly describes a homogeneous material?
- Unknown — density — 3 layers
 - Unknown — filtration — two substances
 - Unknown — magnet — two substances
 - Unknown — boiling — one temperature
25. A student was given a mixture of iron filings and sulphur in the ratio 1 : 2 by weight. He was then asked to heat the mixture over a flame and to observe the colour change. The student will observe that the mixture becomes:
- Black
 - Grey
 - Yellow
 - Orange
26. Filtration can be used to separate
- solids from solids
 - liquids from solids
 - liquids from liquids
 - liquids from gases
27. One common method used to separate dyes is
- filtration
 - distillation
 - chromatography
 - conductivity
28. Melting points can separate materials because
- substances melt at different temperatures
 - molecules vibrate rapidly when heated
 - heat causes molecules to disintegrate
 - many substances fuse at the melting point
29. Distillation is a good separation technique for
- solids
 - liquids
 - solid alloys
 - gases
30. Solubility is a good separation technique for
- pure metals
 - noble gases
 - different salts
 - metallic alloys
31. Magnetism is most beneficial for separating
- gases and non-metallic liquids
 - magnetic solids and solids such as sulfur
 - non-metallic solids and solids such as sulfur
 - non-magnetic solids from non-magnetic liquids
32. The metalloids include the elements
- Boron, Silicon
 - Arsenic, antimony
 - Germanium, Tellurium
 - All the above
33. Antimony and arsenic can be classified as
- Metals
 - Non metals
 - Metalloids
 - Any one of these
34. The best malleable metal is –
- aluminium
 - silver
 - gold
 - lead
35. Pooja took a candle in a vessel and heated the vessel. Even though the candle did not burn, it changed its shape and state. What kind of change is this?
- Irreversible change
 - Physical change
 - Chemical change
 - None of these
36. Liquefied Petroleum Gas (L.P.G.) used as kitchen fuel is supplied in the liquid form in the gas cylinders. When it comes out from cylinder it comes out as a gas which burns. Which of the following statements is correct?
- The process of conversion of liquid to gas involves a chemical change.
 - The process of burning gas is a chemical change.
 - Both the above are correct.
 - None of these
37. On adding a few drops of dilute sulphuric acid to the blue vitriol (copper sulphate) solution, we get a blue coloured solution. When we drop a shaving blade into this blue coloured solution blue colour changes to green after sometime. Which type of change is illustrated by this activity?
- Physical change
 - Chemical change
 - Both of these
 - None of these
38. Three substances given below are kept in the open for a few days and some changes were observed.



Eggs
(P)



Bottle
(Q)



Moth balls
(S)

Which of the substances will show chemical changes?

- Eggs
- Plastic bottle
- Both eggs & moth balls
- All eggs, plastic bottle and moth balls

Is Matter Around us Pure

39. Which of the following is a colloid ?
(a) Sugar solution (b) Urea solution
(c) Silicic acid (d) NaCl solution
40. Smoke is an example of
(a) gas dispersed in liquid
(b) gas dispersed in solid
(c) solid dispersed in gas
(d) solid dispersed in solid
41. Substances whose solutions can readily diffuse through parchment membrane are
(a) colloids (b) crystalloids
(c) electrolytes (d) non electrolytes
42. When dispersed phase is liquid and dispersion medium is gas then the colloidal system is called
(a) smoke (b) clouds
(c) jellies (d) emulsions
43. When a colloidal solution is observed under an ultramicroscope, we can see
(a) light scattered by colloidal particles
(b) size of the particle
(c) shape of the particle
(d) relative size
44. In colloidal state, particle size ranges from
(a) 1 to 10 Å (b) 20 to 50 Å
(c) 10 to 1000 Å (d) 1 to 280 Å
45. Cloud or fog is an example of colloidal system of
(a) liquid dispersed in gas
(b) gas dispersed in gas
(c) solid dispersed in gas
(d) solid dispersed in liquid
46. Which of the following forms a colloidal solution in water ?
(a) NaCl (b) Glucose
(c) Starch (d) Barium nitrate
47. Colloidal solution can be purified by
(a) peptization (b) coagulation
(c) dialysis (d) Bredig Arc method
48. Tyndall effect will be observed in
(a) solution (b) precipitate
(c) sol (d) vapour
49. Small liquid droplets dispersed in another liquid is called
(a) gel (b) suspension
(c) emulsion (d) true solution
50. Butter is a colloid formed when
(a) water is dispersed in fat
(b) milk is dispersed in fat
(c) fat is dispersed in water
(d) milk is dispersed in water
51. Which of the following statements is true about a colloidal system?
(a) It carries a net electric charge
(b) It consists of one phase only
(c) It can be made out of two gases
(d) It is electrically neutral as a whole

52. Choose the correct statement
(a) Latex is an example of solid-in-liquid sol
(b) Latex is a negative sol
(c) Latex is a positive sol
(d) None of the above is correct
53. The most common solvent of Earth is...
(a) gasoline (b) water
(c) turpentine (d) None of the Above
54. Solutions with low concentrations of solutes are
(a) concentrated (b) dilute
(c) solvents (D) None of the above
55. Brownian movement is discovered by
(a) Robert Brown (b) Zsigmondy
(c) Hardy-Schulze (d) Graham
56. Tyndall effect in colloidal solutions is due to
(a) scattering of light
(b) reflection of light
(c) absorption of light
(d) presence of electrically charged particles

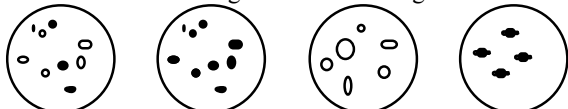
More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

1. Which of the following is/are true for mixtures?
(a) Mixtures can be homogeneous or heterogeneous.
(b) Components in a mixture are present in a fixed ratio.
(c) Components are chemically combined in a mixture.
(d) Components of a mixture can be separated easily by simple physical methods.
2. Which of the following is/are true for a compound?
(a) It is heterogeneous in nature.
(b) A compound contains different elements in a fixed ratio.
(c) Properties of a compound are entirely different from those of the elements present in it
(d) Constituents of a compound cannot be separated by simple physical methods.
3. Which of the following is an example(s) of alloy?
(a) Sugar (b) Brass
(c) Bronze (d) NO₂
4. Which of the following is/are compound(s) ?
(a) Sugar (b) Common salt
(c) Diamond (d) Plaster of paris
5. Which of the following is/are a mixture(s)?
(a) Solution (b) Alloy
(c) Amalgam (d) Ammonia
6. Which of these is/are a mixture?
(a) Oil and water (b) Sand and water
(c) Diet soda (d) Deionized water

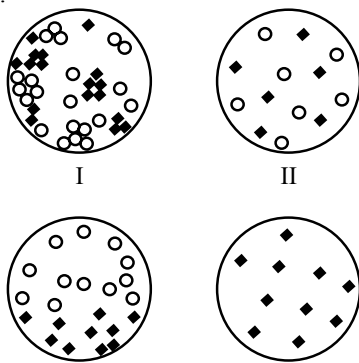
7. Which of the following statements is/are incorrect ?
 (a) A pure substance must contain only one type of atom.
 (b) A mixture containing two compounds must be heterogeneous.
 (c) A heterogeneous mixture must contain at least three elements.
 (d) A homogeneous mixture must be uniform.
8. Mixtures
 (a) can have the same composition throughout
 (b) have physical properties dependent on its components
 (c) have the same physical properties as compounds
 (d) can be separated into their components
9. Which of the following is an example of a heterogeneous substance?
 (a) Mixture of sand and rice grains
 (b) Sodium sulfate
 (c) Aspirin
 (d) Salad dressing

10. Which of the following contains a homogeneous mixture?



- Sample A Sample B Sample C Sample D
 (a) Sample A (b) Sample B
 (c) Sample C (d) Sample D

11. Which of the following diagrams shows a heterogeneous mixture?



- I II
 III IV
 (a) Diagram I (b) Diagram II
 (c) Diagram III (d) Diagram IV

12. Which of the following shows the correct example of a heterogeneous material?
 (a) Mixture — same composition throughout — solution
 (b) Mixture — magnet — two separate substances
 (c) Mixture — filtration — two substances
 (d) Air — wind — one solution of gases
13. Which of the following is/are statements correct?
 (a) Burning of magnesium ribbon is an endothermic reaction.
 (b) Magnesium on burning in air forms ash.
 (c) Ash formed on burning of magnesium ribbon is magnesium dioxide.
 (d) Burning of a petrol is an exothermic reaction.

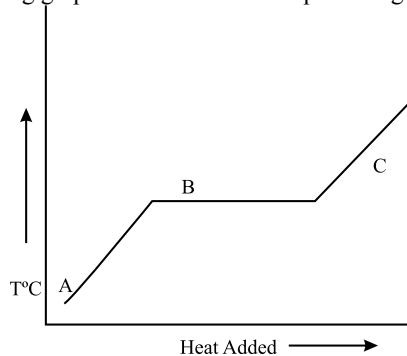
14. Which of the following statements is/are true?
 (a) Bromine is a reddish brown liquid at room temperature.
 (b) Sodium and potassium are soft like wax.
 (c) Non metals are more abundant than metals
 (d) Among all the naturally occurring elements around eighty percent are metals.
15. Aluminium is used for making cooking utensils. Which of the following properties of aluminium are responsible for the same ?
 (a) Good thermal conductivity
 (b) Good electrical conductivity
 (c) Ductility
 (d) High melting point
16. Which one of the following is/are colloid(s)?
 (a) Sodium chloride (b) Starch solution
 (c) Cane sugar (d) Blood
17. Which of the following is/are not a colloid(s) ?
 (a) Chlorophyll (b) Smoke
 (c) Ruby glass (d) Milk
18. Suspensions are
 (a) visible to naked eye
 (b) not visible by any means
 (c) invisible under electron microscope
 (d) visible through microscope
19. Which of the following is/are a colloidal solution(s)?
 (a) Smoke (b) Ink
 (c) Blood (d) Air
20. A colloid always
 (a) contains two phases
 (b) are heterogenous in nature
 (c) contains three phases
 (d) contains only water soluble particle(s)
21. Tyndall phenomena is not exhibited by
 (a) NaCl solution (b) starch solution
 (c) urea solution (d) FeCl₃ solution

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

Passage-1

The following graph was made of two liquid being distilled.



Is Matter Around us Pure

- What observation can be made regarding point B?
 - The first liquid component is boiling
 - The second component has already boiled
 - The mixture is homogeneous
 - Energy is being removed from the liquid
- What observation can be made regarding point C?
 - The second component has already boiled
 - The temperature of first component starts increasing
 - The temperature of second component starts increasing
 - None of the above
- What is the minimum temperature difference required between boiling points of two liquids to be distilled?
 - 10 – 20 K
 - 10°C – 20°C
 - 5 – 15 K
 - 5°C – 15°C

Passage-2

Colloid is a mixture where one of the constituents is dispersed evenly throughout another. The size of the constituent particles which is dispersed is in between 10 nm to 1 μ m and called dispersed phase. The medium in which these particles are dispersed is called dispersion medium. Colloids can be classified based on the physical state of the dispersed phase and dispersed medium.

- What is the physical state of dispersed phase and dispersion medium for cheese respectively.
 - Liquid, Gas
 - Liquid, Solid
 - Solid, Solid
 - Solid, Liquid
- Which of the following is not an example of colloid?
 - Salt solution
 - Starch solution
 - milk
 - Both (a) and (c)
- Which of the following has liquid as both dispersed phase and dispersion medium?
 - Paint
 - Blood
 - Face cream
 - Fog

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - If **Assertion** is **correct** but **Reason** is **incorrect**.
 - If **Assertion** is **incorrect** but **Reason** is **correct**.
- Assertion** : A mixture of He and O₂ is used for respiration for deep sea divers.
Reason : He is soluble in blood.
 - Assertion** : Moving phase is liquid and stationary phase is solid in paper chromatography.
Reason : Paper chromatography is used for analysis of polar organic compounds.
 - Assertion** : Refining of petroleum involves fractional distillation.
Reason : Fractional distillation involves repeated evaporation and distillation.

- Assertion** : Oils are purified by steam distillation.
Reason : The compounds which decompose at their boiling points can be purified by steam distillation.
- Assertion** : Germanium, arsenic and antimony are classified as metalloids.
Reason : Germanium, arsenic and antimony possess some properties of metals and some properties of non-metals.
- Assertion** : Pure substances in which molecules are made up of only one kind of atoms are known as elements.
Reason : Hydrogen, oxygen and nitrogen are elements.
- Assertion** : Conversion of oxygen to ozone is a chemical change.
Reason : Ozone acts as a protective shield and protects us from harmful radiation of the sun.
- Assertion** : The solubility of a gas in a liquid increases with increase of pressure.
Reason : The solubility of a gas in a liquid is directly proportional to the pressure of the gas.
- Assertion** : Colloidal sols scatter light while true solutions do not.
Reason : The particles in the colloidal sol move much slower than that of the true solution.
- Assertion** : Colloidal solution is electrically neutral.
Reason : Due to similar nature of the charge carried by the particles, they repel each other and do not combine to form bigger particles.

Multiple Matching Questions :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

- | 1. | Column-I | Column-II |
|-----|--|-------------------------|
| (A) | Iron fillings and common salt | (p) distillation |
| (B) | Salt and water | (q) magnetic separation |
| (C) | Benzene and toluene | (r) evaporation |
| (D) | Rice grains and alcohol | (s) filtration |
| (a) | A – (q, s); B – (p, r); C – (p); D – (s) | |
| (b) | A – (s); B – (p, r); C – (p); D – (q, s) | |
| (c) | A – (q, s); B – (p); C – (p, r); D – (s) | |
| (d) | A – (p, r); B – (q, s); C – (p); D – (s) | |

2. **Column-I** **Column-II**

- | | |
|--------------------|--------|
| (A) Steel | (p) Fe |
| (B) Copper amalgam | (q) Cr |
| (C) 18 C gold | (r) Hg |
| (D) Iron amalgam | (s) Cu |
- (a) A – (p, q); B – (r, s); C – (s); D – (p, r)
 (b) A – (p, r); B – (r, s); C – (s); D – (p, q)
 (c) A – (r, s); B – (p, q); C – (s); D – (p, r)
 (d) A – (p, q); B – (s); C – (r, s); D – (p, r)

3. **Column-I** **Column-II**

- | | |
|---------------------------------|-------------------------|
| (A) Burning of petrol | (p) physical change |
| (B) Cutting of wood | (q) Reversible change |
| (C) Burning of Magnesium ribbon | (r) Chemical change |
| (D) Melting of ice cream | (s) Irreversible change |

- | | | | | |
|-----|------|------|------|------|
| | A | B | C | D |
| (a) | r, q | p, q | r, s | p, s |
| (b) | r, s | p, s | s, r | p, q |
| (c) | p, q | r, s | r, q | p, s |
| (d) | r, s | p, s | r, q | p, q |

4. Meena is preparing a saturated solution of sugar in water by dissolving 50 g sugar in water. On the basis of given information. Find out the correct answer as per the given directions.

Column-I

- | | |
|--|---|
| A. She added more of the sugar in solution at the same temperature | (p) Sugar crystals are formed and settle down |
| B. She added more of the sugar in solution at high temperature | (q) Solution become unsaturated due to dilution |
| C. She allowed the solution to cool | (r) Sugar does not dissolve any more |
| D. She added more of the water at the same temperature | (s) Sugar get dissolve in water |

Column-II

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | r | s | p | q |
| (b) | r | p | q | s |
| (c) | s | q | r | p |
| (d) | q | s | p | r |

Integer Type Questions :

DIRECTIONS : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

- Consider the following elements : Germanium, sodium, Antimony, Carbon, Hydrogen, Mercury, Oxygen, Potassium, Arsenic, Nitrogen, sulphur. How many of the above elements are metalloids.
- How many of the following substances have liquid dispersion medium whipped cream, clouds, blood, liquid air pollutants, smog, jellies, starch solution.

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1 to 13) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- If 18 g of glucose ($C_6H_{12}O_6$) is present in 1000 gm of an aqueous solution of glucose it is said to be
 - 1 molal
 - 1.1 molal
 - 0.5 molal
 - 0.1 molal
- The amount of $KMnO_4$ required to prepare 100 ml of 0.1 N solution in alkaline medium is
 - 1.58 gm
 - 3.16 gm
 - 0.52 gm
 - 0.31 gm
- 1.0 gm of pure calcium carbonate was found to require 50 ml of dilute HCl for complete reaction. The strength of the HCl solution is given by
 - 4 N
 - 2 N
 - 0.4 N
 - 0.2 N
- The 5.85 g of NaCl and 1 kg of water is added to prepare a solution. What is the strength of NaCl in this solution ? (Molecular weight of NaCl = 58.5)
 - 0.1 Normal
 - 0.1 Molal
 - 0.1 Molar
 - 0.1 Formal
- The molarity of pure water is
 - 55.6
 - 50
 - 100
 - 18
- The number of moles of hydroxide (OH^-) ion in 0.3 litre of 0.005 M solution of $Ba(OH)_2$ is
 - 0.0075
 - 0.0015
 - 0.0030
 - 0.0050
- Mole fraction of glycerine, $C_3H_5(OH)_3$ in a solution containing 36 gm of water and 46 gm of glycerine is
 - 0.46
 - 0.40
 - 0.20
 - 0.36
- Which one of the following is not affected by temperature?
 - Molarity
 - Molality
 - Normality
 - Mole fraction
- How many gm of CH_3OH would have to be added to water prepare 150 ml of a solution that is 2 M CH_3OH ?
 - 9.6×10^3 gm
 - 9.6 gm
 - 9.6×10^2 gm
 - 24 gm
- Number of moles of a solute per kilogram of a solvent is called
 - molarity
 - normality
 - molality
 - formality
- Normal solution is :
 - inert solution
 - acidic solution
 - one litre containing one equivalent
 - basic solution
- The extra stability of lyophilic colloids is due to
 - charge on their particles
 - a layer of medium of dispersion on their particles
 - the smaller size of their particles
 - the large size of their particles
- Which of the following statements is *not* true about the oil-in-water type emulsion?
 - On addition of small amount of water, no separate layer of water appears
 - On addition of oil, separate layer of oil is formed
 - Addition of an electrolyte causes the conductivity of the emulsion to increase
 - Addition of small amount of oil soluble dye renders the entire emulsion coloured.

DIRECTIONS (Qs. 14 to 16) : This section contains 17 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- The colloidal solution of gelatin is known as
 - lyophilic sol
 - reversible sol
 - hydrophilic sol
 - All the above
- Which of the following is/are characteristic(s) of hydrophobic sols?
 - They are highly susceptible to coagulation by addition of electrolytes
 - They have nearly the same surface tension and viscosity as that of dispersion medium
 - Their stability is due to both the electric charge and solvation of the particles
 - Sol particles can be seen under ultramicroscope
- Which of the following statement(s) is/are true in respect of hydrophilic sols?
 - They are quite stable and are not easily coagulated
 - There is considerable interaction between dispersed phase and dispersion medium (water)
 - The sol particles are hydrated
 - They are very sensitive to coagulation by a small concentration of electrolytes

DIRECTIONS (Qs. 17 to 19) : Study the given paragraph(s) and answer the following questions.

Passage-1

An emulsion is a system in which the dispersion medium and the dispersed phase both are liquids. The liquids must be immiscible. Emulsions are classified into two types :

- oil in water emulsion
- water in oil emulsion

A stable emulsion usually requires an emulsifying agent, which forms film in the oil-water interface and thus prevents coalescence of the droplets.

17. The example of oil in water emulsion is
 (a) milk (b) opal
 (c) cod-liver oil (d) None of these
18. The example of water in oil emulsion is
 (a) jellies (b) opal
 (c) cod-liver oil (d) whipped cream
19. The emulsifying agent generally used is/are
 (a) soap
 (b) gelatin
 (c) both soap and gelatin
 (d) soap, gelatin and other hydrophilic sols

DIRECTIONS (Qs. 20 to 22) : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

20. **Assertion** : Lyophilic colloids are called as reversible sols.

Reason : Lyophilic sols are liquid loving.

21. **Assertion** : Milk is an example of water in oil emulsion.

Reason : Emulsion contains liquid dispersed in liquid.

22. **Assertion** : Soaps and detergents belong to the category of macro-molecular colloids.

Reason : The molecules of soaps and detergents are not of big size.

DIRECTIONS (Qs. 23 and 24) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

23.	Column-I	Column-II
(A)	Molarity	(p) w(g)/Eq. wt.
(B)	Normality	(q) n/w (kg)
(C)	Molality	(r) $nA/(nA + nB)$
(D)	Mole fraction	(s) w(g)/Mol. mass
	A B C D	
(a)	s p r q	
(b)	s p q r	
(c)	p s r q	
(d)	p s q r	

24.	Column-I	Column-II
(A)	Protein	(p) Associated colloids
(B)	Arsenic sulphide (As_2S_3)	(q) Macromolecular colloids
(C)	Plastics	(r) Lyophilic colloids
(D)	Micelles	(s) Lyophobic colloids
	A B C D	
(a)	q, p r q s	
(b)	r, s q s p	
(c)	q, r r q p, r, s	
(d)	p q r, s s	

DIRECTIONS (Qs. 25 and 26) : Following are integer based questions Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

25. Calculate the no. of moles of KCl in 1000 ml of 3 M solution is:

26. If 6.8 gm H_2O_2 present in 100 ml of its solution. What is the molarity of solution?

SOLUTIONS

**Brief Explanations
of
Selected Questions**

1 EXERCISE

Fill in the Blanks :

- | | |
|-------------------------------|----------------------------|
| 1. compound | 2. one, any |
| 3. compound, mixture | 4. homogeneous |
| 5. evaporation | 6. M.P & B.P. |
| 7. atoms. | 8. fractional distillation |
| 9. separating funnel | 10. homogeneous |
| 11. solid solution | 12. physical |
| 13. iron | |
| 14. saturated | |
| 15. suspension, heterogeneous | |
| 16. heterogeneous, small | |

True / False :

- | | |
|-----------|-----------|
| 1. True | 2. False |
| 3. False | 4. True |
| 5. True | 6. False |
| 7. False | 8. False |
| 9. False | 10. False |
| 11. False | 12. True |
| 13. False | 14. True |
| 15. True | 16. True |
| 17. True | 18. True |

Match the Following :

- I. $A \rightarrow (r); B \rightarrow (p); C \rightarrow (q); D \rightarrow (s)$
 II. $A \rightarrow (p); B \rightarrow (s); C \rightarrow (q); D \rightarrow (r)$
 III. $A - (p); B - (s); C - (r); D - (q)$

Very Short Answer Questions :

- The difference in boiling point of two miscible liquids enables their separation by fractional distillation.
- Heating the solution to dryness will not remove the soluble impurities and crystals of very poor quality are not obtained.
- Homogeneous mixture – aqueous solution of table salt.
Heterogeneous mixture – milk.
- Separating funnel.
- By checking its melting point whether it agrees with the value given in tables or not. If it agrees then it is pure.
- It is a metal. (Metals are hard and produce tinkling sound.)
- It is mixture, because its composition is variable.
- Alloy is a solid solution.

- Fractional distillation.
- Chromatography.
- On the basis of difference in solubility.
- The given mixture can be separated by magnetic method.
- Solution of I_2 in CCl_4 .
- It is a mixture of 3 parts of HNO_3 and 1 part of HCl .
- A separating funnel is used for this purpose.
- Mercury, Bromine.
- Colloids are mixtures with characteristics intermediate between a solution and a suspension. e.g. fine pottery clay and water.
- Solubility of a solute is defined as the maximum amount that dissolves in a given amount of a solvent at a specified temperature. Solubility increases with increase in temperature.
- It lies in the range 1 nm to 100 nm ($1 \text{ nm} = 10^{-9} \text{ m}$).
- It is a colloidal solution.
- It has *no units*. It is simply a ratio.
- Dispersed phase: solid (dust particles)
Dispersion medium: gas (air)
- Ammonia gas dissolved in water.
- Yes.
- Yes.
- Dispersion medium: gas
Dispersed phase: solid
- During burning of candle physical and chemical change both occurs simultaneously.
- This is because the size of the colloidal particles is such ($10\text{\AA} - 1000\text{\AA}$) that they can scatter light.
- Substances such as metals, their sulphides, etc., which do not mix directly with the dispersion medium to form a colloidal sol.

Short Answer Questions :

- Element:** It is defined as a substance that can not be further reduced to simpler substances.
Compound: It is a pure substance that can be decomposed into simpler substances by some suitable chemical technique. A compound is formed by combination of two or more elements in a definite proportion.
Mixture: A mixture contains more than one substance, elements or compounds. A mixture depicts the properties of its constituting substances.
- A pure substance is one which has constant composition. Lavoisier, a French chemist on the basis of quantitative studies classified pure substances into elements and compounds example of pure substance water, oxygen, nitrogen, etc.
- Metals are those elements which are good conductor of electricity and heat. Metals are ductile and malleable. Two important metals are iron and aluminium.

4. Given mixture can be separated by sublimation. Naphthalene being a sublime substance sublimes off on heating leaving common salt (with high melting point) behind.
5. (i) **Homogeneous mixture** : It is a mixture that has the same composition throughout. e.g., a solution of sugar in water. Such a mixture has two or more components. The composition of such a mixture is variable (e.g., different quantities of sugar are added to the same quantity of water).
Some other examples of homogeneous mixture are common salt in water, copper sulphate solution, etc.
- (ii) **Heterogeneous mixture** : In such a mixture the particles of each component of the mixture remain separate and can be observed as individual grains under a microscope. e.g., mixture of grains and sand. This type of mixtures contain physically distinct parts and have a non-uniform composition. Some other examples of heterogeneous mixture are a mixture of sodium chloride and iron filings, a mixture of salt and sulphur, a mixture of oil and water.
6. (a) element
(b) mixture (heterogeneous mixture)
(c) mixture (homogeneous mixture)
(d) element
(e) compound (CaCO_3)
(f) element
7. (a) Soil is a heterogeneous mixture
(b) Vinegar is a *solution*. It is a homogeneous mixture.
(c) Air is a *solution*. It is a homogeneous mixture.
(d) Wood is *not* a homogeneous mixture so it is *not* a solution.
(e) Soda water is a homogeneous mixture so it is a solution.
[A solution is a homogeneous mixture]
8. Metals :aluminium, thorium, barium, gallium and strontium.
Nonmetals : fluorine, phosphorus and krypton
Metalloids : tellurium
9. (a) Quicklime is calcium oxide, CaO . The elements present in it are : Calcium (Ca) and Oxygen (O).
(b) Hydrogen bromide is HBr . The elements present in it are : Hydrogen (H) and Bromine (Br)
(c) Baking soda is sodium hydrogen carbonate, NaHCO_3 . The elements present in it are : Sodium (Na), Hydrogen (H), Carbon (C) and Oxygen (O).
(d) Potassium sulphate is K_2SO_4 . The elements present in it are : Potassium (K), Sulphur (S) and Oxygen (O)
10. Yes, because there occur a change in physical characteristic of an element when it combines to form compound. For example sodium and chlorine are elements and sodium chloride is compound formed from elements sodium and chlorine and they have difference in physical properties like appearance, melting point, solubility, etc.
11. Water is considered a compound because of the following reasons :
(i) Water cannot be separated into its constituents, hydrogen and oxygen, by the physical methods such as filtration, evaporation, distillation, sublimation, magnet, etc.
(ii) The properties of water are entirely different from those of its constituents, hydrogen and oxygen. For example, water is a liquid whereas hydrogen and oxygen are gases, water does not burn whereas hydrogen burns, water does not support combustion whereas oxygen supports combustion.
12. The chemical properties of the components of a mixture do not change within a mixture. The properties of the whole mixture vary greatly depending on the amount of each component and the specific property being investigated.
13. The components of a mixture are not chemically combined; therefore, the individual chemical and physical properties of the two components remain individual and unique, allowing for easy separation based on these properties.
14. (i) By distillation, salt remains as residue. In case of insoluble salt we can use filtration.
(ii) By magnetic separation. Iron filings are attracted by magnet.
15. Cloud and tap water are homogeneous mixtures. Others are heterogeneous mixtures.
16. In metals we find that *mobile electrons* are present and it accounts for the high electrical conductivity of metals.
17. Main advantages of chromatography are :
(i) Only a small quantity of sample is required.
(ii) The constituents retain their individual characteristics during the process.
(iii) The process can be used for separation of constituents even if they have very similar chemical properties.
18. Tyndall effect is shown by colloidal solution. This is because of the size of colloidal particles (1nm to 100 nm) which can scatter the light falling on them. In true solution the size of particle is very small and they fail to scatter the light falling on them.
19. Solubility of common salt in water increases with increase of temperature. The dissolution of common salt in water is accompanied by absorption of heat (i.e. it is *endothermic*). Due to endothermic nature the solubility increases with increase in temperature.
20. (a) It becomes unsaturated.
(b) Some solute gets deposited at the bottom of solution.
21. This solution contains a solid solute (sugar) dissolved in a liquid solvent (water), so we have to calculate the concentration of this solution in terms of the mass percentage of solute (sugar). We know that :
Concentration of solution = $\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$
Here, Mass of solute (sugar) = 30g

Is Matter Around us Pure

And, Mass of solvent (water) = 370g

So, Mass of solution = Mass of solute + Mass of solvent
 $= 30 + 370 = 400\text{g}$.

Now, putting the values of 'mass of solute' and 'mass of solution' in the above formula, we get :

$$\text{Concentration of solution} = \frac{30}{400} \times 100 = \frac{30}{4} = 7.5\%$$

22. Percentage by mass

$$= \frac{\text{Mass of solute}}{\text{Mass of solute} + \text{Mass of solvent}} \times 100$$

$$= \frac{2}{2+100} \times 100 = \frac{2}{102} \times 100 = 1.96\%$$

23. Volume of solute (ethanol) = 40 ml

Volume of solvent (water) = 100 ml

Volume of the solution = 40 + 100 = 140 mL

volume percentage of solution

$$= \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

$$= \frac{40}{140} \times 100 = 28.6$$

24. Here, mass of solute (salt) = 110g

And, mass of solution = 550g

Now, we know that :

$$\text{Concentration of solution} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

$$= \frac{110}{550} \times 100 = \frac{100}{5} = 20 \text{ percent (or 20\%)}$$

Thus, the concentration of this salt solution is 20 percent (or it is a 20% salt solution).

Long Answer Questions :

1.

Mixture	Compounds
A mixture can be separated into its constituents by physical processes (filtration, evaporation, sublimation, distillation).	A compound cannot be separated into its constituents by physical processes. It can be separated by chemical means.
A mixture shows the properties of its constituents.	A compound has a new set of properties different from its constituents.
Composition of a mixture varies and the constituents are present in any proportion by weight. It does not have definite formula.	The composition of a compound is fixed and the constituents are present in fixed proportions by weight. It has a definite formula.
The constituents do not react chemically, thus no energy changes take place. A mixture does not have a fixed melting point or boiling point Examples: air, sand and salt.	Chemical reactions take place and energy changes in the form of heat and light are involved. A compound has a fixed melting point and boiling point Examples: H_2O (water), FeS (iron sulphide).

2. Chromatography is the collective term for a set of laboratory techniques for the separation of mixtures. The mixture is dissolved in a fluid called the "mobile phase", which carries it through a structure holding another material called the "stationary phase". The various constituents of the mixture travel at different speeds, causing them to separate. The separation is based on differential partitioning between the mobile and stationary phases.

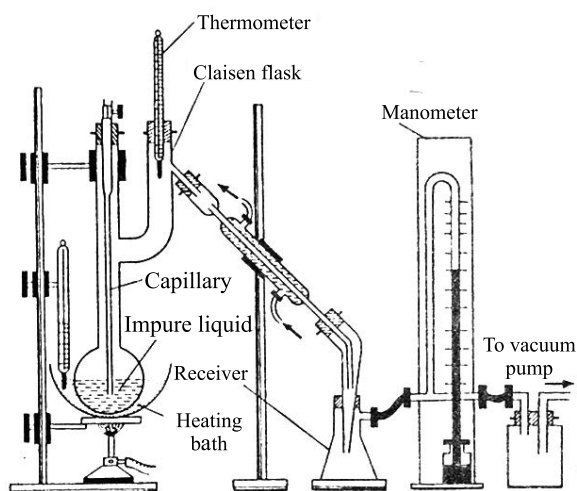
Applications of chromatography

- Toxicology is an area where chromatography is used Separating and identifying different drugs of abuse.
- In sports medicine, any illegal drugs will be picked up using chromatographic techniques. (e.g. Gas chromatography)
- A magnet is moved several times over the mixture containing sulphur, sand, sugar and iron filings. The iron filings are attracted by the magnet, they stick to the magnet and get separated.

The remaining mixture of sugar, sulphur and sand is stirred with water when sugar dissolves in it. On filtration sugar solution is obtained as a filtrate from which sugar can be recovered by evaporation. The residue contains sulphur and sand.

Sulphur and sand mixture is shaken with carbon disulphide when sulphur dissolves in it but sand remains undissolved. On filtration, sulphur is obtained as a filtrate from which solid sulphur can be recovered by evaporating carbon disulphide. Sand is left behind as a residue on the filter paper.

4. **Vacuum distillation or distillation under reduced pressure:** This technique of distillation is applied to liquids which decompose on heating to their boiling point. We know that the boiling point of a liquid varies with atmospheric pressure *i.e.*, at reduced pressure the boiling point of liquid is also reduced and thus liquid distils at low temperature. The apparatus involves a vacuum pump connected with receiver as shown in figure to carryout distillation under reduced pressure.



Distillation under reduced pressure.

5. (i) Chemical change (oxide of magnesium is formed)
 (ii) Chemical change (oxide of sulphur formed)
 (iii) Chemical change (hydrogen and oxygen are formed)
 (iv) Physical change (no new substance formed)
 (v) Physical change
 (vi) Chemical change (a new substance is formed)
 (vii) Physical change
 (viii) Chemical change (carbon dioxide formed formation of new product indicated chemical change)

6. Complete the following table based on separation techniques. The first one example is done for you.

Mixture	Types of mixture	Separation technique	Principle
(i) Sulphur + carbon disulphide	Homogeneous	Evaporation	Difference in boiling point
(ii) Iron + sulphur	Heterogeneous	Magnetic Separation	Iron is magnetic in nature
(iii) Pigments of flower	Homogeneous	Chromatography	Difference in adsorption levels
(iv) Petrol + water	Heterogeneous	Separating funnel	Organic compounds get dissolved in organic solvents
(v) Sand + water	Heterogeneous	Filtration	Particles of larger size will left as residue on filter paper.

7.

Property	True solution	Colloidal Solution	Suspension
1. Appearance	They are clear and transparent	They are translucent .	They are opaque.
2. Visibility	Particles are not visible to naked eye	Scattering of light by particle is observed under ultra-microscope.	Particles are visible to naked eye.
3. Particle size	It is less than 1 nm (nano meter)	It is between 1nm and 100 nm.	Particles size is more than 100 nm.
4. Diffusion	They diffuse quickly.	They diffuse slowly	They do not diffuse.
5. Setting	They do not settle	They do not settle	They settle down on standing
6. Filterability	They can pass through ordinary filter paper as well as animal membrane.	They pass through filter paper but not through animal membrane.	They do not pass through filter paper or membrane.

8. Depending on the Amount of Solute in Solution: on the basis of the amount of the solute present in solution, solutions are classified as

- (1) **Dilute Solution** : The solution in which quantity of solute in a given mass of solvent is very small is called a dilute solution.
 (2) **Concentrated Solution** : The solution in which quantity of solute in a given mass of solvent is very large is called a concentrated solution.
 (3) **Saturated Solution** : If a solution contains dissolved solute equal to its solubility then it is called saturated solution.

- (4) **Unsaturated Solution** : A solution in which amount of solute dissolved is less than its solubility at that temperature and more solute can be dissolved in it is called unsaturated solution.
- (5) **Supersaturated Solution** : Sometimes it is also possible to dissolve more solute than its solubility. Such solutions are known as supersaturated solutions.

2 EXERCISE

Text - Book Questions :

1. A pure substance is a single substance or matter which cannot be separated into other kind of matter by any physical process. All elements and compounds are pure substances.

	Homogeneous mixture	Heterogeneous mixture
1.	In it the constituents are uniformly mixed and there are no boundaries of separation.	In it constituents may not be uniformly mixed and there are clear boundaries of separation.
2.	We can not easily see various constituents.	We can easily see different constituents.
3.	We cannot easily separate constituent of such a mixture. We need special methods for separation of constituents of such a mixture.	We can easily separate the constituents of such a mixture by ordinary methods of separation of mixtures.

2. For points of difference refer to Q.2.

Examples : Homogeneous mixture : Air, a solution of salt and water.

Heterogeneous mixture: A mixture of salt and pepper, a mixture of iron fillings and sulphur.

4. Difference between sol (i.e. colloidal solutions), solution (i.e. true solution) and suspension are listed below :

S.No.	Property	Sol	Solution	Suspension
1.	Nature	Appears to be homogeneous but actually heterogeneous	Homogeneous	Heterogeneous
2.	Particle size	1 nm to 100 nm	Less than 1 nm	More than 100 nm
3.	Visibility	Particles are visible through an ultramicro scope but not with naked eye.	Particles are not visible even with microscope.	More than 100 nm Particles are visible with naked eye.
4.	Tyndall effect	Light is scattered	No scattering of light	Light is scattered.
5.	Sedimentation	Settle on centrifugation	Do not settle down	May settle due to gravity
6.	Brownian movement	Show Brownian movement	Do not show Brownian movement	May or may not Show Brownian movement.

5. Mass of common salt dissolved (solute) = 36 g

Mass of water taken (solvent) = 100 g

$$\text{Concentration of solution} = \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 100$$

$$= \frac{36}{100} \times 100 = 36\%$$

6. The components of this mixture can be carried out by process of simple distillation.

[Simple distillation process can be used to separate a mixture of miscible liquids in which the difference in boiling point of components is more than 10°C].

7. (i) Centrifugation
(ii) Evaporation or crystallization
(iii) Sublimation (camphor is sublime and salt is not sub-lime substance).

8. The technique of crystallization can be used to separate components of a mixture in which one component is less soluble in a particular solvent as compared to the other component. i.e. It makes use of the difference in solubility of a substance. It is used to purify substances.

9. (i) Chemical change (It is irreversible)
(ii) Physical change (no new substance formed)
(iii) Chemical change (rust is oxide of iron)
(iv) Physical change (no new substance formed)
(v) Chemical change (Hydrogen and Oxygen are formed)
(vi) Physical change (no new substance formed)
(vii) Physical change (no new substance formed)
(viii) Chemical change (It is irreversible)

10. We have various substances around us. Here we have classified some of these.

Pure substances: Gold, silver, distilled water, sugar, common salt, diamond, graphite etc.

Mixture: Lemonade, ice-cream, cold-creams, steel, bronze, vulcanised rubber, curd, air etc.

Text - Book Exercise :

1. (a) Crystallization: This process is used to separate a pure solid in the form of its crystals from a solution. [Note: This technique is better than evaporation which can also be used to get sodium chloride from its solution in water.]
 (b) Sublimation: Ammonium chloride is a sublime substance and will be collected as sublimate where as sodium chloride is not a sublime substance.
 (c) Filtration: Metal pieces will be collected on filter.
 (d) Chromatography: Pigments are coloured components and they can be separated by chromatographic technique.
 (e) Centrifugation: On centrifugation butter will get separated.
 (f) Separating funnel: Oil and water are immiscible liquids and can be separated using a separating funnel.
 (g) Filtration: Tea leaves (solid) will be collected on filter.
 (h) Magnetic separation: Iron pieces are magnetic in nature hence can be separated from sand (non-magnetic) by use of a magnet. Iron gets attracted towards magnet.
 (i) Sieving: This method can be used to separate wheat grains from husk.
 (j) Sedimentation: During this process mud particles will settle down as precipitate which can then be separated by filtration.
2. In making tea various steps involved are:
 - (i) Put some water (solvent) in a pan and put the pan containing water on a lighted gas burner
 - (ii) After sometime water (solvent) begins to boil, now add appropriate amount of milk and sugar (both milk and sugar are solutes)
 - (iii) Now stir the contents of the pan with the help of a tea spoon. The sugar (solute) being soluble in water will dissolve and milk will become miscible with water (solvent) A solution is thus formed.
 - (iv) On further boiling the solution, solute (sugar) gets completely dissolved.
 - (v) Now add the appropriate amount of tea leaves (solute) to the above solution in the pan. Tea leaves are insoluble in solution. Boil the solution and filter through a sieve. Tea will be collected as filtrate and tea leaves will get collected on sieve as residue.
3. (a) The amount of potassium nitrate present in 100 grams of water in a saturated solution of potassium nitrate = 62 g. (from table)
 Amount of potassium nitrate present in 50 g of water = 31g
 (b) If a saturated solution of potassium chloride, which is prepared at 353 K, is allowed to cool to room temperature (298 K), some of it will separate out and will settle down at the bottom of solution (we can see from the table that the solubility of potassium chloride is less at 298 K than at 353 K). Thus, we would observe a white residue at the bottom of the container.
 (c) From the given table we find solubility (the maximum amount of salt that can be dissolved in 100 g of water) of various salts as solubility at 293 K.
 solubility of potassium nitrate in 100 g of water = 32 g
 solubility of sodium chloride in 100 g of water = 36 g
 solubility of potassium chloride in 100 g of water = 35g
 solubility of ammonium chloride in 100 g of solubility water = 37 g
 Thus, we find that at 293 K the solubility of ammonium chloride in water is maximum (37 g/100 g of water).
 (d) with increase of temperature, the solubility of all the given salts increases and it decreases with a decrease in temperature. [This is quite evident from the data given in table].
4. (a) Saturated solution: A solution in which no more solute can be dissolved at the same temperature is called a saturated solution e.g. a solution of sugar in water. At room temperature if any more sugar added to it, it does not dissolve but gets settled at the bottom.
 (b) Pure substance: A pure substance means a single substance (or matter) which can not be separated into other known kinds of matter by any physical process. The substances, which always have the same colour, taste or texture at a given temperature are pure substances. eg. elements and components, ghee, oil, etc.
 (c) Colloid: It is a type of solution in which the diameter of particles (solute particles) may range from 1 to 100 nm (nm = nanometer = 10^{-9} m). The particles of a colloidal solution do not settle down under the force of gravity even on standing for long time. The particles of a colloid are uniformly spread throughout the solution.
 Due to smaller size particles the solution appears to be homogeneous but actually a colloidal solution is a heterogeneous mixture e.g. milk.
 (d) Suspension: It is a heterogeneous mixture in which the solid particles are spread throughout the liquid without dissolving in it. These particles get settled

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as a precipitate if suspension is left undisturbed for some time, e.g. muddy water, milk of magnesia, some paints etc.

[Note: The particle size of suspended particles in a suspension is more than 10–7m.]

5. (a) Homogeneous mixture
(b) Heterogeneous mixture
(c) Homogeneous mixture
(d) Heterogeneous mixture
(e) Homogeneous mixture
(f) Homogeneous mixture.
6. To confirm whether the colourless liquid given is pure or not we can proceed as follows:
 - (i) Filter the given liquid using a very fine filter paper to see if it contains any suspended impurities. If any thing is left on filter paper the given liquid is impure. Otherwise it is pure.
 - (ii) Evaporate a part of the given liquid in a china dish. If any residue is left it is impure and in case no residue is left it is pure.
 - (iii) Determine the b.p. of the given liquid. If the b.p. is 100°C (373 K) it is pure water.
[Note: B.P. of pure water is 373 K at 1atmospheric pressure]
7. (a) Ice: Pure substance (It is pure water in solid state) It is a component.
(b) Milk: It is a heterogeneous mixture. Mixtures are impure substance. Milk is a colloid.
(c) Iron: Pure substance (It is an elements)
(d) Hydrochloric acid: It is an impure substance. It is a homogeneous mixture. It is a solution of gas in a liquid.
(e) Calcium oxide: It is a pure substance (It is a compound)
(f) Mercury: It is a pure substance (It is an elements)
(g) Brick: Heterogeneous mixture so impure substance.
(h) Wood: Heterogeneous mixture so impure substance.
(i) Air: It is homogeneous mixture so impure substance.
8. (a) Not a solution. It is not a homogeneous mixture
(b) Sea water is a solution. It is a homogeneous mixture.
(c) Air is a solution. It is a homogeneous mixture.
(d) Coal is not a homogeneous mixture so it is not a solution.
(e) Soda water is a homogeneous mixture so it is a solution.
[A solution is a homogeneous mixture]
9. “Tyndall effect” is shown by colloidal solution. So, it will be shown by milk, starch sol and not by salt solution, copper sulphate solution because they are true solutions and not colloidal solutions.

10. (a) element
(b) mixture (heterogeneous mixture)
(c) mixture (homogeneous mixture)
(d) element
(e) compound (CaCO₃)
(f) element
(g) element
(h) mixture (heterogeneous mixture)
(i) mixture (homogeneous mixture)
(j) mixture
(k) Compound (CH₄)
(l) Compound (CO₂)
(m) mixture
11. (a) It is a chemical change.
(b) It is a chemical change
(c) It is a physical change
(d) It is a chemical change
(e) It is a chemical change
(f) It is a physical change
(g) It is a chemical change.

Exemplar Questions :

1. The presence of beads in tube (I) would provide a larger surface area for cooling.
2. It is a pure substance because chemical composition of sugar crystals is same irrespective of its source.
3. Particle size in a suspension is larger than those in a colloidal solution. Also molecular interaction in a suspension is not strong enough to keep the particles suspended and hence they settle down.
4. (a) No.

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solute} + \text{Mass of solvent}} \times 100$$

- (b) Solution made by Ramesh

$$\text{Mass \%} = \left(\frac{10}{10 + 100} \right) 100$$

$$= \frac{10}{110} \times 100 = 9.09\%$$

Solution made by Sarika

$$\text{Mass \%} = \frac{10}{110} \times 100 = 10\%$$

The solution prepared by Sarika has a higher mass % than that prepared by Ramesh.

5. Distillation, since acetone is more volatile it will separate out first.

HOTS Questions :

- To confirm whether the colourless liquid given is pure or not? We can proceed as follows:
 - Filter the given liquid using a very fine filter paper to see if it contains any suspended impurities. If any thing is left on filter paper the given liquid is impure. Otherwise it is pure.
 - Evaporate a part of the given liquid in a china dish. If any residue is left it is impure and in case no residue is left it is pure.
 - Determine the b.p. of the given liquid. If the b.p. is 100°C (373 K) it is pure water.
[Note: B.P. of pure water is 373 K at 1atmospheric pressure]
- Chromatography.
 - Crystallisation
 - Decantation
 - Evaporation
 - Fractional distillation
- Ice: Pure substance* (It is pure water in solid state) It is a component.
 - Milk: It is a heterogeneous mixture. Mixtures are impure substance.*
 - Iron: Pure substance* (It is an element)
 - Hydrochloric acid: It is an impure substance. It is a homogeneous mixture. It is a solution of gas in a liquid.*
 - Calcium oxide: It is a pure substance.* (It is a compound)
 - Mercury: It is a pure substance.* (It is an element)
 - Brick: Heterogeneous mixture so impure substance.*
 - Wood: Heterogeneous mixture so impure substance.*
 - Air: It is homogeneous mixture so impure substance.*
- Ice is less dense than water. An ice water mixture is pure but heterogeneous in nature.
- Heterogeneous mixture
 - Homogeneous mixture
 - Pure substance
 - Homogeneous mixture.
- a solution (a solid dissolved in a liquid)
 - heterogeneous mixture (probably a solid suspended in a liquid such as dirty water)
 - element
 - compound
 - solution (two liquids)
- C is a compound; it contains carbon and oxygen. A is a compound as it decomposes to give B which contains carbon and oxygen and C. B is not defined by the data given; it is probably a compound because few elements exist as white solids.
- All mixtures of gases are homogeneous. In case of liquids similar liquid molecules form homogeneous mixtures, while very dissimilar molecules form heterogeneous mixtures.
- Ethyl alcohol and water mixture is considered as homogeneous mixture because in such a mixture there is no boundary of separation between two components (i.e., ethyl alcohol and water) and it exists in a single phase.
In case of oil and water mixture we can see the boundary of separation between oil and water and there are two separate layers, one of water and other of oil is formed thus it is heterogeneous mixture.
- Alloys contain two or more metals and have a variable composition so they are homogeneous mixtures. e.g., brass contains copper and zinc. Copper is a crystalline solid i.e. in copper atoms are closely packed to form a crystal lattice. When brass is formed some of these atoms of copper have been replaced by atoms of zinc (i.e., zinc atoms have substituted some copper atoms) and due to this brass is considered as *substitutional solid solution*.
- The fractionating column obstructs the upward movement of vapours of the liquids and helps in absorptions of heat energy released by high boiling liquid to be taken up by low boiling liquid. Due to this the lower boiling liquid remains in vapour phase and the high boiling liquid condenses back to liquid phase and this condensed liquid falls back in the distillation flask. In this way fractionating column helps in the separation of two liquids.
- C is a compound
Both the above observations are observed during formation of a compound.
- No, only elements and compounds are pure substance. Air is a mixture and so not pure substance.
- A compound is a single substance with a fixed composition and so it fulfills the condition required for a pure substance. The m.p. or b.p. of a compound is fixed. These conditions are not fulfilled by a mixture so it is not considered as a pure substance.
- During chemical change substances may undergo decomposition or addition that is they themselves change. Therefore, separation of mixture by taking advantage of difference in their chemical properties is less convenient.
- The particles of iron can be separated by magnet.
- Both fog and clouds are *liquid in gas* type of colloids. They differ in the fact that clouds are formed in the upper atmosphere but fog is formed in the region of atmosphere that is close to earth.
- Particle size in a suspension is larger than those in a colloidal solution. Also molecular interaction in a suspension is not strong enough to keep the particles suspended and hence they settle down.
- Milk is a colloid and would show Tyndall effect.
 - Salt solution is a true solution and would not scatter light.
 - Detergent solution, sulphur solution.

20. The process of dissolution of sodium chloride in water is an *endothermic process* (i.e. energy is absorbed in the process). In such a process solubility increases with temperature.

The process of dissolution of lithium carbonate in water is an *exothermic process* (i.e. heat is evolved in the process). In such a process the solubility of solute decreases with increase in temperature.

21. (a) No, the solubility do not increases uniformly in every case.
(b) (i) Potassium nitrate
(ii) Potassium chlorate
(iii) Sodium chloride
(iv) Calcium acetate.

22. This solution contains a liquid solute (alcohol) mixed with a liquid solvent (water), so we have to calculate the concentration of this solution in terms of volume percentage of solute (alcohol). Now, we know that :

$$\text{Concentration of solution} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

Here, Volume of solute (alcohol) = 50 mL

And, Volume of solvent (water) = 150 mL

So, Volume of solution = Volume of solute + Volume of solvent = 50 + 150 = 200 mL

Now, putting these values of 'volume of solute' and 'volume of solution' in the above formula we get :

$$\text{Concentration of solution} = \frac{50}{200} \times 100 = \frac{50}{2} = 25\%$$

23. Here we have been given that 75 grams of water dissolves 12 grams of potassium sulphate. We have to find how much potassium sulphate will dissolve in 100 grams of water. Now,

75g of water dissolves = 12g of potassium sulphate

So, 100g of water will dissolve = $\frac{12}{75} \times 100$ g of potassium sulphate = 16g of potassium sulphate

Thus, the solubility of potassium sulphate in water is 16g at 60°C.

24. Generally colloids can not be separated by filtration since the size of the dispersed particles is smaller than that of the pores present in the filter paper.

25. Volume of water = $\frac{\text{mass}}{\text{density}} = \frac{56\text{g}}{1.0\text{gcm}^{-3}} = 56\text{ cm}^3$
= 56 mL

Volume of ethanol = 0.17 L = 0.17 × 1000 mL = 170 mL

∴ Volume of solution = (56 + 170) mL = 226 mL

∴ Concentration percent by volume

$$= \frac{56}{226} \times 100 = 24.78\%$$

3 EXERCISE

Multiple Choice Questions :

- (d) Air is a mixture of different gases like N₂, O₂, CO₂ etc. Its general composition is N₂ = 78%, O₂ = 21% and traces of few other gases but there may be variation in its composition from place to place and at different height.
- (d) Hydrogen sulphide (H₂S) is a compound of hydrogen and sulphur stainless steel and bronze are alloys whereas graphite is allotropic form of element carbon.
- (c) As oil being less denser than water it forms upper layer. Thus mixture of oil and water can be separated by using separating funnel.
- (a)
- (b) Sulphur will left behind. As in given mixture only sulphur gets dissolved in carbon disulphide.
- (a)
- (d) Camphor being a sublime substance can be purified by sublimation.
- (d)
- (b) Every pure solid has a fixed melting point.
- (c) Mixture of ethanol and water can be separated by fractional distillation as they have different boiling points.
- (c) Sea water is a solution of salt and water. During evaporation water gets evaporated off and salt left as a residue.
- (d) Homogeneous mixture is a solution having uniform composition and properties throughout.
- (b)
- (b) Mixture of methyl alcohol and acetone can be separated by fractional distillation as there is only small difference between boiling points of methyl alcohol and acetone.
- (d) Sulphur gets dissolved in CS₂ and separates out as a filtrate. While sand gets collected on filter paper as residue.
- (b)
- (b) Sodium chloride being compound is a pure substance. Granite, muddy water and milk of magnesia all are mixtures.
- (d) Physical properties of mixtures are same as of its components.
- (d)
- (a) As they have uniform composition throughout they are considered as homogeneous mixture. Both samples are mixture of two metals (gold and palladium) thus are alloys.
- (c)
- (d) Candle is a heterogeneous mixture of wax and threads. Copper is element while bottled water and table salt are compounds.

23. (b) It is a compound.
 24. (d) Homogeneous liquid have a particular boiling point.
 25. (a) 26. (b)
 27. (c) Chromatography is based on the difference in adsorption of different substances on the surface of a solid medium. This method is frequently used to separate coloured mixture at commercial scale also in laboratories.
 28. (a) Different pure solid substances melts at a different temperatures.
 29. (b) Distillation is a separation techniques used for separation of miscible liquids having different boiling point.
 30. (c) Different salts have different solubility in a particular solvent. Thus on this basis mixture of different salts can be separated.
 31. (b) Magnetism is useful for separation of magnetic and non-magnetic substances.
 32. (a) Both boron and silicon are metalloids.
 33. (c)
 34. (c)
 35. (b) Since no new substance was formed it is physical change.
 36. (b) The process in statement (a) is a physical change as the liquid changes to a gaseous state and no new substance is formed.
 In statement (b), LPG burns to form a new substance so it is a chemical change.
 37. (b) Copper sulphate (blue) reacts with iron (in shaving blade) to form iron sulphate (green).
 38. (a) Eggs will rotten to give foul smell. Bottle will remain as such whereas moth balls will sublime which is a physical change.
 39. (c) Those substances which can not pass through membrane are termed as colloids eg silicic acid. Whereas sugar solution, urea and NaCl can pass through the membrane.
 40. (c) Smoke consists of carbon particles (solid) dispersed in air (gas).
 41. (b) Crystalloids can easily be obtained in the crystalline form and their solution can diffuse rapidly through a vegetable or animal membrane [i.e., parchment membrane]; e.g. sugar, urea, salts, acids, etc.
 42. (b) Cloud consists of fine droplets of water suspended in air.
 43. (a) Scattering of light from the surface of colloidal particles—tyndall effect is observed. When a colloidal solution is observed under an ultramicroscope.
 44. (c) Size of particles in colloidal state ranges from 10 to 1000 Å
 Size of particles $\xrightarrow{\text{True solution}}$ 10Å $\xrightarrow{\text{particles in colloidal state}}$ 1000Å $\xrightarrow{\text{Suspension}}$
 45. (a) Fog is a colloidal system consisting water droplets dispersed in air.
 46. (c) Starch molecules have colloidal dimensions whereas NaCl, glucose and $\text{Ba}(\text{NO}_3)_2$ are crystalloids and soluble in water.
 47. (c) Collidal solutions can be purified by dialysis. It may be defined as the process of separating a crystalloid from a colloid by diffusion or filtration through a fine membrane. The process of dialysis can be quickened by using hot water (hot dialysis) or by applying an electric field (electrodialysis).
 48. (c) Sol is a colloidal solution so, it will show Tyndall effect.
 49. (c) When the dispersed phase and dispersion medium both are liquid, the colloidal system is called as an emulsion like milk, vashishing cream, etc.
 Option (c) is correct.
 50. (a) Butter is obtained from milk which itself is an emulsion of fat dispersed in water, reverse is true for butter.
 Option (a) is correct.
 51. (d)
 52. (b) *Latex* is a negative sol. *Latex* is an *emulsion* in which the rubber particles forming emulsion are stabilized by various proteins.
 53. (b)
 54. (b)
 55. (a) Robert Brown, an English botanist observed that the particles in colloidal solution were in constant random motion, traversing no definite set paths, but travelling in zig-zag directions all over the field of view. This movement of colloidal particles is called Brownian movement.
 56. (a) When a beam of light is passed through a colloidal solution, the path of the beam is illuminated and becomes visible as a blue bright streak when seen in a direction perpendicular to the incident beam. This phenomenon is called as Tyndall Effect. This phenomenon is due to scattering of light by the sol particles.

More than One Option Correct :

1. (a, d) Mixtures are not pure substances thus composition of its components vary in different proportion. Components of a mixture can be separated easily by physical method like filtration, magnetism, solubility, distillation, etc.
 2. (b, c, d)
 3. (b, c) Alloys are homogeneous mixture of metals. Brass is an alloy of copper and zinc whereas bronze is an alloy of copper and tin.
 4. (a, b, d) Diamond is an allotrope of element carbon where as rest all are compounds.
 5. (a, b, c) Ammonia is a compound whereas solution, alloy and amalgam are mixtures.
 6. (a,b,c) Deionized water is a compound with fixed composition of its substituents whereas rest all are mixtures.
 7. (a,b,c) A homogeneous mixture must have uniform composition and properties.

Is Matter Around us Pure

8. (b, d) Mixtures have same physical properties as of their components and can be separated into its components by various physical methods.
9. (a, d) Glass and deionized water have uniform composition throughout.
10. (b, c) As it is clear from given figure that sample B and C contains mixture with uniform composition.
11. (a, b, c) Diagram I, II and III represents heterogeneous mixture with non-uniform composition and properties.
12. (b, c)
13. (b, c, d) Burning of magnesium ribbon is an exothermic reaction. Ash formed on burning of magnesium ribbon is magnesium oxide (MgO)
14. (a, b, d) There are fewer non metals out of all the elements known.
15. (a, d)
16. (b, d) Starch molecules have colloidal dimensions. Blood is a -vely charged colloidal system. Rest of the compounds, i.e., NaCl, urea & cane sugar form true solution in water.
17. (b, c, d) Chlorophyll. Smoke is an example of solid-gas colloid system Ruby glass is an example of solid-solid colloid system. Milk is an liquid-liquid colloid system.
18. (a, d) Suspension particles are visible under a microscope and sometimes even to a naked eye.
19. (a, b, c) Air is a homogeneous mixture of gases, mainly nitrogen and oxygen.
20. (a, b) A collidal solution is biphasic and heterogeneous. It is composed of two phases :
1. Dispersed phase; 2. Dispersion medium.
21. (a, c, d) Scattering of light (Tyndall effect) is exhibited by a colloidal solution. Among the given solutions starch solution is a colloid whereas other are example of true solution.
2. (b)
3. (b) As different fractions of petroleum have very less difference in their boiling points. Thus refining of petroleum is done via fractional distillation.
4. (c) Essential oils and other oils are separated by method of steam distillation. The compounds which decompose at their boiling points are purified by vaccum distillation.
5. (a) They possess some properties of metals and some properties of non-metals.
6. (b) Hydrogen, oxygen and nitrogen are elements as they all are made up of similar kind of atoms only.
7. (b) Ozone absorbs the harmful radiations of the sun.
8. (a) This is according to Henry's law which states that the solubility of a gas in given volume of a liquid at a particular temperature is directly proportional to the pressure of gas above the liquid. $m \propto p$, $m = kp$ where k = Henry's constant.
9. (b) The size of the colloidal particles is large enough to scatter light while particles of a true solution are too small to scatter light.
10. (b) The particles of the colloids are electrically charged and carry positive or negative charge but colloidal solution as whole is neutral. The dispersion medium has an equal and opposite charge making the system neutral as a whole.

Multiple Matching Questions :

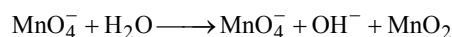
1. (a) (A)→(q, s); (B)→(p, r); (C)→(p); (D)→(s)
2. (a) (A)→(p, q); (B)→(r, s); (C)→(s); (D)→(p, r)
3. (b) (A)→(r, s); (B)→(p, s); (C)→(s, r); (D)→(p, q)
4. (a) (A)→(r); (B)→(s); (C)→(p); (D)→(q)

Integer Based Questions :

1. 3
Germanium, Antimony and Arsenic are metalloids.
2. 3
Whipped cream, blood, starch solution have liquid dispersion medium.

4 EXERCISE

1. (d) Molality = $\frac{18 \times 1000}{180 \times 1000} = 0.1$.
2. (a) In alkaline medium the reduction of KMnO_4 takes place as



$$\text{Eq. mass of KMnO}_4 = \frac{\text{Mol. mass}}{1} = 158$$

Wt. of KMnO_4 needed

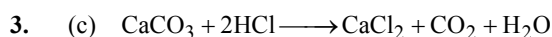
$$= \frac{0.1 \times 100}{1000} \times 158 = 1.58 \text{ g.}$$

Passage Based Questions :

1. (a) The constant temperature at point B indicates boiling point of first liquid out of two.
2. (c) When first component of mixture is boiled off the temperature of second component starts increasing.
3. (a)
4. (b) Cheese has liquid dispersed phase and solid dispersion medium.
5. (a) Salt solution is a true solution. Starch solution and milk are colloids.
6. (c) Face cream is an emulsion therefore has liquid as both dispersed phase and dispersion medium.

Assertion & Reason :

1. (c) Sea diver must be supplied air at a pressure equal to that of water surrounding the upper limit of oxygen pressure above which the oxygen becomes biochemically toxic. So, it is necessary to include something to dilute the oxygen in a diver's breathing gas by a non-toxic diluent N_2 , He, etc.



$$E_{\text{CaCO}_3} = \frac{\text{Mol. mass}}{2} = 50$$

$$N \times V = \frac{1}{50} \quad \text{or,} \quad \frac{50 \times N}{1000} = \frac{1}{50};$$

$$\text{or } N = 0.4$$

4. (b) Molality = $\text{mol kg}^{-1} = \frac{5.85}{58.5} = 0.1$

5. (a) 1 l of water weight 1 kg (at 4°C).

Hence, molarity of pure water

$$= \frac{1000}{18} = 55.55$$

6. (c) Molarity = $\frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$

Number of moles of hydroxide (OH^-) ions

= Molarity of $\text{OH}^- \times$ Volume of solutions (l)

$$= 2 \times 0.005 \times 0.3 = 0.003.$$

7. (c) $n_{\text{H}_2\text{O}} = \frac{36}{18} = 2;$

$$n_{\text{glycerine}} = \frac{46}{92} = 0.5$$

$$\text{Hence, } x_{\text{glycerine}} = \frac{0.5}{0.5 + 2} = 0.2$$

8. (b) Molality of a solution is the number of moles of solute dissolved in 1000 g or 1 kg of solvent. It is independent of volume term. Therefore, it is not affected by rise or fall of temperature.

Hence, option (b) is correct.

9. (b) Molarity of a solution (M) = 2 M
Volume of a solution (V) = 150 ml
Molecular weight of CH_3OH = 32
Molarity of a solution (M)

$$= \frac{\text{No. of moles of solute}}{\text{Volume of solution in litres}}$$

$$= \frac{\frac{w}{m}}{\frac{V}{1000}} = n_{\text{H}_2\text{O}} = \frac{36}{18} = 2;$$

According to question,

$$w = \frac{M \times m \times V}{1000}$$

$$= \frac{2 \times 32 \times 150}{1000} = 9.6 \text{ gm.}$$

Hence, 9.6 gm of CH_3OH would have to be added to water.

10. (c) Molality is the number of moles of solute present in 1000 gm or 1 kg of solvent.
Therefore, option (c) is correct.

11. (c) The solution which has one gram equivalent in one litre is called normal solution.

12. (b) The stability of a lyophilic sol is due to both the charge and solvation of particles, but largely due to the solvation factor. As in lyophilic colloids particles of dispersed phase has more affinity for dispersion medium.

13. (d) Since, oil is the inner (dispersed) phase, the added dye will not be dissolved in the emulsion.

14. (a, b) Gelatin is a lyophilic solution (lyo-liquid philic -love). Lyophilic colloids are also known as reversible colloids since on evaporating the dispersion medium (i.e., water), the residue can again be easily reconverted into colloidal state simply by addition of the liquid i.e., water.

15. (a, b, d) The stability is attributed to electric charge on the sol particles.

16. (a, b, c) Hydrophilic sols have extra stability due to both the charge on the particles and their solvation.

17. (a) Milk is an emulsion in which small drops of liquid fat are dispersed in water.

18. (c) Cod-liver oil is an emulsion in which particles of water are dispersed in oil.

19. (d) Soap, gelatin and other hydrophilic sols are useful emulsifying agents.

20. (a) If the dispersion medium is separated from the dispersed phase, the lyophilic sol can be obtained by simply remixing the disperse phase with the dispersion medium. That is why these sols are also called reversible sols.

21. (d) Milk is an oil in water emulsion i.e., liquid fat is dispersed in water. Emulsions are colloids in which both dispersed phase and dispersion medium are liquids.

22. (d) Soaps and detergents are colloidal electrolytes and can not be regarded as macromolecules as they have individual molecules of very big size.

23. (b) (A)-(s), (B)-(p), (C)-(q), (D)-(r)

24. (c) (A)-(q, r), (B)-(r), (C)-(q), (D)-(p, r, s)

Micelles are associated colloids. They have both a lyophilic and lyophobic part.

25. 3

Molarity of KCl (m)

$$= \frac{\text{No. of moles of solute}}{\text{volume of solution in litre}}$$

$$3 = \frac{\text{No. of moles of solute KCl}}{1 \text{ litre}}$$

$$\text{No. of moles of (KCl)} = 3$$

26. 2

$$\therefore \text{Weight of } \text{H}_2\text{O}_2 \text{ in 100 ml of } \text{H}_2\text{O}_2 \text{ solution} = 6.8 \text{ gm}$$

$$\therefore \text{Weight of } \text{H}_2\text{O}_2 \text{ in 1000 ml of its solution} = 6.8 \times 10 = 68 \text{ g}$$

$$\text{Molecular weight of } \text{H}_2\text{O}_2 = 34$$

$$\text{Then, Molarity} = 2 \text{ M}$$

Atoms and Molecules

INTRODUCTION

The smallest unit of any element is called an atom. The atoms of different elements combine with one another to form new substances called compounds. The compounds so formed are neutral in character, i.e., they have no electric charges on them. All matter is made by the combination of atoms of different elements combined together in some fixed ratio. In a way, atoms are the basic building blocks of the matter.

There is a beautiful description of atomic theory in the treatise “Vaisheshik Darshan” written by Kanad, in which he has proposed that the whole universe is created by atoms. According to this theory when atoms are separated from each other then there is a state of catastrophe and when they combine with each other to form different substances then creation occurs. By the late 1700s, scientists began to realize that the concept of atoms provided an explanation for many experimental observations about the nature of matter.

By the early 1800s, the law of Conservation of Matter and the Law of Definite Proportions were both accepted as general descriptions of how matter behaves. John Dalton, an English school teacher, tried to explain why matter behaves in such simple and systematic ways as those expressed above. In 1808, he published the first “modern” ideas about the existence and nature of atoms. He summarized and expanded the fabulous concepts of early philosophers and scientists, more importantly, his ideas were based on reproducible experimental results of measurements by many scientists.

All matter is made up of tiny atoms, so how do we get larger substances, like water, sugar, or iron? These very small atoms can bond together into bigger compounds, either ions or molecules. Molecules have different shapes, depending on the types of atoms bonded together.

LAWS OF CHEMICAL COMBINATION

One of the most important aspects of the subject of chemistry is the study of chemical reactions. These chemical reactions take place according to the certain laws, called the 'laws of chemical combination'.

- (A) Law of conservation of mass (B) Law of constant proportions (C) Law of multiple proportions
(D) Law of reciprocal proportions (E) Law of combining volumes (Gas Lussac's Law of Gaseous volumes)

The laws of chemical combination are the experimental laws which led to the idea of atoms being the smallest unit of matter. The laws of chemical combination played a significant role in the development of Dalton's atomic theory of matter.

The first four laws deal with the mass relationships whereas the fifth law deals with the volumes of the reacting gases.

(A) Law of Conservation of Mass or Matter :

This law was given by Antoine Lavoisier. The law of conservation of mass means that in a chemical reaction, the total mass of products is equal to the total mass of the reactants. There is no change in mass during a chemical reaction. Suppose we carry out a chemical reaction between A and B. If the products formed are C and D then.

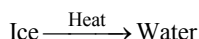


$$\text{Mass of (A + B) = Mass of (C + D)}$$

or mass of reactants = mass of products.

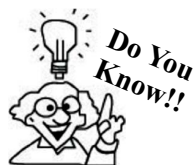
Example :

- (i) A piece of ice (solid water) is taken in a small conical flask. It is well corked and weighed. The flask is now heated gently to melt the ice (solid) into water (liquid).



The flask is again weighed. It is found that there is no change in the weight though a physical change has taken place.

- (ii) When we burn a sample of metallic magnesium in the air, the magnesium combines with oxygen from the air to form magnesium oxide, a white powder. This chemical reaction is accompanied by the release of large amounts of heat energy and light energy. When we weigh the product of the reaction, magnesium oxide, we find that it is heavier than the original piece of magnesium. The increase in mass of the solid is due to the combination of oxygen with magnesium to form magnesium oxide. Now when we conduct same experiment in a closed container we find that now the mass of the magnesium oxide is exactly the sum of the masses of magnesium and oxygen that combined to form it. Similar statements can be made for all chemical reactions.



Thus according to this law, there is no increase or decrease in the total mass of matter during a chemical or a physical change.

ILLUSTRATION : 1

90 g of KClO_3 when heated produce 1.92 g of oxygen and the residue (KCl) left behind weighs 2.96 g. Show that these results illustrate the law of conservation of mass.

SOLUTION :

Mass of KClO_3 taken = 4.90 g

Total mass of the products ($\text{KCl} + \text{O}_2$)

$$= 2.96 + 1.92 = 4.88 \text{ g}$$

Difference between the mass of the reactant and the total mass of the products

$$= 4.90 - 4.88$$

$$= 0.02 \text{ g}$$

This small difference may be due to experimental error.

Thus, law of conservation of mass holds good within experimental errors.

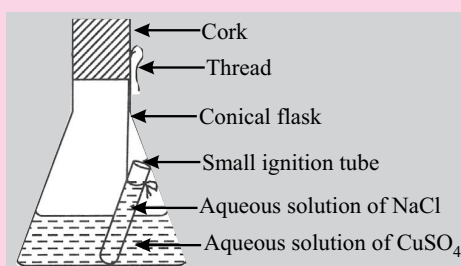


idea box

From your school chemistry lab collect the following apparatus

- Ignition tube • Conical flask • Thread • Rubber cork
- Aqueous solution of NaCl and CuSO₄

By using above apparatus make an arrangement as shown in figure below



Now first weigh the apparatus shown in figure and record your observation then tilt the ignition tube containing aqueous NaCl. Swirl the conical flask slowly for few minutes so that solution of both salts get mixed properly. After mixing weigh apparatus again record your observation. On the basis of your observation draw a conclusion.

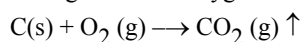
(B) Law of Constant (Definite) Proportion

Most elements can interact with other elements to form compounds. Hydrogen gas, for example, burns in oxygen gas to form water. Conversely, water can be decomposed into its component elements by passing an electrical current through it. Pure water, regardless of its source, consists of 11% hydrogen and 89% oxygen by mass. This macroscopic composition corresponds to the molecular composition, which consists two hydrogen atoms combined with one oxygen atom.

The observation that the element composition of a pure compound is always the same is known as the law of constant composition (or the laws of definite proportion). It was first put forth by the French chemist Joseph Louis Proust (1754-1826) in 1799. Although this law has been known for 200 years, the general belief persists among some people that a fundamental difference exists between compounds prepared in the laboratory and the corresponding compounds found in nature. However, a pure compound has the same composition and properties regardless of its source. Both chemists and nature must use the same elements and operate under the same natural laws. When two materials differ in composition and properties, we know that they are composed of different compounds or that they differ in purity.

For example, carbon dioxide can be obtained by any of the following chemical reactions.

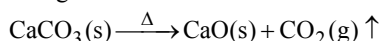
(i) by burning carbon in oxygen



(ii) by heating sodium bicarbonate



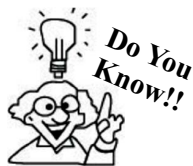
(iii) by heating calcium carbonate



(iv) by reacting calcium carbonate with hydrochloric acid



whatever sample of carbon dioxide is taken. It is found that the ratio of carbon and oxygen is always same which is 12 : 32 or 3 : 8.



When same elements combine in the same proportion it is not necessary that same compound will be formed. Sometimes different compounds under different experimental conditions are formed e.g., combination of carbon, hydrogen and oxygen in the ratio 12 : 3 : 8 may give either ethyl alcohol (C_2H_5OH) or dimethyl ether (CH_3OCH_3) under different experimental conditions.

ILLUSTRATION : 2

Copper oxide was prepared by the following methods:

- In one case, 1.75 g of the metal were dissolved in nitric acid and igniting the residual copper nitrate yielded 2.19 g of copper oxide.
- In the second case, 1.14 g of metal dissolved in nitric acid were precipitated as copper hydroxide by adding caustic alkali solution. The precipitated copper hydroxide after washing, drying and heating yielded 1.43 g of copper oxide.
- In the third case, 1.45 g of copper when strongly heated in a current of air yielded 1.83 g of copper oxide. Show that the given data illustrate the law of constant composition.

SOLUTION :

In the first experiment.

2.19 g of copper oxide contained 1.75 g of Cu.

∴ 100 g of copper oxide contained

$$= \frac{1.75}{2.19} \times 100 = 79.91 \text{ g}$$

In the second experiment.

1.43 g of copper oxide contained 1.14 g of copper

∴ 100 g of copper oxide contained

$$= \frac{1.14}{1.43} \times 100 = 79.72 \text{ g}$$

In the third experiment.

1.83 g of copper oxide contained 1.46 g of copper

∴ 100 g of copper oxide contained

$$= \frac{1.46}{1.83} \times 100 \\ = 79.78 \text{ g}$$

Thus, the percentage of copper in copper oxide derived from all the three experiments is nearly the same.

Hence, the above data illustrate the law of constant composition.

CHECK Point

- ☛ Is the law of constant composition true for all types of compounds? Explain why or why not.

SOLUTION

No, law of constant composition is not true for all types of compounds. It is true only for the compounds obtained from one isotope. For example, carbon exists in two common isotopes, ^{12}C and ^{14}C . When it forms CO_2 from ^{12}C , the ratio of masses is 12 : 32 = 3 : 8 but from ^{14}C , the ratio will be 14 : 32 = 7 : 16 which is not same as in the first case.

CONNECTING TOPIC

In addition to these two laws there are some other laws, of chemical combination.

(i) Law of Multiple Proportions :

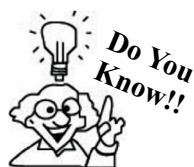
Whenever two elements form more than one compound, the different masses of one element that combine with the same mass of the other element are in the ratio of small whole numbers. This law was given by Dalton in 1804.

For example, sulfur and oxygen form two different compounds which we call sulfur dioxide and sulfur trioxide. If we decompose a 2.00g sample of sulfur dioxide, we find it contains 1.00g of S and 1.00 g of O. If we decompose a 2.50g sample of sulfur trioxide, we find it also contains 1.00g of S, but this time the mass of O is 1.50g. This is summarized in the following table.

Compound	Sample mass	Mass of sulfur	Mass of oxygen
Sulfur dioxide	2.00 g	1.00 g	1.00 g
Sulfur trioxide	2.50 g	1.00 g	1.50 g

First, notice that sample masses aren't the same, they were chosen so that each has the same mass of sulfur. Second, the ratio of the masses of oxygen in the two samples is one of small whole numbers.

$$\frac{\text{mass of oxygen in sulfur trioxide}}{\text{mass of oxygen in sulfur dioxide}} = \frac{1.50\text{g}}{1.00\text{g}} = \frac{3}{2}$$



Similarly, N and O form as many five stable oxides, N_2O , NO , N_2O_3 , N_2O_4 and N_2O_5 . In these oxides, amount of oxygen which react with 28 g of N_2 are in the ratio 16 : 32 : 48 : 64 : 80 i.e., 1 : 2 : 3 : 4 : 5.

Similar observations are made when we study other elements that form more than one compound with each other, and these observations form the basis of the law of multiple proportions.

ILLUSTRATION : 3

The following data were collected for several compounds of nitrogen and oxygen :

Mass of nitrogen that combines with 1g of oxygen

Compound I 1.750g

Compound II 0.8750g

Compound III 0.4375g

Show how these data illustrate the law of multiple proportions.

SOLUTION :

For the law of multiple proportions to hold true, the ratios of the masses of nitrogen combining with 1 gram of oxygen in each pair of compounds should be small whole numbers. We therefore get the ratios as follows :

$$\frac{\text{I}}{\text{II}} = \frac{1.750}{0.875} = \frac{2}{1}$$

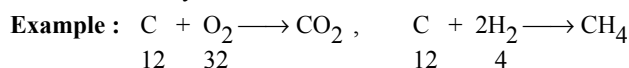
$$\frac{\text{II}}{\text{III}} = \frac{0.875}{0.4375} = \frac{2}{1}$$

$$\frac{\text{I}}{\text{III}} = \frac{1.750}{0.4375} = \frac{4}{1}$$

These results support the law of multiple proportions.

(ii) The Law of Reciprocal Proportions or Law of Equivalent Proportion :

This law was given by "Richer" in 1792–94. The law states that the ratio of the masses of two elements A and B which combine separately with a fixed mass of the third element C is either the same or some simple multiple of the ratio of the masses in which A and B combine directly with each other.



C combines with O to form CO_2 and with H to form CH_4 . In CO_2 , 12 gm of C reacts with 32 gm of O, whereas in CH_4 , 12 gm of C reacts with 4 gm of H. Therefore when O combines with H, they should combine in the ratio of 32 : 4 (i.e. 8 : 1) or in simple multiple of it. The same ratio is found to be true in H_2O molecules, the ratio of weight of H and O in H_2O is 1 : 8.

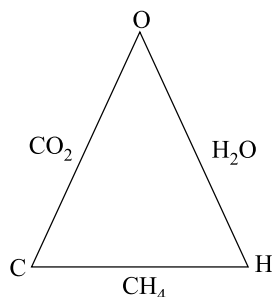


ILLUSTRATION : 4

Ammonia contains 82.35% of nitrogen and 17.65% of hydrogen. Water contains 88.90% of oxygen and 11.10% of hydrogen. Nitrogen trioxide contains 63.15% of oxygen and 36.85% of nitrogen. Show that these data illustrate the law of reciprocal proportions.

SOLUTION :

In NH_3 , 17.65 g of H combine with N = 82.35 g

$$\therefore 1 \text{ g of H combine with N} = \frac{82.35}{17.65} \text{ g} = 4.67 \text{ g}$$

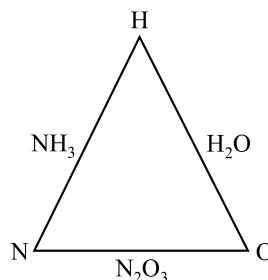
In H_2O , 11.10 g of H combine with O = 88.90 g

$$\therefore 1 \text{ g of H combine with O} = \frac{88.90}{11.10} \text{ g} = 8.01 \text{ g}$$

$\therefore$ Ratio of the masses of N and O which combine with fixed mass (=1g) of H = 4.67 : 8.01 = 1 : 1.72

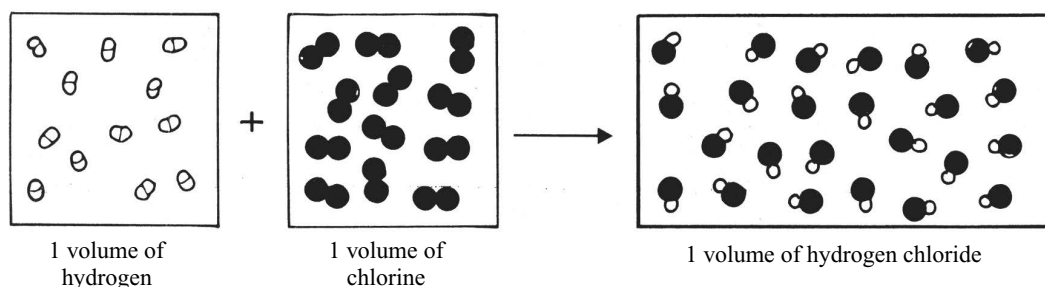
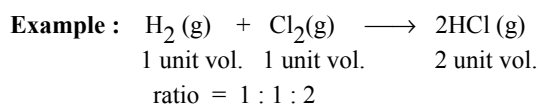
In N_2O_3 , ratio of masses of N and O which combine with each other = 36.85 : 63.15 = 1 : 1.72

Since, the two ratios are the same. Hence, it illustrates the law of reciprocal proportions.

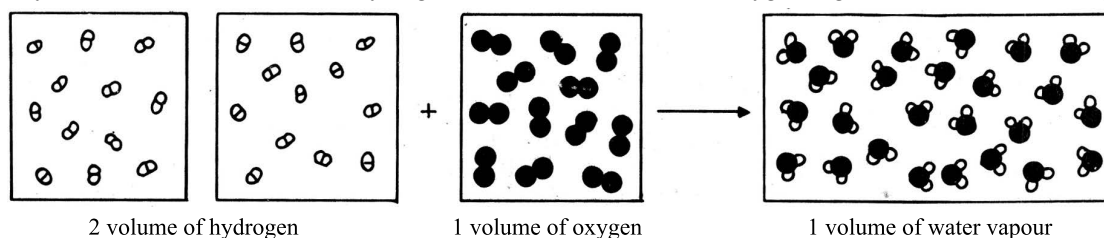


(iii) The Law of Gaseous Volume :

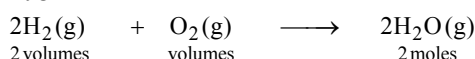
This law was given by “Gay lussac” in 1808. The law states that “Whenever gases react together, the volumes of the reacting gases as well as the products if they are gases, bear a simple whole number ratio, provided all the volumes are measured under similar conditions of temperature & pressure.



Similarly, we observe that 2 volumes of hydrogen combine with 1 volume of oxygen to give 2 volumes of water vapour as



Thus, the volumes of hydrogen and oxygen which combine to form water is in the ratio : 2 : 1 : 2.



DALTON'S ATOMIC THEORY

To describe the structure of matter which could explain the experimental facts known at that time about elements, compounds and mixtures and also the laws of chemical combination, John Dalton in 1808 put forward a theory known as Dalton's atomic theory. The main points of this theory are as follows:

- (i) Each element is composed of extremely small particles called atoms.
- (ii) All atoms of a given element are identical, the atoms of different elements are different and have different properties (including different masses).
- (iii) Atoms are neither created nor destroyed in chemical reactions.
- (iv) Compounds are formed when atoms of more than one element combine. A given compound always has the same relative number and kind of atoms.
- (v) When atoms combine with one another to form compound, they do so in simple whole number ratios, such as 1 : 1, 2 : 1, 2 : 3 and so on.
- (vi) Atoms of two elements may combine in different ratios to form more than one compound. For example, sulphur combines with oxygen to form sulphur dioxide and sulphur trioxide, the combining ratios being 1 : 2 and 1 : 3 respectively.
- (vii) An atom is the smallest particle that takes part in a chemical reaction.

Knowledge ENHANCER

Dalton's atomic theory provides a simple explanation for the laws of chemical combination and was the first modern attempt to describe the behaviour of matter in terms of atoms. For example, the postulate of Dalton's atomic theory that "atoms can neither be created nor destroyed" was a result of the law of conservation of mass. And the postulates of Dalton's atomic theory that "the elements consist of atoms having fixed mass, and that the number and kind of atoms in a given compound is fixed," came from the law of constant proportions. Dalton used his theory to deduce the law of multiple proportions.

Law of Reciprocal Proportions. As atoms combine with each other in simple ratio, therefore all the ratios involved are simple which may be same or some simple multiple of each other.

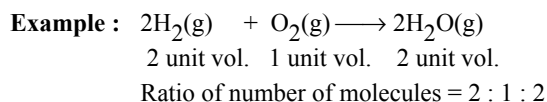
Drawbacks of Dalton's Atomic Theory :

Some of the drawbacks of the Dalton's atomic theory of matter are given below :

- (i) According to Dalton's atomic theory, atoms were thought to be indivisible. But, it is now known that under circumstances, atoms can be further divided into still smaller particles called electrons, protons and neutrons.
- (ii) Dalton's atomic theory said that all the atoms of an element have exactly the same mass. But, it is now known that atoms of the same element can have slightly different masses.
- (iii) Dalton's atomic theory said that atoms of different elements have different masses. But, it is now known that even atoms of different elements can have the same mass.
- (iv) It could explain the laws of chemical combination by mass but failed to explain the law of gaseous volumes.
- (v) Why do atoms of the same or different elements combine at all to form molecules?
- (vi) What is the nature of binding force between atoms and molecules which accounts for the existence of matter in three states i.e., solids, liquids and gases?
- (vii) It makes no distinction between the ultimate particles of an element or a compound.

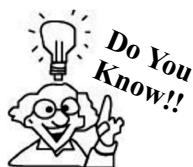
THE AVOGADRO LAW

This law states that “equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules”.



Applications of Avogadro's Law

- (1) This law helped to remove anomaly between Dalton's atomic theory and Gaylussac's law of volume by making a clear distinction between atoms and molecules.
- (2) It reveals that common elementary gases like Hydrogen, Nitrogen, Oxygen etc. are diatomic.
- (3) It provides a method to determine the molecular weights of gaseous elements.



It provides a relationship between vapour density and molecular weight of substances.

$$\text{Vapour density (VD)} = \frac{\text{Mass of same volume of gas at STP}}{\text{Mass of same volume of H}_2 \text{ at STP}}$$

or molecular weight of gas = 2 × vapour density.

CHECK Point

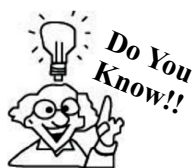
- ☛ 1 L of a gas at S.T.P. weighs 1.97 g. What is the vapour density of the gas?

SOLUTION

22.4 L of the gas at S.T.P. will weigh = $1.97 \times 22.4 = 44.1$ g i.e. molecular mass = 44.1. Hence vapour density = $44.1 / 2 = 22.05$.

ATOM

Atoms are the building blocks of all matter. It is the smallest particle of an element that maintains its chemical identity throughout all chemical and physical changes. Atoms of one element are different from those of the other elements. Today, we know that atoms are not truly indivisible they are themselves made up of particles (protons, neutrons, electrons, etc.). The size of an atom is indicated by its radius which is measured in nanometres ($= 10^{-9}$ m), $1\text{m} = 10^9$ nm.

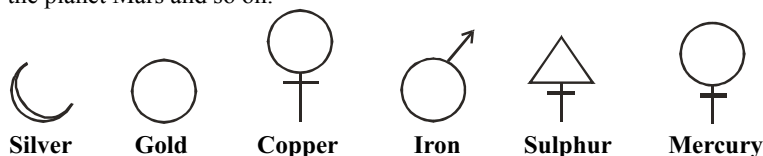


Modern technology has made it possible to take photograph of atoms. The scanning tunneling microscope (STM) is a very sophisticated instrument. It can produce image of the surfaces of the elements which shows the individual atoms.

Experiments have demonstrated that the STM probe can be used to move individual atoms or molecules.

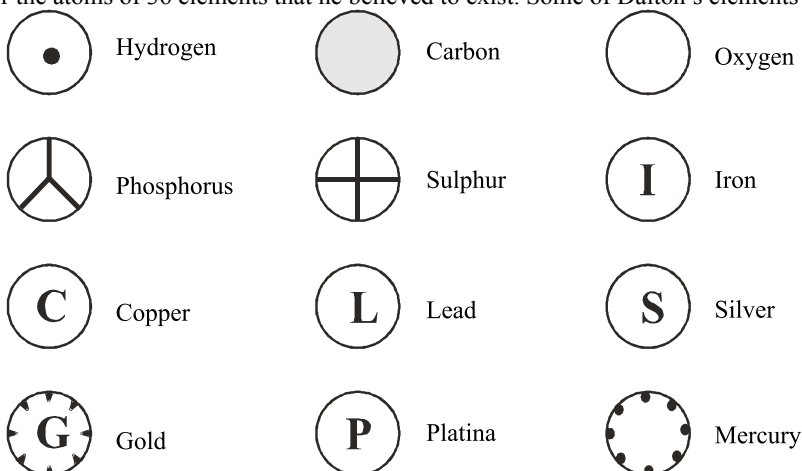
Atomic Symbols

For the sake of convenience, the chemists instead of writing the full and lengthy names of elements preferred the use of notations or symbols. In chemistry the history of symbols is very old. Alchemists used the symbol system. They represented gold by the sun, silver by the moon, iron by the planet Mars and so on.



Atoms and Molecules

Dalton's symbols for the atoms of 36 elements that he believed to exist. Some of Dalton's elements shown below



Berzilius proposed that the symbols of elements be made from one or two letters of the name of the element. Now-a-days, IUPAC (International Union of Pure and Applied Chemistry) approves names of elements. Many of the symbols are the first one or two letters of the element's name in English. The first letter of a symbol is always written as a capital letter (uppercase) and the second letter as a small letter (lowercase).

Sometimes when the names of two elements start with same two letters. We use completely different letter to represent one of them like in case of potassium and polonium where potassium is represented by letter K and polonium as Po.

CHECK Point

☛ What could be possible symbols for elements platinum and palladium?

SOLUTION

Symbol for platinum is 'Pt' and for palladium is 'Pd'.

Elements Named after Places

Scandium (Sc) - found in Scandinavia

Thulium (Tm) - named after Thule the earlier name of Scandinavia.

Europium (Eu) - after the continent Europe.

Polonium (Po) - named after Curies after their home town.

Named after Planets

Selenium (Se) - 'Seles' Greek name for the moon

Plutonium (Pu) - Neptunium (Np) Uranium (U)

Mercury was named after a planet but derives its symbol Hg from the Latin word 'Hydragyrum' meaning liquid silver.

Named after Scientists

Curium (Cm) after Pierre and Marie Curie

Fermium (Fm) after Enrico Fermi

Einsteinium (Es) after Albert Einstein

Mendelevium (Md) after Dimitri Mendeleev

Dalton was the first scientist to use the symbols to represent elements in a very specific sense.

Table : Symbols for some element

Element	Symbol	Element	Symbol	Element	Symbol
Aluminium	Al	Copper	Cu	Nitrogen	N
Argon	Ar	Fluorine	F	Oxygen	O
Barium	Ba	Gold	Au	Potassium	K
Boron	B	Hydrogen	H	Silicon	Si
Bromide	Br	Iodine	I	Silver	Ag

Calcium	Ca	Iron	Fe	Sodium	Na
Carbon	C	Lead	Pb	Sulphur	S
Chlorine	Cl	Magnesium	Mg	Uranium	U
Cobalt	Co	Neon	Ne	Zinc	Zn

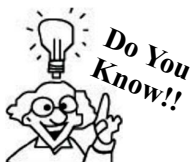


idea box

Go through the concept of symbols of elements thoroughly and ask your friends to do the same. Call your friends at your home and divide them in two groups A and B and you also become a part of one group. Now ask your father to conduct a quiz in which he first write symbol of an element at black board. First group A have to answer the name of element then again different symbol should be written for group B. 10 points will be given for right answer and 5 points should be deducted for wrong answer. Let's see which group wins?

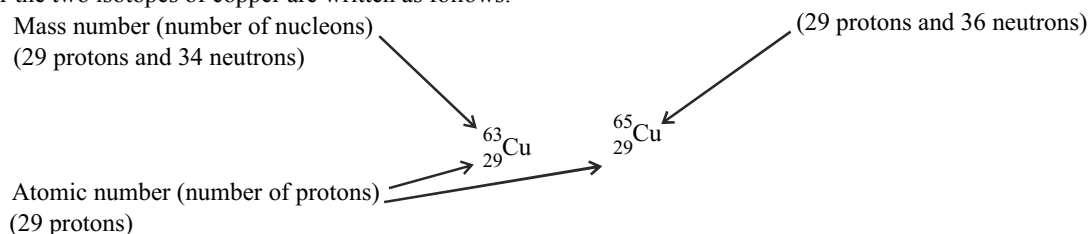
Atomic Mass

In Dalton's original atomic theory, he suggested that all atoms of an element are the same. But if we look at a number of atoms of most elements we find that this statement is not exactly true. For example, consider the atoms of the element copper. A typical atom is composed of a nucleus containing a total of 63 nucleons, of which 29 are protons and 34 are neutrons. The atom also contains 29 electrons that exactly balance the positive charge of the protons. Thus the atom is neutral. The number of protons in the nucleus (which is equal to the total positive charge) is referred to as the atom's atomic number. The total number of nucleons is called the mass number. Therefore, this particular copper atom has an atomic number of 29 and a mass number of 63. There are other copper atoms that are not exactly the same, however. Some atoms of copper have a mass number of 65 rather than 63. This means that these atoms have 36 neutrons as well as 29 protons. Atoms having the same atomic number but different mass numbers are known as isotopes.



Most elements that are present in nature exist as a mixture of isotopes. It is the atomic number, however, that distinguishes one element from another. Any atom with an atomic number of 29, regardless of any other consideration, is an atom of copper. If the atomic number is 28, the element is nickel; if it is 30, the element is zinc.

Specific isotopes are written in a form known as the isotopic notation. In isotopic notation, the mass number is written as a superscript to the left of the element. Sometimes, the atomic number is written as a subscript also on the left. The indication of the atomic number is strictly a convenience, however, since the atomic number determines the identity of the element and vice versa. The isotopic notation for the two isotopes of copper are written as follows.



The convention for verbally naming specific isotopes is to give the element followed by the mass number. For example ${}^{63}\text{Cu}$ is called "copper-63" and ${}^{65}\text{Cu}$ is called "copper-65."

Now, another important consideration will be how the mass of one element compares to another. The mass of the electrons is extremely small compared to the masses of the protons and neutrons, so it is not included in the mass of an isotope. Thus the mass number of an isotope is a convenient but rather imprecise measure of its mass. It is imprecise not only because electrons are not included but also because the proton and neutron do not have exactly the same mass. A more precise measure of the mass of one isotope relative to another is known as the *isotopic mass*.



Atoms and Molecules

But how to express this mass, Dalton suggested that by considering the weight of hydrogen as standard, the comparative weight of atoms of other elements can be calculated. According to hydrogen standard.

$$\text{Atomic weight of element} = \frac{\text{Weight of one atom of element}}{\text{Weight of one atom of hydrogen}}$$

Later on in comparison to hydrogen atom the sixteenth part of atomic weight of oxygen (16) atom was considered as standard unit because oxygen forms more stable compounds in comparison to hydrogen.

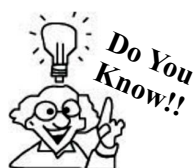
$$\text{Atomic weight of element} = \frac{\text{Weight of one atom of element}}{\text{Weight of } 1/16^{\text{th}} \text{ part of oxygen atom}}$$

These values were based on oxygen 16. But if all the isotopes of oxygen (${}^8\text{O}^{16}$, ${}^8\text{O}^{17}$, ${}^8\text{O}^{18}$) are included, then the atomic weight of oxygen comes out to be 16.0044. To overcome this problem in 1961 twelfth part of C-12 was considered as standard atomic mass unit (amu) at the International level according to it. Thus

$$\text{Atomic weight of element} = \frac{\text{Mass of an atom of element}}{\frac{1}{12} \times \text{Mass of an atom of C-12}}$$

$$1 \text{ amu or u} = 1.66056 \times 10^{-24} \text{ g.}$$

It has been found even theoretically, that a substance can't be obtained in absolutely pure form and similarly the naturally occurring elements contains more than one isotopes. Isotopes are the atoms of same element having same atomic number but different atomic masses. In such cases the atomic mass reported for an isotopic mixture of an element is actually **average atomic mass**.



It can be calculated as follows :

$$\text{Average atomic mass} = \frac{\sum (\% \text{abundance} \times \text{isotopic mass})}{100} \text{ or}$$

$$\text{Average atomic mass} = \sum (\text{Isotopic mass} \times \text{Fractional abundance})$$

CHECK Point

☛ Why atomic masses are the average values?

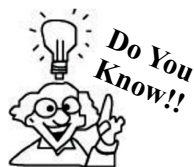
SOLUTION

Most of the elements exist in different isotopes i.e. atoms with different masses, e.g., Cl has two isotopes with mass numbers 35 and 37 existing in the ratio 3 : 1. Hence, average value is taken.

Gram Atomic Mass or Gram Atomic Weight

The atomic mass of an element when expressed in grams is known as **gram atomic mass** or simply as **gram atom (g-atom)**. Thus one g-atom of carbon (C-12) weighs 12.0 g. The number of gram-atoms of any element in its certain weight is given by:

$$\text{Number of gram-atom} = \frac{\text{Mass of the element in grams}}{\text{Atomic mass}}$$



One gram atomic mass of any element contains the same number of atom of that element as there are carbon atoms in exactly 12 g of carbon-12. This number is Avogadro's number 6.022×10^{23} atoms/g.

Thus

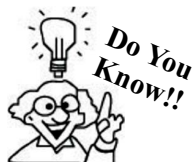
One g-atom of carbon = 12.000 g carbon = mass of N_A carbon atoms.

$N_A = 6.022 \times 10^{23}$, the Avogadro's number

One gram atom = Atomic mass in gram = 6.022×10^{23} atoms

$$\text{No. of g-atoms} = \frac{\text{Mass in gram}}{\text{gram atomic mass}}$$

and $\text{No. of atoms in given mass} = \text{No. of gram atoms} \times 6.022 \times 10^{23}$



In accurate determination, atomic mass has been found to be fractional (e.g., $\text{Cl} = 35.5$, etc.), but here in common practice, its nearby whole number has been used.

Thus, one atomic mass unit is a mass of exactly $1/12$ of the mass of ^{12}C . For example, precise measurements show that the mass of ^{10}B is 0.83442 times the mass of ^{12}C , which means it has an isotopic mass of 10.013 amu. From similar calculations we find that the atomic mass of ^{11}B is 11.009 amu.

CHECK Point

👉 **Why are the atomic masses of most of the elements fractional?**

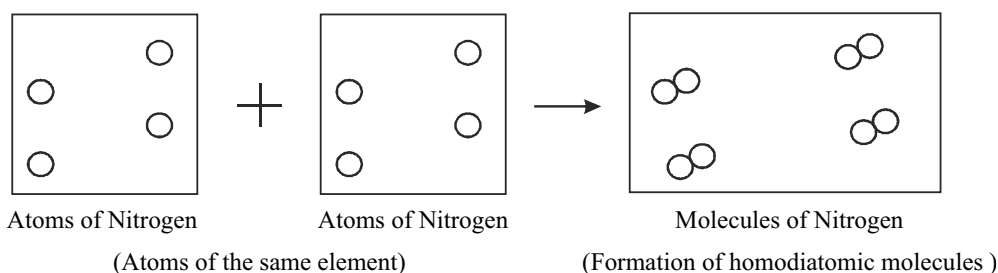
SOLUTION

This is because atomic masses are the relative masses of atoms as compared with an atom of C-12 isotope taken as 12.

MOLECULE

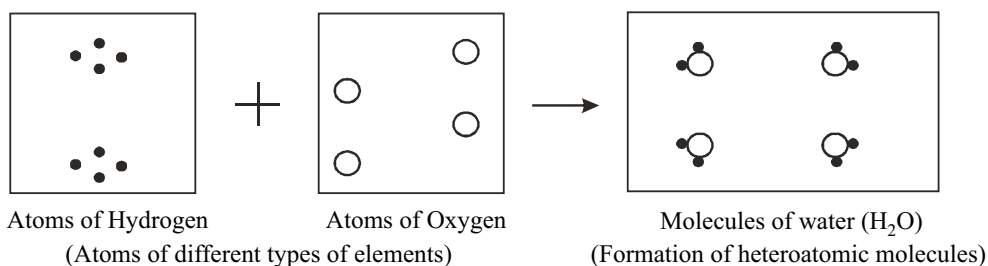
The smallest particle of an element or compound which can exist independently in nature is termed as molecule. Molecules are formed by the combination of two or more atoms in a constant ratio. If same type of atoms are present in molecule then it is termed as Homoatomic molecule.

For example : N_2 (Nitrogen molecule), O_2 (Oxygen molecule), Cl_2 (Chlorine molecule)



If two or more than two types of atoms are present in any molecule then it is termed as heteroatomic molecule.

For example – H_2O (water), CO_2 (Carbon dioxide), H_2O_2 (Hydrogen peroxide) etc.



Atoms and Molecules

Molecular Mass or Molecular Weight

The molecular mass of a substance (element or compound) is the number of times the molecule of the substance is heavier than $1/12$ th the mass of an atom of carbon-12 isotope.

Molecular weight is calculated by adding the atomic weights of all the constituent atoms present in a molecule. For example,

Molecular weight of a molecule of hydrogen (H_2) = $2 \times$ atomic weight of hydrogen = $2 \times 1 = 2$ amu

Similarly, molecular weight of some compounds are:

Water (H_2O) = $2 \times 1 + 1 \times 16 = 2 + 16 = 18$ amu.

Carbon dioxide (CO_2) = $1 \times 12 + 2 \times 16 = 12 + 32 = 44$ amu

Molecular weight is expressed in amu.

ILLUSTRATION : 5

Calculate the molecular mass of glucose ($C_6H_{12}O_6$) molecule.

SOLUTION :

Molecular mass of glucose ($C_6H_{12}O_6$)
 = $6 (12.011 \text{ amu}) + 12 (1.008 \text{ amu}) + 6 (16.00 \text{ amu})$
 = $72.066 \text{ amu} + 12.096 \text{ amu} + 96.000 \text{ amu} = 180.162 \text{ amu}$.

ILLUSTRATION : 6

Calculate the molecular masses of the following compounds :

(a) copper sulphate crystals, $CuSO_4 \cdot 5H_2O$

(b) cane sugar, $C_{12}H_{22}O_{11}$

SOLUTION :

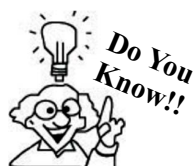
(a) The relative molecular mass of $CuSO_4 \cdot 5H_2O = 63.5 + 32 + (16 \times 4) + 5 (2 + 16) = 159.5 + 90 = 249.5$

(b) The relative molecular mass of $C_{12}H_{22}O_{11} = 12 \times 12 + 1 \times 22 + 16 \times 11 = 144 + 22 + 176 = 342$

Gram Molecular Mass

The molecular mass of a substance when expressed in grams is known as **gram-molecular mass** or simply **gram-mole (g-mole)** for the sake of convenience it is expressed simply as **mole**. Thus one **mole** (or **g-mole**) of water weighs 18g (molecular weight of water = 18)

$$\text{Number of moles} = \frac{\text{Mass of substance in grams}}{\text{Molecular weight}}$$



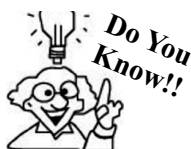
According to definition one mole is the amount of substance that contains as many particles or entities as there are atoms in exactly 12.00 g of C-12.

IONS

Imagine that we have a piece of copper metal. Would it be easier to change the number of electrons on some of the copper atoms in this piece of metal or to change the number of protons ?

It is much easier to change the number of electrons on an atom than the number of protons in the nucleus. The best evidence for this is the fact that copper can conduct an electric current. Since the number of protons determines the identity of an atom and the copper wire does not change to some other element while conducting electricity, the moving charged particles must be electrons not protons. *The electrically charged particles formed when electrons are added to or removed from a neutral atom are called ions.* Neutral atoms are turned into positively charged ions called cations by removing one or more electrons. By removing an electron from a sodium atom—which contains 11 electrons and 11 protons a Na^+ ion is produced that has 10 electrons and 11 protons. Ions with larger positive charges can be produced by removing more electrons. A neutral aluminum atom, for example, has 13 electrons and 13 protons. If we remove three electrons from this atom, we get a positively charged Al^{3+} ion that has 10 electrons and 13 protons, for a net charge of +3.

Atoms that gain extra electrons become negatively charged ions called anions. A neutral chlorine atom, for example, has 17 protons and 17 electrons. By adding one more electron to this atom, a Cl^- ion is produced that has 18 electrons and 17 protons, for a net charge of -1 .



The gain or loss of electrons by an atom to form negative or positive ions has an enormous impact on the chemical and physical properties of the atom. Sodium metal, which consists of neutral sodium atoms, bursts into flame when it comes in contact with water. But positively charged Na^+ ions are so unreactive with water they are essentially inert. Neutral chlorine atoms instantly combine to form Cl_2 molecules, which are so reactive that entire communities are evacuated when trains carrying chlorine gas derail. However, chloride ions do not react with one another.

CHECK Point

- ☛ Is there any differences between property of atom and ion ? Explain with example.

SOLUTION

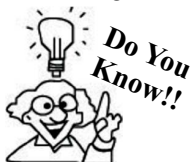
The identity of elements is not altered when it changed into ion, however the properties of ion differ from those of atoms e.g., Magnesium metal is appropriately reactive but its divalent ion is inert in nature because of stable electronic configuration.

CONNECTING TOPIC

Size of Ions

Cations are formed by the loss of electron from the metal atom. In cation, number of protons are more than the number of electrons. As a result of it, the attraction of positively charged nucleus is more on the electrons. So electron cloud will squeeze and its size would be smaller, therefore size of cation is smaller than its corresponding neutral atom.

Anions are formed by taking up electrons by the non metal atom. In anion number of electrons are more than the number of protons. As a result repulsion would increase between the electron cloud and electron cloud will expand and its size would be large. Size of atom is larger than its corresponding neutral atom.

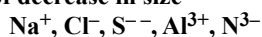


Mutual relationship between the various ions formed by the atom 'A' would be as follows :



CHECK Point

- ☛ Arrange the following ions in a correct order of decrease in size



SOLUTION

The correct decreasing order of size is



Polyatomic Ions :

Simple ions, such as Mg^{2+} and N^{3-} ions, are formed by adding or subtracting electrons from neutral atoms. Polyatomic ions are electrically charged substances composed of more than one atom. There are only two polyatomic cations that you will commonly encounter. These are NH_4^+ and H_3O^+ . There are many more polyatomic anions. A few of the more common anions are –

-1 ions			
HCO_3^-	hydrogen carbonate	OH^-	hydroxide
CH_3CO_2^-	acetate	ClO_4^-	perchlorate
NO_3^-	nitrate	ClO_3^-	chlorate
NO_2^-	nitrite	ClO_2^-	chlorite
MnO_4^-	permanganate	ClO^-	hypochlorite
CN^-	cyanide		

-2 ions			
CO_3^{2-}	carbonate	O_2^{2-}	peroxide
SO_4^{2-}	sulfate	CrO_4^{2-}	chromate
SO_3^{2-}	sulfite	$\text{Cr}_2\text{O}_7^{2-}$	dichromate
$\text{S}_2\text{O}_3^{2-}$	thiosulfate		
-3 ions			
PO_4^{3-}	phosphate	AsO_4^{3-}	arsenate
BO_3^{3-}	borate		



idea box

Make a power point presentation on formation of ion from neutral atom. Your presentation should clearly differentiate between the ions of opposite charges. Also mention about simple and polyatomic ions in your presentation with some examples.

CHEMICAL FORMULA

The composition of a molecular compound can be represented by a chemical formula. The subscripts in a chemical formula represent the relative numbers of atoms present in the compound. When there is no subscript, as in the case of carbon in CO_2 , a value of one is assumed. Thus the formula CO_2 represents a molecule that contains one carbon atom and two oxygen atoms. The same elements can combine in different ratios to give completely different compounds. Carbon dioxide, CO_2 , is the gas that results from respiration (we exhale CO_2). Carbon and oxygen can combine to also form carbon monoxide, CO , a very toxic gas. In this compound carbon and oxygen are in a 1 : 1 ratio.

We use coefficients in front of chemical formulas to represent the number of units of a compound present. If we wish to describe the three molecules of carbon dioxide, we would write 3 CO_2 .

How many atoms are in a molecule of AlSO_4 (aluminum sulfate)? There is one aluminum (Al), one sulfur (S), and four oxygen (O) for a total of six atoms. What is the formula for ammonium chloride? Since it contains one nitrogen (N), four hydrogen (H), and one chlorine (Cl), the chemical formula is NH_4Cl .

Have you ever wondered what to call a chemical compound, such as $\text{Ca}(\text{OH})_2$? There are a few general rules that can help you. Start with the name of the first atom, Ca. That's the element calcium. Then, take the next atom and replace the ending with "ide." Oxygen (O) becomes "oxide". As with many rules, though, there are exceptions! In this case, the first part of the name for the last atom, hydrogen (H), is tacked on so that the correct chemical name is calcium hydroxide.

Prefixes, the part added to the beginning of a word, can help you figure out names and formulas, too. Mono means one, di means two. So a compound with two of the same molecule (represented by a subscript 2) would have "di" before that molecule name. What would you call CO_2 ? (Carbon dioxide) How many oxygen atoms are in carbon monoxide? (one)

One carbon (C) and four chlorine (Cl) atoms combine to give carbon tetrachloride.

(1) Carbon + (4) Chlorine = Carbon tetrachloride.

HOW TO WRITE CHEMICAL FORMULA OF A COMPOUND ?

The formation of chemical compound (say AB) takes place by the combination of positive radical (say A) and negative radical (say B) to give a neutral molecule. Thus total positive charge due to A will be equal to the total negative charge due to B.

The chemical formula of a compound can be written in following steps –

Step 1. Write the symbols or formulae of the positive and negative radicals side by side. Positive radical should be written to the left and negative radical to the right. If necessary write the formulae of the radicals in the brackets.

Step 2. Write the valency of each radical at the top. If necessary divide the valencies by H. C. F. to get a simple ratio.

Step 3. Now apply the rule of criss-cross by shifting the valencies cross-wise to the lower right of the radicals.

The above method of writing a chemical formula can be illustrated by the following examples :

Compound	Symbol with valencies	Criss-cross	Molecular Formula
Sodium Chloride	$\begin{array}{cc} 1 & 1 \\ \text{Na} & \text{Cl} \end{array}$	$\begin{array}{cc} 1 & 1 \\ \text{Na} & \text{Cl} \end{array}$	NaCl
Sodium Oxide	$\begin{array}{cc} 1 & 2 \\ \text{Na} & \text{O} \end{array}$	$\begin{array}{cc} 1 & 2 \\ \text{Na} & \text{O} \end{array}$	Na ₂ O
Chromium Nitrate	$\begin{array}{cc} 3 & 1 \\ \text{Cr} & (\text{NO}_3) \end{array}$	$\begin{array}{cc} 3 & 1 \\ \text{Cr} & (\text{NO}_3) \end{array}$	Cr(NO ₃) ₃

When the valencies of the radicals are divisible by a common factor they are first reduced to a simple ratio. Then criss-cross is applied. Examples :

Compound	Symbols	Simple ratio	Criss-cross	Formula
Calcium sulphate	$\begin{array}{cc} 2 & 2 \\ \text{Ca} & (\text{SO}_4) \end{array}$	$\begin{array}{cc} 1 & 1 \\ \text{Ca} & (\text{SO}_4) \end{array}$	$\begin{array}{cc} 1 & 1 \\ \text{Ca} & (\text{SO}_4) \end{array}$	CaSO ₄
Stannic sulphate	$\begin{array}{cc} 4 & 2 \\ \text{Sn} & (\text{SO}_4) \end{array}$	$\begin{array}{cc} 2 & 1 \\ \text{Sn} & (\text{SO}_4) \end{array}$	$\begin{array}{cc} 2 & 1 \\ \text{Sn} & (\text{SO}_4) \end{array}$	Sn(SO ₄) ₂

Some more examples are given in the table below :

Compound	Symbol with valencies	Criss-Cross	Formula
1. Phosphorous trichloride	$\begin{array}{cc} 3 & 1 \\ \text{P} & \text{Cl} \end{array}$	$\begin{array}{cc} 3 & 1 \\ \text{P} & \text{Cl} \end{array}$	PCl ₃
2. Ferrous sulphate	$\begin{array}{cc} 2 & 2 \\ \text{Fe} & \text{SO}_4 \end{array}$	$\begin{array}{cc} 2 & 2 \\ \text{Fe} & \text{SO}_4 \end{array}$	FeSO ₄
3. Ferric sulphate	$\begin{array}{cc} 3 & 2 \\ \text{Fe} & \text{SO}_4 \end{array}$	$\begin{array}{cc} 3 & 2 \\ \text{Fe} & \text{SO}_4 \end{array}$	Fe ₂ (SO ₄) ₃
4. Ferric chloride	$\begin{array}{cc} 3 & 1 \\ \text{Fe} & \text{Cl} \end{array}$	$\begin{array}{cc} 3 & 1 \\ \text{Fe} & \text{Cl} \end{array}$	FeCl ₃
5. Barium nitrate	$\begin{array}{cc} 2 & 1 \\ \text{Ba} & \text{NO}_3 \end{array}$	$\begin{array}{cc} 2 & 1 \\ \text{Ba} & \text{NO}_3 \end{array}$	Ba(NO ₃) ₂
6. Ammonium phosphate	$\begin{array}{cc} 1 & 3 \\ (\text{NH}_4) & (\text{PO}_4) \end{array}$	$\begin{array}{cc} 1 & 3 \\ (\text{NH}_4) & (\text{PO}_4) \end{array}$	(NH ₄) ₃ PO ₄

7. Bismuth Carbonate	$\overset{3}{\text{Bi}} \quad \overset{2}{(\text{CO}_3)}$	$\begin{array}{c} 3 \quad 2 \\ \swarrow \searrow \\ \text{Bi} \quad (\text{CO}_3) \end{array}$	$\text{Bi}_2(\text{CO}_3)_3$
8. Potassium permanganate	$\overset{1}{\text{K}} \quad \overset{1}{(\text{MnO}_4)}$	$\begin{array}{c} 1 \quad 1 \\ \swarrow \searrow \\ \text{K} \quad \text{MnO}_4 \end{array}$	KMnO_4
9. Aluminium phosphate	$\overset{3}{\text{Al}} \quad \overset{3}{(\text{PO}_4)}$	$\begin{array}{c} 3 \quad 3 \\ \swarrow \searrow \\ \text{Al} \quad (\text{PO}_4) \end{array}$	$\text{Al}_3(\text{PO}_4)_3 \quad \text{Al PO}_4$
10. Stannous chloride	$\overset{2}{\text{Sn}} \quad \overset{1}{\text{Cl}}$	$\begin{array}{c} 2 \quad 1 \\ \swarrow \searrow \\ \text{Sn} \quad \text{Cl} \end{array}$	SnCl_2

Formulae of some compounds

Negative radical	Chloride	Nitrate	Sulphate	Carbonate	Hydroxide	Phosphate
	Cl^-	NO_3^-	SO_4^{2-}	CO_3^{2-}	OH^-	PO_4^{3-}
Positive radical						
H^+	HCl	HNO_3	H_2SO_4	H_2CO_3	HOH or H_2O	H_3PO_4
Na^+	NaCl	NaNO_3	Na_2SO_4	Na_2CO_3	NaOH	Na_3PO_4
Mg^{2+}	MgCl_2	$\text{Mg}(\text{NO}_3)_2$	MgSO_4	Mg_2CO_3	$\text{Mg}(\text{OH})_2$	$\text{Mg}_3(\text{PO}_4)_2$
Al^{3+}	AlCl_3	$\text{Al}(\text{NO}_3)_3$	$\text{Al}_2(\text{SO}_4)_3$	$\text{Al}_2(\text{CO}_3)_3$	$\text{Al}(\text{OH})_3$	AlPO_4
NH_4^+	NH_4Cl	NH_4NO_3	$(\text{NH}_4)_2\text{SO}_4$	$(\text{NH}_4)_2\text{CO}_3$	NH_4OH	$(\text{NH}_4)_3\text{PO}_4$
Fe^{2+} (ous)	FeCl_2	$\text{Fe}(\text{NO}_3)_2$	FeSO_4	FeCO_3	$\text{Fe}(\text{OH})_2$	$\text{Fe}_3(\text{PO}_4)_2$
Fe^{3+} (ic)	FeCl_3	$\text{Fe}(\text{NO}_3)_3$	$\text{Fe}_2(\text{SO}_4)_3$	$\text{Fe}_2(\text{CO}_3)_3$	$\text{Fe}(\text{OH})_3$	FePO_4
Ba^{2+}	BaCl_2	$\text{Ba}(\text{NO}_3)_2$	BaSO_4	BaCO_3	$\text{Ba}(\text{OH})_2$	$\text{Ba}_3(\text{PO}_4)_2$

CHECK Point

- ☛ Calcium carbonate is the major component in chalk, limestone, and marble. It is composed of calcium ions, Ca^{2+} and the polyatomic anion carbonate, CO_3^{2-} . Predict the chemical formula of calcium carbonate.

SOLUTION

The total charge from the positive ions must balance the charge from the negative ions. Since the calcium ion has a charge of +2 and the carbonate ion has a charge of -2, the charge of one calcium ion cancels the charge of one carbonate ion. The ionic formula of calcium carbonate is CaCO_3 .

Formula Mass and Gram Formula Mass

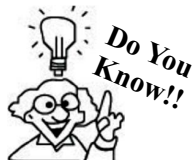
Ionic compounds such as NaCl , KNO_3 , Na_2CO_3 etc. do not consist of molecules, i.e., single entities but exist as ions. Each ion is surrounded by a number of oppositely charged ions, e.g., in NaCl , each Na^+ ion is surrounded by six Cl^- ions and vice versa. Hence, in such cases, the formula is used to calculate the formula mass instead of molecular mass. Just like the molecular mass, it is found by adding the atomic masses of the atoms present in one formula unit, e.g.,

Formula mass of NaCl = Atomic mass of Na + Atomic mass of Cl = 23.0 u + 35.5 u = 58.5 u

Gram formula mass represents the formula mass expressed in grams.

MOLE CONCEPT

We use units for counting many small things in large numbers. For example we use 2 dozen pencils instead of 24 pencils. We use one kilometre for 1000 metres. A relation exist between the atomic weight of an element and number of atoms. In 12 grams of carbon $12\text{ (C}^{12}\text{)}$ number of atoms of carbon are 6.02×10^{23} . In the honour of scientist *Amdio Avogadro* this number is called *Avogadro's number*.

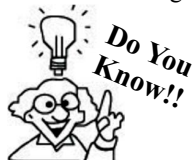


In one gram atomic weight of any element 6.02×10^{23} atoms of that element are present. This quantity of the element is known as one mole.

One mole of any substance is that quantity in which the chemical units present, are same in number as the number of atom in the 12 grams of the pure carbon-12.

One mole of any element is the value of atomic mass in grams. For any compound or substance it is the value of molecular mass in grams.

Example : Atomic mass of sodium atom is 23, so 23 grams of sodium is equal to one mole. Molecular weight of hydrochloric acid is 36.5. So 36.5 gram hydrochloric acid is equal to one mole.



It is pertinent to note that mole is not a number but it is the quantity of substance which is represented in terms of weight or volume.

The mole concept would become more clear with the following examples.

One mole of magnesium (Mg) = 6.02×10^{23} magnesium atoms = 24 gram of Mg^{24}

One mole of atomic nitrogen (N^{14}) = 6.02×10^{23} nitrogen atoms = 14 gram of N^{14} .

One mole of molecular nitrogen (N_2) = 6.02×10^{23} N_2 molecules = 28 gram of N_2 molecules.

One mole of molecular oxygen (O_2) = 6.02×10^{23} O_2 molecules = 32 gram of O_2 molecules.

One mole of nitrate (NO_3^-) ion = 6.02×10^{23} NO_3^- ion = 62 gram of NO_3^- ion.

One mole of sulphate (SO_4^{2-}) ion = 6.02×10^{23} SO_4^{2-} ion = 96 gram of SO_4^{2-} ion

One mole of water (H_2O) = 6.02×10^{23} H_2O molecules = 18 gram of water .

One mole of salt (NaCl) = 6.02×10^{23} NaCl molecules = 58.5 gram of salt.

From the above examples, it becomes clear that one mole of a substance is the quantity in which 6.02×10^{23} chemical units of that substance molecule, atom or ions are present.

CHECK Point

- Two bulbs B_1 and B_2 of equal capacity contain 10g oxygen (O_2) and ozone (O_3) respectively. Which bulb will have greater number of O-atoms and which will have greater number of molecules?

SOLUTION

$$10\text{ g O}_2 = \frac{10}{32}\text{ mole} = \frac{10}{32} \times 6.02 \times 10^{23}\text{ molecules} = 1.88 \times 10^{23}\text{ molecules}$$

$$= 2 \times 1.88 \times 10^{23}\text{ atoms} = 3.76 \times 10^{23}\text{ atoms}$$

$$10\text{ g O}_3 = \frac{10}{48}\text{ mole} = \frac{10}{48} \times 6.02 \times 10^{23}\text{ molecules}$$

$$= 1.254 \times 10^{23}\text{ molecules}$$

$$= 3 \times 1.254 \times 10^{23}\text{ atoms}$$

$$= 3.76 \times 10^{23}\text{ atoms}$$

Thus, both contain the same number of atoms but bulb B_1 contains more number of molecules.

ILLUSTRATION : 7

How many atoms and molecules of sulphur are present in 64.0 g of sulphur (S_8)?

SOLUTION :

Molecular formula of sulphur = S_8

$\therefore$ Molecular mass of sulphur (S_8) = $32 \times 8 = 256.0$ u

1 mole of sulphur molecules = 256 g

= 6.022×10^{23} molecules of sulphur

Now, 256 g of sulphur contain 6.022×10^{23} molecules

$\therefore$ 64 g of sulphur will contain = $\frac{6.022 \times 10^{23} \times 64}{256} = 1.506 \times 10^{23}$ molecules.

1 molecule of sulphur (S_8) contains 8 atoms of sulphur

$\therefore$ 1.506×10^{23} molecules of sulphur will contain sulphur atoms = $8 \times 1.506 \times 10^{23} = 1.2048 \times 10^{24}$ atoms.

Mole and Gram Atomic Mass

Since 1 mole of any atomic substance is equal to its gram atomic mass and contains 6.023×10^{23} atoms of that element therefore gram atomic mass of an element is defined as the mass of Avogadro number of atoms [i.e., mass of 6.023×10^{23} atoms]. For example,

Mass of 6.023×10^{23} atoms of hydrogen = 1 g

Mass of 6.023×10^{23} atoms of oxygen = 16 g

Mass of 6.023×10^{23} atoms of potassium = 39 g

Mole and Gram Molecular Mass

One gram molecular mass of any molecular substance is its gram molecular mass and contains 6.023×10^{23} molecules of that substance. Therefore, gram molecular mass of any substance is defined as the mass of Avogadro's number of molecules (i.e., mass of 6.023×10^{23} molecules) of that substance. For example,

Mass of 6.023×10^{23} molecules of oxygen (O_2) = 32 g

Mass of 6.023×10^{23} molecules of nitrogen (N_2) = 28 g

Mass of 6.023×10^{23} molecules of ozone (O_3) = 48 g

The mass of one mole of any substance is normally termed as molar mass which is equal to the molecular mass expressed in grams. Since the molar mass is the mass of Avogadro's number (6.023×10^{23}) of molecules, the absolute mass of single molecule may be obtained by dividing the molar mass by Avogadro's number. Similarly, the absolute mass of a single atom is obtained by dividing the atomic mass by Avogadro's number.

$$\text{Mass of a single atom} = \frac{\text{Atomic mass (g mole}^{-1}\text{)}}{6.023 \times 10^{23} \text{ atom}}$$

$$\text{Mass of a single molecule} = \frac{\text{Molar mass (g mole}^{-1}\text{)}}{6.023 \times 10^{23} \text{ molecules}}$$

ILLUSTRATION : 8

Calculate the mass of (i) an atom of silver (ii) a molecule of carbon dioxide.

SOLUTION :

(i) 1 mole of Ag atoms = 108 g

($\therefore$ Atomic mass of silver = 108 u)

= 6.022×10^{23} atoms

6.022×10^{23} atoms of silver have mass = 108 g

$\therefore$ Mass of one atom of silver = $\frac{108}{6.022 \times 10^{23}} = 1.793 \times 10^{-22}$ g

(ii) 1 mole of CO_2 = 44 g

($\therefore$ Molecular mass of CO_2 = $1 \times 12 + 2 \times 16 = 44$ u)

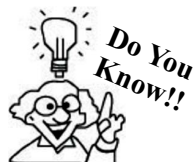
= 6.022×10^{23} molecules

Thus, 6.022×10^{23} molecules of CO_2 have mass = 44g

$\therefore$ 1 molecule of CO_2 has mass = $\frac{44}{6.022 \times 10^{23}} = 7.307 \times 10^{-23}$ g

Mole and Volume

We know from Avogadro's law that equal volumes of all gases contains equal number of molecules under similar conditions of temperature and pressure i.e., under similar conditions, different samples of gases containing equal number of molecules and occupy same volume. Since one mole of all gases contain 6.02×10^{23} molecules, hence one mole of each gas occupy same volume which is 22.4 litres at N.T.P.



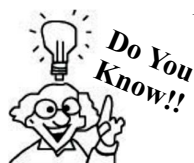
The volume occupied by one gram mole of a gas at NTP is known as gram molecular volume (GMV) or molar volume.

Therefore, the relationship of mole and volume of a gas may be defined as the amount of gas which occupies a volume of 22.4 litres at 273 K and 760 mm pressure is one mole of the gas. i.e.,

1 mole of N_2 gas = 22.4 litres at NTP

1 mole of CO_2 gas = 22.4 litres at NTP

1 mole of helium gas (He) = 22.4 litres at NTP



In case of gases, a mole is defined as that amount of the gas which has a volume of 22.4 litres at STP.

ILLUSTRATION : 9

Calculate the volume at STP occupied by (i) 14 g of nitrogen, (ii) 1.5 moles of carbon dioxide and (iii) 10^{21} molecules of oxygen.

SOLUTION :

(i) Molecular mass of nitrogen = 28 u

1 mole of nitrogen 22.4 litres at STP

i.e., 28 g of nitrogen occupy = 22.4 litres at STP

$$\therefore 14 \text{ g of nitrogen occupy} = \frac{22.4}{28} \times 14 = 11.2 \text{ litres at STP}$$

(ii) 1 mole of carbon dioxide = 22.4 litres at STP

$$\therefore 1.5 \text{ moles of carbon dioxide will occupy} = \frac{22.4}{1} \times 1.5 = 33.6 \text{ litres at STP}$$

(iii) 1 mole of O_2 molecules = 6.022×10^{23} molecules = 22.4 litres at STP

i.e., 6.022×10^{23} molecules of oxygen occupy 22.4 litres at STP

10^{21} molecules of oxygen will occupy

$$= \frac{22.4}{6.022 \times 10^{23}} \times 10^{21} \text{ litres at STP}$$

$$= 3.72 \times 10^{-2} \text{ litres at STP}$$

$$= 3.72 \times 10^{-2} \times 10^3 \text{ cm}^3 \text{ at STP}$$

$$= 37.2 \text{ cm}^3 \text{ at STP}$$

Mole Concept and Ionic Compounds

The mole concept can also be applied to ionic compounds although these are made up of positively and negatively charged ions and not of molecules. The formula of an ionic compound does not represent a molecule of that compound but represent the relative ratio between the constituent ions. However, the term mole may also be applied to ionic compounds with little modification. For ionic compounds, the formula mass is used in place of molecular mass and the formula unit is used in place of molecules. Hence, a mole of an ionic compound is that mass of the substance which contains Avogadro's number (6.02×10^{23}) of formula unit or mass equal to gram formula mass of the compound (i.e., the formula mass expressed in grams).

Atoms and Molecules

For example,

$$\begin{aligned}\text{One mole of sodium chloride (NaCl)} &= 6.02 \times 10^{23} \text{ formula units of NaCl} \\ &= 6.02 \times 10^{23} \text{ Na}^+ \text{ ions} + 6.02 \times 10^{23} \text{ Cl}^- \text{ ions}\end{aligned}$$

Since 1 mole of NaCl = 23 + 35.5 = 58.5 g of NaCl

$$\therefore 58.5 \text{ g NaCl} = 6.02 \times 10^{23} \text{ Na}^+ \text{ ions} + 6.02 \times 10^{23} \text{ Cl}^- \text{ ions}$$

Similarly, 1 mole of MgCl_2 has mass equal to formula mass [i.e., $24 + (2 \times 35.5) = 95 \text{ g}$]. Therefore,

$$\begin{aligned}1 \text{ mole of MgCl}_2 &= 95 \text{ g} = 6.02 \times 10^{23} \text{ formula units of MgCl}_2 \\ &= 6.02 \times 10^{23} \text{ Mg}^{2+} \text{ ions} + 2 \times 6.02 \times 10^{23} \text{ Cl}^- \text{ ions}\end{aligned}$$

ILLUSTRATION : 10

Calculate the number of atoms of the constituent elements in 53 g of Na_2CO_3 .

SOLUTION :

By mole concept,

1 mole of Na_2CO_3

$$= \text{Gram formula mass of Na}_2\text{CO}_3 = 2 \times 23 + 12 + 16 \times 3 = 106 \text{ g}$$

Now, 106 g of $\text{Na}_2\text{CO}_3 = 1 \text{ mole}$

$$\therefore 53 \text{ g of Na}_2\text{CO}_3 = \frac{1 \times 53}{106} = 0.5 \text{ mole of Na}_2\text{CO}_3$$

But 1 mole of Na_2CO_3 contains 2 moles of Na^+ ion or $2 \times 6.022 \times 10^{23} \text{ Na}^+$ ions.

$\therefore$ 0.5 mole of Na_2CO_3 will contain

$$2 \times 6.022 \times 10^{23} \times 0.5 \text{ Na}^+ \text{ ions}$$

$$= 6.022 \times 10^{23} \text{ Na}^+ \text{ ions}$$

Again, 1 mole of Na_2CO_3 contains 1 mole of carbon atoms = 6.022×10^{23} carbon atoms

$\therefore$ 0.5 mole of Na_2CO_3 will contain

$$= 6.022 \times 10^{23} \times 0.5 \text{ carbon atoms}$$

$$= 3.011 \times 10^{23} \text{ carbon atoms}$$

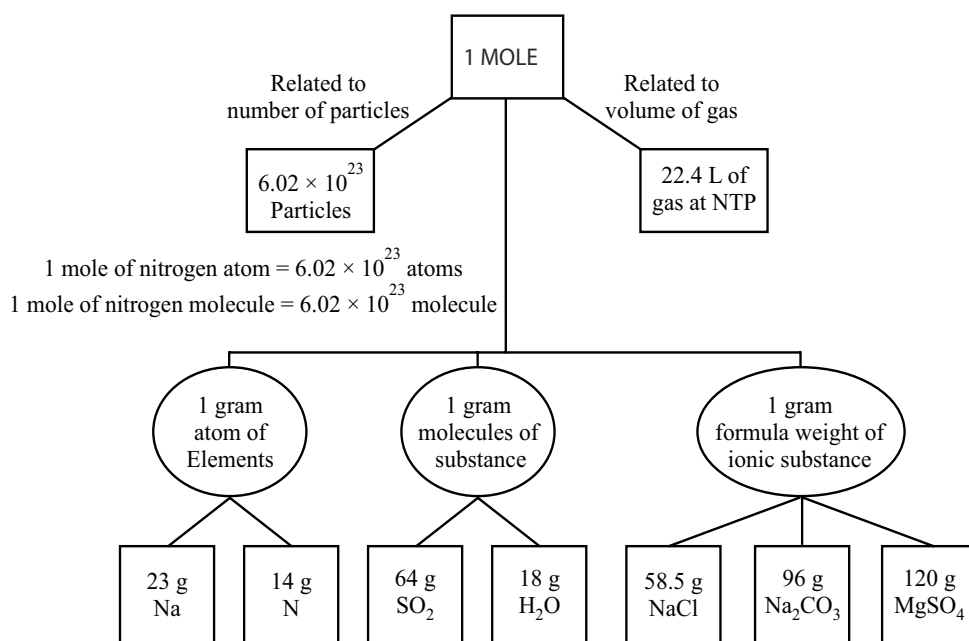
Further, 1 mole of Na_2CO_3 contains 3 moles of oxygen atoms or $3 \times 6.022 \times 10^{23}$ oxygen atoms

$\therefore$ 0.5 mole of Na_2CO_3 will contain

$$3 \times 6.022 \times 10^{23} \times 0.5 \text{ oxygen atoms}$$

$$= 9.033 \times 10^{23} \text{ oxygen atoms}$$

Summing up the above discussed various relationships of mole, a mole may be illustrated as follows :



CONNECTING TOPIC

DETERMINATION OF EMPIRICAL AND MOLECULAR FORMULAE

The chemical formula of a substance is symbolic representation indicating the number and kinds of atoms of different elements present in one molecule of the substance. For example, chemical formula of water is H_2O and that of ammonia is NH_3 . Similarly, chemical formula of sulphur dioxide SO_2 and that of sodium carbonate is Na_2CO_3 . In order to arrive at the chemical formula of any compound, following two analysis have to be carried out.

- Elemental analysis of the substance i.e., determination of the types of elements.
- Determination of relative number of each type of elements.

This data is then used to determine first the empirical formula and then molecular formula of the substance.

(a) Empirical Formula :

It expresses the whole number ratio of the atoms of various elements present in one molecule of the compound. For example, the empirical formula of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is CH_2O . This shows that C, H and O are present in the simple ratio of 1 : 2 : 1.

Steps Taken to Arrive at the Empirical Formula :

- Determine the percentage of each element : Percentage of each element is determined quantitatively. However, the percentage of oxygen is obtained by subtracting the sum of all other percentages from 100 i.e., % of O = [100 – sum of % of all other constituent elements].
- Divide the percentage of each element by its atomic mass to obtain the relative numbers of atoms (or atomic ratio).

$$\text{Atomic ratio} = \frac{\text{Percentage of an element}}{\text{Atomic mass of the same element}}$$

- Divide the atomic ratio by smallest quotient to get the simplest ratio of various elements.
- Convert the simplest ratio to the whole number ratio either by
 - rounding them off to whole number, if the value is quite close to whole number e.g., 1.98, 2.99, 3.95 are rounded off as 2, 3 and 4 respectively.
 - Multiply the figures by suitable integer (2, 3, or 4, etc.)
- Write the empirical formula of the compound by writing the symbols of the various elements side by side. Now insert the whole number ratio of each element as the subscripts to the lower right hand corner of each symbol.

ILLUSTRATION : 11

An inorganic salt gave the following percentage composition —Na = 29.11, S = 40.51 and O = 30.38 Calculate the empirical formula of the salt.

SOLUTION :

Calculation of empirical formula

Element	Symbol	Percentage of elements	At. mass of elements	Moles of the element = $\frac{\text{Percentage}}{\text{At. mass}}$ (Relative no. of moles)	Simplest molar ratio	Simplest whole no. molar ratio
Sodium	Na	29.11	23	$\frac{29.11}{23} = 1.266$	$\frac{1.266}{1.266} = 1$	2
Sulphur	S	40.51	32	$\frac{40.51}{32} = 1.266$	$\frac{1.266}{1.266} = 1$	2
Oxygen	O	30.38	16	$\frac{30.38}{16} = 1.897$	$\frac{1.89}{1.266} = 1.5$	3

Thus, the Empirical Formula is $\text{Na}_2\text{S}_2\text{O}_3$.

ILLUSTRATION : 12

2.38 g of uranium was heated strongly in a current of air. The resulting oxide weighed 2.806 g. Determine the empirical formula of the oxide. (At. mas U = 238; O = 16).

SOLUTION :

Step 1. To calculate the percentage of uranium and oxygen in the oxide.

2.806 g of the oxide contain uranium = 2.38 g.

$$\therefore \text{Percentage of uranium} = \frac{2.38}{2.806} \times 100 = 84.82$$

Hence, the percentage of oxygen in the oxide = 100.00 – 84.82 = 15.18.

Step 2. To calculate the empirical formula

Element	Symbol	Percentage of elements	At. mass of elements	Moles of the element = Percentage At. mass (Relative no. of moles)	Simplest molar ratio	Simplest whole no. molar ratio
Uranium	U	84.82	238	$\frac{84.82}{238} = 0.3562$	$\frac{0.3562}{0.3562} = 1$	3
Oxygen	O	15.18	16	$\frac{15.18}{16} = 0.94875$	$\frac{0.94875}{0.3562} = 2.666$	8

Hence, the empirical formula of the oxide is U_3O_8 .

CHECK Point

- Write the formulae and names of three compounds containing same percentage composition of C, H and O.

SOLUTION

Compounds with the same percentage composition of C, H and O will have the same empirical formula. The compounds with the empirical formula CH_2O can be

HCHO
Formaldehyde

CH_3COOH
Acetic acid

$\text{C}_6\text{H}_{12}\text{O}_6$
Glucose

(b) Molecular Formula :

It is that formula of the compound which gives the actual number of atoms of various elements in a molecule of that compound. For example, molecular formula of glucose is $\text{C}_6\text{H}_{12}\text{O}_6$. Molecular formula of a compound is related to empirical formula as being a simple whole number multiple of empirical formula. Thus,

$$\text{Molecular formula} = n \times \text{Empirical formula} \quad [\text{where } n = 1, 2, 3, \dots, \text{etc.}]$$

The value of n is obtained by the following relation

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$

When $n = 1$, Molecular Formula = Empirical Formula

The molecular mass of a volatile compound can be determined by Victor Meyer's method (based on the principle that 22.4 L of vapours of a substance at STP have mass equal to gram molecular mass) or by employing the relation

$$\text{Molecular mass} = 2 \times \text{Vapour density.}$$

Empirical formula mass can however, be obtained from its empirical formula simply by adding the atomic masses of the various atoms present in it. Thus, for glucose (E.F. = CH_2O), the empirical formula mass can be calculated as follows:

$$\begin{aligned} \text{Empirical formula mass} &= \text{At. mass of carbon} + 2 \times \text{At. mass of hydrogen} + \text{At. mass of oxygen} \\ &= 12.0 \text{ u} + 2 \times 1.0 \text{ u} + 16.0 \text{ u} \\ &= 30.0 \text{ u.} \end{aligned}$$

Hence following steps have to be carried out to arrive at the molecular formula of the compound.

1. Determine the empirical formula mass by adding the atomic masses of all the atoms represented in the empirical formula e.g., empirical formula of glucose is CH_2O .
 $\therefore$ Empirical formula mass = $12 + 2 \times 1 + 16 = 30$
2. Determine the molecular mass from the data obtained by a suitable experiment.
3. Divide the molecular mass by the empirical formula mass and find out the value of n .
4. Multiply empirical formula by n to get the molecular formula.

ILLUSTRATION : 13

An organic substance containing carbon, hydrogen and oxygen gave the following percentage composition.

C = 40.68%; H = 5.085% and O = 54.228%

The vapour density of the compound is 59. Calculate the molecular formula of the compound.

SOLUTION :

Calculate the empirical formula of the compound.

Element	Symbol	Percentage of element	At. mass of elements	Moles of the element = Percentage At. mass (Relative no. of moles)	Simplest molar ratio	Simplest whole no. molar ratio
Carbon	C	40.687	12	$\frac{40.687}{12} = 3.390$	$\frac{3.390}{3.389} = 1$	2
Hydrogen	H	5.085	1	$\frac{5.085}{1} = 5.085$	$\frac{5.085}{3.389} = 1.5$	3
Oxygen	O	54.228	16	$\frac{54.228}{16} = 3.389$	$\frac{3.389}{3.389} = 1$	2

$\therefore$ Empirical formula is $\text{C}_2\text{H}_3\text{O}_2$

Calculate the empirical formula mass. The empirical formula of the compound is $\text{C}_2\text{H}_3\text{O}_2$.

$\therefore$ Empirical formula mass = $(2 \times 12) + (3 \times 1) + (2 \times 16) = 59$.

To calculate the molecular mass of the salt. The vapour density of the compound = 59. (Given)

Using the relation between vapour density and molecular mass—

Molecular mass = $2 \times$ vapour density = $2 \times 59 = 118$.

To calculate the value of ' n '

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{118}{59} = 2$$

To calculate the molecular formula of the salt.

Molecular formula = $n \times$ Empirical formula = $2 \times \text{C}_2\text{H}_3\text{O}_2 = \text{C}_4\text{H}_6\text{O}_4$

Thus, the molecular formula is $\text{C}_4\text{H}_6\text{O}_4$.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- During a chemical reaction, the sum of the of the reactants and products remains unchanged.
- Clusters of atoms that act as an ion are called ions.
- In ionic compounds, the charge on each ion is used to determine the of the compound.
- The Avogadro constant is defined as the number of atoms in exactly of carbon-12.
- The abbreviation used for lengthy names of elements are termed as their
- A chemical formula is also known as a
- Those ions which are formed from single atoms are called.....
- Ionic compounds are formed by the combination between and
- The valency of an ion isto the charge on the ion.
- Mole is a link between theand
- The SI unit of amount of a substance is

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- Formula mass of Na_2O is 62 amu.
- Those particles which have more or less electrons than the normal atoms are called ions.
- Formula for sulphur dioxide is SO_3 .
- Molar mass of ethyne (C_2H_2) is 26 g/mol.
- 22gm of CO_2 consists of 1 mole.
- Number of molecules in 32 gram of oxygen is 6.02×10^{23} .
- Water is an atom.
- Mass of 1 mole of a substance is called its formula mass.
- In a pure chemical compound, elements are always present in a definite proportion by mass.

Match the Following :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in Column I have to be matched with statements (p, q, r, s) in Column II.

- Column II gives molecular mass in amu for substance in column I match them correctly.

Column I

- (A) H_2O
(B) HNO_3
(C) NaCl
(D) CaCl_2

Column II

- (p) 58.5
(q) 111
(r) 18
(s) 63

- Column II gives no. of moles for description given in column I match them correctly.

Column I

- (A) 52 g of He
(B) 12.044×10^{23} He atoms
(C) 8 of gm O_2
(D) 28 of gm N_2

Column II

- (p) 2
(q) 13
(r) 1/4
(s) 1

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Define term mole.
- What is Avogadro's number ?
- What is gram atomic weight ?
- What is the difference between CO_2 and 2CO_2 ?
- Name the following compounds PCl_3 and SO_2 .
- In what smallest whole-number ratio must N and O atoms combine to make dinitrogen tetroxide N_2O_4 ? What is the mole ratio of the elements in this compound ?
- How many moles of sodium atoms correspond to 1.56×10^{21} atoms of sodium ?
- What is the relationship between the formula weight of a substance and its molar mass ?
- How many grams of silver are in 0.263 mol of Ag ?
- How many atoms are 1.00×10^{-9} g of lead ?
- How many grams of iron are needed to combine with 25.6g of O to make Fe_2O_3 ?
- What is the mass of 4 moles of aluminium atoms ? (Atomic mass of Al = 27u)
- Calculate the mass of 6.022×10^{22} atoms of He.
- Calculate the number of moles in 3.011×10^{22} molecules of carbon dioxide.
- A sample of 45.8g of H_2SO_4 contains how many moles of H_2SO_4 ?
- What is the mass in grams of 5 moles of Fe ?
- How many moles of NaCl are present in 20 gm of the substance?

18. How many grams of O is present in 50 gm of CaCO_3 ?
19. How many grams of CO_2 are present in 0.1 mole CO_2 ?
20. Describe the difference between the mass of a mole of oxygen atoms (O) and the mass of a mole of oxygen molecules (O_2).
21. What do you understand by the term Formula unit?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Convert into moles :
 - (a) 12 g of oxygen gas
 - (b) 22 g of carbon dioxide
2. What is molecular weight? Explain with example.
3. Write formulas for ionic compounds formed from
 - (a) Na and F
 - (b) Na and O
 - (c) Mg and F
 - (d) Al and C.
4. Write formulas for
 - (a) aluminium sulfide
 - (b) strontium fluoride
 - (c) titanium (IV) oxide
 - (d) calcium bromide.
5. How many moles of Al atoms are needed to combine with 1.58 mole of O atoms to make aluminium oxide, Al_2O_3 ?
6. What is the volume of 32g of sulphur dioxide measured at STP?
7. How many grams of sodium will have the same number of atoms as 6 grams of magnesium? (Na = 23, Mg = 24).
8. How many grams of Chromium (molar mass Cr = 52 g/mol) are there in 85 g of Cr_2S_3 ?
9. Which has more number of atoms : 100 grams of sodium or 100 grams of iron (Given atomic mass of Na = 23 u and Fe = 56 u)
10. A compound of nitrogen and oxygen has the formula NO . In this compound there are 1.143g of oxygen for each 1.000g of nitrogen. A different compound of nitrogen and oxygen has the formula NO_2 . How many grams of oxygen would be combined with each 1.000g of nitrogen in NO_2 ?
11. How many moles of nitrogen atoms are combined with 8.60 mol of oxygen atoms in dinitrogen pentoxide, N_2O_5 ?
12. Calculate the volume occupied by 2.8 g of N_2 at STP.
13. How much weight of carbon dioxide is present in same volume which is occupied by 4.0 grams of oxygen? Volumes are measured at normal temperature and pressure. (NTP)
14. What is the mass of 0.5 mole of water (H_2O). (Atomic masses : H = 1u, O = 16u)
15. How many moles of sulphur atoms and oxygen atoms are present in one mole each of H_2SO_4 , H_2SO_3 and SO_2 ?
16. What is the percentage composition of the elements in ammonia, NH_3 ? (at. mass: H = 1, N = 14)
17. Calculate the theoretical percentage composition of N_2O_3 .
18. 10 grams of CaCO_3 on heating gave 4.4g of CO_2 and 5.6 of CaO . Show that these observations are in agreement with the law of conservation.
19. In which of the following, the number of hydrogen atoms is more (a) 3 moles of H_2O or (b) 10 moles of HCl ?
20. If we decompose 50g of water by passing electricity through it, how many grams of oxygen and hydrogen we obtain?
21. Hydrogen and oxygen combine in the ratio of 1 : 8 by mass to form water. What mass of oxygen gas would be required to react completely with 3g of hydrogen gas?
22. Calculate the number of molecules in 32 grams of oxygen gas and 14 grams of nitrogen.
23. Calculate the mass of a sample of iron metal that would contain 0.250 moles of iron atoms.
24. Calculate the number of atoms in a 0.123-gram sample of aluminium foil.

Long Answer Questions :

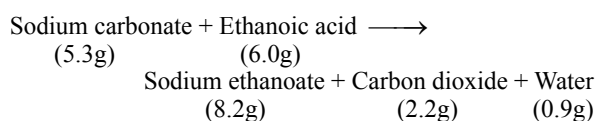
DIRECTIONS : Give answer in four to five sentences.

1. Calculate the formula mass of each of the following and round your answer to the nearest 0.1 u.
 - (a) NaHCO_3
 - (b) $\text{K}_2\text{Cr}_2\text{O}_7$
 - (c) $(\text{NH}_4)_2\text{CO}_3$
 - (d) $\text{Al}_2(\text{SO}_4)_3$
 - (e) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
2. Calculate the molar mass of the following substances.
 - (i) Ethyne, C_2H_2
 - (ii) Sulphur molecule, S_8
 - (iii) Phosphorus molecule, P_4 (atomic mass of phosphorous = 31)
 - (iv) Hydrochloric acid, HCl
 - (v) Nitric acid, HNO_3
3. Calculate the number of aluminium ions present in 0.051g of aluminium oxide (Al_2O_3). (Atomic mass : Al = 27 u, O = 16u)
4. (I) Write the chemical formulas for the following compounds: (a) copper (I) oxide, (b) potassium peroxide (c) mercury (I) bromide, (d) iron (III) carbonate, (e) sodium hydrobromite.
(II) Give the name for each of the following acids
 - (a) HBrO_3 ,
 - (b) HBr ,
 - (c) H_3PO_4 ,
 - (d) HClO
 - (e) HIO_3
5. (I) Calculate both the average mass of a single molecule of carbon dioxide and glucose and the molecular weight of these compounds.
(II) Determine the number of carbon atoms in 0.500 grams of carbon dioxide, CO_2 .
6. (i) Distinguish between an atom and a molecule.
(ii) Differentiate between atom and ion.

2 EXERCISE

Text - Book Questions :

1. In a reaction 5.3g of sodium carbonate reacted with 6g of ethanoic acid. The products were 2.2g of carbon dioxide 0.9g of water and 8.2g of sodium ethanoate. Show that these observation are in agreement with the law of conservation of mass.



2. Hydrogen and oxygen combine in the ratio of 1: 8 by mass to form water. What mass of oxygen gas would be required to react completely with 3g of hydrogen gas?
3. Which postulate of Dalton's atomic theory is the result of the law of conservation of mass?
4. Which postulate of Dalton's atomic theory can explain the law of definite proportions?
5. Define atomic mass unit.
6. Why is it not possible to see an atom with naked eye?
7. Write down the formula of
 - (i) Sodium oxide
 - (ii) Aluminium chloride
 - (iii) Sodium sulphide
 - (iv) Magnesium hydroxide.
8. Write down the names of the following compounds.
 - (1) $\text{Al}_2(\text{SO}_4)_3$
 - (ii) CaCl_2
 - (iii) K_2SO_4
 - (iv) KNO_3
 - (v) CaCO_3
9. What is meant by the term chemical formula?
10. How many atoms are present in a
 - (i) H_2S molecule
 - (ii) PO_4^{3-} ion
11. Calculate the molecular masses of H_2 , O_2 , Cl_2 , CO_2 , CH_4 , C_2H_6 , C_2H_4 , NH_3 , CH_3OH .
12. Calculate the formula unit mass of ZnO , Na_2O , K_2CO_3 . Given atomic masses of $\text{Zn} = 65\text{U}$, $\text{Na} = 23\text{U}$, $\text{K} = 39\text{U}$, $\text{C} = 12\text{U}$ and $\text{O} = 16\text{U}$.
13. If one mole of carbon atom weights 12 grams, what is the mass (in gram) of 1 atom of carbon?
14. Which has more number of atom, 100g of sodium or 100 g of iron?
[Given, Atomic mass of $\text{Na} = 23\text{U}$, $\text{Fe} = 56\text{U}$]
2. When 3.0g of carbon is burnt in 8.0 g of oxygen, 11.0g of carbon dioxide is formed. What mass of carbon dioxide will be formed, when 3.0g of carbon is burnt in 50.0g of oxygen? Which law of chemical composition will govern your answer?
3. What are polyatomic ions? Give examples.
4. Write the chemical formula of the following :
 - (a) Magnesium chloride
 - (b) Calcium oxide
 - (c) Copper nitrate
 - (d) Aluminium chloride
 - (e) Calcium carbonate
 - (f) Zinc sulphate
5. Give the names of the elements present in the following compounds :
 - (a) Quick lime
 - (b) Hydrogen bromide
 - (c) Baking powder
 - (d) Potassium sulphate
6. Calculate the molar mass of the following substances.
 - (a) Ethyne, C_2H_2
 - (b) Sulphur molecule, S_8
 - (c) Phosphorus molecule, P_4
 - (d) Nitric acid (HNO_3)
 - (e) Hydrochloric acid (HCl)
7. What is the mass of
 - (a) 1 mole of nitrogen atom?
 - (b) 4 moles of aluminium atoms?
 (atomic mass of aluminum atom = 27)
 - (c) 10 moles of sodium sulphite (Na_2SO_3)
8. Convert into moles :
 - (a) 12g of oxygen gas
 - (b) 20g of water
 - (c) 22g of carbon dioxide
9. What is the mass of :
 - (a) 0.2 mole of oxygen atoms?
 - (b) 0.5 moles of water molecules?
10. Calculate the number of molecules of sulphur (S_8) present in 16g of solid sulphur.
11. Calculate the number of aluminium ions in 0.051g of aluminium oxide (Al_2O_3)

Exemplar Problems :

Text - Book Exercise :

1. 0.24 g of sample of a compound of oxygen and boron was found by analysis to contain 0.096g of boron and 0.144 g of oxygen. Calculate the % age composition of the compound by weight.
2. Write the molecular formulae of all the compounds that can be formed by the combination of following ions.
 Cu^{2+} , Na^+ , Fe^{3+} , Cl^- , SO_4^{2-} , PO_4^{3-}
2. Calculate the number of moles of magnesium present in a magnesium ribbon weighting 12g. Molar atomic mass of magnesium is 24g mol^{-1} .

- State the number of atoms present in each of the following chemical species
 (a) CO_3^{2-} (b) PO_4^{3-}
 (c) P_2O_5 (d) CO
- Which of the following represents a correct chemical formula? Name it.
 (a) CaCl (b) BiPO_4
 (c) NaSO_4 (d) NaS
- Calcium chloride when dissolved in water dissociates into its ions according to the following equation.
 $\text{CaCl}_2(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$
 Calculate the number of ions obtained from CaCl_2 when 222 g of it is dissolved in water.
- The mass of one steel screw is 4.11 g. Find the mass of one mole of these steel screws. Compare this value with the mass of the Earth (5.98×10^{24} kg). Which one of the two is heavier and by how many times?
- Raunak took 5 moles of carbon atoms in a container and Krish also took 5 moles of sodium atoms in another container of same weight. (a) Whose container is heavier? (b) Whose container has more number of atoms?
- A sample of ethane (C_2H_6) gas has the same mass as 1.5×10^{20} molecules of methane (CH_4). How many C_2H_6 molecules does the sample of gas contain?
- A gold sample contains 90% of gold and the rest copper. How many atoms of gold are present in one gram of this sample of gold?
- Calculate the weight of (i) one atom of oxygen, and (ii) one molecule of oxygen.
 G.A.W. of $\text{O}_2 = 16.0\text{g}$
- Why can't we say, 'an atom of water'?
- How does Dalton's atomic theory account for the fact that when 1.000g of water is decomposed into its elements, 0.111g of hydrogen and 0.889g of oxygen are obtained regardless of the source of the water?
- Calculate gram molecular weights of the following gases: N_2 (if 360 cm^3 at STP weighs 0.45g)
- How many atoms of oxygen are present in 300 g limestone?
- One sample of CaC_2 contains 0.150 mol of carbon. How many moles and how many grams of calcium are also in the sample?
- An atom is 200 times heavier than $\frac{1}{12}$ of the mass of an atom of carbon (C-12). What is its mass in amu (or U)?
- Calculate the number of moles, molecules and atoms in 11.2 L of H_2 at NTP
- Find the percentage of nitrogen in urea (H_2NCONH_2)
- How many molecules of water of crystallisation are present in 24.95 mg of hydrated copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)?
- Sodium carbonate reacts with ethanoic acid to form sodium ethanoate, carbon dioxide and water. In an experiment, 5.3 g of sodium carbonate reacted with 6 g of ethanoic acid to form 8.2 g of sodium ethanoate, 2.2 g of carbon dioxide and 0.9g of water. Show that this data verifies the law of conservation of mass.
- 1.375 gm of pure cupric oxide was reduced by heating in a current of pure dry hydrogen and the mass of copper that remained 1.0980 gms. In another experiment, 1.179 g of pure copper was dissolved in pure HNO_3 and the resulting copper nitrate converted into cupric oxide by ignition. The mass of copper oxide formed was 1.476 g. Show that the results illustrate to law of constant composition with in the limits of experimental error.
- Write the chemical formula for each substance mentioned in the following word descriptions (a) Zinc carbonate can be heated to form zinc oxide and carbon dioxide. (b) On treatment with hydrofluoric acid, silicon dioxide forms silicon tetrafluoride and water. (c) Sulfur dioxide reacts with water for form sulfurous acid. (d) The substance hydrogen phosphide, commonly called phosphine, is a toxic gas. (e) Perchloric acid reacts with cadmium to form cadmium (II) perchlorate.

HOTS Questions :

- Write formulas for the chlorides and oxides formed by (a) chromium and (b) copper.
- The formula for arsenic acid is H_3AsO_4 . What is the name of the salt Na_3AsO_4 ?
- Sucrose (table sugar) has the formula $\text{C}_{12}\text{H}_{22}\text{O}_{11}$. In this compound, what is the
 (a) atom ratio of C to H ?
 (b) atom ratio of H to O ?
 (c) mole ratio of C to O ?
 (d) mole ratio of H to O ?
- A sample of ascorbic acid (vitamin C) is synthesized in the laboratory. It contains 1.50g of carbon and 2.00g of oxygen. Another sample of ascorbic acid isolated from citrus fruits contains 6.35g of carbon. How many grams of oxygen does it contain ? Which law are you assuming in answering this question ?

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Which of the following has largest number of particles?
(a) 8g of CH_4
(b) 4.4g of CO_2
(c) 34.2g of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
(d) 2g of H_2
- The number of molecules in 16.0g of oxygen is
(a) 6.02×10^{23}
(b) 6.02×10^{-23}
(c) 3.01×10^{-23}
(d) 3.01×10^{23}
- The percentage of hydrogen in H_2O is –
(a) 8.88
(b) 11.12
(c) 20.60
(d) 80.0
- Find the mass of oxygen contained in 1 kg of potassium nitrate (KNO_3)
(a) 475.5 g
(b) 485.5 g
(c) 475.2 g
(d) 485.2 g
- How many moles of electron weigh one kilogram?
(a) 6.023×10^{23}
(b) $\frac{1}{9.108} \times 10^{31}$
(c) $\frac{6.023}{9.108} \times 10^{54}$
(d) $\frac{1}{9.108 \times 6.023} \times 10^8$
- The mass of sodium in 11.7 g of sodium chloride is
(a) 2.3 g
(b) 5.02 g
(c) 6.9 g
(d) 7.1 g
- The formula of a chloride of a metal M is MCl_3 , the formula of the phosphate of metal M will be
(a) MPO_4
(b) M_2PO_4
(c) M_3PO_4
(d) $\text{M}_2(\text{PO}_4)_3$
- Which of the following contains the largest number of molecules?
(a) 0.2 mol H_2
(b) 8.0 g H_2
(c) 17 g of H_2O
(d) 6.0 g of CO_2
- One gram of which of the following contains largest number of oxygen atoms?
(a) O
(b) O_2
(c) O_3
(d) All contains same
- The percentage by weight of O in CaSO_4 (O = 16, S = 32, Ca = 40) is –
(a) 64
(b) 28.2
(c) 47.2
(d) 16.2
- The percentage by weight of Zn in white vitriol, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (Zn = 65, S = 32, O = 16, H = 1), is approximately
(a) 23
(b) 33
(c) 43
(d) 13
- The law of definite proportions was given by –
(a) John Dalton
(b) Humphry Davy
(c) Proust
(d) Michael Faraday
- Molecular mass is defined as the
(a) mass of one atom compared with the mass of one molecule
(b) mass of one atom compared with the mass of one atom of hydrogen
(c) mass of one molecule of any substance compared with the mass of one atom of C-12
(d) None of the above
- 0.001g of C is required to write a letter with a graphite pencil. The total number of C atoms used in writing the letter is
(a) 5.00×10^{12}
(b) 5×10^{19}
(c) 5.0×10^{24}
(d) 6.023×10^{23}
- One mole of a gas occupies a volume of 22.4 L. This is derived from
(a) Berzelius' hypothesis
(b) Gay-Lussac's law
(c) Avogadro's law
(d) Dalton's law
- The mass of one C atom is –
(a) 6.023×10^{23} g
(b) 1.99×10^{-23} g
(c) 2.00 g
(d) 12g
- The chemical symbol for barium is
(a) B
(b) Ba
(c) Be
(d) Bi
- The chemical symbol P stands for
(a) phosphorus
(b) potassium
(c) polonium
(d) promethium
- A group of atoms chemically bonded together is a (an)
(a) molecule
(b) ion
(c) salt
(d) element
- Adding electrons to an atom will result in a (an)
(a) molecule
(b) anion
(c) cation
(d) salt
- When an atom loses electrons, it is called a (an) _____ and has a _____ charge.
(a) anion, positive
(b) cation, positive
(c) anion, negative
(d) cation, positive

22. The molecular formula P_2O_5 means that
- a molecule contains 2 atoms of P and 5 atoms of O
 - the ratio of the mass of P to the mass of O in the molecule is 2:5
 - there are twice as many P atoms in the molecule as there are O atoms
 - the ratio of the mass of P to the mass of O in the molecule is 5 : 2
23. The correct symbol for silver is
- Ag
 - Si
 - Ar
 - Al
24. Aspartame, an artificial sweetener, has the molecular formula $C_{14}H_{18}N_2O_5$. What is the mass in grams of one molecule? (Atomic weights: C = 12.01, H = 1.008, N = 14.01, O = 16.00).
- 4.89×10^{-21}
 - 2.24×10^{-21}
 - 3.85×10^{-22}
 - 4.89×10^{-22}
25. The percentage of copper and oxygen in samples of CuO obtained by different methods were found to be the same. The illustrate the law of
- constant proportions
 - conservation of mass
 - multiple proportions
 - reciprocal proportions
26. The total number of atoms represented by the compound $CuSO_4 \cdot 5H_2O$ is
- 27
 - 21
 - 5
 - 8
27. The weight of a molecule of the compound $C_{60}H_{122}$ is
- 1.4×10^{-21} g
 - 1.09×10^{-21} g
 - 5.025×10^{23} g
 - 16.023×10^{23} g
28. The mass of a molecule of water is
- 3×10^{-26} kg
 - 3×10^{-25} kg
 - 1.5×10^{-26} kg
 - 2.5×10^{-26} kg
29. The number of atoms in 4.25g of NH_3 is approximately
- 1×10^{23}
 - 2×10^{23}
 - 4×10^{23}
 - 6×10^{23}
30. Volume of a gas at STP is 1.12×10^{-7} cc. Calculate the number of molecules in it –
- 3.01×10^{20}
 - 3.01×10^{12}
 - 3.01×10^{23}
 - 3.01×10^{24}
31. The number of molecules of CO_2 present in 44g of CO_2 is
- 6.02×10^{23}
 - 3×10^{23}
 - 12×10^{23}
 - 3×10^{10}
32. The volume occupied by 4.4g of CO_2 at STP is
- 22.4 L
 - 2.24 L
 - 0.224 L
 - 0.1 L
33. How many molecules are present in one gram of hydrogen?
- 6.02×10^{23}
 - 3.01×10^{23}
 - 2.5×10^{23}
 - 1.5×10^{23}

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- Which of the following has same mass?
 - 4g of He
 - 6.023×10^{23} atoms of He
 - 1 atom of He
 - 1 mole atoms of He
- Which of the following is/are the best example of law of conservation of mass?
 - 12g of carbon combines with 32g of oxygen to form 44g of CO_2
 - When 12g of carbon is heated in a vacuum there is no change in mass
 - A sample of air increases in volume when heated at constant pressure but its mass remains unaltered.
 - 2 g of hydrogen combines with 16 g of oxygen to form 18 g of water.
- Which one of the following pairs of gases contains the same number of molecules?
 - 16 g of O_2 and 14 g of N_2
 - 8 g of O_2 and 22 g of CO_2
 - 28 g of N_2 and 22 g of CO_2
 - 8 g of O_2 and 7 g of N_2
- Which of the following represents a polyatomic ion?
 - Sulphite
 - Chloride
 - Sulphate
 - Phosphate
- Which of the following statements is /are correct?
 - An atom is the smallest particle of matter according to Dalton's theory
 - An atom is the smallest particle of an element
 - An atom is the smallest indivisible particle of an element that can take part in a chemical change
 - An atom is the radioactive emission
- Which of the following symbols does not represent an elements?
 - CO
 - Ar
 - K
 - NO
- Which of the following element(s) has a symbol having two letters?
 - Tin
 - Uranium
 - Carbon
 - Aluminium
- Which of the following is / are not a correct symbol for an element(s)?
 - Ng
 - Fi
 - Bk
 - Zc

Atoms and Molecules

9. 0.220 g of a gas occupies a volume of 112 ml at a pressure of 1 atm and temperature of 273 K. The gas can be
(a) nitrogen dioxide (b) nitrous oxide
(c) carbon dioxide (d) propane
10. Which of the following contains the same number of molecules?
(a) 1g of O_2 , 2 g of SO_2
(b) 1g of CO_2 , 1g of N_2O
(c) 112 ml of O_2 at STP, 224 ml of He at 0.5 atm and 273K
(d) 1g of oxygen, 1g of ozone
11. SO_2 gas is slowly passed through an aqueous suspension containing 12 g $CaSO_3$ till the milkiness just disappears. What amount of SO_2 would be required ?
(a) 12.8 g (b) 6.4 g
(c) 0.1 mole (d) 0.2 mole
12. 8g of O_2 has the same number of molecules as
(a) 7g CO (b) 14g N_2
(c) 11g CO_2 (d) 16g SO_2
13. A vessel contains 4.4 g of CO_2 . It means that it contains
(a) 0.1 mole of CO_2
(b) 6.02×10^{22} molecules of CO_2
(c) 8.8g atoms of oxygen
(d) 1120 mL of CO_2 S.T.P.
- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.
1. **Assertion** : Molecular weight of oxygen is 16.
Reason : Atomic weight of oxygen is 16.
2. **Assertion** : One mole of SO_2 contains double the number of molecules present in one mole of O_2 .
Reason : Molecular weight of SO_2 is double to that of O_2 .
3. **Assertion** : Pure water obtained from different sources such as, river, well, spring, sea etc. always contains hydrogen and oxygen combined in the ratio 1 : 8 by mass.
Reason : A chemical compound always contains elements combined together in same proportion by mass, it was discovered by French chemist, Joseph Proust (1799).
4. **Assertion** : 1 amu equals to 1.66×10^{-24} g
Reason : 1.66×10^{-24} g equal to $\frac{1}{12}$ the of mass of a C^{12} atom.
5. **Assertion** : 1 mol of H_2 and O_2 each occupy 22.4 L of volume at $0^\circ C$ and 1 bar pressure.
Reason : Molar volume for all gases at the same temperature and pressure has the same volume.

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

Ashok is conducting an experiment in laboratory involving compound x. One mole of compound x weighs 35 g. If he consumed 15 g of x in his experiment. Then

1. The no. of moles of x consumed in above experiment is
(a) 1 (b) 2
(c) 3 (d) 4
2. The name of compound x is
(a) NH_4OH (b) NaOH
(c) NaCl (d) Na_2O_2
3. If in another experiment only 70g of that substances is required. Then what will be the number of moles now ?
(a) 3 (b) 2
(c) 4 (d) 1

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

Multiple Matching Questions :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

1.	Column I	Column II		
(A)	1.5 mole of $\text{CO}_2(\text{g})$	(p) 33600 ml at STP		
(B)	3.0 g of H_2	(q) Total number of atoms = $4.5 \times N_A$		
(C)	1.5 mole of $\text{O}_3(\text{g})$	(r) Weighs 72 g		
(D)	1 mole of oxygen	(s) Weighs 32 g		
	A	B	C	D
(a)	p, q, r	p	p, q	s
(b)	p, q	p	p, q, r	s
(c)	p, q	s	p, q, r	p
(d)	p	p, q, r	p, q	s

Integer Type Questions :

DIRECTIONS : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

1. Calculate the number of gram molecules of oxygen in 6.02×10^{24} CO molecules.
2. If the mass of a molecule of water is $x \times 10^{-26}$. Find the value of x.

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1 to 20): This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- 25.4g of iodine and 14.2g of chlorine are made to react completely to yield a mixture of ICl and ICl_3 . Calculate the ratio of moles of ICl and ICl_3
(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2
- Two samples of lead oxide were separately reduced to metallic lead by heating in a current of hydrogen. The weight of lead from one oxide was half the weight of lead obtained from the other oxide. The data illustrates –
(a) law of reciprocal proportions
(b) law of constant proportions
(c) law of multiple proportions
(d) law of equivalent proportions
- In compound A, 1.00g of nitrogen unites with 0.57g of oxygen. In compound B, 2.00g of nitrogen combines with 2.24g of oxygen. In compound C, 3.00g of nitrogen combines with 5.11g of oxygen. These results obey the following law
(a) law of constant proportion
(b) law of multiple proportion
(c) law of reciprocal proportion
(d) Dalton's law of partial pressure
- Which one of the following pairs of compounds illustrate the law of multiple proportions ?
(a) H_2O and Na_2O (b) MgO and Na_2O
(c) Na_2O and BaO (d) SnCl_2 and SnCl_4
- The molecular weight of O_2 and SO_2 are 32 and 64 respectively. At 15°C and 150 mm Hg pressure, one litre of O_2 contains 'N' molecules. The number of molecules in two litres of SO_2 under the same conditions of temperature and pressure will be :
(a) $\text{N}/2$ (b) N (c) 2N (d) 4N
- Among the following pairs of compounds, the one that illustrates the law of multiple proportions is
(a) NH_3 and NCl_3 (b) H_2S and SO_2
(c) CS_2 and FeSO_4 (d) CuO and Cu_2O
- In a compound C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular weight of compound is 108. Molecular formula of compound is
(a) $\text{C}_2\text{H}_6\text{N}_2$ (b) $\text{C}_3\text{H}_4\text{N}$ (c) $\text{C}_6\text{H}_8\text{N}_2$ (d) $\text{C}_9\text{H}_{12}\text{N}_3$
- The simplest formula of a compound containing 50% of element X (atomic mass 10) and 50% of element Y (atomic mass 20) is
(a) XY (b) XY_3 (c) X_2Y (d) X_2Y_3
- Empirical formula of hydrocarbon containing 80% carbon and 20% hydrogen is :
(a) CH_3 (b) CH_4 (c) CH (d) CH_2
- The empirical formula of an acid is CH_2O_2 , the probable molecular formula of acid may be :
(a) $\text{C}_3\text{H}_6\text{O}_4$ (b) CH_2O (c) CH_2O_2 (d) $\text{C}_2\text{H}_4\text{O}_2$
- An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of the compound is :
(a) $\text{C}_3\text{H}_5\text{O}_2$ (b) $\text{C}_4\text{H}_{10}\text{O}_2$
(c) $\text{C}_6\text{H}_{10}\text{O}_4$ (d) $\text{C}_3\text{H}_{10}\text{O}_2$
- The empirical formula of a compound is CH_2 . One mole of this compound has a mass of 42 grams. Its molecular formula is :
(a) C_3H_6 (b) C_3H_8 (c) CH_2 (d) C_2H_2
- A compound contains 54.55 % carbon, 9.09% hydrogen, 36.36% oxygen. The empirical formula of this compound is:
(a) $\text{C}_3\text{H}_5\text{O}$ (b) $\text{C}_4\text{H}_8\text{O}_2$
(c) $\text{C}_2\text{H}_4\text{O}_2$ (d) $\text{C}_2\text{H}_4\text{O}$
- If two compounds have the same empirical formula but different molecular formula, they must have
(a) Different percentage composition
(b) Different molecular weights
(c) Same viscosity
(d) Same vapour density
- An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C, 38.71% and H, 9.67%. The empirical formula of the compound would be :
(a) CH_3O (b) CH_2O (c) CHO (d) CH_4O
- A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g. of CO_2 . The empirical formula of the hydrocarbon is :
(a) C_2H_4 (b) C_3H_4
(c) C_6H_5 (d) C_7H_8
- The formation of SO_2 and SO_3 explain
(a) the law of conservation of mass
(b) the law of multiple proportions
(c) the law of definite properties
(d) Boyle's law

Atoms and Molecules

18. Which one of the following is smallest in size ?
 (a) N^{3-} (b) O^{2-}
 (c) Na^+ (d) F^-
19. Which one of the following has largest size?
 (a) Al (b) Al^{3+}
 (c) Al^+ (d) Al^{2+}
20. The correct sequence which shows decreasing order of the ionic radii of the elements is
 (a) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$
 (b) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$
 (c) $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$
 (d) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

DIRECTIONS (Qs. 21 to 24): This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

21. Which of the following pairs of substances illustrate the law of multiple proportions?
 (a) CO and CO_2 (b) H_2O and D_2O
 (c) SO_2 and SO_3 (d) MgO and $\text{Mg}(\text{OH})_2$
22. Which of the following pairs of compound illustrate the law of multiple proportions.
 (a) CO and CO_2 (b) NH_3 and NCl_3
 (c) H_2O and H_2O_2 (d) CO_2 and CH_4
23. Which of the following set of compounds illustrates the law of reciprocal proportions?
 (a) HCl, HBr, HI (b) CO_2 , CH_4 , H_2O
 (c) NH_3 , N_2O_5 , H_2O (d) NH_3 , NCl_3 , N_2O_3
24. Which of the following show correct order to of ionic radii
 (a) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+$ (b) $\text{Ca}^{++} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$
 (c) $\text{K}^+ > \text{Ca}^{++}$ (d) $\text{N}^{3-} > \text{O}^{--} > \text{f}^-$

DIRECTIONS (Qs. 25 to 27): Study the given paragraph(s) and answer the following questions.

Passage 1

An empirical formula gives the smallest whole-number ratio of atoms of each element in a compound. NO_2 is an empirical formula showing that nitrogen and oxygen atoms are present in a 1 : 2 ratio. N_2O_4 is not an empirical formula. It shows a 2 : 4 ratio of nitrogen atoms to oxygen atoms. The 2 : 4 ratio has a common divisor, 2, and can therefore be reduced to 1 : 2, just as you reduce a fraction to lowest terms. Therefore, the empirical formula of N_2O_4 is NO_2 .

Not all empirical formulas represent real compounds. For example, the molecular formula for hydrogen peroxide is H_2O_2 , which gives HO as its empirical formula. There is no known stable substance with the formula HO.

The distinction between empirical and molecular formulas may be summarised as follows:

If the subscripts of a formula can be divided evenly by a whole number, it is a molecular formula. If the subscripts of a formula cannot be reduced further, it is an empirical formula. It may also be a molecular formula.

25. What is the empirical formula of the compound whose molecular formula is P_4O_{10} ?
 (a) PO (b) PO_2
 (c) P_2O_5 (d) P_8O_{20}
26. A compound is 86% carbon and 14% hydrogen by mass. What is the empirical formula for this compound?
 (a) CH (b) CH_2
 (c) CH_3 (d) CH_4
27. What is the empirical formula of a compound that contains 85% Ag and 15% Cl by mass?
 (a) AgF (b) Ag_2F
 (c) AgF_2 (d) Ag_2F_2

DIRECTIONS (Qs. 28 to 30): Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) **Assertion** is True, **Reason** is True, **Reason** is a **correct explanation** for **Assertion**
 (b) **Assertion** is True, **Reason** is True ; **Reason** is NOT a **correct explanation** for **Assertion**
 (c) **Assertion** is True, **Reason** is False
 (d) **Assertion** is False, **Reason** is True
28. **Assertion** : Volume of a gas is inversely proportional to the number of moles of gas.
Reason : The ratio by volume of gaseous reactants and products is in agreement with their mole ratio.
29. **Assertion** : One mole of SO_2 contains double the number of molecules present in one mole of O_2 .
Reason : Molecular weight of SO_2 is double to that of O_2 .
30. **Assertion**: The empirical mass of ethene is half of its molecular mass.
Reason: The empirical formula represents the simplest whole number ratio of various atoms present in a compound.

DIRECTIONS (Q. 31) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

31.	Column I				Column II
	Empirical formula				Molecular formula
	(A)	CH			(p) N_2H_4
	(B)	NH_2			(q) C_2H_4
	(C)	CH_2O			(r) C_2H_2
	(D)	CH_2			(s) C_6H_6
					(t) $\text{C}_2\text{H}_4\text{O}_2$
	A	B	C	D	
(a)	r, s	p	t	q	
(b)	p	r, s	t	q	
(c)	r, s	t	p	q	
(d)	q	p	t	r, s	

DIRECTIONS (Qs. 32 to 36) : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

32. Number of moles of NaOH present in 2 litre of 0.5 M NaOH _____ is
33. A compound was found to contain 55.2 percent xenon and 44.8 percent chlorine. If the the empirical formula of the compound is XeCl_n . Calculate the value of n.
34. A 10g sample of a compound contains 4.00g C, 0.667g H, and 5.33g O. if the the ratio between empirical and molecular formulas is n:1 find the value of n. The MW is 180 amu.

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- masses
- polyatomic
- chemical formula
- 6.022×10^{23} , 12g
- symbol.
- molecular formula.
- simple ions.
- metal and non-metals.
- equal.
- mass of atoms & number of atoms.
- mole

True / False :

- True
- True
- False
- True
- False
- True
- False
- False
- True

Match the Column :

- $A \rightarrow (r), B \rightarrow (s), C \rightarrow (p), D \rightarrow (q),$
- $A \rightarrow (q), B \rightarrow (p), C \rightarrow (r), D \rightarrow (s)$

Very Short Answer Questions :

- In one gram atomic weight of any element 6.02×10^{23} atoms of that element are present. This quantity of the element is known as one mole.
- One gram atomic mass of any element contains the same number of atom of that element as there are carbon atoms in exactly 12 g of carbon-12. This number is Avogadro's number i.e., 6.023×10^{23} .
- The atomic mass of an element when expressed in grams is known as **gram atomic mass** or simply as **gram atom (g-atom)**. Thus one g-atom of carbon (C-12) weighs 12.0 g. The number of gram-atoms of any element in its certain weight is given by:

$$\text{Number of gram-atom} = \frac{\text{Mass of the element in grams}}{\text{Atomic mass}}$$
- CO_2 represents one molecule of triatomic carbon dioxide while the notation 2CO_2 represents two molecules of carbon dioxide.
- Phosphorus trichloride and sulfur dioxide.
- 1 : 2
- 2.59×10^{-3} mole of Na atoms.
- The formula weight of a substance in amu has the same numerical value as the molar mass expressed in grams.

- No. of grams of Ag = $(0.263 \text{ mol of Ag}) \frac{107.9 \text{ g of Ag}}{1 \text{ mol of Ag}}$
 $= 28.4 \text{ g of Ag}$
- No. of atoms of Pb = $(1.00 \times 10^{-9} \text{ g of Pb}) \frac{1 \text{ mol of Pb}}{207.2 \text{ g of Pb}}$
 $\left(\frac{6.022 \times 10^{23} \text{ atoms of Pb}}{1 \text{ mol of Pb}} \right) = 2.906 \times 10^{12} \text{ atoms of Pb}$
- No. of g of Fe
 $= (25.6 \text{ g of O}) \left(\frac{1 \text{ mol of O}}{16.0 \text{ g mol of O}} \right) \left(\frac{2 \text{ mol of Fe}}{3 \text{ mol of O}} \right) \left(\frac{55.8 \text{ g of Fe}}{1 \text{ mol of Fe}} \right)$
 $= 59.5 \text{ g of Fe}$
- The atomic mass of aluminium is given to be 27u. This means that 1 mole of aluminium atoms has a mass of 27 grams.
 Now, 1 mole of aluminium atoms = 27g
 So, 4 moles of aluminium atoms = $27 \times 4 \text{ g} = 108 \text{ g}$
- 6.022×10^{23} atom of He weighs 4 g
 $\therefore 6.022 \times 10^{22}$ atoms of He will weighs

$$\frac{4 \times 6.022 \times 10^{22}}{6.022 \times 10^{23}} = 0.4 \text{ g}$$
- 6.022×10^{23} molecules of $\text{CO}_2 = 1 \text{ mole of CO}_2$
 $\therefore 3.011 \times 10^{22}$ molecules of CO_2

$$= \frac{3.011 \times 10^{22}}{6.022 \times 10^{23}} = 0.05 \text{ moles of CO}_2$$
- No. of mol H_2SO_4
 $= (45.8 \text{ g of H}_2\text{SO}_4) \left(\frac{1 \text{ mol of H}_2\text{SO}_4}{98.1 \text{ g of H}_2\text{SO}_4} \right) = 0.467 \text{ mol of H}_2\text{SO}_4$
- The atomic mass of Fe = 56 gm
 1 mole of Fe = 56 gm
 5 moles of Fe = $5 \times 56 \text{ gm} = 280 \text{ gm}$
- 1 mole of NaCl = Molecular mass of NaCl in grams
 $= 60 \text{ gm}$
 60 gm of NaCl = 1 mole
 20 gm of NaCl will have 0.33 mole of NaCl.
- Molecular mass of CaCO_3 is 100 gm. This has 48 gm of O. Thus 50 gm of CaCO_3 will have 24 gm of O.
- Molecular weight of $\text{CO}_2 = 12 + 32 = 44$
 $\therefore$ Quantity of CO_2 present in 1 mole CO_2 is 44 grams
 $\therefore$ Quantity present in 0.1 mole of CO_2 would be

$$\frac{44 \times 0.1}{1} = 4.4 \text{ gm}$$

20. Because the atomic weight of oxygen is 15.999 amu, a mole of oxygen atoms has a mass of 15.999 grams. Each O_2 molecule has two atoms, however, so the molecular weight of O_2 molecules is twice as large as the atomic weight of the atom.
- 1 mol O = 15.999 g, 1 mol O_2 = 31.998g
21. It is the smallest repeating formula in the structure of ionic compound.

Short Answer Questions :

- (a) 32 g O_2 = 1 mole
 $\therefore 12 \text{ g } O_2 = \frac{12}{32} = 0.375 \text{ mole}$
 (b) 44 g CO_2 = 1 mole
 $\therefore 22 \text{ g } CO_2 = 0.5 \text{ mole}$
- The molecular mass of a substance (element or compound) is the number of times the molecule of the substance is heavier than 1/12th the mass of an atom of carbon-12 isotope.
 Molecular weight is calculated by adding the atomic weights of all the constituent atoms present in a molecule.
 For example.
 Molecular weight of a molecule of hydrogen
 $(H_2) = 2 \times \text{atomic weight of hydrogen} = 2 \times 1 = 2 \text{ amu}$
- (a) NaF (b) Na_2O
 (c) MgF_2 (d) Al_4C_3
- (a) Al_2S_3 (b) SrF_2
 (c) TiO_2 (d) $CaBr_2$
- According to formula of Al_2O_3 . In one mole of aluminium oxide
 3 moles of oxygen combines with 2 moles of aluminium.
 1 mole of oxygen combines with $\frac{2}{3}$ moles of aluminium.
 1.58 mole of oxygen combines with $\frac{2}{3} \times 1.58$ moles of aluminium.
 1.58 mole of oxygen combines with 1.053 moles of aluminium.
- Molecular weight of $SO_2 = 32 + 2 \times 16 = 64$
 $\therefore 64 \text{ g of } SO_2 \text{ at STP occupies } 22.4 \text{ litres}$
 Hence 32g of SO_2 at STP will occupy
 $\frac{22.4}{64} \times 32 = 11.2 \text{ litres}$
- Number of moles of magnesium = $\frac{2.8}{24} = \frac{1}{4} \text{ mol}$
 $\therefore$ Number of moles of sodium should also be $\frac{1}{4}$ and
 1 mol of Na = 23 g
 $\therefore \frac{1}{4} \text{ mol of Na will be } 23 \times \frac{1}{4} = 5.75 \text{ g}$

8. Molar mass of $Cr_2S_3 = 52 \times 2 + 32 \times 3 = 200 \text{ g}$
 $\therefore 200 \text{ g of } Cr_2S_3 \text{ contains } 2 \text{ mol of}$
 $Cr \text{ or } 2 \times 52 = 104 \text{ g of } Cr$
 $\therefore 85 \text{ g of } Cr_2S_3 \text{ contain } \frac{104}{200} \times 85 = 44.2 \text{ g of } Cr.$

9. Number of atoms of Na
 $= \frac{\text{Given mass of sodium}}{\text{Molar mass of sodium}} \times \text{Avogadro number}$
 $= \frac{100}{23} \times 6.022 \times 10^{23} = 2.618 \times 10^{24}$

Number of atoms of Fe

$$= \frac{100}{56} \times 6.022 \times 10^{23} = 1.075 \times 10^{24}$$

We find that 100 g of Na contains more number of atoms than 100 g of Fe.

10. The amount of oxygen per gram of nitrogen in NO_2 should be exactly twice as that of NO , as required by the formulas of the two substances. Therefore, 2.285g oxygen would combine with 1.000g nitrogen.
11. No. of moles of N

$$= (8.60 \text{ mol of O}) \left(\frac{2 \text{ mol of N}}{5 \text{ mol of O}} \right) = 3.44 \text{ mol of N}$$

12. Molecular weight of $N_2 = 2 \times 14 = 28 \text{ g}$
 28 g of N_2 at STP occupies = 22.4L
 2.8 g of N_2 at STP = ?

$$\frac{22.4 \times 2.8 \text{ g}}{28 \text{ g}} = 2.24 \text{ L}$$

2.8 g of N_2 at STP occupies a volume of 2.24 L.

13. $\therefore$ At NTP weight of 22.4 litres of one mole of gas is equal to its molecular weight. So volume of 32 grams of Oxygen at NTP = 22.4 litre.
 $\therefore$ Volume of 4.0 gram of oxygen at NTP is

$$= \frac{22.4}{32} \times 4.0 = 2.8 \text{ litres}$$

$\therefore$ Weight of 22.4 litres of CO_2 at NTP = 44 grams.

$\therefore$ Weight of 2.8 litres of CO_2 at NTP

$$= \frac{44 \times 2.8}{22.4} = 5.5 \text{ gram}$$

14. In order to solve this problem, we should know the mass of 1 mole of water. This can be obtained by using the given values of the atomic masses of hydrogen and oxygen as follows :

1 mole of water (H_2O)

= Molecular mass of H_2O in grams

= Mass of 2H atoms + Mass of O atom

= $2 \times 1 + 16 = 2 + 16 = 18 \text{ grams}$

Thus, the mass of 1 mole of water is 18 grams.

So, Mass of 0.5 mole of water = $18 \times 0.5 \text{ g} = 9 \text{ g}$

Thus, the mass of 0.5 mole of water (H_2O) is 9 grams

Atoms and Molecules

15. One moles of H_2SO_4 contains 1 mole of sulphur atoms and 4 mole of oxygen atoms.
One mole of H_2SO_3 has 1 mole of sulphur atoms and 3 moles of oxygen atoms.
One mole of SO_2 has 1 mole of sulphur atoms and 2 moles of oxygen atoms.
16. Molar mass of $\text{NH}_3 = 14 \times 1 + 1 \times 3 = 17 \text{ g/mol}$
Percentage of nitrogen

$$= \frac{\text{Molar mass of N in } \text{NH}_3}{\text{Molar mass of } \text{NH}_3} \times 100$$

$$= \frac{14}{17} \times 100 = 82.35\%$$
Percentage of hydrogen $= \frac{3}{17} \times 100 = 17.65\%$
17. We first determine the number of grams of each element that are present in one mole of sample :
 $2 \text{ mol of N} \times 14.01 \text{ g/mol} = 28.02 \text{ g of N}$
 $3 \text{ mol of O} \times 16.00 \text{ g/mol} = 48.00 \text{ g of O}$
The percentages by mass are then obtained by using the formula mass of the compound (76.02 g) :

$$\% \text{ N} = \frac{28.02}{76.02} \times 100 = 36.86\% \text{ of N}$$

$$\% \text{ O} = \frac{48.00}{76.02} \times 100 = 63.14\% \text{ of O}$$
18. Mass of the reactants = 10g
Mass of the products = 4.6 + 5.6g = 10g
Since the mass of the reactants is equal to the mass of the products, the observations are in agreement with the law of conservation of mass.
19. 1 mole of H_2O molecules has 2 moles of H atoms.
3 moles of H_2O molecules will have 6 moles of H atoms.
1 mole of HCl molecules will have 1 mole of H atoms.
10 moles of HCl will have 10 moles of H atoms.
Thus 10 moles of HCl will have more atoms than 3 moles of H_2O .
20. Water (H_2O) contains hydrogen and oxygen in the ratio of 1 : 8 (by weight). It means if 9g of water is decomposed it will produce 1g of hydrogen and 8g of oxygen. Thus on decomposing 50g of water.
Weight of hydrogen obtained $= \frac{1}{9} \times 50 \text{ g} = 5.56 \text{ g}$
Weight of oxygen obtained $= \frac{8}{9} \times 50 \text{ g}$

$$= \frac{400}{9} \text{ or } 44.44\text{g}$$
21. Here we have been given that hydrogen and oxygen always combine in the fixed ratio of 1 : 8 by mass. This means that :
1g of hydrogen gas requires = 8g of oxygen gas
So, 3g of hydrogen gas requires
 $= 8 \times 3 \text{ g of oxygen gas} = 24 \text{ g of oxygen gas}$
Thus, 24 grams of oxygen gas would be required to react completely with 3 grams of hydrogen gas.
22. As number of molecules of a substance present in one gram mole is 6.02×10^{23} .
So number of molecules present in 32 grams of oxygen is 6.02×10^{23} .
 $\therefore$ Number of molecules present in 28 grams of nitrogen is 6.02×10^{23} .
So number of molecules present in 14 grams of nitrogen is

$$= \frac{6.02 \times 10^{23} \times 14}{28} = 3.01 \times 10^{23}$$
23. The molar mass of iron is 55.85 g/mol.
In order to convert from moles to grams we need the unit factor that tells us how many grams of iron can be found in one mole of this metal.

$$0.250 \text{ mol Fe} \times \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} = 14.0 \text{ g}$$
24. Before we can do anything else, we need to know the number of moles of a aluminium metal in the sample. This can be calculated from the mass of the sample and the molar mass of aluminium, which is 26.982 g/mol.

$$0.123 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.982 \text{ g Al}} = 4.56 \times 10^{-3} \text{ mol Al}$$

Now, we can use Avogadro's number to calculate the number of atoms in the sample.

$$4.56 \times 10^{-3} \text{ mol Al} \times \frac{6.02 \times 10^{23} \text{ Al atoms}}{1 \text{ mol Al}} = 2.75 \times 10^{21} \text{ Al atoms}$$

Long Answer Questions :

1. (a) $\text{NaHCO}_3 = 1 \text{ Na} + 1 \text{ H} + 1 \text{ C} + 3 \text{ O}$
 $= (22.99) + (1.01) + (12.01) + (3 \times 16.00)$
 $= 84.0 \text{ g/mole}$
- (b) $\text{K}_2\text{Cr}_2\text{O}_7 = 2 \text{ K} + 2 \text{ Cr} + 7 \text{ O}$
 $= (2 \times 39.10) + (2 \times 52.00) + (7 \times 16.00)$
 $= 294.2 \text{ g/mole}$
- (c) $(\text{NH}_4)_2\text{CO}_3 = 2 \text{ N} + 8 \text{ H} + 1 \text{ C} + 3 \text{ O}$
 $= (2 \times 14.01) + (8 \times 1.01) + (12.01) + (3 \times 16.00)$
 $= 96.1 \text{ g/mole}$
- (d) $\text{Al}_2(\text{SO}_4)_3 = 2 \text{ Al} + 3 \text{ S} + 12 \text{ O}$
 $= (2 \times 26.98) + (3 \times 32.07) + (12 \times 16.00)$
 $= 342.2 \text{ g/mole}$

$$\begin{aligned}
 \text{(e) } \text{CuSO}_4 \cdot 5\text{H}_2\text{O} &= 1\text{Cu} + 1\text{S} + 9\text{O} + 10\text{H} \\
 &= 63.55 + 32.07 + (9 \times 16.00) + (10 \times 1.01) \\
 &= 249.7 \text{ g/mole}
 \end{aligned}$$

2. (a) (i) Molar mass of Ethyne
 $= 2 \times \text{Molar mass of C atom} + 2 \times \text{Molar mass of H atom}$
 $= 2 \times 12.011 + 2 \times 1 \text{ g} = 26.022 \text{ g}$
- (ii) Molar mass of S_8 molecule
 $= 8 \times \text{Molar mass of S atom}$
 $= 8 \times 32.066 = 256.53 \text{ g}$
- (iii) Molar mass of P_4 molecule
 $= 4 \times \text{molar mass of P atom} = 4 \times 31 \text{ g} = 124 \text{ g}$
- (iv) Molar mass of HCl
 $= \text{Molar mass of H atom} + \text{Molar mass of Cl atom}$
 $= 1 + 35.453 \text{ g} = 36.453 \text{ g}$
- (v) Molar mass of HNO_3
 $= \text{Molar mass of H atom} + \text{Molar mass of N atom} + 3 \times \text{Molar mass of O atom}$
 $= 1 + 14.007 + 3 \times 15.999 \text{ g}$
 $= 1 + 14.007 + 47.997 \text{ g}$
 $= 63.004 \text{ g}$

3. This problem involves aluminium ions. Please note that the mass of an aluminium ion is the same as that of an aluminium atom. In order to solve this problem, first of all we have to find out the mass of aluminium atoms in 0.051g of aluminium oxide (which will give us the mass of aluminium ions) This can be done as follows :

$$\begin{aligned}
 1 \text{ mole of } \text{Al}_2\text{O}_3 &= \text{Formula mass of } \text{Al}_2\text{O}_3 \text{ in grams} \\
 &= \text{Mass of Al} \times 2 + \text{Mass of O} \times 3 \\
 &= 27 \times 2 + 16 \times 3 = 54 + 48 = 102 \text{ grams}
 \end{aligned}$$

Now, 1 mole of Al_2O_3 contains 2 moles of Al.

So, Mass of Al in 1 mole of

$$\text{Al}_2\text{O}_3 = \text{Mass of Al} \times 2 = 27 \times 2 = 54 \text{ grams}$$

Now, 102 g of aluminium oxide contains = 54 g of Al

Now, 0.051 g aluminium oxide contains

$$= \frac{54}{102} \times 0.051 \text{ g Al} = 0.027 \text{ g of Al}$$

The atomic mass of aluminium is given to be 27u. This means that 1 mole of aluminium atoms (or aluminium ions) has a mass of 27 grams, and it contains 6.022×10^{23} aluminium ions.

Now, 27g of aluminium has ions = 6.022×10^{23}

So, 0.027g of aluminium has ions

$$= \frac{6.022 \times 10^{23}}{27} \times 0.027 = 6.022 \times 10^{20}$$

Thus, the number of aluminium ions (Al^{3+}) in 0.051 gram of aluminium oxide is 6.022×10^{20} .

4. (I)
- (a) Cu_2O (b) K_2O_2
 (c) Hg_2Br_2 (d) $\text{Fe}_2(\text{CO}_3)_3$
 (e) NaBrO

(II)

- (a) Bromic acid (b) Hydrobromic acid
 (c) Phosphoric acid (d) Hypochlorous acid
 (e) Iodic acid

5. (I) The average mass of a molecule of carbon dioxide would be equal to the sum of the atomic weights of the three atoms in a CO_2 molecule.

Mass of a single CO_2 molecule :

$$1 \text{ C atom} = 1 (12.011 \text{ amu}) = 12.011 \text{ amu}$$

$$2 \text{ O atom} = 2 (15.999 \text{ amu}) = 31.998 \text{ amu}$$

$$\underline{\hspace{1.5cm}} \\ 44.009 \text{ amu}$$

The mass of a mole of carbon dioxide would be 44.009 grams.

The average mass of a molecule of glucose is equal to the sum of the atomic weights of the 24 atoms in a $\text{C}_6\text{H}_{12}\text{O}_6$ molecule.

Mass of a single $\text{C}_6\text{H}_{12}\text{O}_6$ molecule :

$$6 \text{ C atom} = 6 (12.011 \text{ amu}) = 72.066 \text{ amu}$$

$$12 \text{ H atoms} = 12 (1.0079 \text{ amu}) = 12.095 \text{ amu}$$

$$6 \text{ O atoms} = 6 (15.999 \text{ amu}) = 95.994 \text{ amu}$$

$$\underline{\hspace{1.5cm}} \\ 180.155 \text{ amu}$$

The molecular weight of the compound is therefore 180.155 grams/mol.

- (II) The first in this calculation involves converting the mass of the sample into the number of moles of CO_2 .

$$0.500 \text{ g of } \text{CO}_2 \times \frac{1 \text{ mol of } \text{CO}_2}{44.009 \text{ g of } \text{CO}_2}$$

$$= 1.14 \times 10^{-2} \text{ moles of } \text{CO}_2$$

Once we know the number of moles, we can use Avogadro's number to calculate the number of CO_2 molecules in the sample.

$$1 \text{ mole of } \text{CO}_2 = 6.023 \times 10^{23} \text{ molecules}$$

$$1.14 \times 10^{-2} \text{ mol of } \text{CO}_2$$

$$= 6.023 \times 10^{23} \times 1.14 \times 10^{-2} \text{ } \text{CO}_2 \text{ molecules}$$

$$= 6.86 \times 10^{21} \text{ } \text{CO}_2 \text{ molecules}$$

We can now use the chemical formula for carbon dioxide to determine the number of carbon atoms in the sample. The formula suggests that there is a single carbon atom for each CO_2 molecule.

$$6.86 \times 10^{21} \text{ } \text{CO}_2 \text{ molecules} = 6.86 \times 10^{21} \text{ C atoms}$$

6. (i)

	Atom		Molecule
1.	It is the smallest unit of an element	1.	It is the smallest unit of an element or a compound.
2.	It may or may not be capable of free existence but it is the smallest unit that takes part in a chemical reaction.	2.	It is capable of free existence.

(ii)

	Atom		Ion
1.	It is neutral and carries no charge.	1.	It is a charged particle and carries a positive or negative charge.
2.	The number of electrons in an atom is equal to its atomic number.	2.	In cation (ion with positive charge) the number of electron are less than that present in corresponding atom. In an anion (ion with negative charge) the number of electrons are more than that present in corresponding atom.
3.	It is quite reactive.	3.	It is quite stable.

2 EXERCISE

Text - Book Questions :

- Here, the mass of reactants = $(5.3 + 6.0) \text{ g} = 11.3 \text{ g}$
Mass of products = $(8.2 + 2.2 + 0.9) \text{ g} = 11.3 \text{ g}$
Since, the mass of reactants is equal to mass of products. So, these observations are in agreement with the law of conservation of mass.
- Hydrogen + Oxygen = Water
The ratio in which hydrogen and oxygen combine is 1 : 8 i.e., 1 g of hydrogen requires 8 g of oxygen for complete reaction.
Thus, mass of oxygen required by 1 g of hydrogen = 8 g
 $\therefore$ Mass of oxygen required by 3 g of hydrogen
 $= \frac{8}{1} \times 3 \text{ g} = 24 \text{ g}$
- The law of conservation of mass is the result of the following postulate of Dalton's atomic theory.
Atoms are indivisible and they can neither be created nor be destroyed
- The following postulates of Dalton's atomic theory can explain the law of definite proportions
Atoms of various element combine in a simple whole number but fixed ratio to form compound atom (molecules).
- One atomic mass unit (a.m.u. or 'U') is exactly equal to one-twelfth $\left(\frac{1}{12}\text{th}\right)$ of the mass of one atom of carbon – 12

$1 \text{ U} = \frac{1}{12} \text{ th of mass of 1 atom of C-12}$
- We cannot see an atom with naked eye because of its *very small size*. The radius of an atom is the order of 10^{-10} m and such a small object cannot be seen with a naked eye.
- Sodium Oxide
 $\text{Na}^+ \text{O}^{2-}$
 Na_2O , or Na_2O (neglect the subscript 1)
 - Aluminium Chloride
 $\text{Al}^{3+} \text{Cl}^-$
 Al_1Cl_3 or AlCl_3
 - Sodium Sulphide
 $\text{Na}^+ \text{S}^{2-}$
 Na_2S_1 or Na_2S
 - Magnesium Hydroxide
 $\text{Mg}^{2+} (\text{OH})^{-1}$
 $\text{Mg}_1(\text{OH})_2$ or $\text{Mg} (\text{OH})_2$
- Aluminium sulphate
 - Calcium chloride
 - Potassium Sulphate
 - Potassium nitrate
 - Calcium carbonate
- The chemical formula of a compound represents the composition of a molecule of the compound in terms of symbols of the elements present in it.
- H_2S molecules consists of 2 atom of hydrogen and one atom of sulphur. So the total number of atoms present in a molecule of H_2S is 3 ($2 + 1 = 3$). Thus, the atomicity of H_2S is 3.
 - PO_4^{3-} ion contains 1 atom of phosphorus and 4 atoms of oxygen. So, the total number of atom in it is 5 ($1 + 4 = 5$).
- Molecular mass of $\text{H}_2 = 2 \times 1 = 2 \text{ U}$ (or 2 amu)
Molecular mass of $\text{O}_2 = 2 \times 16 = 32 \text{ U}$
Molecular mass of $\text{Cl} = 2 \times 35.5 = 71 \text{ U}$
Molecular mass of $\text{CO}_2 = 12 + 2 \times 16$
 $= 12 + 32 = 44 \text{ U}$
Molecular mass of $\text{CH}_4 = 1 \times 12 + 4 \times 1$
 $= 12 + 4 = 16 \text{ U}$
Molecular mass of $\text{C}_2\text{H}_6 = 2 \times 12 + 6 \times 1$
 $= 24 + 6 = 30 \text{ U}$
Molecular mass of $\text{C}_2\text{H}_4 = 2 \times 12 + 4 \times 1$
 $= 24 + 4 = 28 \text{ U}$
Molecular mass of $\text{NH}_3 = 1 \times 14 + 3 \times 1$
 $= 14 + 3 = 17 \text{ U}$
Molecular mass of CH_3OH
 $= 1 \times 12 + 3 \times 1 + 1 \times 16 + 1 \times 1$
 $= 12 + 3 + 16 + 1 = 32 \text{ U}$
- Formula unit mass of $\text{Zn} = 1 \times 65 + 1 \times 16$
 $= 65 + 16 = 81 \text{ U}$
Formula unit mass of $\text{Na}_2\text{O} = 2 \times 23 + 1 \times 16$
 $= 46 + 16 = 62 \text{ U}$
Formula unit mass of K_2CO_3
 $= 2 \times 39 + 1 \times 12 + 3 \times 16$
 $= 78 + 12 + 48 = 138 \text{ U}$
- Mass of 1 mole carbon atom = 12 g (given)
 $\therefore$ Mass of 6.023×10^{23} atoms of carbon
 $= 12 \text{ g}$ (1 mole = 6.023×10^{23} atom)
 $\therefore$ Mass of 1 atom of carbon = $\frac{12}{6.023 \times 10^{23}} \text{ g}$
 $= 1.99 \times 10^{-23} \text{ g}$

14. Number of atom

$$= \frac{\text{Mass in grams}}{\text{Atomic Mass}} \times 6.023 \times 10^{23}$$

$$\therefore \text{Number of Na atom} = \frac{100}{23} \times 6.023 \times 10^{23}$$

$$= 2.62 \times 10^{24} \text{ atom}$$

$$\text{Number of Fe atoms} = \frac{100}{56} \times 6.023 \times 10^{23}$$

$$= 1.07 \times 10^{24} \text{ atoms}$$

Thus, number of atom is more in 100 g of sodium.

Text - Book Exercise :

- Mass of the compound = 0.24g
Mass of boron in the compound = 0.096 g
Mass of oxygen in the compound = 0.144g
Percentage of boron (by weight)

$$= \frac{\text{Mass of boron}}{\text{Mass of compound}} \times 100$$

$$= \frac{0.096}{0.24} \times 100 = 40$$
 Percentage of oxygen (by weight)

$$= \frac{\text{Mass of oxygen}}{\text{Mass of compound}} \times 100$$

$$= \frac{0.144}{0.24} \times 100 = 60$$
- The reaction between carbon and oxygen to form carbon dioxide can be represented by the following chemical equation.

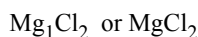
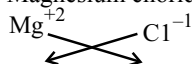
$$\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$$

$$(2 \times 16)$$

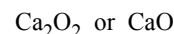
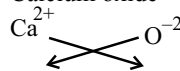
12g	32g	12 + 32 = 44g
or 3g	8g	11g

From the above data we find that 3g of carbon combines with 8g of oxygen to form 11g of carbon dioxide.
This is also the data given in first case.
In second case when 3.0 g of carbon is burnt in 50.0g of oxygen, only 8.0g of oxygen will combine with 3.0 g of carbon to produce 11.0g of CO₂ and the remaining oxygen will remain as such (i.e., unreacted)
Thus, mass of carbon dioxide formed will be 11g
In second case also the mass of unreacted oxygen = (50 - 8.0) = 42.0g
[In this case carbon that is present in smaller amounts and is completely consumed in the limiting reagent. The answer of this problem is based on laws of constant proportions.]
- Polyatomic ions are the group of atoms that carry either same positive charge (cation) or same negative charge (anion) e.g. NH₄⁺, CO₃²⁻, NO₃⁻, PO₄³⁻

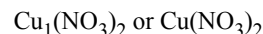
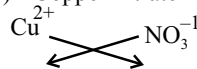
4. (a) Magnesium chloride



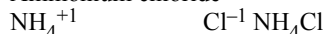
- (b) Calcium oxide



- (c) Copper nitrate



- (d) Ammonium chloride



- (e) Calcium + carbonate



- (f) Zinc sulphate



- Quick lime is the commercial name of calcium oxide (CaO). The elements present in it are calcium and oxygen.
 - Hydrogen bromide has the chemical formula HBr and the elements present in it are hydrogen and bromine.
 - Baking Powder is commercial name of sodium hydrogen carbonate (NaHCO₃). The elements present in it are sodium, hydrogen, carbon and oxygen.
 - Potassium sulphate has the formula K₂SO₄. The elements present in it are potassium, sulphur and oxygen.
- Ethyne, (C₂H₂)
Molar mass = 2 × atomic mass of C + 2 × atomic mass of H.

$$= (2 \times 12 + 2 \times 1) \text{ U} = 24 + 2 = 26 \text{ U}$$
 (Note : U is the unit used for atomic masses)
 - Sulphur molecule, (S₈)
Molar mass = 8 × atomic mass of sulphur

$$= 8 \times 32 \text{ or } 256 \text{ U}$$
 - Phosphorus molecule, (P₄)
Molar mass = 4 × atomic mass of phosphorus

$$= (4 \times 31) \text{ U} = 124 \text{ U}$$
 - Nitric acid (HNO₃)
Molar mass = 1 × atomic mass of H + 1 × atomic mass of N + 3 × atomic mass of O

$$= (1 \times 1 + 1 \times 14 + 3 \times 16) \text{ U}$$

$$= (1 + 14 + 48) \text{ U or } 63 \text{ U}$$
 - Hydrochloric acid (HCl)
Molar mass = 1 × atomic mass of H + 1 × atomic mass of Cl

$$= (1 \times 1 + 1 \times 35.5) \text{ U}$$

$$= (1 + 35.5) \text{ U or } 36.5 \text{ U}$$
- 1 mole of nitrogen atom
Mass of 1 mole of nitrogen atom
= gram atomic mass of nitrogen = 14 g
 - 4 moles of aluminium atoms
Mass of 4 moles of aluminium atoms
= 4 × gram atomic mass of Al = 4 × 27 g
= 108 g
 - 10 moles of sodium sulphite (Na₂SO₃)
Mass of 10 moles of sodium sulphite
= 10 × gram molecular mass of Na₂SO₃.

Atoms and Molecules

Molecular mass of $\text{Na}_2\text{SO}_3 = 2 \times \text{atomic mass of Na} + 1 \times \text{atomic mass of S} + 3 \times \text{atomic mass of O}$
 $= (2 \times 23 + 1 \times 32 + 3 \times 16) \text{ U}$
 $= (46 + 32 + 48) \text{ U}$
 $= 126 \text{ U}$

$\therefore$ Gram molecular mass of $\text{Na}_2\text{SO}_3 = 126 \text{ g}$

$\therefore$ Mass of 10 moles of

$\text{Na}_2\text{SO}_3 = 10 \times 126 = 1260 \text{ g}$

8. (a) 12 g of oxygen gas (O_2)

Molar mass of oxygen (O_2) = $2 \times 16 \text{ g}$

= 32 g

$\therefore$ 32 g of oxygen gas = 1 mole

Thus, 12 g of oxygen gas = $\frac{1}{32} \times 12 \text{ mole}$
 $= 0.375 \text{ mole}$

- (b) 20 g of water (H_2O)

Molar mass of water = $2 \times 1 + 1 \times 16$

= 18 g

Thus, 18 g of water = 1 mole

$\therefore$ 20 g of water = $\frac{1}{18} \times 20 \text{ mole} = 1.11$

- (c) 22 g of carbon dioxide (CO_2)

Molar mass of $\text{CO}_2 = 1 \times 12 + 2 \times 16$

= $12 + 32 = 44 \text{ g}$

Thus 44 g of $\text{CO}_2 = 1 \text{ mole}$

$\therefore$ 22 g of $\text{CO}_2 = \frac{1}{44} \times 22 = 0.5 \text{ mole}$

9. (a) 0.2 mole of oxygen atoms

Mass of 1 mole of oxygen atom

= gram atomic mass of oxygen

= 16 g.

$\therefore$ Mass of 0.2 mole of oxygen atom

= $(0.2 \times 16) \text{ g} = 3.2 \text{ g}$

- (b) 0.5 moles of water

Mass of 1 mole of water = gram molecular mass of water = 18 g

$\therefore$ Mass of 1 mole water = 18 g

Mass of 0.5 mole of water

= $0.5 \times 18 \text{ g} = 9.0 \text{ g}$

10. Gram molecular mass of sulphur (S_8)

= $8 \times 32 \text{ g}$

= 256 g

Thus, number of S_8 molecules in 256 g of sulphur = 1 mole molecules

= 6.023×10^{23} molecules

$\therefore$ Number of S_8 molecules in 16 g of sulphur

= $\frac{16}{256} \times 6.023 \times 10^{23}$

= 3.76×10^{22} molecules

11. Calculate the number of moles of Al_2O_3 in 0.051 g of (Al_2O_3)

Gram molecular mass of Al_2O_3

= $(2 \times 27 + 3 \times 16) \text{ g}$

= $(54 + 48) \text{ g} = 102 \text{ g}$

Thus, 102 g of $\text{Al}_2\text{O}_3 = 1 \text{ mole } \text{Al}_2\text{O}_3$

$\therefore$ 0.051 g of $\text{Al}_2\text{O}_3 = \frac{0.051}{102} \times 1 \text{ mole } \text{Al}_2\text{O}_3$
 $= 0.0005 \text{ mole}$

Calculation of number of aluminium atom in 0.0005 mole of Al_2O_3

Number of Al atom in 1 mole of $\text{Al}_2\text{O}_3 = 2 \times 6.023 \times 10^{23}$

$\therefore$ Number of Al atom in 0.0005 moles of Al_2O_3

= $0.0005 \times 2 \times 6.023 \times 10^{23}$

= 6.023×10^{21} atoms

In Al_2O_3 , the valency of Aluminum is + 3

$\therefore$ Number of aluminum ions (Al^{3+}) present is equal to the number of aluminium atoms present

Hence, Number of Al^{3+} ions = 6.023×10^{21}

Exemplar Problems :

1. CuCl_2 / CuSO_4 / $\text{Cu}_3(\text{PO}_4)_2$

NaCl / Na_2SO_4 / Na_3PO_4

FeCl_3 / $\text{Fe}_2(\text{SO}_4)_3$ / FePO_4

2. Number of moles = $\frac{12}{24} = 0.5 \text{ mol}$

3. (a) 4 (b) 5

(c) 7 (d) 2

4. (b) BiPO_4 — Both ions are trivalent
Bismuth phosphate

5. 1 mole of calcium chloride = 111 g

$\therefore$ 222 g of CaCl_2 is equivalent to 2 moles of CaCl_2

Since 1 formula unit CaCl_2 gives 3 ions, therefore, 1 mol of CaCl_2 will give 3 moles of ions

2 moles of CaCl_2 would give $3 \times 2 = 6$ moles of ions.

No. of ions = No. of moles of ions $\times$ Avogadro number

= $6 \times 6.022 \times 10^{23}$

= 36.132×10^{23}

= 3.6132×10^{24} ions

6. One mole of screws weigh = $2.475 \times 10^{24} \text{ g}$

= $2.475 \times 10^{21} \text{ kg}$

$\frac{\text{Mass of the Earth}}{\text{Mass of 1 mole of screws}} = \frac{5.98 \times 10^{24} \text{ kg}}{2.475 \times 10^{21} \text{ kg}} = 2.4 \times 10^3$

Mass of earth is 2.4×10^3 times the mass of screws

The earth is 2400 times heavier than one mole of screws.

7. (a) Mass of sodium atoms carried by Krish = $(5 \times 23) \text{ g}$
= 115 g

While mass of carbon atom carried by Raunak

= $(5 \times 12) \text{ g} = 60 \text{ g}$

Thus, Krish's container is heavy.

- (b) Both the bags have same number of atoms as they have same number of moles of atoms.

8. Mass of 1 molecule of $\text{CH}_4 = \frac{16\text{g}}{N_A}$
 Mass of 1.5×10^{20} molecules of methane = $\frac{1.5 \times 10^{20} \times 16}{N_A} \text{g}$
 Mass of 1 molecule of $\text{C}_2\text{H}_6 = \frac{30}{N_A} \text{g}$
 Mass of molecules of C_2H_6 is = $\frac{1.5 \times 10^{20} \times 16}{N_A} \text{g}$
- $\therefore$ Number of molecules of ethane
 $= \frac{1.5 \times 10^{20} \times 16}{N_A} \times \frac{N_A}{30} = 0.8 \times 10^{20}$
9. One gram of gold sample will contain
 $\frac{90}{100} = 0.9\text{g}$ of gold
- Number of moles of gold = $\frac{\text{Mass of gold}}{\text{Atomic mass of gold}}$
 $= \frac{0.9}{197} = 0.0046$
- One mole of gold contains N_A atoms = 6.022×10^{23}
 $\therefore$ 0.0046 mole of gold will contain = $0.0046 \times 6.022 \times 10^{23}$
 $= 2.77 \times 10^{21}$

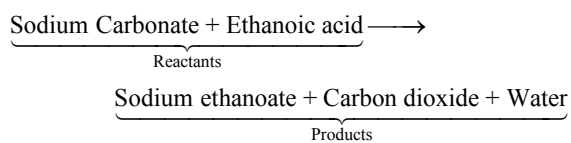
HOTS Questions :

- CrCl_3 and CrCl_2 , Cr_2O_3 and CrO
 - CuCl , CuCl_2 , Cu_2O and CuO
 - Sodium arsenate
 - 6 : 11
 - 2 : 1
 - 12 : 11
 - 2 : 1
 - In first sample of ascorbic acid
 1.50 g of carbon combines with 2.00 g of oxygen.
 1 g of carbon combines with $\frac{2}{1.5} \text{g}$ of oxygen.
 In second sample
 6.35 g of carbon combines with $\frac{2 \times 6.35}{1.5} \text{g}$ of oxygen
 $\therefore$ 6.35 g of carbon combine with 8.47 g of oxygen.
- | | |
|----------|-------------|
| Sample I | Sample II |
| 2 : 1.5 | 8.47 : 6.35 |
| | 2 : 1.5 |
- Thus, the above calculations supports the law of constant composition.
- Number of oxygen atoms in 16.0g of oxygen
 $= 6.023 \times 10^{23}$ atoms
 $\therefore$ Weight of one atom of oxygen = $\frac{16.0}{6.023 \times 10^{23}}$
 $= 2.657 \times 10^{-23} \text{g}$
 - G.M.W. of oxygen = 32.0g
 Number of O_2 molecules in 32.0 g of O_2
 $= 6.023 \times 10^{23}$ molecules
 $\therefore$ Weight of one molecule of oxygen
 $= \frac{32.0}{6.023 \times 10^{23}} = 5.314 \times 10^{-23} \text{g}$
 - Water molecule consists of three interconnected particles or atoms. Each water molecule divides into two hydrogen atoms and one oxygen atom. Molecules of a substance can be destroyed by a chemical reaction, whereas atoms remain unchanged. For this reason, we cannot say 'an atom of water' but we say 'a molecule of water'.
 - The atomic theory states that the relative number and kinds of atoms in a compound are constant, regardless of the source. Therefore, 1.0g of pure water should always contain the same relative amounts of hydrogen and oxygen, no matter where or how the sample is obtained.
 - 22.4L of gas = 1 gram molecular weight
 22.4L = 22,400 cm^3 . (1L = 1000 cm^3)
 360 cm^3 of N_2 = 0.45g
 22,400 cm^3 of N_2 = ?
 $\frac{0.45\text{g} \times 22400\text{L}}{360\text{L}} = 28\text{g}$
 Gram molecular weight of N_2 is 28 g.
 - The molecular formula of limestone is CaCO_3
 Its molar mass is 100 g/mol
 $\therefore$ 100 g CaCO_3 = 1 mol.
 $\therefore$ 300 g CaCO_3 = 3 mol.
 1 mol CaCO_3 contains 3 mol of O
 or $3 \times 6.022 \times 10^{23}$ O atom.
 $\therefore$ 3 mol CaCO_3 would contain
 $3 \times 3 \times 6.022 \times 10^{23} = 5.42 \times 10^{24}$ of O atom.
 - The formula CaC_2 indicates that there is 1 mole of Ca for every 2 moles of C. Therefore, if there are 0.150 moles of C there must be 0.0750 moles of Ca.
 No. of moles of calcium $\times$ molar mass of Ca = grams of calcium in the sample
 $= 0.0750 \times 40.078 \text{g of Ca} = 3.01 \text{g of Ca}$.
 - Its mass is **200 amu** (or U).
 - We know, 22.4 L of H_2 at NTP = 1 mole of H_2
 $\therefore$ 11.2 L of H_2 at NTP = $\frac{11.2}{22.4}$ mole of H_2
 $= 0.5$ mole of H_2
 Number of molecules present in one mole of H_2
 $= 6.023 \times 10^{23}$
 $\therefore$ Number of molecules present in 0.5 mole of H_2 (or 11.2L of Hydrogen at NTP)
 $= 0.5 \times 6.023 \times 10^{23}$
 $= 3.011 \times 10^{23}$ molecules
 Number of hydrogen atoms in one molecules of H_2 = 2
 $\therefore$ Number of hydrogen atoms in 3.011×10^{23} molecules of H_2 = $2 \times 3.011 \times 10^{23} = 6.022 \times 10^{23}$ atoms
 - Molecular mass of urea (H_2NCONH_2)
 $= 2 \times 1 + 1 \times 14 + 1 \times 12 + 1 \times 16 + 1 \times 14 + 2 \times 1$
 $= 2 + 14 + 12 + 16 + 14 + 2 = 60 \text{U}$
 Mass of nitrogen in 1 molecule of urea
 $= 1 \times 14 + 1 \times 14$
 $= 14 + 14 = 28\text{U}$
 $\therefore$ % age of nitrogen in urea
 $= \frac{28}{60} \times 100 = 46.6\%$

Atoms and Molecules

14. Molecular mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 $= 63.5 + 32 + 64 + 5 \times 18$
 $= 63.5 + 96 + 90 = 249.5$
 Number of moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in 24.95 mg
 $= \frac{24.95 \times 10^{-3}}{249.5} = 10^{-4}$ moles
 Molecules of water of crystallisation present in 1 mole of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} = 5 \times 6.023 \times 10^{23}$
 $\therefore 10^{-4}$ moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ contains molecules of water of crystallisation
 $= 5 \times 6.023 \times 10^{23} \times 10^{-4}$
 $= 30.115 \times 10^{19}$ molecules
 $= 3.01 \times 10^{20}$ molecules.

15. All that we have to do in this problem is to calculate the mass of reactants and products separately, and then compare the two. If the two masses are equal, then the law of conservation of mass gets verified. The given reaction can be written as :



- (i) Sodium carbonate and ethanoic acid are reactants.
 So, mass of reactants
 $= \text{Mass of sodium carbonate} + \text{Mass of ethanoic acid}$
 $= 5.3 + 6 = 11.3 \text{ g}$
 (ii) Sodium ethanoate, carbon dioxide and water are products
 So, mass of products
 $= \text{Mass of sodium ethanoate} + \text{Mass of carbon dioxide} + \text{Mass of water} = 8.2 + 2.2 + 0.9 = 11.3 \text{ g}$
 We find that the mass of reactants is 11.3 g and the mass of products is also 11.3 g. Since the mass of products is equal to the mass of reactants, the given data verifies the law of conservation of mass.

16. 1.375 gm of pure cupric oxide gave 1.098 gm of Cu and hence,

$$\text{percentage of Cu in the oxide} = \frac{1.098}{1.375} \times 100 = 79.85\%$$

in another experiment, 1.179 gm of pure copper gave 1.476 gm of the oxide and hence

$$\text{percentage of Cu in the oxide} = \frac{1.179}{1.476} \times 100 = 79.87\%$$

Since, both the oxides have almost the same percentage of Cu and hence, of oxygen, the result is obeying the law of constant composition.

17. (a) ZnCO_3 , ZnO , CO_2
 (b) HF , SiO_2 , SiF_4 , H_2O
 (c) SO_2 , H_2O , H_2SO_3
 (d) H_3P (or PH_3)
 (e) HClO_4 , Cd , $\text{Cd}(\text{ClO}_4)_2$

3 EXERCISE

Multiple Choice Questions :

1. (d) (a) $8 \text{ g of CH}_4 = \frac{8}{16} = \frac{1}{2}$ mole of CH_4
 $= \frac{1}{2} \times 6.023 \times 10^{23}$
 (b) $4.4 \text{ g of CO}_2 = \frac{4.4}{44} = \frac{1}{10}$ mole of CO_2
 $= \frac{1}{10} \times 6.023 \times 10^{23}$
 (c) $34.2 \text{ g of C}_{12}\text{H}_{22}\text{O}_{11} = \frac{34.2}{342} = \frac{1}{10}$ mole of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
 $= \frac{1}{10} \times 6.023 \times 10^{23}$
 (d) $2 \text{ g of H}_2 = 2 \text{ moles of H}_2 = 2 \times 6.023 \times 10^{23}$
 $\therefore 32 \text{ gm or 1 mole of O}_2 = 6.023 \times 10^{23}$
 $\therefore 16 \text{ gm of O}_2 = \frac{6.023 \times 10^{23} \times 16}{32} = 3.011 \times 10^{23}$
 2. (d) Percentage of hydrogen
 $= \frac{\text{Mass of hydrogen in water}}{\text{Molar mass of water}} \times 100$
 $= \frac{2}{18} \times 100 = 11.12\%$
 3. (b) Molar mass of $\text{KNO}_3 = 39 + 14 + 48 = 101$
 $\% \text{ of O in KNO}_3 = \frac{48}{101} \times 100 = 0.475$
 $\therefore \text{Mass of oxygen in 1000 gm} = 0.4752 \times 1000$
 $= 475.2 \text{ gm}$
 4. (c) $\therefore 58.5 \text{ gm of sodium chloride contains}$
 $= 23 \text{ gm of sodium}$
 $\therefore 11.7 \text{ gm of sodium chloride contains}$
 $= \frac{23}{58.5} \times 11.7 = 5.02 \text{ g}$

7. (a) $\text{MCl}_3 \rightarrow \text{M}^{3+} + 3\text{Cl}^-$
 $\text{M}^{3+} + \text{PO}_4^{3-} \rightarrow \text{MPO}_4$
8. (b) (a) $0.2 \text{ mol H}_2 \rightarrow 0.2 \times 6.023 \times 10^{23} \text{ molecules}$
 (b) $8.0 \text{ gm H}_2 = 8 \text{ mol H}_2 = 8 \times 6.023 \times 10^{23} \text{ molecules}$
 (c) $17 \text{ gm H}_2\text{O} = \frac{17}{18} \text{ mol H}_2\text{O} = \frac{17}{18} \times 6.023 \times 10^{23} \text{ molecules}$
 (d) $6 \text{ g of CO}_2 = \frac{6}{44} \text{ mol CO}_2 = \frac{6}{44} \times 6.023 \times 10^{23}$
9. (c) O_3 - contain 3 atoms of oxygen.
10. (c) Molar mass of $\text{CaSO}_4 = 40 + 32 + 16 \times 4 = 136$
 $\% \text{ of O}_2 \text{ in CaSO}_4 = \frac{64}{136} \times 100 = 47\%$
11. (a)
12. (c) 13. (c)
14. (b) $\therefore 12 \text{ g of carbon contains} = 6.023 \times 10^{23} \text{ atoms}$
 $\therefore 0.001 \text{ of carbon contains} = \frac{6.023 \times 10^{23}}{12} \times 0.001$
 $= 5 \times 10^{19}$
15. (c)
16. (b) The mass of 6.023×10^{23} atoms of carbon = 12 g
 $\therefore$ The mass of 6.023×10^{23} atoms of carbon
 $= \frac{12}{6.023 \times 10^{23}} \text{ g} = 1.99 \times 10^{-23} \text{ g}$
17. (b) 18. (a)
19. (a) 20. (b) 21. (b) 22. (a)
23. (a)
24. (d) Molar mass of Aspartam =
 $14 \times \text{C} + 18 \times \text{H} + 14 \times \text{N} + 16 \times \text{O} = 294$
 $\therefore$ Mass of 6.023×10^{23} molecules = 294 gm
 $\therefore$ Mass of 1 molecules = $\frac{294}{6.023 \times 10^{23}}$
 $= 4.89 \times 10^{-22}$
25. (a) Constant proportions according to which a pure chemical compound always contains same elements combined together in the same definite proportion of weight.
26. (b) 1 atom of Cu + 1 atom of sulphur + 9 atoms of oxygen + 10 atoms of hydrogen. Total number of atoms in compound is 21.
27. (a) Molecular weight of $\text{C}_{60}\text{H}_{122} = 12 \times 60 + 122 \times 1$
 $= 720 + 122 = 842$
 $\therefore 6 \times 10^{23}$ molecules of $\text{C}_{60}\text{H}_{122}$ has mass = 842 gm.
 $\therefore$ 1 molecule of $\text{C}_{60}\text{H}_{122}$ has mass
 $= \frac{842}{6 \times 10^{23}} = 140.333 \times 10^{-23} \text{ gm} = 1.4 \times 10^{-21} \text{ gm}$
28. (a) 6×10^{23} molecule has mass = 18 gm
 1 molecules has mass
 $= \frac{18}{6 \times 10^{23}} = 3 \times 10^{-23} \text{ gm} = 3 \times 10^{-26} \text{ kg}$
29. (d) $\therefore 17 \text{ gm NH}_3$ contains 6×10^{23} molecules of NH_3
 $\therefore 4.25 \text{ gm NH}_3$ contains
 $= \frac{6 \times 10^{23}}{17} \times 4.25 \text{ molecules of NH}_3$
 $\therefore$ No. of atoms = $\frac{6 \times 10^{23} \times 4.25}{17} \times 4 = 6 \times 10^{23}$
30. (b) $\therefore 22400 \text{ cc of gas at STP}$ has 6×10^{23} molecules
 $\therefore 1.12 \times 10^{-7}$ of gas at STP has
 $\frac{6 \times 10^{23} \times 1.12 \times 10^{-7}}{22400} = 0.03 \times 10^{14} = 3 \times 10^{12}$
31. (a) wt. of $\text{CO}_2 = 44$, mol wt. of $\text{CO}_2 = 44$
 No. of molecule = $\frac{\text{wt. of CO}_2}{\text{mol wt. of CO}_2} \times 6.02 \times 10^{23}$
 $= \frac{44}{44} \times 6.02 \times 10^{23} = 6.02 \times 10^{23}$
32. (b) 44g CO_2 occupies 22.4 L at STP
 4.4g CO_2 occupies $\frac{22.4}{44} \times 4.4 = 2.24 \text{ L}$
33. (b) $\therefore 2 \text{ gm of hydrogen} = 6.02 \times 10^{23}$ molecules
 $\therefore 1 \text{ gm of hydrogen}$
 $= \frac{6.02 \times 10^{23}}{2} = 3.01 \times 10^{23} \text{ molecule}$

More than One Option Correct :

1. (b,d)
 2. (a,d)
 3. (a,d) 16 g O_2 has no. of moles = $\frac{16}{32} = \frac{1}{2}$
 14 g N_2 has no. of moles = $\frac{14}{28} = \frac{1}{2}$

Atoms and Molecules

No. of moles are same, so no. of molecules are same.

$$8 \text{ g of O}_2 \text{ has no. of moles} = \frac{8}{32} = \frac{1}{4}$$

$$7 \text{ g of O}_2 \text{ has no. of moles} = \frac{7}{28} = \frac{1}{4}$$

4. (a,c,d) Sulphite (SO_3^{2-}) Sulphate (SO_4^{2-}) and phosphate (PO_4^{3-}) ions are polyatomic as they contain more than one ion.

5. (a, b, c)

6. (a, d)

7. (a,d) Symbol of tin and aluminium are Sn and Al respectively.

8. (a, b, d) Bk is Berkelium.

9. (b,c,d) Molar mass of gas = $\frac{0.220}{112} \times 22400 = 44$

10. (a, b, c)

11. (b, c) $\text{CaSO}_{3(s)} + \text{H}_2\text{O}_{(l)} + \text{SO}_{2(g)} \rightarrow \text{Ca}(\text{HSO}_3)_2$

(soluble)

mol of SO_2 required = mole of CaSO_3

$$\frac{12}{120} = 0.1; \text{ mass of } \text{SO}_2 = 0.1 \times 64 = 6.4 \text{ g}$$

12. (a,c,d) $8 \text{ g O}_2 = \frac{8}{32} \text{ mole} = 0.25 \text{ mole}$

$$7 \text{ g CO} = \frac{7}{28} = 0.25 \text{ mole}, 14 \text{ g N}_2 = \frac{14}{28} = 0.5 \text{ mole}$$

$$16 \text{ g SO}_2 = \frac{16}{64} = 0.25 \text{ mole.}$$

$$11 \text{ g CO}_2 = \frac{11}{44} = 0.25 \text{ mole}$$

Equal moles contain equal number of molecules.

13. (a, b) $4.4 \text{ g of CO}_2 = \frac{4.4}{44} \text{ mole} = 0.1 \text{ mole}$
 $= 0.1 \times 6.02 \times 10^{23} \text{ molecules}$
 $= 6.02 \times 10^{22} \text{ molecules.}$

Passage Based Questions :

1. (c) 2. (a) 3. (b)

Since one of mole of a compound weighs 35g. Hence the molar mass of the compound is 35. Which is possible only in case of NH_4OH .

$$\text{No. of moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{70}{35} = 2$$

Assertion & Reason :

1. (d) Molecular weight of oxygen is 32
 2. (d) Molecular weight of SO_2 is 64 and of O_2 is 32.
 3. (a)
 4. (a) 6.023×10^{23} atom of $\text{C}^{12} = 12 \text{ g}$

$$1 \text{ atom of } \text{C}^{12} = \frac{12}{6.023 \times 10^{23}} \text{ g}$$

$$\frac{1}{12} \text{ of one atom of } \text{C}^{12} = \frac{1}{6.023 \times 10^{23}} \text{ g}$$

$$[1 \text{ amu} = \frac{1}{12^{\text{th}}} \text{ of one atom of } \text{C}^{12}]$$

$$= 0.166 \times 10^{-23} \text{ g} = 1.66 \times 10^{-24} \text{ g}$$

5. (a)

Multiple Matching Questions :

1. (b) A - (p, q); B - (p); C - (p, q, r); D - (s)

$$1.5 \text{ mole of CO}_2 (\text{g}) = 1.5 \times 22400 \text{ mL at NTP}$$

$$= 33600 \text{ ml at NTP}$$

$$\text{Total number of atoms in one molecule of CO}_2$$

$$= 1 + 2 = 3$$

$$\text{Total number of atoms in 1.5 mole of CO}_2$$

$$= 1.5 \times 3 \times N_A = 4.5 \times N_A$$

$$3.0 \text{ g of H}_2 = \frac{3}{2} \times 22400 \text{ mL of H}_2 \text{ at NTP}$$

$$= 33600 \text{ ml at NTP}$$

$$\text{Number of mole of H}_2 \text{ in } 3.0 \text{ g H}_2 = \frac{3}{2} \text{ or } 1.5 \text{ moles}$$

$$\text{Number of atoms in 1 molecule of H}_2 = 2$$

$$\text{Number of atoms in 1.5 mole H}_2 = 2 \times 1.5 \times N_A$$

$$= 3.0 \times N_A$$

$$\text{Volume of 1.5 moles of ozone at NTP}$$

$$= 22400 \times 1.5 \text{ ml} = 33600 \text{ ml}$$

$$\text{Number of atoms in one molecule of O}_3 = 3$$

$$\text{Total number of atoms in 1.5 mole of O}_3$$

$$= 3 \times 1.5 \times N_A = 4.5 \times N_A$$

$$\text{Weight of 1 mole of O}_3 = 48 \text{ g}$$

$$\text{Weight of 1.5 mole of O}_3 = 48 \times 1.5 \text{ g} = 72 \text{ g}$$

$$\text{Weight of 1 mole of oxygen (O}_2) = 32 \text{ g}$$

Integer Type Questions :

1. (5) 6.02×10^{23} molecules of CO = 1 mole of CO
 6.02×10^{24} CO molecules = 10 moles CO
 $= 10 \text{ g atoms of O} = 5 \text{ g molecules of O}_2$

2. (3) Mass of one molecule of Water
 $= \frac{18}{6.023 \times 10^{23}} = 3 \times 10^{-23} \text{ g} = 3 \times 10^{-26} \text{ Kg}$
 $\therefore x = 3$

4 EXERCISE

1. (a) Moles of $I_2 = 25.4 \text{ g}/254 = 0.1$
 Moles of $Cl_2 = 14.2/70.9 = 0.2$
 $I_2 + 2Cl_2 \rightarrow ICl + ICl_3$
 The ratio between I_2 and Cl_2 is 1 : 2
 We have 0.100 moles I_2 and .200 moles of Cl_2 so the ratio is the same and no reactants is the excess the ratio between I_2 and ICl (or ICl_3) is 1 : 1. So starting with 0.100 moles of I_2 we get 0.100 moles of ICl and 0.100 moles of ICl_3 .
2. (c) Law of multiple proportions.
3. (b) Law of multiple proportion. As the ratio of oxygen which combine with fix weight of 1 g of nitrogen bears a simple whole number ratio
 $0.57 : 1 : 12 : 1.7031 : 2 : 3$
4. (d) $SnCl_2$ $SnCl_4$
 $119 : 2 \times 35.5$ $119 : 4 \times 35.5$
 Chlorine ratio in both compounds is
 $= 2 \times 35.5 : 4 \times 35.5 = 1 : 2$
5. (c) According to Avogadro's law "equal volumes of all gases contain equal number of molecules under similar conditions of temperature and pressure". Thus if 1 L of one gas contains N molecules, 2 L of any other gas under the same conditions of temperature and pressure will contain 2N molecules.
6. (d) In CuO and Cu_2O the O : Cu is 1 : 1 and 1 : 2 respectively. This is law of multiple proportion.

7. (c)	Percentage	R.N.A	Simplest ratio
C	9	$\frac{9}{2} = \frac{3}{4}$	3
H	1	$\frac{1}{1} = 1$	4
N	3.5	$\frac{3.5}{14} = \frac{1}{4}$	1

Empirical formula = C_3H_4N
 $(C_3H_4N)_n = 108$
 $(12 \times 3 + 4 \times 1 + 14)_n = 108$
 $(54)_n = 108$
 $n = \frac{108}{54} = 2$
 $\therefore$ molecular formula = $C_6H_8N_2$

8. (c) 50% of X (Atomic mass 10), 50% of Y (Atomic mass 20). Relative number of atoms of X = $\frac{50}{10} = 5$ and than
 $Y = \frac{50}{20} = 2.5$

Simple Ratio 2 : 1. Formula X_2Y

9. (a)	Element	%	Atomic mass	Relative no. of atoms	Simple ratio of atoms
	C	80	12	$\frac{80}{12} = 6.66$	$\frac{6.66}{6.66} = 1$
	H	20	1	$\frac{20}{1} = 20.0$	$\frac{20.0}{6.66} = 3$

$\therefore$ Empirical formula is CH_3

10. (c) The acid with empirical formula CH_2O_2 is formic acid, $H-COOH$.

11. (c)	Element	%	Relative no. of atoms	Simplest ratio of atoms
	C	49.3	$49.3/12 = 4.1$	$4.1/2.74 = 1.5$ $1.5 \times 2 = 3$
	H	6.84	$6.84/1 = 6.84$	$6.84/2.74 = 2.5$ $= 2.5 \times 2 = 5$
	O	43.86	$43.86/16 = 2.74$	$2.74/2.74 = 1$ $1 \times 2 = 2$

$\therefore$ Empirical formula = $C_3H_5O_2$

Empirical formula mass

$$= (3 \times 12) + (5 \times 1) + (2 \times 16) = 36 + 5 + 32 = 73$$

Molecular mass = $2 \times$ Vapour density

$$= 2 \times 73 = 146$$

$$n = \frac{\text{molecular mass}}{\text{empirical formula mass}} = 146/73 = 2$$

Molecular formula = Empirical formula $\times$ 2

$$= (C_3H_5O_2) \times 2 = C_6H_{10}O_4$$

12. (a) Empirical formula of compound = CH_2
 Molecular mass of the compound = 42

$$\therefore n = 42/14 = 3$$

$\therefore$ Hence molecular formula = C_3H_6

13. (d)	C	54.55	$54.55/12 = 4.5$	$4.5/2.27 = 2$
	H	9.09	$9.09/1 = 9.09$	$9.09/2.27 = 4$
	O	36.36	$36.36/16 = 2.27$	$2.27/2.27 = 1$

Hence empirical formula of the compound = C_2H_4O

Atoms and Molecules

14. (b)

15. (a)	Element	Percentage	Atomic weight	Atomic ratio	Simple ratio
	C	38.71	12	$\frac{38.71}{12} = 3.23$	$\frac{3.23}{3.23} = 1$
	H	9.67	1	$\frac{9.67}{1} = 9.67$	$\frac{9.67}{3.23} = 3$
	O	$100 - (38.71 + 9.67) = 51.62$	16	$\frac{51.62}{16} = 3.23$	$\frac{3.23}{3.23} = 1$

$\therefore$ Hence the empirical formula of this compound is CH_3O

16. (d) $\therefore$ 18 gm, H_2O contains = 2 gm H

$$\therefore 0.72 \text{ gm } \text{H}_2\text{O} \text{ contains} = \frac{2}{18} \times 0.72 \text{ gm} = 0.08 \text{ gm H}$$

$$\therefore 44 \text{ gm } \text{CO}_2 \text{ contains} = 12 \text{ gm C}$$

$$\therefore 3.08 \text{ gm } \text{CO}_2 \text{ contains} = \frac{12}{44} \times 3.08 = 0.84 \text{ gm C}$$

$$\therefore \text{C} : \text{H} = \frac{0.84}{12} : \frac{0.08}{1}$$

$$= 0.07 : 0.08 = 7 : 8$$

$$\therefore \text{Empirical formula} = \text{C}_7\text{H}_8$$

17. (b)

18. (c) They are isoelectronic species.

	N^{3-}	O^{2-}	Na^+	F^-
No. of electrons	10	10	10	10
No. of protons	7	8	11	9

$\therefore$ Attractive forces are highest in Na^+ .

$\therefore \text{Na}^+$ is smallest in size.

19. (a) A cation is always smaller in size as compared to corresponding neutral atom. Greater the magnitude of charge, smaller will be size of ion. Following is the correct order of decreasing size $\text{Al}^{3+} < \text{Al}^{2+} < \text{Al}^+ < \text{Al}$.

$\therefore \text{Al}$ has largest size.

20. (d) All the given species contains $10 e^-$ each i.e. isoelectronic.

For isoelectronic species anion having high negative charge is largest in size and the cation having high positive charge is smallest.

21. (a,c) Here the amount of oxygen which combines with fixed amount of C and S in their oxides will be in a simple whole number ratio.

22. (a,c) Ratio of Masses of oxygen in CO and $\text{CO}_2 = 16 : 32 = 1 : 2$

Ratio of Masses of oxygen in H_2O and $\text{H}_2\text{O}_2 = 16 : 32 = 1 : 2$

23. (b,c)

24. (a,b,d) Size of isoelectronic cations decreases with increase in magnitude of nuclear charge where as size of isoelectronic anion increases with increase in magnitude of negative charge.

25. (c) To find its empirical formula reduce P_4O_{10} to the smallest whole number ratio of atoms. P_4O_{10} reduced by 2 equals P_2O_5 .

26. (b) Moles of C = $86 \text{ g} / 12 \text{ g per mole} = 7$ moles of C; Moles of H = $14 / 1 \text{ g} = 14$ moles of H.

hence the ratio of moles of C and H to each other: $14 / 7$ is 2.

In other words there are 2 H moles for every C mole or CH_2 . This is the empirical formula.

27. (a) The molecular mass of Ag is 108 g/mol ; molecular mass of F is 19 g/mol .

$$\text{moles of Ag} = 85 \text{ g} / 108 \text{ g mol} = 0.79 \text{ moles}$$

$$\text{Moles of F} = 15 \text{ g} / 19 \text{ g mol} = 0.79 \text{ moles}$$

$$\text{Or the whole number ratio is } 0.79 / 0.79 = 1$$

Hence the empirical formulae is AgF

28. (d) We know that from the reaction $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ that the ratio of the volume of gaseous reactants and products is in agreement with their molar ratio. The ratio of $\text{H}_2 : \text{Cl}_2 : \text{HCl}$ volume is $1 : 1 : 2$ which is the same as their molar ratio. Thus volume of gas is directly related to the number of moles. Therefore, the assertion is false but reason is true.

29. (d) One mole of any substance corresponding to 6.023×10^{23} entities in respective of its weight.

$$\text{Molecular weight of } \text{SO}_2 = 32 + 2 \times 16 = 64 \text{ gm.}$$

$$\text{Molecular weight of } \text{O}_2 = 16 \times 2 = 32 \text{ gm.}$$

$\therefore$ Molecular weight of SO_2 is double to that of O_2

30. (a)

31. (a) (A)-r, s, (B)-p, (C)-t, (D)-q

32. 1 Given $V = 2 \text{ L}$, Molarity = 0.5 M , Moles = ?

$$\text{Molarity} = \frac{\text{No. of moles of solute}}{V \text{ of solution in L}} \text{ or } 0.5 = \frac{\text{Moles}}{2}$$

$$\therefore \text{Moles} = 2 \times 0.5 = 1.0$$

$$33. \quad 3 \quad \text{moles Xe} = \frac{55.2 \text{ g}}{131 \text{ g/mole}} = 42 \text{ moles}$$

$$\text{moles Cl} = \frac{44.8 \text{ g}}{35.5 \text{ g/mole}} = 1.26 \text{ moles}$$

$$\frac{\text{moles Cl}}{\text{moles Xe}} = \frac{1.26}{.42} = 3$$

$\therefore$ empirical formula = XeCl_3

Hence the value of n is 3

$$34. \quad 6 \quad \text{moles C} = \frac{4 \text{ g}}{12 \text{ g/mole}} = .333 \text{ moles}$$

$$\text{moles H} = \frac{.667 \text{ g}}{1 \text{ g/mole}} = .667 \text{ moles}$$

$$\text{moles O} = \frac{5.33}{16 \text{ g/mole}} = .333 \text{ moles}$$

$$\frac{\text{moles H}}{\text{moles C}} = \frac{.667}{.333} = 2$$

$$\frac{\text{moles C}}{\text{moles O}} = \frac{.333}{.333} = 1$$

$$\frac{\text{moles O}}{\text{moles O}} = \frac{.333}{.333} = 1$$

$\therefore$ empirical formula = CH_2O

$$\text{EW} = 12 + 2(1) + 16 = 30 \text{ grams}$$

$$\frac{\text{MW}}{\text{EW}} = \frac{180}{30} = 6$$

$\therefore$ molecular formula = $6(\text{CH}_2\text{O}) = \text{C}_6\text{H}_{12}\text{O}_6$

Hence the value of n is 6

Chapter

4

Structure of Atom

INTRODUCTION

600 B.C. Kanaad – Indian philosopher and saint conceived of matter as being composed of smallest individual discrete particles. He called them “paramaanu”. He didn’t tell anyone about ‘Paramaanu’ due to fear of misuse. Around 400 BC, the Greek philosopher Democritus suggested that all matter is composed of tiny, discrete, indivisible particles that he called atoms. His ideas, based entirely on philosophical speculation rather than experimental evidence.

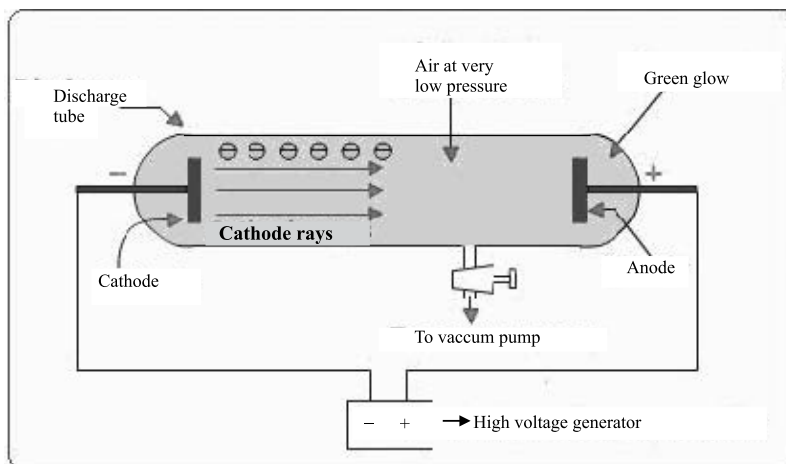
But experiments in late 1800s and early 1900s have shown that atoms are composed of three principal kinds of subatomic particles : protons, neutrons and electrons. Experiments also revealed that at the center of an atom there exists a very tiny, extremely dense core called the nucleus, which is where an atom’s protons and neutrons are presents they are also called nucleons. The electrons in an atom surround the nucleus and fill the remaining volume of the atom.

Protons carry a single unit of positive charge, which is one type of electrical charge. Electrons carry one unit of the opposite charge, a negative charge. These electrical charges have the property that like charges repel each other and opposite charges attract, so negatively charged electrons are attracted to positively charged protons. In fact, it is this attraction that holds the electrons around the nucleus. Neutrons have no charge and are said to be electrically neutral.

Matter as we generally find it in nature appears to be electrically neutral, which means that it contains equal numbers of positive and negative charges. Therefore, in a neutral atom, the number of electrons must be equal to the number of protons.

DISCOVERY OF ELECTRON

In 1879, William Crookes studied the conduction of electricity through gases at low pressure. Gas at 0.01 mm mercury pressure was filled in 60 cm long glass discharge tube and two electrodes were placed, at both ends of discharge tube. High potential difference of the 10,000 volts to 30,000 volts was applied across the electrodes then rays are emitted from cathode and collide with the opposite, wall. They produce illumination of green colour.



Properties of Cathode Rays

From the various experiments carried out by J.J. Thomson (1897) and others, the cathode rays have been found to possess the following properties.

- (i) **Cathode rays travel in straight line.** When an opaque object like a metal cross is placed in the path of cathode rays in a discharge tube, a shadow of the metal cross is formed at the end opposite to the cathode.

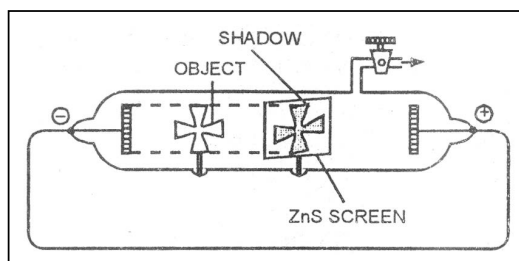


Fig. Production of a shadow of the solid object by cathode rays

- (ii) **Cathode rays produce mechanical effect.** On placing a light paddle in the path of cathode rays in a discharge tube the blades of the paddle wheel rotate. This shows that cathode rays are a beam of particles having mass and possessing kinetic energy.

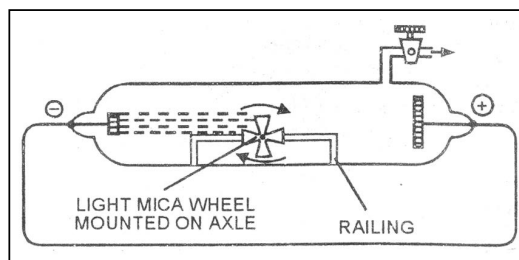


Fig. Rotation of light paddle wheel by cathode rays

- (iii) **Cathode rays are negatively charged particles.** When an electric field is applied in the path of cathode rays, they are deflected towards the positive plate of the electric field, which shows cathode rays are made up of negatively charged particles.

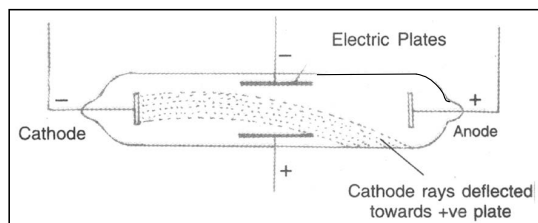
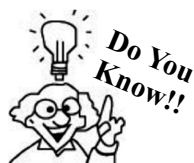
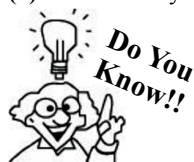


Fig. Deflection of cathode rays in an electric field indicating negative charge on its particles.



Similarly, when a magnetic field is applied on the cathode rays, these rays are deflected in a direction which shows that they carry negative charge.

- (iv) Cathode rays travel with high speed approaching that of light.
- (v) Cathode rays cause fluorescence.



The television picture tube is a cathode ray tube in which a picture is produced due to fluorescence on the television screen coated with suitable material. Similarly, fluorescent light tubes are also cathode ray tubes coated inside with suitable materials which produce visible light on being hit with cathode rays.

- (vi) Cathode rays heat the object on which they fall due to transfer of kinetic energy to the object.
 - (vii) When cathode rays fall on solids such as copper, X-rays are produced.
 - (viii) Cathode rays possess ionizing power i.e. they ionize the gas through which they pass.
 - (ix) The cathode rays affect the photographic plates.
 - (x) They penetrate through thin metallic sheets.
 - (xi) The nature of these rays does not depend upon the nature of gas & the cathode material used in discharge tube.
- The negatively charged material particles constituting the cathode rays are called electrons.

CHECK Point

- ☛ Why cathode rays are produced only when the pressure of the gas inside the discharge tube is very low ?

SOLUTION

At higher pressure (e.g., at 1 atmospheric pressure), no electric current flows through the tube because gases are poor conductor of electricity.

CONNECTING TOPIC

DETERMINATION OF CHARGE AND MASS OF ELECTRON

After discovery of electron some other important experiments are carried out by different scientists to determine charge and mass of electron

(i) Determination of Charge to Mass Ratio (e/m) of Electron (i.e., J.J. Thomson Experiment) :

The apparatus used by J.J. Thomson (1897) for the determination of $\frac{e}{m}$ ratio of electron is shown in figure.

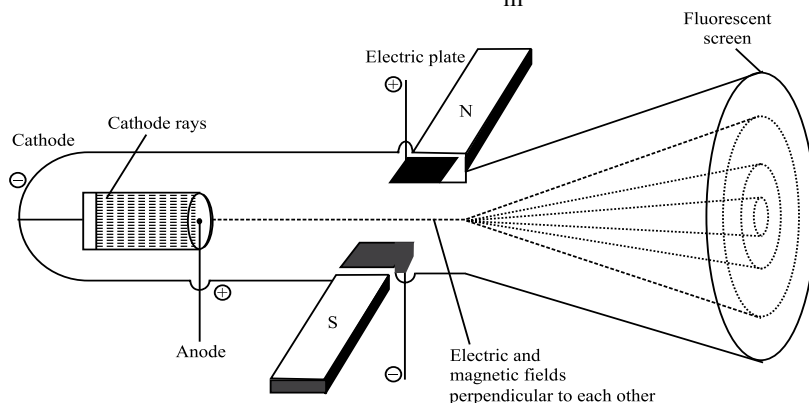


Fig. The apparatus used for determining the ratio of electric charge (e) to mass (m) of electrons.

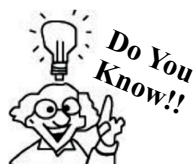
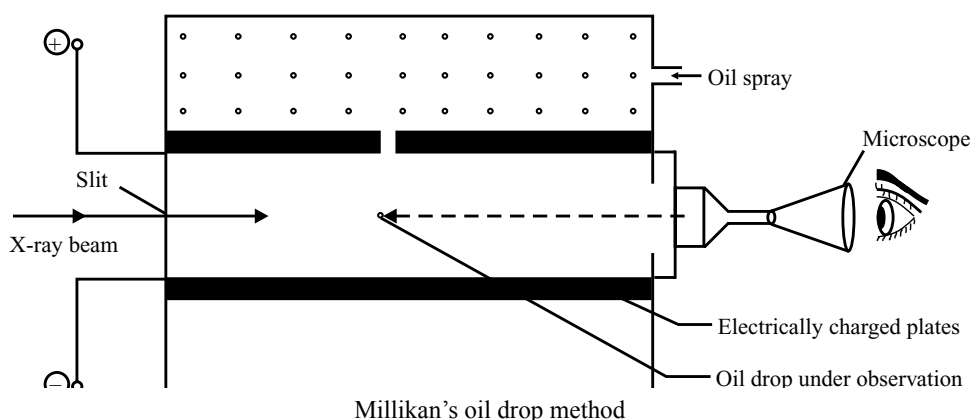
He studied the deflection of cathode rays under the simultaneous influence of electric and magnetic fields perpendicular to each other. By adjusting the two fields in such a way so that the cathode rays strike the fluorescent screen at the same place as they do in the absence of either field. From the magnitude of the applied magnetic and electric fields, Thomson calculated the e/m ratio of the electrons.

J. J. Thomson found the ratio of charge to mass (e/m) constant by using different discharge tubes fitted with electrodes of different metals. He also used different gases in the discharge tube. The value of e/m was found to be 1.76×10^8 coulombs/g. Two important conclusions can be drawn from this experiment.

- (i) The electrons are universal constituents of all matter.
- (ii) For an electron e/m value is 1.76×10^8 coulomb/g or $1.76 \times 10^{11} \text{ Ckg}^{-1}$

(ii) Determination of Charge of the Electron – Millikan's Oil Drop Method :

The first accurate determination of the charge of an electron (e) was made by R.A. Millikan (1909) by his famous oil drop experiment. The apparatus used by him is shown in figure. Small oil droplets are formed by sprayer and these droplets are allowed to fall under gravitational force between two charged metal plates. The movement of a single droplet is observed by means of a microscope. He irradiated the space between two metal plates with X-rays. These X-rays ionised electrons from some of the air molecules which in turn were caught by oil droplets. By applying the electrical field on the droplet under study, which balances the downward gravitational force, the droplet becomes stationary. From the magnitude of the charge applied on the plate and the mass of the droplet the charge on the droplet was determined.



After gaining electrons from ionised air the oil droplets becomes negatively charged. Thus it is necessary that upper plate should be positively charged and lower plate negative.

The mass of the droplet was determined by its fall freely under gravity. The charge on the electron is determined to be 1.6×10^{-19} coulombs. Since this is the smallest charge carried by any charged particle, it is referred to as one unit charge.

(iii) Determination of Mass of Electron

By knowing the value of $\frac{e}{m}$ ratio (by J.J. Thomson's experiment) and charge on the electron (by Millikan's oil drop method), it is possible to calculate the exact mass of a electron.

$$\frac{e}{m} = 1.76 \times 10^8 \text{ coulombs/g}$$

$$e = 1.60 \times 10^{-19} \text{ coulombs}$$

$$\begin{aligned} \therefore \text{Mass of the electron, } m &= \frac{e}{e/m} = \frac{1.60 \times 10^{-19} \text{ coulombs}}{1.76 \times 10^8 \text{ coulombs/g}} \\ &= 9.1 \times 10^{-28} \text{ g} \\ &= 9.1 \times 10^{-31} \text{ kg} \end{aligned}$$

Thus, an electron is that subatomic fundamental particle which carries one unit negative charge (1.6×10^{-19} coulombs) and has a mass of 9.1×10^{-28} g.

Knowledge ENHANCER

Mass of electron relative to hydrogen atom :

The mass of the electron (9.1×10^{-28} g) is approximately $\frac{1}{1837}$ th the mass of a hydrogen atom as shown below. However,

the mass of the electron is so small that for all practical purposes, its value is considered negligible.

Mass of 1 gm atoms of hydrogen = 1.008 g

$\therefore$ Mass of 6.02×10^{23} atoms of hydrogen = 1.008 g

So, mass of one atom of hydrogen = $\frac{1.008}{6.02 \times 10^{23}} = 1.67 \times 10^{-24}$ g

Now, mass of electron = 9.1×10^{-28} g.

$\therefore \frac{\text{Mass of one atom of hydrogen}}{\text{Mass of an electron}} = \frac{1.67 \times 10^{-24} \text{ g}}{9.1 \times 10^{-28} \text{ g}} = 0.1837 \times 10^4 = 1837$

or, Mass of an electron = $\frac{1}{1837} \times \text{Mass of one atom of hydrogen}$

Determination of Avogadro's number :

We know, 1 Faraday = 96500 coulombs or 9.65×10^4 coulombs.

Charge on one electron = 1.60×10^{-19} coulombs

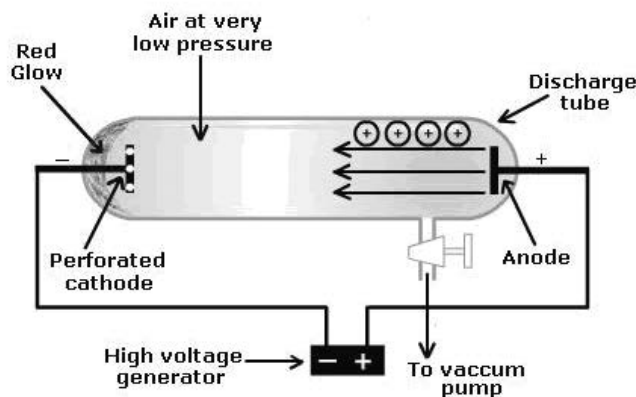
$\therefore N_0 = \text{Avogadro's number} = \frac{9.65 \times 10^4}{1.60 \times 10^{-19}} = 6.02 \times 10^{23} \text{ electrons}$

So, from the above discussed experiments, we conclude

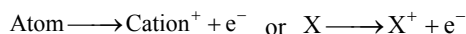
- (i) All atoms contain electrons
- (ii) An electron is a subatomic particles which carries a negative charge of 1.6×10^{-19} C.
- (iii) The mass of an electron is 9.1×10^{-28} g or 9.1×10^{-31} kg.
- (iv) Mass of an electron is $1/1837$ of the mass of one hydrogen atom.

DISCOVERY OF PROTON

In 1886 Eugen Goldstein first observed that a cathode ray tube also generates a stream of positively charged particles that moves towards the cathode. These were called canal rays because they were observed occasionally to pass through a channel, or "canal", drilled in the negative electrode.



These positive rays, or positive ions, are created when cathode rays knock electrons from the gaseous atoms in the tube, forming positive ions by processes such as



These rays were attracted towards the negative plate in the electric field as shown in figure. This means that these rays consist of positively charged particles and were also named positive rays or anode rays.

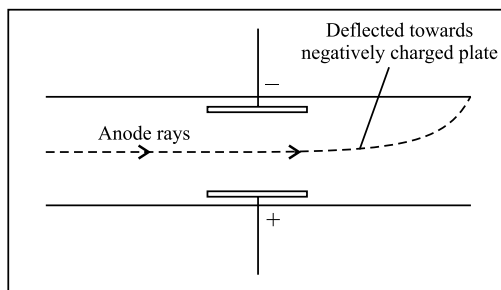
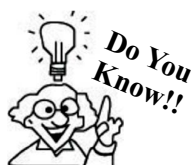
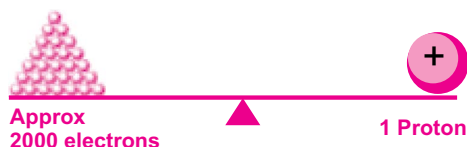


Fig. Deflection of canal rays in an electric field



Different elements give positive ions with different e/m ratios. The regularity of the e/m values for different ions lead to the idea that there is a unit of positive charge and that resides in the proton. The proton is a fundamental particle with a charge equal in magnitude but opposite in sign to the charge on the electron.



CHECK Point

What is the difference in the origin of cathode rays and anode rays ?

SOLUTION

Cathode rays originate from the cathode whereas anode rays are not produced from the anode. They are produced from the gaseous atoms by knock out of the electrons by high speed cathode rays.

Properties of Anode Rays :

- (i) Anode rays travel in straight lines.
- (ii) Anode rays are material particles.
- (iii) Anode rays are positively charged.
- (iv) Anode rays are affected by electric and magnetic field.
- (v) Anode rays affect the photographic plate.



idea box

Make a power point presentation on cathode rays. Use one slide for showing the apparatus used for cathode rays generation. You should use one slide for each property of cathode rays. You can make your presentation better by using different colour, fonts, style etc. Your presentation should also include comparison between cathode and anode rays.

CONNECTING TOPIC

After discovery of proton some other important experiments were carried out by different scientists to determine charge and mass of electrons.

Determination of Charge and Mass of Particles Constituting Anode Rays :

Using the same experimental methods, as described earlier for cathode rays, the charge to mass ratio (e/m) of the particles constituting anode rays have been determined for different gases. Following are the observations.

- (i) e/m ratio is not constant and is dependent on the nature of the gas taken in discharge tube.
- (ii) e/m value for particle of anode rays is considerably smaller than that for electron.
- (iii) e/m ratio is maximum for hydrogen gas in the discharge tube.

These observations have indicated that positive ions formed from hydrogen are lightest. These lightest positively charged particles have been named **protons** e/m ratio for a proton is determined to be 9.58×10^4 coulombs/g. The charge on proton is equal to electron in magnitude i.e., 1.6×10^{-19} coulombs but opposite.

Thus, the mass of each proton is calculated.

$$\text{Mass of proton} = \frac{e}{e/m} = \frac{1.6 \times 10^{-19}}{9.58 \times 10^4} = 1.67 \times 10^{-24} \text{ g}$$

This value is nearly the same as that of hydrogen atom. Hence proton is 1837 times heavier than electron.

Thus, a proton may be defined as a subatomic fundamental particle which carries one unit positive charge (1.6×10^{-19} coulombs) and has mass (1.67×10^{-27} kg) almost equal to that of an atom of hydrogen.

ILLUSTRATION : 1

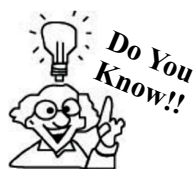
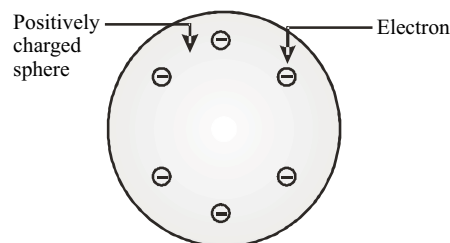
In one discharge tube, H_2 gas is taken and in the other O_2 gas is taken. Will the electrons and positive ions in the cathode rays and in the anode rays be same or different ?

SOLUTION :

Electrons will be same but positive ions will not be same.

THOMSON ATOMIC MODEL

J.J. Thomson proposed an atomic model which is called **plum pudding model of atom**. J.J. Thomson (1904) proposed a simple model of an atom showing the arrangement of electrons and protons in it. According to J.J. Thomson, the positive charge is spread over a sphere of the size of the atom (i.e., radius of sphere is of the order 10^{-8} cm). A sufficient number of electrons as to neutralise the positive charge are embedded over the sphere. This model of the atom is known as “**Plum pudding model**”. He explained the stability of the atom due to balance of the repulsive forces between the electrons and the forces of attraction of electrons towards positive sphere. Although this model explained some of the known properties of the atom but failed to explain the results of scattering experiment and hence Thomson model of atom was discarded.



This model is compared with a water melon in which seeds are embedded or with a cake or pudding in which raisins (dried grapes/kishmish) are embedded. That is why this model is sometimes called raisin pudding model or watermelon model.

RUTHERFORD'S MODEL OF THE ATOM**Discovery of Nucleus :**

Rutherford (1911) performed scattering experiment by bombarding fast moving α -particles, emitted from a radioactive substance, on thin foil (4×10^{-5} cm thick) of the metals like silver, gold, copper, platinum, etc. To detect the α -particles after scattering, a movable circular screen coated with zinc sulphide is placed around the gold or anyother metal foil. When an α -particle strikes the ZnS screen, the flashes of light (i.e., scintillation) is produced. These can be detected.

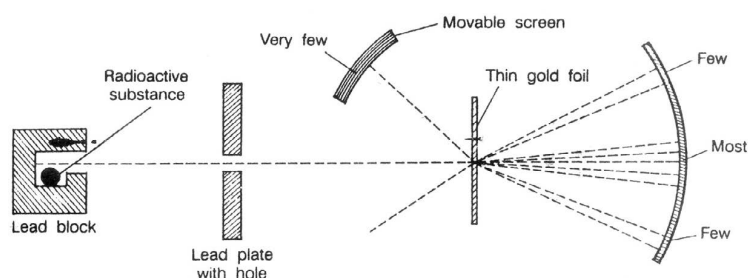


Fig. Rutherford's α -particle scattering experiment.

The observations led to the following conclusions.

- Most of the α -particles (nearly 99%) passed through the metal foil (Au, Ag, Pt, etc.) undeflected.
- Some of the α -particles underwent deflection by small angles.
- Very few α -particles (1 in 20,000) returned back suffering a deflection of 180° .

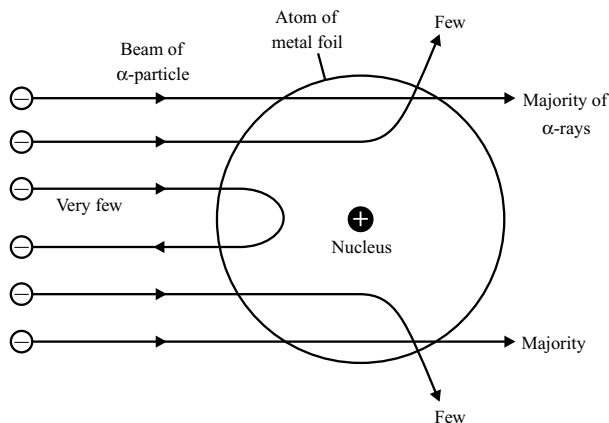
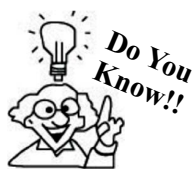


Fig. Scattering of α -particles

This experiment led Rutherford to conclude the main features as follows :

- Most of the space in the atom is largely empty as most of the α -particles pass straight through the atom.
- Centre of the atom has a heavy positively charged body, called **nucleus**, which repel positively charged α -particles and thus explains the scattering phenomenon.
- Whole of the atomic mass is concentrated in the nucleus i.e., the central nucleus is rigid and hence α -particles which strike on it are thrown back.
- Since very few α -particles are deflected back, the size of the nucleus must be very small (radius nearly 10^{-13} cm) compared to the total volume of the atom (radius nearly 10^{-8} cm). It shows that nucleus is 1/100000 in size compared to the total size of the atom.



Thus, according to Rutherford's model of atom, the atom consists of two parts.

- Nucleus** : It is a very small positively charged part of the atom. It is situated at the centre and carries almost the entire mass of the atom. Since electrons have negligible mass, the mass of the atom is mainly due to protons i.e., the protons are present in the nucleus. The positive charge on the nucleus is also because of the presence of protons in the nucleus.
- Extranuclear part** : This is the space around the nucleus in which the electrons revolve at a high speed in a fixed path called orbits or shells. This extra nuclear part provides volume to an atom and thus called atomic volume.

CHECK Point

- ☛ Why Rutherford's model of atom is also called planetary model of atom ?

SOLUTION

This model of atom is similar to our solar system where the nucleus is like the sun and the electrons, are like the planets. That is why these electrons are also called planetary electrons.

CONNECTING TOPIC

In order to understand that Rutherford's model of atom suffered from a serious drawback it is necessary to learn about following theories first.

ELECTROMAGNETIC WAVE THEORY

This theory was put forward by James Clark Maxwell in 1864. The main points of this theory are following

- The energy is emitted from any source (like the heated rod or the filament of a bulb through which electric current is passed) continuously in the form of radiations (or waves) and is called the radiant energy.
- The radiations consist of electric and magnetic fields oscillating perpendicular to each other and both perpendicular to the direction of propagation of the radiation.
- The radiations possess wave character and travel with the velocity of light (i.e., nearly 3×10^8 m/sec).
- These waves do not require any material medium for propagation. For example, rays from the sun reach us through space which is a non-material medium.

Electromagnetic Spectrum

It is defined as the arrangement of various types of electromagnetic radiations in terms of increasing (or decreasing) wavelengths (or frequency). The complete range of electromagnetic waves is called electromagnetic spectrum. The wavelengths of various waves increases in the following order.

Cosmic rays < γ -rays < X-rays < UV rays < Visible < IR rays < Microwaves < Radiowaves

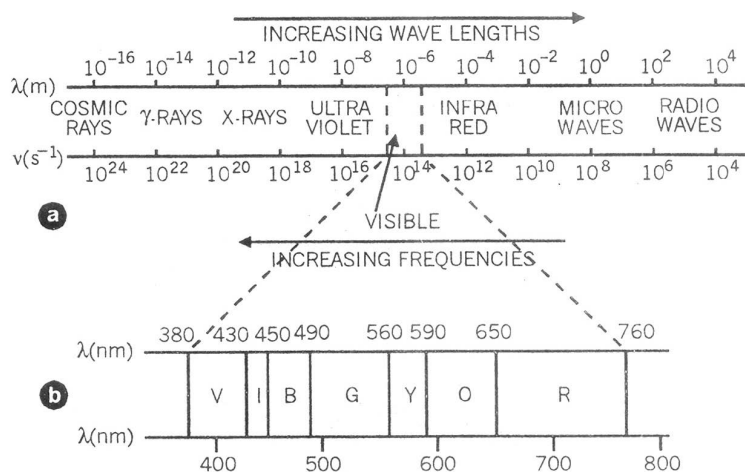


Fig. (a) Complete electromagnetic spectrum (b) Spectrum of visible light.

CHECK Point

- ☛ Arrange X-rays, cosmic rays and radiowaves according to frequency.

SOLUTION

Cosmic rays > X-rays > Radiowaves

Limitations of electromagnetic wave theory

Electromagnetic wave theory successfully explains the properties of light such as interference and diffractions but it could not explain the phenomena of photoelectric effect and black body radiations.

PLANCK'S QUANTUM THEORY

In order to explain the phenomenon of photoelectric effect and black body radiations, Max Planck (1900) postulated quantum theory of radiations, which was further extended by Einstein (1905). The main features of Planck's quantum theory are :

- The radiant energy is not emitted or absorbed continuously but discontinuously in the form of small packets of energy called quantum. For light the quantum of energy is termed as photon.

- (ii) The energy associated with each quantum is directly proportional to the frequency of the radiation

i.e., $E \propto \nu$ or $E = h\nu$ (i)

where h is called Planck's constant.

Numerically $h = 6.62 \times 10^{-34}$ Joules sec or 6.62×10^{-27} erg sec.

CHECK Point

- ☛ What is the difference between a quantum and a photon ?

SOLUTION

The smallest packet of energy of any radiation is called a quantum whereas that of light is called photon.

- (iii) The total amount of energy emitted or absorbed by a body will be some whole number multiple of quantum by an integer n i.e., $E = nh\nu$ where n is an integer e.g., $n = 1, 2, 3, \dots$, etc.

This implies that a body can emit or absorb energy equal to $h\nu, 2h\nu, 3h\nu, 4h\nu, \dots$ etc., but not in fractions i.e., $0.5 h\nu, 1.2 h\nu$. Further we know that

$$\nu = \frac{c}{\lambda}$$

$\therefore$ Equation (i) becomes $E = \frac{hc}{\lambda}$

This equation represent that a wave of higher frequency or lower wavelength will be more energetic and vice versa. e.g., violet light of high frequency has more energy while the red light of low frequency is associated with less energy.

ILLUSTRATION : 2

Calculate the frequency and energy of a photon of radiation having wavelength 6000 Å.

SOLUTION :

(i) Frequency $\nu = \frac{c}{\lambda}$

Substituting $c = 3 \times 10^8 \text{ ms}^{-1}$

$\lambda = 6000 \text{ Å} = 6000 \times 10^{-10} \text{ m}$

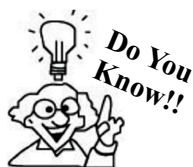
we get $\nu = \frac{3 \times 10^8}{6000 \times 10^{-10}} = 5 \times 10^{14} \text{ s}^{-1}$

(ii) Energy of the photon $E = h\nu$

Substituting $h = 6.625 \times 10^{-34} \text{ Js}$

$\nu = 5 \times 10^{14} \text{ s}^{-1}$

we get $E = 6.625 \times 10^{-34} \times 5 \times 10^{14}$
 $= 3.3125 \times 10^{-19} \text{ J.}$



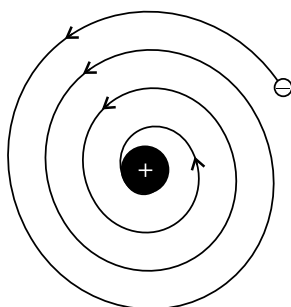
Some properties of light (or in general any radiation) like interference, diffraction etc. can be explained only if we consider light to have wave nature whereas some other properties of light or any other radiation such as black body radiation and photoelectric effect can be explained if we consider light to have particle nature. Thus, light is said to have a dual nature, i.e., it behaves as a wave as well as a stream of particles.

DRAWBACKS OF THE RUTHERFORD THEORY

According to electromagnetic theory, a charged particle revolving in a circular path continuously emits energy. Thus, it will continuously lose energy.

An electron revolving around the nucleus in a circular path will continuously lose energy and fall into the nucleus. However, the nucleus is found to be quite stable. Thus, Rutherford could not explain the stability of the nucleus in the light of continuous loss of the energy of the electron.

The stability of the atom is explained by Bohr's theory of atom in terms of orbits of fixed energies.



BOHR'S MODEL OF THE ATOM

To overcome the drawback of Rutherford's model of atom and to explain the line spectrum of hydrogen, Neils Bohr, a Danish physicist in 1913 proposed a new model of atom based upon Planck's quantum theory. This new model is called Bohr's model of atom. Bohr proposed the following postulates for revising the Rutherford model.

- (i) Electrons move around the nucleus in stationary circular orbits of fixed energy. They are termed as energy levels or shells.
- (ii) Electrons move around only in those circular orbits in which its angular momentum is integral multiple of $\frac{h}{2\pi}$. Where h = Planck's constant.
- (iii) Energy is neither absorbed nor emitted when electron is moving in an orbit. But energy is absorbed when it jumps from lower orbit to higher orbit whereas energy is emitted when it jumps from higher orbit to lower orbit.
- (iv) Energy is absorbed or emitted in the form of quanta or bundles, so various energy levels of atom are quantised.

Under normal conditions, the single electron in a hydrogen atom will occupy the lowest energy level, which is the orbit closest to the nucleus. This is called the ground state. When energy is supplied to a hydrogen atom, as when it is heated, the electron can absorb the appropriate amount of this energy to "jump" from the ground state (i.e., the $n = 1$ level) to a higher energy levels (i.e., $n = 2, 3, 4$, etc.). Because of its new position in a higher energy level, the electron now has potential energy just like a weight suspended above the ground. Energy levels higher than the ground state are called excited states. What is most significant according to Bohr was what happened when the electron fell back down to lower excited states or all the way to the ground state. He suggested that the electron gives up this energy in the form of light when it falls back. Since energy levels are quantized, the difference in energy between any two levels is quantized. Thus, when an electron falls back to a lower energy level, it must emit a discrete amount of energy. Since this energy is emitted as light, the light would have a discrete energy, a discrete wavelength, and a discrete color (if the light is in the visible region of the spectrum).

CHECK Point

- ☛ Why are Bohr's orbits called stationary states ?

SOLUTION

This is because the energies of the orbits in which the electrons revolve are fixed.

CONNECTING TOPIC

APPLICATIONS OF BOHR'S THEORY

- (a) The energies of the different stationary states in case of hydrogen atom are given by the expression (called Bohr formula) :

$$E_n = -\frac{2\pi^2 me^4}{n^2 h^2}$$

Substituting the values of m (mass of the electron), e (charge on the electron) and h (Planck's constant), we get

$$\begin{aligned} E_n &= -\frac{21.8 \times 10^{-19}}{n^2} \text{ J/atom} \\ &= -\frac{13.6 \times 10^{-19}}{n^2} \text{ eV/atom} \quad (\text{As } 1 \text{ eV} = 1.6022 \times 10^{-19} \text{ J}) \\ &= -\frac{1312}{n^2} \text{ kJ mol}^{-1} \end{aligned}$$

Where $n = 1, 2, 3, \dots$ etc., stand for 1st, 2nd, 3rd, etc., level respectively.

For H-like particle, e.g., He^+ , Li^{2+} etc. (containing one electron only), the expression for energy is

$$E_n = -\frac{2\pi^2 m Z^2 e^4}{n^2 h^2} = -\frac{1312 Z^2}{n^2} \text{ kJ mol}^{-1}$$

where Z is the atomic number of the element (For He^+ , $Z = 2$; for Li^{2+} , $Z = 3$).

- (b) The *radii* of the stationary states of the hydrogen atom are given by the expression.

$$r_n = a_0 n^2$$

where $a_0 = 52.9 \text{ pm}$ is the radius of the first stationary state and is called *Bohr radius*.

For H-like particles, the radii of the stationary states are given by the expression.

$$r_n = \frac{a_0 n^2}{Z}$$

- (c) The *velocities* of the electrons in different orbits are given by the expression.

$$v_n = v_0 \times \frac{Z}{n}$$

where $v_0 = 2.188 \times 10^8 \text{ cm s}^{-1}$ is the velocity of the electron in the first orbit of hydrogen atom.

- (d) Like energy, the angular momentum of an electron in an atom can have certain definite or discrete values and not any value of its own.

$$mvr = \frac{nh}{2\pi}$$

i.e., angular momentum of the electron is an integral multiple of $h/2\pi$. Here, m is the mass of the electron, v is the tangential velocity of the revolving electron, r is the radius of the orbit, h is the Planck's constant and n is any integer. In other words, the angular momentum of the electron can be $h/2\pi$, $2h/2\pi$, $3h/2\pi$ etc. This means that like energy, the angular momentum of an electron in an atom is also quantized.

ILLUSTRATION : 3

Calculate the velocity of electron in the first Bohr orbit of hydrogen atom. Given that Bohr radius = $0.529 \text{ \AA}$,

Planck's constant, $h = 6.626 \times 10^{-34} \text{ J s}$,

mass of electron = $9.11 \times 10^{-31} \text{ kg}$ and

$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$.

SOLUTION :

$$mvr = \frac{nh}{2\pi} \text{ or } v = \frac{nh}{2\pi mr}$$

$$= \frac{(1)(6.626 \times 10^{-34} \text{ J s})}{2 \times 3.14 \times (9.11 \times 10^{-31} \text{ kg}) \times (0.529 \times 10^{-10} \text{ m})}$$

$$= 2.189 \times 10^6 \text{ Js kg}^{-1} \text{ m}^{-1}$$

But $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$

Hence, $v = 2.189 \times 10^6 \text{ ms}^{-1}$

ILLUSTRATION : 4

The ionisation energy of He^+ is $8.72 \times 10^{-18} \text{ J atom}^{-1}$. Calculate the energy of the first stationary state of Li^{2+} .

SOLUTION :

$$E_n = -\frac{2\pi^2 m Z^2 e^4}{n^2 h^2} = -K \frac{Z^2}{n^2} \quad (K = \text{constant})$$

$$\text{I.E. of } \text{He}^+ = E_\infty - E_1$$

$$= 0 - \left(-K \frac{2^2}{1^2} \right) = 4K$$

Hence, $4K = 8.72 \times 10^{-18} \text{ J atom}^{-1}$ (Given)

or $K = 2.78 \times 10^{-18} \text{ J atom}^{-1}$

For Li^{2+} , $Z = 3$ and for 1st stationary state, $n = 1$

$$E_1 = -K \frac{Z^2}{n^2} = -2.18 \times 10^{-18} \times \frac{3^2}{1^2}$$

$$= -19.62 \times 10^{-18} \text{ J atom}^{-1}$$



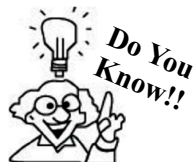
idea box

Make a project report on "Models of Atom – Comparison". Project report should include merits and demerits of each model of atom. You can make your project more effective by using colored pictures, diagrams, or illustrations etc. You can refer to science Journals, science magazines and internet to enhance the quality of content of your project.

DISCOVERY OF NEUTRON

So far we have studied by Rutherford's atomic model that the mass of the atom is mainly due to the protons and it is concentrated in the nucleus. **Chadwick** (1932), while studying the bombardment of light elements, such as beryllium, boron and lithium, by fast moving α -particles obtained highly penetrating radiations. These radiations were found to have high ionising power and were not deflected by electric or magnetic field. These neutral particles were found to have mass $1.675 \times 10^{-24} \text{ g}$ and were named neutrons.

Thus, a neutron is defined as a sub-atomic particle which has mass almost equal to that of a proton ($1.675 \times 10^{-27} \text{ kg}$) but has no charge.



*This discovery led to the modification of Rutherford atomic model. In the modified atomic model the nucleus consists of protons and neutrons and these are collectively termed as **nucleons**. The entire mass of the atom is due to the number of nucleons present in the atom.*

ILLUSTRATION : 5

Complete the following table :

Name of particle	Charge	Mass (kg)
Electron	-1	
Proton		
Neutron		1.67×10^{-27}

SOLUTION :

Name of particle	Charge	Mass (kg)
Electron	-1	9.1×10^{-31}
Proton	+1	1.67×10^{-27}
Neutron	0	1.67×10^{-27}

ARRANGEMENT OF ELECTRONS IN AN ATOM – BOHR-BURY RULE

Orbit :

An orbit or shell is a circular path around the nucleus in which the electron revolves. These are represented by the symbol K, L, M and N.

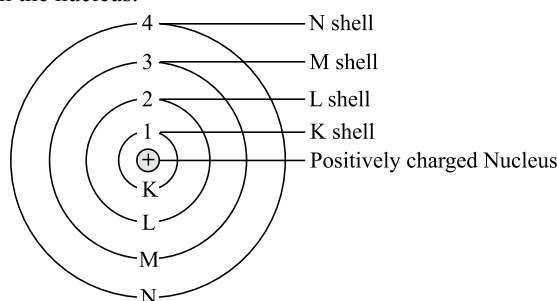
Orbital :

An orbital is the region around the nucleus within which the probability of finding an electron is maximum. These are represented by the symbols s, p, d and f each orbital contains maximum 2 electrons. It represents three dimensional motion of an electron around the nucleus.

Distribution of electrons revolving around the nucleus of an atom, in different orbits (energy levels or shells) is called electronic configuration. This scheme was proposed by two scientists Bohr and Bury.

Bohr-Bury Scheme of Electronic Configuration

- (i) The maximum number of electrons which can be present in any shell of an atom is given by the formula $2n^2$, where n is the number of shells as counted from the nucleus.



Thus, according to this scheme the distribution of the electrons in the different shells is shown in the table given below.

Shell number or energy level	Maximum number of electrons
First shell or -K-shell	$2(n)^2 = 2(1)^2 = 2$
Second shell or L-shell	$2(n)^2 = 2(2)^2 = 8$
Third shell or M-shell	$2(n)^2 = 2(3)^2 = 18$
Fourth shell or N-shell	$2(n)^2 = 2(4)^2 = 32$

- (ii) The outermost shell (also called valence shell) cannot have more than 8 electrons and the shell next to it cannot have more than 18 electrons.
- (iii) It is not necessary for a given shell to complete itself, before another shell starts forming. As a rule, a new shell is formed as soon as the shell attains 8 electrons.
- (iv) An atom becomes stable (i.e. it stops reacting with other elements), when its outermost shell has 8 electrons or it has only one shell containing 2 electrons.)

CHECK Point

- ☛ Comment on the spacing between the energy levels.

SOLUTION

The spacing between the energy levels decreases as we move outwards from the nucleus.

VALENCY

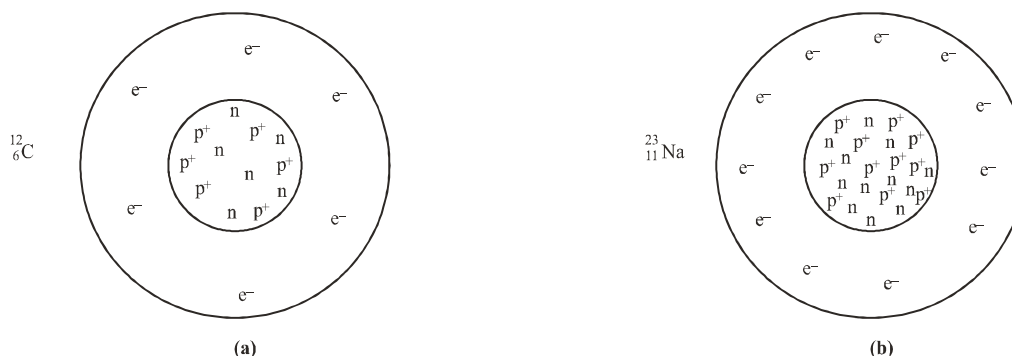
During the formation of molecules of compounds, atoms combine in certain fixed proportions. This is because of the fact that different atoms have different combining capacities. The combining capacity of an atom or radical is known as its **valency**.

Outermost shell of an atom is called valence shell. Valence shell can accommodate a maximum of 8 electrons. Valency can also be explained as an attempt to attain a fully-filled outermost shell. Number of electrons are gained, lost or shared to obtain fully-filled outermost shell. Outermost shell gives the valency of that element.

ATOMIC NUMBER AND MASS NUMBER

Each element has been assigned an atomic number (Z) that describes the number of protons in the nucleus of an atom of that element. i.e., atomic number (Z) = number of proton = number of electron (in a neutral atom). The number of protons in the nucleus of an atom determines the identity of the atom. Every carbon atom (Z = 6) has six protons in the nucleus of the atom, whereas sodium atoms (Z = 11) have eleven. Since atoms are electrically neutral, the nucleus of the carbon atom must be surrounded by six electrons, and the sodium atom must contain 11 electrons.

Structure of Atom

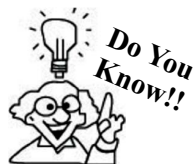


The nucleus of an atom is also described by a mass number (A), which is the sum of the number of protons and neutrons in the nucleus. The difference between the mass number and the atomic number of an atom is therefore equal to the number of neutrons in the nucleus of that atom.

$$\text{Mass number (A)} = \text{Number of protons} + \text{Number of neutrons}$$

or $\text{Number of neutrons} = A - Z$

The carbon atom has a mass number of 12 because it contains six protons and six neutrons. The sodium atom has a mass number of 23 because it contains 11 protons and 12 neutrons.



A shorthand notation has been developed to describe the number of neutrons and protons in the nucleus of an atom. The atomic number is written in the bottom left corner of the symbol for the element and the mass number is written in the top left corner: A_ZX . The atoms in figure would therefore be given the symbol ${}^{12}_6\text{C}$ and ${}^{23}_{11}\text{Na}$.

Table : Atomic numbers and mass numbers of the first twenty elements

Element	Atomic number Z	Mass number A	Number of protons Z	Number of neutrons $A-Z$	Number of electrons
Hydrogen	1	1	1	0	1
Helium	2	4	2	2	2
Lithium	3	7	3	4	3
Beryllium	4	9	4	5	4
Boron	5	11	5	6	5
Carbon	6	12	6	6	6
Nitrogen	7	14	7	7	7
Oxygen	8	16	8	8	8
Fluorine	9	19	9	10	9
Neon	10	20	10	10	10
Sodium	11	23	11	12	11
Magnesium	12	24	12	12	12
Aluminium	13	27	13	14	13
Silicon	14	28	14	14	14
Phosphorus	15	31	15	16	15
Sulphur	16	32	16	16	16
Chlorine	17	35	17	18	17
Argon	18	40	18	22	18
Potassium	19	39	19	20	19
Calcium	20	40	20	20	20

CHECK Point

- ☛ Calculate the number of protons, neutrons and electrons in ${}^{80}_{35}\text{Br}$.

SOLUTION

Here, $Z = 35$, $A = 80$

$\therefore$ No. of protons = Atomic No. = 35

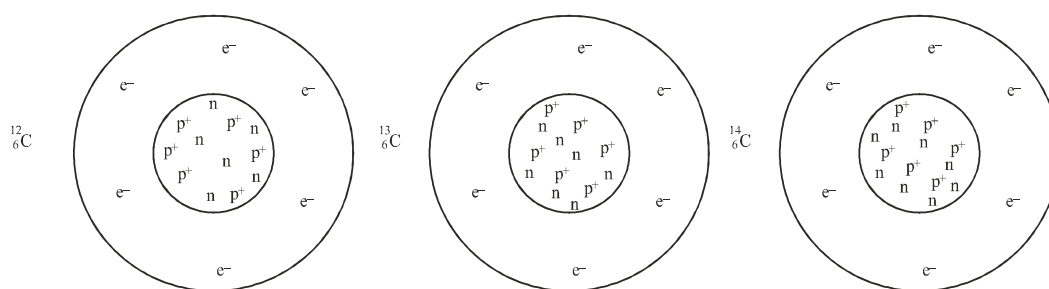
No. of neutrons = $A - Z = 80 - 35 = 45$

As the atom is neutral, No. of electrons = No. of protons = 35.

Isotopes :

The number of protons in the nucleus of an atom determines the identity of the atom. As a result, all atoms of an element must have the same number of protons. But they don't have to contain the same number of neutrons.

Atoms with the same atomic number but different mass numbers are called isotopes. Carbon, for example, has the three naturally occurring isotopes shown in Figure : ^{12}C , ^{13}C and ^{14}C . Carbon-12 (^{12}C) has six protons and six neutrons, ^{13}C has six protons and seven neutrons, and ^{14}C has six protons and eight neutrons.



The characteristics of isotopes are:

1. They have different atomic masses (mass number)
2. They have the same atomic number.
3. They have the same electronic configuration.
4. They have the same valence electrons.
5. They have the same chemical properties.
6. They have slightly different physical properties.

The fractional atomic masses of elements are due to existence of their isotopes having different masses.

Example : The chlorine isotope $^{35}_{17}\text{Cl}$ has a mass of 35u and its abundance (or proportion) in nature is 75%. The chlorine isotope $^{37}_{17}\text{Cl}$ has a mass of 37 u and its abundance (or proportion) in nature is 25%.

$$\text{Average atomic mass of chlorine} = 35 \times \frac{75}{100} + 37 \times \frac{25}{100} = \frac{2625}{100} + \frac{925}{100} = 26.25 + 9.25 = 35.5 \text{ u}$$

Hence, the average atomic mass of chlorine is 35.5 u.

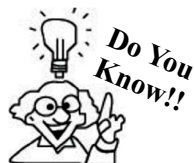


Table : Isotopes of some elements and their nuclear compositions

Element	Isotopes	Mass number	Nuclear composition number of		
			protons	neutrons	electrons
Hydrogen	Protium, ^1_1H	1	1	0	1
	Deuterium, ^2_1H or D	2	1	1	1
	Tritium, ^3_1H or T	3	1	2	1
Carbon	Carbon-12, $^{12}_6\text{C}$	12	6	6	6
	Carbon-13, $^{13}_6\text{C}$	13	6	7	6
	Carbon-14, $^{14}_6\text{C}$	14	6	8	6
Chlorine	Chlorine-35, $^{35}_{17}\text{Cl}$	35	17	18	17
	Chlorine-37, $^{37}_{17}\text{Cl}$	37	17	20	17

Structure of Atom

Isotone :

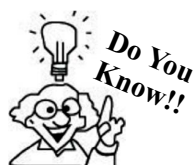
Atoms of different element, in which number of electrons and protons are different, but their number of neutrons are same, are called isotones.

For example, in Radium and Actinium, despite having same numbers of neutrons (138), they show different properties because number of protons present in them are different.

Isotones	Atomic number	Mass number	Electrons	Protons	Neutrons
$_{11}\text{Na}^{23}$	11	23	11	11	$23 - 11 = 12$
$_{12}\text{Mg}^{24}$	12	24	12	12	$24 - 12 = 12$

Isobars :

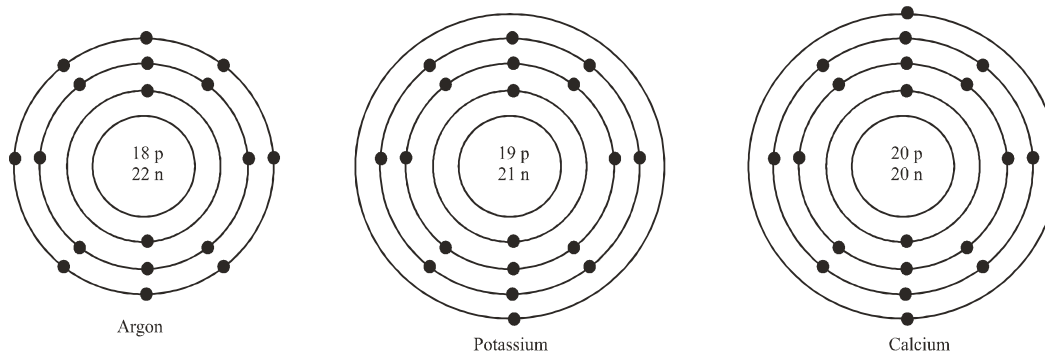
Atoms whose atomic numbers are different but mass numbers are same. Such atoms are called isobars.



Chemical reactions of various atoms having different atomic numbers are different and these atoms are called atoms of different elements.

Some important examples of Isobars are as follows :

Argon $_{18}\text{Ar}^{40}$, Potassium $_{19}\text{K}^{40}$ and Calcium $_{20}\text{Ca}^{40}$ are Isobars. Their structure are as follows :



The characteristics of isobars are:

1. They have the same mass number.
2. They have different atomic numbers.
3. They have different number of protons.
4. They have different electronic configurations.
5. They have different number of valence electrons.
6. They have different chemical properties.

CHECK Point

Give one example of each of the following :

(i) Isotope of $^{35}_{17}\text{Cl}$ (ii) Isobar of $^{40}_{18}\text{Ar}$ (iii) Isotone of $^{14}_6\text{C}$

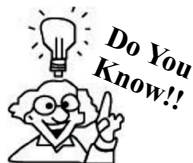
SOLUTION

(i) $^{37}_{17}\text{Cl}$ (ii) $^{40}_{19}\text{K}$ (iii) $^{16}_8\text{O}$

CONNECTING TOPIC

QUANTUM NUMBERS

We have studied that orbital is that area around the nucleus in which the probability of finding the electron is maximum. However, an atom contains a large number of orbitals, which are distinguished from each other by their size, shape and orientation (direction) in space. The parameters by which the orbitals are distinguished can be expressed by a set of numbers known as quantum numbers.



Thus, quantum numbers may be defined as set of numbers which display complete information about size, shape and orientation of the orbital.

These are designated as principal quantum number (n), azimuthal quantum number (l) and magnetic quantum number (m). Thus, an orbital is designated by three quantum numbers. The fourth quantum number designated as spin quantum number (s), represent the spin of electron i.e., rotation of electron about its own axis.

1. Principal Quantum Number (n) :

This is an important quantum number which gives the following information about the electron.

- This refers to the average distance of the electron from the nucleus i.e., it relates to the size of the electron cloud.
- It denotes the energy level to which the electron belongs.
- It gives information about the maximum number of electrons that can be accommodated in any shell. Number of electrons in any shell is given by expression $2n^2$.

2. Azimuthal Quantum Number or Angular Momentum Quantum Number (l) :

This quantum number is denoted by ' l ' and gives the following information about the electron.

- It tells about the number of subshells within a given principal energy shell to which the electrons belong. For a given value of principal quantum number, ' n ', the azimuthal quantum number, ' l ', may have all integral values from 0 to $(n-1)$, each of which represent a different subenergy level or subshell. These subshells are designated by letters s, p, d and f for which $l = 0, 1, 2$ and 3 respectively.

n	1	2	3	4
l	0	0, 1	0, 1, 2	0, 1, 2, 3
Subshell	s	s, p	s, p, d	s, p, d, f
Designation of sub shell	1s	2s, 2p	3s, 3p, 3d	4s, 4p, 4d, 4f

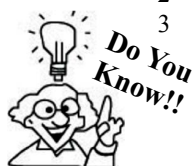
Thus, the total number of subshells for a given principal shell is equal to the value of n .

- It tells about the relative energies of subshells belonging to same shell. The energies of different subshells present within the same principal shell are in the following order

$$\frac{s < p < d < f}{\text{Increasing energy} \rightarrow}$$

- The value of l refers to the shape of the subshells. For example,

l	Subshell	Shape of subshell
0	s	Spherical
1	p	Dumbbell
2	d	Double dumbbell
3	f	Complex shape



$$\text{Angular momentum of the electron} = \sqrt{l(l+1)} \frac{h}{2\pi} = \sqrt{l(l+1)} \hbar$$

Note that it depends only on the value of l and not on the value of n . Thus, 1s, 2s, 3s etc. electrons will have the same angular momentum. Same applies to all p or d electrons.

CHECK Point

- ☛ What is the difference between the notations l and L ?

SOLUTION

' l ' represents azimuthal quantum number which can have values 0, 1, 2 etc. L represents second Bohr orbit for which $n = 2$ always.

3. **Magnetic Quantum Number (l)** : This quantum number describes the behaviour of electron in a magnetic field. This quantum number provides following information about the electron.

- (i) Magnetic quantum number gives the number of permitted orientations of subshells. For example, for a given value of ' l ', the possible values of ' m ' range from $-l$ through 0 to $+l$. Each value of m corresponds to one atomic orbital. For example,

For s -subshell,	$l = 0$	$m = 0$	i.e., s -subshell has one orbital
For p -subshell,	$l = 1$	$m = -1, 0, +1$	i.e., p -subshell has three orbitals
For d -subshell,	$l = 2$	$m = -2, -1, 0, +1, +2$	i.e., d -subshell has five orbitals
For f -subshell,	$l = 3$	$m = -3, -2, -1, 0, +1, +2, +3$	i.e., f -subshell has seven orbitals

- (ii) It explained successfully the splitting of spectral lines in the magnetic field i.e., Zeeman effect.

4. **Spin Quantum Number (s)** : It has actually been observed that the electron in an atom is not only revolving around the nucleus but is also spinning about its own axis. Spin quantum number accounts for the spinning orientation of the electron.

The electron in an orbital can have only two types of spins i.e., in clockwise and anticlockwise direction. Therefore, the spin quantum number can have only two values i.e., $+\frac{1}{2}$ or $-\frac{1}{2}$. The two spins of the electrons in an orbital are usually represented

by arrows pointing in the opposite directions i.e., $\uparrow$ and $\downarrow$ respectively.

The value of spin quantum number is independent of the values of other three quantum numbers.

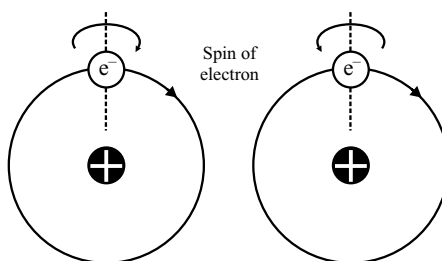


Fig. Clockwise and anticlockwise spins of an electron

ILLUSTRATION : 6

What designation is given to an orbital having

- (i) $n = 2, l = 1$, (ii) $n = 3, l = 0$, (iii) $n = 5, l = 3$ and (iv) $n = 4, l = 2$?

SOLUTION :

- (i) $n = 2, l = 1$ means 2p-orbital
 (ii) $n = 3, l = 0$ means 3s-orbital
 (iii) $n = 5, l = 3$ means 5f-orbital
 (iv) $n = 4, l = 2$ means 4d orbital.

PAULI'S EXCLUSION PRINCIPLE

According to this principle "no two electrons in an atom can have the same set of all the four quantum numbers". In other words,

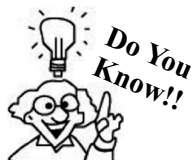
- (a) an orbital cannot have more than two electrons.
 (b) if an orbital has two electrons, they must have opposite spin i.e., they must be paired electrons.

AUFBAU PRINCIPLE

In the ground state of the atoms, the orbitals are filled in order of their increasing energies. In other words, electrons first occupy the lowest energy orbital available to them and then enter into higher energy orbitals only when the lower energy orbitals are filled.

The order in which the energies of the orbitals increase and hence the order in which the orbitals are filled is as follows :

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p,



In neutral isolated atom, the lower the value of $(n + l)$ for an orbital, lower is its energy. However, if the two different types of orbitals have the same value of $(n + l)$, the orbital with lower value of n has lower energy.

This trend of increasing energies of various orbitals may be remembered in the form of following diagram.

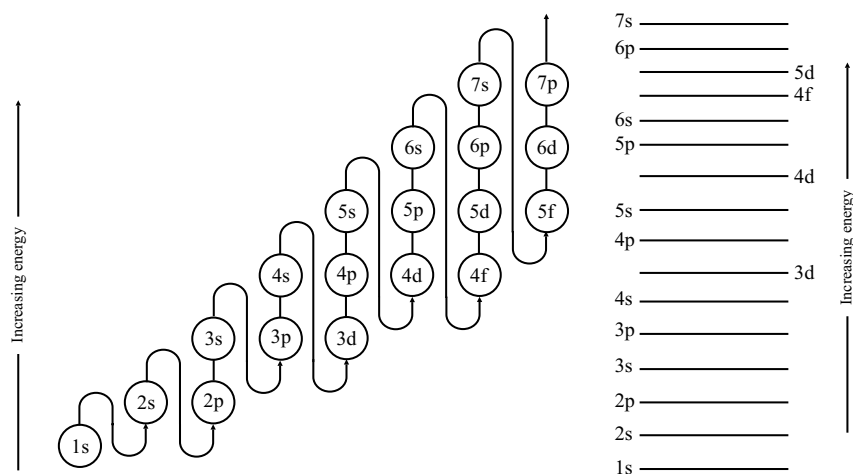


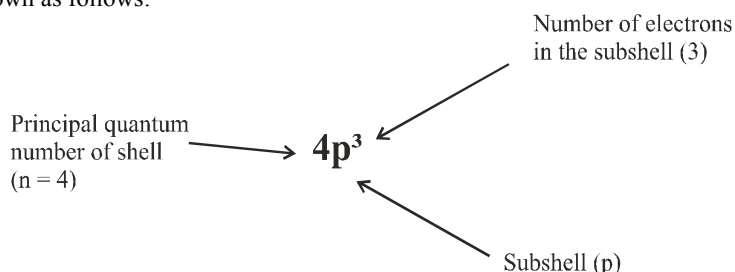
Fig. Increasing order of energy of various orbitals

HUND'S RULE OF MAXIMUM MULTIPLICITY

This rule deals with the filling of electrons into degenerate (equal energy) orbitals of the same sub-shell (p, d and f). According to this rule, Electron pairing in p, d and f orbitals cannot occur until each orbital of a given subshell contains one electron each or is singly occupied.

ELECTRONIC CONFIGURATION

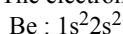
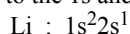
The assignment of all of the electrons in an atom into specific shells and subshells is known as the element's electronic configuration. A shell is designated by the principal quantum number (n), the subshell by the azimuthal quantum number (l), and the number of electrons in that subshell by the appropriate superscript number. For example, the existence of three electrons in the p subshell of the fourth shell ($n = 4$) is shown as follows:



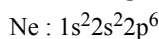
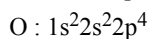
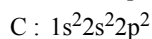
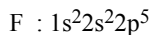
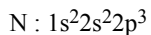
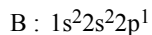
We begin with the ground state of the one electron in the simplest and first element, hydrogen. The electron is in the $n = 1$ shell, which has an s subshell. The electron configuration for H is $1s^1$.

The next element, helium, has two electrons. The 1s subshell has a capacity of two electrons, so He has the configuration $1s^2$.

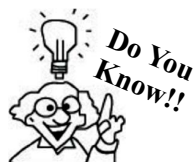
Next comes Li with three electrons. The first two electrons fill the first shell, but the third is assigned to the second shell ($n = 2$). The second shell has two subshells, the s and the p, but the s is lower in energy so it fills first. The next element is Be. Its four electrons are also assigned to the 1s and 2s subshells. The electron configurations of Li and Be are



The next element is B. Four of its five electrons have the same configuration as Be, but the fifth electron begins the filling of the next subshell, the 2p. The next five elements, C to Ne, complete the filling of the 2p subshell.



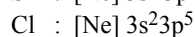
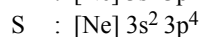
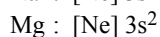
With the element neon, all of the orbitals in the second shell are full. We now continue with the element sodium, which has 11 electrons. The first 10 electrons fill the first and second shells, so the 11th electron is assigned to the third shell. The third shell has three subshells, the 3s, 3p, and 3d. The lowest in energy is the 3s subshell, so the 11th electron is in the 3s subshell.



A shorthand method of writing configurations is to represent all of the filled subshells of a noble gas by the symbol of that noble gas in brackets (e.g., [Ne]). Thus, in the following electron configurations, we will assume



We choose a noble gas to use in our shorthand notation for a good reason. As, electrons in noble gas cores are not involved in bonding so we can treat them as a group. Using the [Ne] core symbolism, the electron configurations of the next eight elements after neon are shown as follows.



CHECK Point

- Find the number of unpaired electrons present in phosphorous (atomic no. 15), chromium (atomic no. 24) and copper (atomic no. 29) after writing their orbital electronic configurations.

SOLUTION

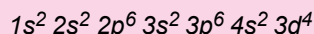
Knowledge ENHANCER

Exceptional Configuration – Stability of Completely Filled and Exactly Half-filled Orbitals.

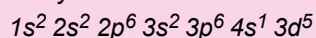
However in certain elements when the two subshells differ slightly in their energies, an electron may shift from a subshell of lower energy to a subshell of higher energy only if such a shift results in the symmetrical distribution (either completely filled or exactly half-filled) of the electrons in the various orbitals of the subshell of higher energy.

This is due to the following two reasons :

- (i) **Symmetrical distribution** : It is well known fact that symmetry leads to stability. Thus the electronic configuration in which all the orbitals of the same subshell are either completely filled or are exactly half filled are more stable because of symmetrical distribution of electrons. For example, the expected electronic configuration of chromium ($Z = 24$) is



But if one of the 4s electrons shifts to the vacant 3d-orbital, the distribution of the electrons will become more symmetrical and this will impart extra stability.



- (ii) **Exchange energy** : The electrons with parallel spins present in the degenerate orbitals tend to exchange their position. The energy released during this exchange is called exchange energy. The number of exchanges that can take place is maximum when the degenerate orbitals are exactly half-filled or completely filled. As a result, the exchange energy is maximum and so is the stability.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. According to Maharishi Kanad, the tiniest to tiny particle of a pure substance is called
2. An atom is the smallest unit of an element which takes part in a
3. The K-shell of any atom cannot have more than electrons.
4. Isotopes are the atoms of element, having same atomic number but different mass numbers.
5. An atom of an element has 11 protons, 11 electrons and 12 neutrons. The atomic mass of the atom is
6. Almost all the mass of an atom is concentrated in a small region of space called the
7. The subatomic particle not present in a hydrogen atom is
8. Cathode rays are a beam of fast moving
9. and more or less completely make up the mass of an atom.
10. The number of neutrons in the nucleus of an atom can be calculated by the atomic number its mass number.
11. The isotopes of an element do not differ in the number of but do differ in the number of

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. Thomson proposed that the nucleus of an atom contains protons and neutrons.
2. J.J. Thomson proposed that the nucleus of an atom contains only nucleons.
3. An electron has a mass that is much less than a proton.
4. There is no particle of matter smaller than an atom.
5. Atoms of an element may have more or less neutrons or electrons than other atoms of the same element.
6. The innermost atomic shell can hold a maximum of 18 electrons.
7. α -particles are same thing as helium atoms.

Match the Following :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in Column I have to be matched with statements (p, q, r, s) in Column II.

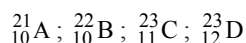
1. Column-II gives name of scientist associated with particle given in Column I match them correctly.

Column-I	Column-II
(A) Electron	(p) Chadwick
(B) Proton	(q) J.J. Thomson
(C) Nucleus	(r) Goldstein
(D) Neutron	(s) Rutherford

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

1. Name the scientist who discovered proton.
2. What is the approximate ratio of diameters of atom to that of its nucleus?
3. What are nucleons?
4. What are isotopes?
5. Neutrons can be found in all atomic nuclei except in one case. Name this atomic nucleus.
6. What did Rutherford's alpha-particle scattering experiment prove?
7. An atom contains five protons and six neutrons. What is its atomic number and mass number?
8. If the number of electrons in an atom 8 and number of protons are also 8. Their atomic number of atom will be eight or zero?
9. Write the symbol of an element 'X' having atomic number 'Z' and mass number 'A'.
10. For which element is the atomic mass is same as that of a proton?
11. Who discovered the neutral particles present in an atom?
12. What conclusion would you derive from the observation, that in Rutherford's scattering experiment, most of the alpha-particles passed through gold foil without any deflection?
13. Which shell is closest to the nucleus of an atom?
14. Select the isotopes from the following?

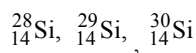


Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- How is atomic mass related to the number of protons and number of neutrons?
- What are valence electrons?
- What happens when cathode rays strike the glass walls of the cathode ray tube?
- State Bohr-Bury rules for the distribution of electrons in different energy levels.
- What experiment results into failure of Thomson's model of the atom as an intimate mixture of negative and positive particles.
- Which experiment led to the discovery of neutrons?
- From amongst **K**, **L**, **M** and **N** shells, which is associated with lowest energy and which one is associated with highest energy?
- A element 'A' has mass number 24 and contain 12 neutrons. What is the valency of 'A'?
- What are the similarities between various isotopes?

- For each isotope give the composition of its nucleus.



Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- Write 4 points of difference between cathode rays and anode rays.
- Write conclusion of Rutherford's scattering experiment. Also mention drawbacks of Rutherford's model of atom.
- Name the model of atom proposed to explain the objections raised for Rutherford's model. Explain this model of atom in detail.
- A neutral atom of an element has 2K, 8L, and 5M electrons. Find out the following from the data :
 - Atomic number
 - Total number of s-electrons
 - Total number of p-electrons
 - Number of protons in the nucleus and
 - Valency of the element

2 EXERCISE

Text - Book Questions :

- What are canal rays?
- If an atom contains one electron and one proton, will it carry any charge or not ?
- On the basis of Thomson's model of an atom, explain how atom is neutral as a whole.
- On the basis of Rutherford's model of an atom, which sub-atomic particle is present in the nucleus of an atom.
- Draw a sketch of Bohr's model of an atom with three shells.
- What do you think would be the observation if the alpha (α -) particle scattering experiment is carried out using a foil of metal other than gold?
- Name the three sub-atomic particles of an atom.
- Helium atom has an atomic mass of 4U and 2 protons in its nucleus. How many neutrons does it have?
- Write the distribution of electrons in carbon and sodium atoms.
- If K and L shells of an atom are full, then what would be the total number of electrons in the atom?
- How will you find the valency of chlorine, sulphur and magnesium.
- If the number of electrons in an atom is 8 and number of protons is also 8.
 - What is the atomic number of the atom ?
 - What is the charge on the atom ?
- With the help of table 4.1 in your ncert find out mass number of oxygen and sulphur atom.
- For the symbol H, D and T, tabulate three sub-atomic particles found in each of them.
- Write the electronic configuration of any one pair of isotopes and isobars.
- The average atomic mass of a sample of element 'X' is 16.2U. What are the percentage of isotopes $^{16}_8\text{X}$ and $^{18}_8\text{X}$ in the sample?
- If $Z = 3$, what would be the valency of the element? Also name the element.
- Composition of the nuclei of two atomic species X and Y are given as under

	X	Y
Proton =	6	6
Neutron =	6	8

 Give the mass numbers of 'X' and 'Y'. What is the relation between the two species?
- For following statement, write 'T' for true and 'F' for False.
 - J.J. Thomson proposed that the nucleus of an atom contains only nucleus.
 - A neutron is formed by an electron and proton combining together. Therefore, it is neutral
 - The mass of an electron is about $\frac{1}{2000}$ times that of a proton
 - An isotopes of iodine is used for making Tincture of iodine, which is used as a medicine.
- Rutherford's alpha-particle scattering experiment was responsible for the discover of
 - Atomic nucleus
 - Electron
 - Proton
 - Neutron
- Isotopes of element have :
 - The same physical properties
 - different chemical properties
 - different number of neutrons
 - different atomic numbers
- Number of valence electrons in Cl^- ions are :
 - 16
 - 8
 - 17
 - 18
- Which one of the following is a correct electronic configuration of sodium?
 - 2, 8
 - 2, 8, 1
 - 2, 1, 8
 - 2, 8, 1

Text - Book Exercise :

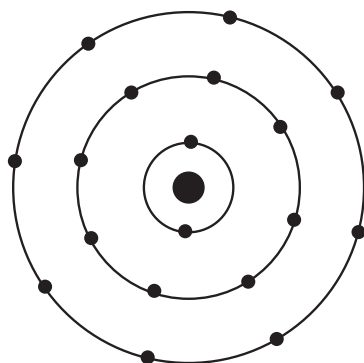
- Compare the properties of electrons, protons and neutrons
- What are the limitations of J.J. Thomson's model of an atom?
- What are the limitations of Rutherford's model of an atom?
- Describe Bohr's model of an atom.
- Compare all the proposed models of an atom given in this chapter
- Summarise the rules for writing the distribution of electrons in various shells for first eighteen elements.
- Describe valency by taking examples of silicon and oxygen.
- Explain with examples (i) Atomic number (ii) Mass number (iii) Isotopes (iv) Isobars. Give any two uses of the isotopes.
- Na^+ ion has completely filled K and L shells. Explain.
- If the element bromine is in the form of two isotopes which are $^{79}_{35}\text{Br}$ (49.7%) and $^{81}_{35}\text{Br}$ (50.3%), then calculate the average atomic mass of bromine.

19. Complete the following table :

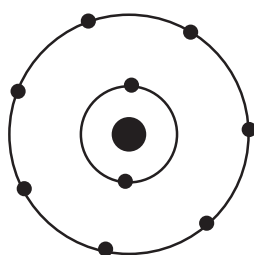
Atomic Number	Mass Number	Number of Neutrons	Number of Protons	Number of Electrons	Name of Atomic species
9	—	10	—	—	—
16	32	—	—	—	Sulphur
—	24	—	12	—	—
—	2	—	1	—	—
—	1	0	1	0	—

Exemplar Questions:

1. Why did Rutherford select a gold foil in his α -ray scattering experiment?
2. Find out the valency of the atoms represented by the Fig. (a) and (b).

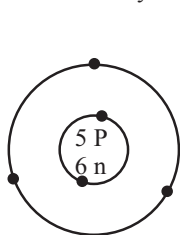


(a)

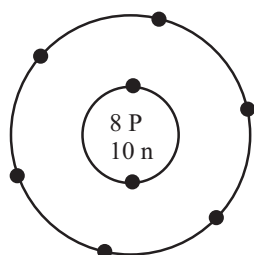


(b)

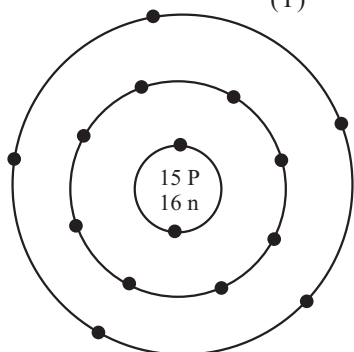
3. What information do you get from the Fig. 4.4 about the atomic number, mass number and valency of atoms X, Y and Z? Give your answer in a tabular form.



(X)



(Y)



(Z)

4. Calculate the number of neutrons present in the nucleus of an element X which is represented as ${}_{15}^{31}\text{X}$
5. The atomic number of calcium and argon are 20 and 18 respectively, but the mass number of both these elements is 40. What is the name given to such a pair of elements?
6. Why do Helium, Neon and Argon have a zero Valency?

HOTS Questions :

1. The composition of nuclei of two atomic species 'X' and 'Y' is given below

	'X'	'Y'
Number of protons	6	6
Number of neutrons	6	8

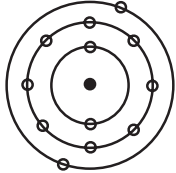
- (i) What is the mass number of 'X' and 'Y'?
 - (ii) How are 'X' and 'Y' related?
2. Nuclear radius is of the order of 10^{-13} cm while atomic radius is of the order of 10^{-8} cm. Assuming the nucleus and the atom to be spherical, what fraction of the atomic volume is occupied by the nucleus?
 3. What is the evidence that cathode rays are part of all matters?
 4. The atomic number of uranium is 92. How many electrons are there in the neutral atom? How many electrons and protons are there in U^{+2} ion?
 5. If an element 'X' has atomic weight 24 and its atomic number is 12, then calculate the number of neutron present in it
 6. The relative atomic mass of boron is 10.8. Calculate the % age of its isotopes ${}_{5}^{10}\text{B}$ and ${}_{5}^{11}\text{B}$ occurring in nature.
 7. Define **charge number**. How is it related to the number of electrons for a neutral atom?
 8. Chlorine consists of isotopes ${}_{17}^{35}\text{Cl}$, ${}_{17}^{37}\text{Cl}$. Their relative abundance of number of atoms is 754 : 246. Find out the relative atomic mass of natural chlorine.
 9. Neon has atomic weight 20.20. An atom of neon contains 10 protons and 10 neutrons. How can you account for the value of atomic weight of neon?

3 EXERCISE

Multiple Choice Questions :

DIRECTIONS : This section contains each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- What is an electron ?
(a) A wave
(b) A particle
(c) Either of two, depending on how it is observed
(d) Neither of these
- The angular momentum of the electron in first excited energy state of hydrogen atom is
(a) $\frac{h}{\pi}$
(b) $\frac{h}{2\pi}$
(c) $\sqrt{2(2+1)}\frac{h}{2\pi}$
(d) None of these
- The cathode ray experiment was done for the first time by
(a) J.J. Thomson
(b) John Dalton
(c) Goldstein
(d) Rutherford
- The charge on an electron is
(a) 1.6×10^{-6} coulombs
(b) 1.6×10^{-20} coulombs
(c) 1.6×10^{-19} coulombs
(d) 1.6×10^{-16} coulombs
- The nucleus of an atom contains
(a) protons
(b) electrons
(c) protons and neutrons
(d) neutrons
- By whom was neutron discovered?
(a) Bohr
(b) Chadwick
(c) Rutherford
(d) Dalton
- In an atom valence electrons are present in
(a) outermost orbit
(b) next to outermost orbit
(c) first orbit
(d) any one of its orbit
- The maximum number of electrons that can be accommodated in third shell ($n = 3$) is
(a) 2
(b) 8
(c) 18
(d) 10
- In an atom, the constituent electrons
(a) do not move
(b) are uniformly distributed
(c) move around the nucleus in fixed energy levels.
(d) move around the nucleus in a random way.
- What is the number of valence electrons of Al?
(a) 1
(b) 2
(c) 3
(d) 4
- Which of the of the following arrangements of electrons represent magnesium (Mg)?
(a) 2, 8, 1
(b) 2, 8, 2
(c) 2, 8, 3
(d) 2, 8, 4
- Which of the following statements is *incorrect* for cathode rays?
(a) They move in straight line
(b) Their nature depends upon the nature of gas present in the discharge tube.
(c) They cast shadow of solid objects placed in their path
(d) They get deflected towards positive charge.
- The isotopes of an element have
(a) same number of neutrons
(b) same atomic number
(c) same mass number
(d) None of these
- Which of the following statements is **not** correct for Bohr's model of an atom?
(a) The nucleus of an atom is situated at its centre
(b) The electrons move in circular orbits
(c) Electrons jumps from one orbit to another
(d) An electron neither loses nor gains energy it jumps from one orbit to another.
- The atomic number of an element is 11 and its mass number is 23. The correct order representing the number of electrons, protons and neutrons respectively in this atom is
(a) 11, 11, 12
(b) 11, 12, 11
(c) 12, 11, 11
(d) 23, 11, 23.
- Which of the following pairs are isotopes?
(a) Oxygen and ozone
(b) Ice and steam
(c) Nitric oxide and nitrogen dioxide
(d) Hydrogen and deuterium.
- Which of the following have equal number of neutrons and protons?
(a) Hydrogen
(b) Deuterium
(c) Fluorine
(d) Chlorine
- The number of electrons in an element with atomic number X and atomic mass Y will be
(a) $(X - Y)$
(b) $(Y - X)$
(c) $(X + Y)$
(d) X

19. The relative atomic masses of many elements are not whole numbers because
- they cannot be determined accurately
 - the atoms ionize during determination of their masses
 - existence of isotopes
 - presence of impurities
20. The diagram given below represent an atom of
- 
- Chlorine
 - Magnesium
 - Calcium
 - Helium
21. Which of the following has a charge of +1 and a mass of 1 amu ?
- A neutron
 - A proton
 - An electron
 - A helium nucleus
22. Which of the following describes an isotope with a mass number of 99 that contains 56 neutrons in its nucleus ?
- $^{99}_{56}\text{Ba}$
 - $^{43}_{56}\text{Ba}$
 - $^{99}_{43}\text{Tc}$
 - $^{56}_{43}\text{Tc}$
23. Which of the following isotopes is used as the standard for atomic mass ?
- ^{12}C
 - ^{16}O
 - ^{13}C
 - ^1H
24. Which of the following is not a basic particle of an element?
- An atom
 - A molecule
 - An ion
 - None of these
25. Which would be the electrical charge on a sulphur atom containing 18 electrons ?
- 2-
 - 1-
 - 0
 - 2+
26. Which of the following pairs are isotopes ?
- Oxygen and ozone
 - Ice and steam
 - Nitric oxide and nitrogen dioxide
 - Hydrogen and deuterium
27. Members of which of the following have similar chemical properties ?
- Isotope
 - Isobars
 - Allotropes
 - Both isotopes and allotropes
28. Aluminium has a valence of 3 and sulphate has valence of 2. Therefore, the correct formula for aluminium sulphate is –
- $\text{Al}_2\text{S}_2\text{O}_4$
 - $\text{Al}_2(\text{SO}_4)_3$
 - $\text{Al}_3(\text{SO}_4)_2$
 - AlSO_4
29. The correct representation of 3 molecules of chlorine is–
- 6 Cl
 - 3 Cl_2
 - 2 Cl_3
 - Cl_6
30. While performing cathode ray experiments, it was observed that there was no passage of electric current under normal conditions. Which of the following can account for this observation ?
- Dust particles are present in air
 - Carbon dioxide is present in air
 - Air is a poor conductor of electricity under normal conditions
 - None of the above
31. The fluorescence on the walls of discharge tube is due to
- cathode rays
 - anode rays
 - canal rays
 - none of the above
32. Which of the following electronic configurations is wrong?
- Be (3) = 2, 1
 - O(8) = 2, 6
 - S(16) = 2, 6, 8
 - P(15) = 2, 8, 5
33. $^{55}_{25}\text{Mn}^{++}$ has
- 25 protons and 30 neutrons
 - 25 neutrons and 25 protons
 - 25 electrons and 40 protons
 - None of the above
34. Which one of the following statement is not true ?
- Most of the space in an atom is empty
 - The total number of neutrons and protons is always equal in a neutral atom
 - The total number of electrons and protons in an atom is always equal
 - The total number of electrons in any energy level can be calculated by the formula $2n^2$
35. $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$ are examples of
- isobars
 - isotopes
 - isoelectrons atoms
 - None of the above
36. How many electrons in a Hydrogen (H) atom
- One
 - Two
 - Three
 - Four
37. Which of the following statements about Rutherford's model of atom are correct?
- Considered the nucleus as positively charged
 - Established that the α -particles are four times as heavy as a hydrogen atom
 - Can be compared to solar system
 - Was in agreement with Thomson's model
- (i) and (iii)
 - (ii) and (iii)
 - (i) and (iv)
 - only (i)

38. In the Thomson's model of atom, which of the following statements are correct?
- The mass of the atom is assumed to be uniformly distributed over the atom
 - The positive charge is assumed to be uniformly distributed over the atom
 - The electrons are uniformly distributed in the positively charged sphere
 - The electrons attract each other to stabilise the atom
- (a) (i), (ii) and (iii) (b) (i) and (iii)
(c) (i) and (iv) (d) (i), (iii) and (iv)
39. Rutherford's α -particle scattering experiment showed that
- electrons have negative charge
 - the mass and positive charge of the atom is concentrated in the nucleus
 - neutron exists in the nucleus
 - most of the space in atom is empty
- Which of the above statements are correct?
- (a) (i) and (iii) (b) (ii) and (iv)
(c) (i) and (iv) (d) (iii) and (iv)
4. Which is/are *not* correct amongst the following statement about cathode rays :
- They are deflected by electric and magnetic fields
 - e/m ratio of the particles forming them depends upon the gas filled in the discharge tube
 - e/m ratio is constant, independent of the gas filled and cathode material
 - particles forming them possess the same mass and same charge.

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

Passage - I

Elements from A to F have in them the distribution of electrons, neutrons and protons as follows :

Atoms/ ions	Numbers of Electrons	Number of Neutrons	Number of Protons
A	4	4	3
B	10	12	11
C	17	18	17
D	17	20	17
E	18	22	18
F	19	21	19

Making use of these data, find

- A pair of ions
(a) A and B (b) B and C
(c) D and E (d) E and F
- A pair of isobars
(a) A and B (b) C and D
(c) E and F (d) C and E
- A pair of isotopes
(a) A and B (b) C and D
(c) E and F (d) C and E

Assertion & Reason :

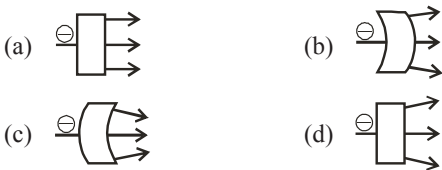
DIRECTIONS : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - If **Assertion** is **correct** but **Reason** is **incorrect**.
 - If **Assertion** is **incorrect** but **Reason** is **correct**.
1. **Assertion :** It is **not** possible to find an electron present in the nucleus.
Reason : Velocity of electron-wave is less as compared to velocity of light.

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- For which of the following species, the expression for the energy of electron in n th orbit

$$\left(E_n = -\frac{13.6Z^2}{n^2} \text{eV atom}^{-1} \right)$$
 has the validity?
 (a) He^{2+} (b) Li^{2+}
 (c) Deuterium (d) Tritium
- The possible path(s) of cathode rays that are ejecting from cathode surface can be shown as

- According to Bohr's theory
 (a) When a required amount of energy is supplied to an electron in an atom it jumps from lower orbit to higher orbit and remains there
 (b) When a required amount of energy is supplied to an electron in an atom it jumps from lower orbit to higher orbit and remains there for very short interval of time and returns back to lower orbit, radiating energy
 (c) The angular momentum of an electron is proportional to its quantum number, n
 (d) The angular momentum of an electron is independent of its quantum number, n

Structure of Atom

2. **Assertion :** ${}_1\text{H}^1$, ${}_1\text{H}^2$ and ${}_1\text{H}^3$ are isotopes of hydrogen.
Reason : Nuclides of the same element of different mass numbers are called isotopes of that element.
3. **Assertion :** Atom is electrically neutral.
Reason : A neutral particle, neutron is present in the nucleus of atom.
4. **Assertion :** Cathode rays get deflected towards the positive plate of electric field.
Reason : Cathode rays consist of negatively charged particles known as electrons.
5. **Assertion :** A few positively charged α -particles are deflected in Rutherford experiments.
Reason : Most of the space in the atom is empty.
6. **Assertion :** Isotopes are electrically neutral.
Reason : Isotopes are species with same mass number but different atomic number.
7. **Assertion :** Isobars are identical in chemical properties.
Reason : Isobars have same atomic number.

Multiple Matching Questions :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

- | 1. | Column I | Column II |
|-----|-----------------------------|---------------------------|
| (A) | Electron | p. Negatively charged |
| (B) | Proton | q. Neutral |
| (C) | Atom | r. Positively charged |
| (D) | Neutron | s. Sub atomic particle |
| | | t. Can be further divided |
| | (A) (B) (C) (D) | |
| (a) | (p, s) (r, s) (q) (q, s) | |
| (b) | (p, s) (r, s) (q, t) (q, s) | |
| (c) | (s) (r) (p) (q) | |
| (d) | (p) (r) (q) (s) | |

Integer Based Questions :

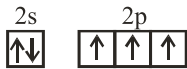
DIRECTIONS : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

1. How many of the following are isobars ?
 ${}_{17}^{37}\text{Cl}$, ${}_{19}^{40}\text{K}$, ${}_{18}^{40}\text{Ar}$, ${}_{32}^{76}\text{Ge}$, ${}_{20}^{40}\text{Ca}$
2. What is the number of valence electron in oxygen atom (O)?
3. What is the number of neutrons in ${}_{8}^{16}\text{O}$?

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1 to 12) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The orbital angular momentum of $2p$ electron is
(a) $\sqrt{3}h$ (b) $\sqrt{6}h$
(c) Zero (d) $\sqrt{2} \frac{h}{2\pi}$
- The orbital diagram in which both the Pauli's exclusion principle and Hund's rule are violated, is
(a)  (b) 
(c)  (d) 
- The four quantum numbers that could identify the third $3p$ electron in sulphur are
(a) $n = 3, l = 0, m = +1, s = +\frac{1}{2}$
(b) $n = 2, l = 2, m = -1, s = +\frac{1}{2}$
(c) $n = 3, l = 2, m = +1, s = -\frac{1}{2}$
(d) $n = 3, l = 1, m = -1, s = +\frac{1}{2}$
- The atomic number of the element with maximum number of unpaired electrons is
(a) 23 (b) 33
(c) 15 (d) 26
- Which of the following orbitals is more close to the nucleus?
(a) $5f$ (b) $6d$
(c) $7s$ (d) $7p$
- The number of d -electrons in Fe^{2+} ($Z = 26$) is not equal to that of
(a) s -electrons in Mg (b) p -electrons in Ne
(c) d -electrons in Fe (d) p -electrons in Cl
- Which of the following orbital diagram violates the Pauli's exclusion principle?
(a)  (b) 
(c)  (d) 
- The correct set of quantum numbers for the unpaired electron of scandium ($Z = 21$) is
(a) $4, 2, 2, \pm\frac{1}{2}$ (b) $3, 2, -2, \pm\frac{1}{2}$
(c) $3, 1, 1, \pm\frac{1}{2}$ (d) $3, 2, 3, \pm\frac{1}{2}$
- The electronic configuration of Mn^{2+} is
(a) $[\text{Ar}] \begin{array}{c} 3d \\ \uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow \end{array} \begin{array}{c} 4s \\ \uparrow\downarrow \end{array}$
(b) $[\text{Ar}] \begin{array}{c} 3d \\ \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \end{array} \begin{array}{c} 4s \\ \uparrow\downarrow \end{array}$
(c) $[\text{Ar}] \begin{array}{c} 3d \\ \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \end{array} \begin{array}{c} 4s \\ \uparrow\downarrow \end{array}$
(d) $[\text{Ar}] \begin{array}{c} 3d \\ \uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow \end{array} \begin{array}{c} 4s \\ \uparrow\downarrow \end{array}$
- The shortest wavelengths of radiation emitted when an electron from infinity falls into the ionized H-atom ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)
(a) 91 nm (b) 192 nm
(c) 406 nm (d) $1.9 \times 10^{-8} \text{ nm}$
- From amongst the following chemical species :
(i) $^{39}_{18}\text{Ar}$ (ii) $^{40}_{19}\text{K}^+$
(iii) $^{41}_{20}\text{Ca}^{++}$ (iv) $^{42}_{20}\text{Ca}^+$
The one having identical electronic configurations are
(a) (i) and (ii) (b) (ii) and (iv)
(c) (iii) and (iv) (d) (i), (ii) and (iii)
- Consider the following statements :
In $^{90}_{38}\text{Sr}^{2+}$:
(i) atomic number is 36
(ii) number of electrons is 38
(iii) number of neutrons is 52
(iv) number of protons is 38
Which of these are correct ?
(a) (i) and (ii) (b) (ii) and (iii)
(c) (iii) and (iv) (d) (i) and (iv)

DIRECTIONS (Qs. 13 to 21) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- Which of the following sets of quantum numbers is/are not allowed?
(a) $n = 3, l = 2, m = -1$ (b) $n = 2, l = 3, m = -1$
(c) $n = 3, l = 0, m = 1$ (d) $n = 6, l = 2, m = -1$

14. Which of the following statements is/are true about the quantum numbers n , l , m and s ?
- l gives an idea of the shape of the orbital
 - m gives the information of the energy of electron in a given orbital
 - n gives the idea of the size of the orbital
 - s gives the direction of spin of electron in an orbital

15. The angular momentum of electron can have the value(s)

- $0.5 \frac{h}{\pi}$
- $\frac{h}{\pi}$
- $1.5 \frac{h}{\pi}$
-

16. Ground state electronic configuration of P atom can be represented as

- [Ne] $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow & \uparrow \\ \hline \end{array}$
- [Ne] $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow \\ \hline \end{array}$
- [Ne] $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow & \downarrow \\ \hline \end{array}$
- [Ne] $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \downarrow & \downarrow \\ \hline \end{array}$

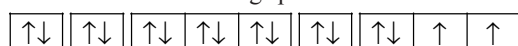
17. The possible correct set(s) of quantum numbers for the unpaired electron of oxygen atom in the orbitals of p-subshell is/are

- $n = 2, l = 1, m = 0$
- $n = 2, l = 1, m = -1$
- $n = 2, l = 1, m = +1$
- $n = 2, l = 1, m = -2$

18. Select the pair of atoms having the same number of electrons in their outermost shell.

- Na, Ca
- Mg, Fe
- As, Bi
- Pb, Sb

19. The diagram illustrates a possible electronic configuration of which of the following species?



- Cl^+ ion
- S atom
- Ar^{+2} ion
- P^{-2} ion

20. Which of the following statement(s) is/are correct ?

- The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$ (Atomic no. of Cr = 24)
- The magnetic quantum number may have a negative value
- Weight of electrons contribute to weight of atom.
- The oxidation state of nitrogen in HN_3 is -3

21. Ground state electronic configuration of nitrogen atom can be represented by

- $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow \\ \hline \end{array}$
- $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow \\ \hline \end{array}$
- $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \downarrow \\ \hline \end{array}$
- $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \downarrow \\ \hline \end{array}$

DIRECTIONS (Qs. 22 to 24) : Study the given paragraph(s) and answer the following questions.

Passage - I

Quantum numbers are defined as a set of numbers which gives complete information about size shape and orientation of the orbital. These are denoted by n , l and m and called principal, azimuthal and magnetic quantum numbers respectively. n assumes integral values (1, 2, 3, 4, etc.), l assumes values 0 to $n - 1$ (all integral) and m assumes $-l$ to $+l$ through zero (all integral). The principal quantum number describes the size and the energy of the orbital, l describes the shape and m the orientation of the orbital in the applied magnetic field. Spin quantum number can take up two values $+\frac{1}{2}$ and $-\frac{1}{2}$ which describe the spin angular momentum of the electron in clockwise and anticlockwise directions.

22. The valence shell electron of sodium has the quantum numbers (n, l, m, s)

- $2, 1, -1, -\frac{1}{2}$
- $3, 0, 0, +\frac{1}{2}$
- $3, 2, -2, -\frac{1}{2}$
- $3, 2, 2, +\frac{1}{2}$

23. Magnetic quantum number specifies

- size of the orbitals
- shape of orbitals
- orientation of orbitals
- nuclear stability

24. Which of the following element outermost orbit's last electron has magnetic quantum number $m = 0$?

- Na
- O
- Cl
- N

DIRECTIONS (Qs. 25 to 30) : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

25. **Assertion :** Angular momentum of an electron in any orbit is given by angular momentum $= \frac{n \cdot h}{2\pi}$, where n is the principal quantum number.

Reason : The principal quantum number, n , can have any integral value.

26. **Assertion :** Electromagnetic radiations around 10^{15} Hz are called as visible light.

Reason : This is the only part of electromagnetic radiation which is visible to eyes.

27. **Assertion :** The main shell with principal quantum number $n = 2$ has four orbitals present in it.

Reason : Number of orbitals present in a shell is given by n^2 .

28. **Assertion :** Magnetic quantum number can have the value $l = 0, \dots, (n - 1)$.

Reason : Magnetic quantum number specifies the number of orbitals.

29. **Assertion :** $5s$ orbital has greater energy than $4s$.

Reason : Energy of the orbital depends on the azimuthal quantum number.

30. **Assertion :** Total number of electrons in a subshell is designated by azimuthal quantum number l is $2l + 1$

Reason : l can have value $0, 1, 2, \dots, n - 1$, where n is principal quantum number.

DIRECTIONS (Q. 31) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

Column - I

- (A) $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow \uparrow$
 (B) $[\text{Ne}] 3s^2 3p^5$
 (C) $[\text{He}] 2s^2 2p^4$
 (D) $[\text{Ne}] \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow$

Column - II

- (p) Aufbau principle
 (q) Oxygen atom
 (r) N^- ion
 (s) Cl atom
- | | | | |
|---------------|--------|--------|-----------|
| (A) | (B) | (C) | (D) |
| (a) (p, q) | (s, p) | (r, s) | (p, s) |
| (b) (p, q, r) | (s) | (q, r) | (p, s) |
| (c) (r, p) | (q, s) | (q, r) | (p, q, s) |
| (d) (p, s) | (q, r) | (s) | (p, q) |

DIRECTIONS (Qs. 32 to 33) : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

32. What is the value of Azimuthal quantum number for $4f$ orbital
 33. The total number of electrons that can be accommodated in all orbitals having principal quantum number 2 and azimuthal quantum number 1 is

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- anu
- chemical reaction
- two
- same
- 23
- Nucleus
- Neutron
- electrons
- Proton, neutron
- eight
- subtracting, from

True / False :

- | | |
|----------|----------|
| 1. False | 2. False |
| 3. True | 4. False |
| 5. True | 6. False |
| 7. True | |

Match the following

- A – (q); B – (r); C – (s); D – (p)

Very Short Answer Questions :

- Goldstein.
- 10^5
- The protons and neutrons present in atomic nucleus are collectively known as nucleus.
- These are atom of the *same element* (atomic number is same) having different mass numbers e.g., ^1_1H and ^2_1H .
- It is the nucleus of hydrogen atom (^1_1H).
- It proves that atoms contain massive, positively charged centre called nucleus.
- Atomic number = Number of protons = 5
Mass number = Number of protons + number of neutrons = $5 + 6 = 11$
- Atomic number = number of protons = number of electrons = 8
- Mass Number_X or $\frac{A}{Z}X$
- Hydrogen.
- James Chadwick (1932).
- This observation leads to the conclusion that most of the space in an atom is empty.
- First shell ($n = 1$) or **K-shell**.
- $^{21}_{10}\text{A}$ and $^{22}_{10}\text{B}$ are isotopes. They have same atomic number ($Z = 10$) but different mass numbers.

Short Answer Questions :

- Atomic mass or Mass number = Number of protons + Number of neutrons.
- These are the electrons present in the outermost shell.
- When cathode rays strike the glass walls of the discharge tube, they produce a faint greenish, fluorescence.
- An **orbit** (or **energy level** or **shell**) can accommodate a maximum of $2n^2$ electrons, where n is the number of orbit.
 1^{st} orbit (K-shell) can have $2 \times 1^2 = 2$ electrons.
 2^{nd} orbit (L-shell) can have $2 \times 2^2 = 8$ electrons.
 3^{rd} orbit (M-shell) can have $2 \times 3^2 = 18$ electrons.
 - The outermost shell of an atom cannot have more than 8 electrons and the shell next to outermost (second last shell) shell can not have more than 18 electrons.
- Rutherford's experiment with large angle scattering of alpha-particles by a metal foil results into failure of Thomson's model of the atom as an intimate mixture of negative and positive particles.
- Neutrons were discovered in 1932 by **James Chadwick**. Neutrons are produced when we bombard a thin foil of beryllium with fast moving α -particles. Neutrons are neutral particles with a mass of 1.675×10^{-27} kg.
- Lowest energy = K-shell (nearest to nucleus, $n = 1$)
Highest energy = N-shell (far from nucleus $n = 4$)
(**Note** : As we move away from the nucleus, energy of electron increases).
- Valency = 2
Number of proton = $24 - 12 = 12$
Number of electron = 12

K L M

 Electronic configuration = 2, 8, 2
 Number of valence electrons = 2
 $\therefore$ Valency = 2
- Various isotopes of the same element have *similar chemical properties*. All the isotopes have same number of electrons and protons. All the isotopes have same valency.
- The nucleus *contains protons and neutrons*.

Isotope	Number of Protons	Number of neutrons
$^{28}_{14}\text{Si}$,	14	14
$^{29}_{14}\text{Si}$,	14	15
$^{30}_{14}\text{Si}$	14	16

Since, the mass of natural silicon is very close to the Si – 28 isotope, hence Si – 28 isotope will be most abundant in nature.

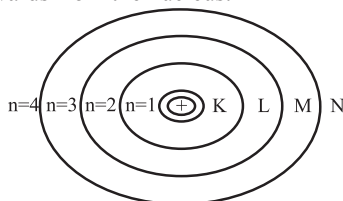
Long Answer Questions :

1. **Anode rays** **Cathode rays**
- | | |
|--|---|
| 1. They carry positive charge | 1. They carry negative charge |
| 2. $\frac{e}{m}$ depends on the gas present in the discharge tube | 2. $\frac{e}{m}$ ratio does not depend on the gas present in discharge tube |
| 3. Mass of the particles of anode rays is variable | 3. These have a definite mass. |
| 4. They generally carry a charge of +1, but sometimes they may have a charge +2, +3 etc. | 4. They always have a charge of -1. |
2. The conclusions are :
- Most of the space inside the atom is hollow or empty.
 - There is a positive centre in the atom which is a charged heavy body and is known as *nucleus*.
 - The total volume occupied by nucleus is very small compared to the total volume of the atom.

Drawbacks of Rutherford's Model :

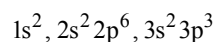
Rutherford's atomic model provides a reasonable view of the atom but it does not convey regarding the positions of electrons relative to each other and to the nucleus or to their state of motion around the positively charged nucleus.

3. In order to explain the objections raised for Rutherford's model. **Neils Bohr** in 1912 gave a new model of atom called **Bohr's model**. According to this model
- An atom consists of a small positively charged **nucleus** situated at its centre.
 - The negatively charged electrons revolve around the nucleus in *certain definite circular paths* called *energy levels*.
 - Each energy level is associated with a *definite amount of energy*.
 - The energy levels are either numbered 1, 2, 3, 4 ($n = 1, 2, 3, 4 \dots$) or designated as *K, L, M, N* outwards from the nucleus.



- The change in energy of an electron takes place only when it jumps from a lower energy level to a higher energy level (gain of energy) or when it jumps from a higher energy level to a lower energy level (Loss of energy). It means that as long as the electrons remain in a given energy level, they neither gain nor lose energy and atom remains stable.

4. The electronic configuration of the element with 2K, 8L and 5M electrons will be

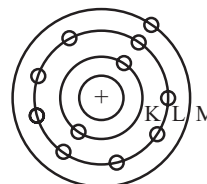


- Total number of electrons = $2 + 8 + 5 = 15$
 $\therefore$ Atomic number of element = 15
- Total number of s-electrons = $2 + 2 + 2 = 6$
- Total number of p-electrons = $6 + 3 = 9$
- Since the atom is neutral,
 $\therefore$ No. of protons = number of electrons
= Atomic number = 15
- Since the element has only three half-filled atomic orbitals, therefore, valency of the element = 3

2 EXERCISE

Text - Book Questions :

- When high voltage is applied between the electrodes of a discharged tube, provided with a perforated cathode, at a pressure below 0.001 mm Hg, some rays are seen coming through the perforation in the cathode. These rays coming through the perforation of a perforated electrode on passing high voltage at a pressure below 0.001 mm Hg are known as **canal rays** or **anode rays** or **positive rays**.
- It will not carry any charge. One unit of negative charge present on electron is completely balanced by one unit of positive charge present on one protons and there will be no resultant charge on atom.
- In Thomson's model atom is considered to be made up of a uniform sphere (radius $\approx 10^{-10}\text{m}$) of positive charge with electron (negatively charged) embedded into it in such a way as to give the most stable electrostatic arrangement. This model is also known as *raisin pudding model*. Since the positive and negative charge present in an atom are fully balanced so the atom as a whole is neutral.
- The nucleus contains protons (sub-atomic particle) which is positively charged and has a mass approximately equal to that of an atom of hydrogen.



It contains three shells $n = 1, n = 2, n = 3$ (or *K, L, M*). It contains 11 electrons as there is only 1 electron in $\frac{3rd}{n}$ shell.

- The similar observations will be recorded.
- The three sub-atomic particles of an atom are :
(i) electron, (ii) proton, (iii) neutron.
- Atomic mass = mass of nucleus = number of protons + number of neutrons
 $\therefore 4 = 2 + \text{number of neutrons}$
or, number of neutrons = $4 - 2 = 2$

Structure of Atom

9. Carbon ($z = 6$) i.e., it contains 6 electrons. Out of these 6 electrons, 2 electrons will be present in **K-shell** ($n = 1$) and remaining 4 in **L-shell** ($n = 2$)

∴ Carbon K L
 2 4

Similarly, for sodium ($z = 11$). The distribution of electrons in various shells is as follows.

Sodium K L M
 2 8 1

10. Maximum possible number of electrons are present when a shell is full. Maximum number of electrons that could be accommodated in **K** and **L** shell are

$$K (n = 1) 2n^2 \text{ or } 2 \times 1^2 = 2$$

$$L (n = 2) 2n^2 \text{ or } 2 \times 2^2 = 8$$

Thus, the total number of electrons in an atom when K and L shells are full is $2 + 8 = 10$.

i.e., it has 10 electrons.

11. First write down the electronic configuration of each of these

	K	L	M	Valency
Magnesium ($z = 12$)	2	8	2	2 (number of valence electrons)
Sulphur ($z = 16$)	2	8	6	2 (Valency = $8 - \text{Valence electrons} = 8 - 6 = 2$)
Chlorine ($z = 17$)	2	8	7	1 (Valency = $8 - \text{valence electrons} = 8 - 7 = 1$)

12. (a) Atomic number = number of protons = 8

(b) Charge on atom = charge on electron + charge on protons

$$= (-1 \times 8) + (1 \times 8) = -8 + 8 = 0$$

i.e., There will be no charge on the atom.

13. Mass number = number of protons + number of neutron

$$\therefore \text{Mass number of oxygen} = 8 + 8 = 16$$

$$\text{Mass number of sulphur} = 16 + 16 = 32$$

14. **Symbol Name Proton Electron Neutron**

H	Hydrogen	1	1	0
D	Deuteron	1	1	1
T	Tritium	1	1	2

15. **Pair of isotopes** is ${}^1_1\text{H}$ and ${}^2_1\text{H}$

These are atoms of the same element and so they contain same number of electrons and protons. They have different number of neutrons.

They have same electronic configuration

 K L
H 1 –

Pair of Isobars : They are atoms of different elements having different atomic number thus different number of protons and electrons but same mass number i.e., same

number of nucleons (protons + neutrons).

Since, they have different number of electrons they will have different electronic configuration.

Example : K- 40 and Ar-40

or ${}^{40}_{19}\text{K}$ and ${}^{40}_{18}\text{Ar}$

The electronic configuration are as follows

Atom	K	L	M	N
Ar	2	8	8	–
K	2	8	8	1

Text - Book Exercise :

1.

Particle (symbol)	Charge	Mass	Location
Electron (e^-)	$1.6 \times 10^{-19}\text{C}$ (– 1 unit)	$9.1 \times 10^{-31}\text{kg}$ (negligible)	Revolve around the nucleus in fixed orbit. i.e. extra nuclear space
Proton (p)	$1.6 \times 10^{-19}\text{C}$ (+1 unit)	$1.67 \times 10^{-27}\text{kg}$ (1 U)	Present in the nucleus of the atom
Neutron (n)	Nil (neutral)	$1.675 \times 10^{-27}\text{kg}$ (1 U)	Present in the nucleus of the atom

2. The limitations of J.J. Thomson's model of an atom are :

- It failed to explain the results of Rutherford's scattering experiment.
- It found no experimental support.

3. Various limitations of Rutherford's model of an atom are :

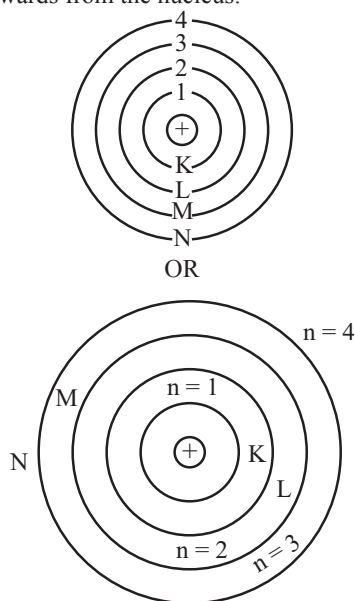
1. *It failed to explain the stability of atom*

According to Clark Maxwell, when charged particle moves under the influence of attractive forces it emits energy. Thus due to *electrostatic force of attraction* between a negatively charged electron and positively charged nucleus, the electron will be attracted by the nucleus and so the revolving electron would lose energy continuously and would finally merge into the nucleus. Thus, the atom should be *highly unstable*. However, it is **not** true because we know that atom is quite stable.

4. This model was proposed in 1913 by **Niels Bohr**. He slightly modified the **Rutherford's** model, to overcome its limitations, retaining its main features. On this model

- An atom consists of small positively charged nucleus situated at its centre.
- The negatively charged electrons revolve around the nucleus in certain definite circular paths called *energy levels* or *energy shells*.

- (iii) Each energy level is associated with a definite amount of energy.
- (iv) The energy levels are either numbered as $n = 1, 2, 3, 4, \dots$ etc. or designated as $K, L, M, N, \dots$ outwards from the nucleus.



- (v) The change in energy of an electron takes place only when it jumps from a lower energy level to a higher energy level (gain of energy) or when it jumps from a higher energy level to a lower energy level (loss of energy)

While revolving in an **orbit** (energy level) the electron neither gains or loses energy i.e., its energy remains stationary. Therefore, those energy states for the electron are known as stationary states.

5. We have come across the three models of atom in this chapter. These are *Thomson's model*, *Rutherford's model* and *Bohr's model*.

The *Thomson's model* also known as **Raisin Puddling model** because this model may be visualized like a **Christmas pudding** or **cake**.

It was the first attempt to predict the picture of an atom. This model regarded atom as *positively charged sphere* in which protons are present and the negatively charged electrons are embedded in this sphere. Since the positive charges due to protons are balanced by electrons so the atom is neutral. Since this model failed to get any experimental evidence so it was rejected.

According to Rutherford's model an atom consists of a positively charged small sized nucleus which is present at the centre of atom and the mass of the atom is mainly due to this part as it contains all the protons and neutrons present in the atom. The nucleus is surrounded by extra nuclear space in which electron revolve around the nucleus in circular path at high speed. This model is quite similar to that of solar system. This model also *failed to explain the stability of atom*.

Bohr's model is a modification of Rutherford's model. It postulated that the electron can revolve around the nucleus *only in certain fixed orbits* out of infinite possible number of such orbits. Moreover, the energy is neither gained nor lost by the revolving electron while revolving in this permitted orbit. In this way he could explain the stability of atom, which was the main draw back of Rutherford's model.

6. The rules for distribution of electrons are :
- The maximum number of electrons which can be present in a particular energy shell of an atom is given by $2n^2$, where n = number of energy shell ($n = 1, 2, 3, \dots$) i.e., maximum number of electrons in **1st** energy level is 2; second energy level = 8 and so on.
 - The outermost *energy shell* of an atom cannot have more than eight electrons even if it has a capacity to take more electrons according to rule (i).
 - Electrons are not accommodated in a given shell unless the earlier shells are filled i.e., stepwise filling of shells is followed.

7. **Valency** : It is the capacity of an element to combine with another element in order to acquire 8 electrons (or 2 electrons in case of hydrogen) in its outermost energy shell.

Valency depends on the number of peripheral electrons which may be involved in a chemical bond and is never greater than 7. The variability of valency depends upon the external electronic configuration.

Valency of Silicon (Si) : Atomic number of silicon (Si) is 14. Thus, it has 14 electrons and these are distributed as 2 (K-shell), 8 (L-shell) and 4 (M-shell) i.e., its electronic configuration is 2, 8, 4. Since, it has 4 electrons in its outermost orbit it has a tendency to lose all these four electrons. So that it is left with 8 electrons in its outermost orbit (2, 8) or it can gain four electrons to have 8 electrons in its outermost orbit (2, 8, 8). Therefore it shows a valency of 4.

Valency of oxygen (O) : Atomic number of oxygen (O) is 8. Its electronic configuration is 2, 6, i.e., it has six electrons in its outermost orbit. It needs two more electrons to have 8 electrons in its outermost orbit. Therefore it shows a valency of 2.

8. (i) **Atomic number (Z)** : Atomic number of an element is equal to the number of protons present in the nucleus of an atom.

$$\text{Atomic number (Z)} = \text{Number of protons}$$

$$= \text{Number of electrons of an atom.}$$

For hydrogen, $Z = 1$, in hydrogen there is 1 proton in the nucleus of hydrogen.

For carbon, $Z = 6$, in carbon atom we have 6 proton in its nucleus.

- (ii) **Mass number (A)** : It refers to total number of protons and neutrons (collectively known as **nucleus**) present in the nucleus of an atom.

$$\text{Mass number (A)} = \text{number of protons}$$

$$+ \text{number of neutrons} = \text{number of nucleons}$$

Mass number of carbon, it is **12** because in carbon atom we have 6 protons and 6 neutron i.e., number of nucleons = $6 + 6 = 12$.

Structure of Atom

Mass number of Aluminium (Al); It is 27. An atom of aluminium has 13 protons and 14 neutrons so the total number of nucleus is **27** ($13 + 14 = 27$)

(iii) **Isotopes** : Atoms of the same element have same atomic number but different mass numbers are known as isotopes. e.g. $^{17}_{35}\text{Cl}$ and $^{37}_{17}\text{Cl}$, $^{12}_6\text{C}$ and $^{14}_6\text{C}$; ^1_1H and ^2_1H etc.

(iv) **Isobars** : Atoms of different elements having same mass number but different atomic numbers are known as isobars.

e.g. K-40 and Ar-40; $^{40}_{18}\text{Ar}$ and $^{40}_{20}\text{Ca}$; $^{22}_{10}\text{Ne}$ and $^{22}_{11}\text{Na}$ etc.

Applications of Isotopes

Isotopes are used in almost all fields, however isotopes of some elements are very useful e.g..

- (i) Isotopes of uranium (U-235) is used as a fuel in nuclear reactors.
- (ii) Isotopes of cobalt (Co-60) is used in treatment of cancer.
- (iii) Isotopes of Iodine (I-131) is used in treatment of goitre.
- (iv) Isotopes of sodium (Na-24) is used for examining the circulation of blood in our body.
- (v) Isotope of carbon (C-14) is used in **radio-carbon dating**. (Give any two of these)

9. The atomic number of sodium is 11. Thus, it contains 11 protons and 11 electrons. The electronic configuration is 2, 8, 1.

When it forms Na^+ ion it loses one electron. The electron lost is one present in outermost shell (i.e., M-shell). The electronic configuration of Na^+ is 2, 8. It clearly shows that in Na^+ we have both **K** and **L** shell complete, because maximum number of electrons that can be accommodated in these shells are 2 and 8 respectively.

K-shell ($n = 1$); $2n^2$ i.e., $2 \times 1^2 = 2$

L-shell ($n = 2$); $2n^2$ i.e., $2 \times 2^2 = 8$

10. % age of Br-79 isotope = 49.7

% age of Br-81 isotope = 50.3

$\therefore$ Average atomic mass of bromine

$$= \frac{79 \times 49.7 + 81 \times 50.3}{(49.7 + 50.3)}$$

$$= \frac{3926.3 + 4074.3}{100} = \frac{8000.6}{100} = 80.006\text{U}$$

19.

Atomic Number	Mass Number	Number of Neutrons	Number of Protons	Number of Electrons	Number of Atomic species
9	(19)	10	(9)	(9)	(Fluorine)
16	32	(16)	(16)	(16)	Sulphur
(12)	24	(12)	12	(12)	(Magnesium)
(1)	2	(1)	1	(1)	(Deuterium)
(1)	1	0	1	0	(Protium or Hydrogen)

11. Let the % age of X-16 isotope = x

Then % age of X-18 isotopes = $(100 - x)$

From this, *calculate the average atomic mass*.

Average atomic mass of 'X'

$$= \frac{x \times 16}{100} + \frac{(100 - x) \times 18}{100}$$

Compare it with given average atomic mass of 'X'

$$\therefore \text{We have, } 16.2 = \frac{x \times 16}{100} + \frac{(100 - x) \times 18}{100}$$

$$\text{or } 16.2 \times 100 = 16x + 1800 - 18x$$

$$\text{or } 2x = 1800 - 1620 \quad \text{or } x = \frac{180}{2} = 90$$

Hence % age of X-16 isotope = **90**

and % age of X-18 isotope = $(100 - 90) = 10$

12. For $Z = 3$, the electronic configuration is 2, 1. Thus, there is one electron in outermost shell so its valency is 1.

The element with $z = 3$ is **Lithium (Li)**

13. Mass number (A) = Number of protons + Number of neutrons

$\therefore$ Mass number of 'X' = $6 + 6 = 12$

Mass number of 'Y' = $6 + 8 = 14$

Since both 'X' and 'Y' contain same number of protons. So, they have same atomic number (z) i.e., **6**

Since they have same atomic number but different mass numbers so they are **isotopes**.

14. (a) F, (b) F, (c) T, (d) F.

15. (a) Atomic nucleus was discovered as a result of Rutherford's alpha-particle scattering experiment

16. (c) **Isotopes** have same atomic numbers but different mass numbers. Thus, they have different number of neutrons. Isotopes are differ in some physical properties but they have same chemical properties.

17. (d) Atomic number of Cl is 17. It has 17 electrons arranged in K, L, M shell as **2, 8, 7**. When Cl form Cl^- it gains 1 electron and its electronic configuration then becomes **2, 8, 8** and hence total number of electrons in $\text{Cl}^- = 18$

18. (d) Atomic number of Na (sodium) is 11. It has 11 electron in K, L, M shells as 2, 8, 1.

For calculations use the relations given below :

Mass number = number of protons + number of neutrons

Atomic number of atom = number of protons

= number of electrons

Number of neutrons = Mass number – Atomic number

Number of protons = Atomic number

= Mass number – number of neutrons

Number of electrons = atomic number = number of protons

Exemplar Questions :

1. Gold has high malleability

2. (a) 0
(b) 1

3.

	Atomic Number	Mass Number	Valency
X	5	11	3
Y	8	18	2
Z	15	31	3, 5

4. Mass number = No. of protons + No. of neutrons = 31

$$\therefore \text{Number of neutrons} = 31 - \text{number of protons} \\ = 31 - 15 \\ = 16$$

5. These are Isobars

6. Helium has two electrons in its only energy shell, while Argon and Neon have 8 electrons in their valence shells. As these have maximum number of electrons in their valence shells, they do not have any tendency to combine with other elements. Hence, they have a valency equal to zero.

HOTS Questions :

1. (i) Mass number of 'X' = 6 + 6 = 12

Mass number of 'Y' = 6 + 8 = 14

- (ii) Because they (X and Y) have same number of protons (i.e., same atomic number) but have different mass numbers so they are **Isotopes**.

2. Fraction of atomic volume occupied by the nucleus

$$= \frac{\text{Nuclear volume}}{\text{Atomic volume}} = \frac{\frac{4}{3}\pi(10^{-13})^3}{\frac{4}{3}\pi(10^{-8})^3} = 10^{-15}$$

3. Cathode rays are particles of matter and they have a definite $\frac{e}{m}$ $\left(\frac{\text{Charge}}{\text{mass}}\right)$ ratio. These rays are independent of the gas present in the discharge tube and also do **not** depend on the kind of material of electrodes. These are the evidences to prove that cathode rays are part of all matter.

4. In a neutral atom,

number of protons = number of electrons

$\therefore$ In Uranium ($z = 92$), we have

Atomic number (z) = number of electrons

= number of protons = 92

In the formation of cation the atom loses electrons. It acquires +1 charge by loss of 1 electron. So in the formation of U^{+2} ion, an atom of Uranium has lost two electrons.

$$\therefore \text{Number of electrons in } U^{+2} \text{ ion} = 92 - 2 = 90$$

There is no change in number of protons

$$\therefore \text{Number of protons in } U^{+2} \text{ ion} = 92$$

5. (i) Atomic weight = 24

Atomic number = 12

Since, atomic number = number of protons

$$\therefore \text{Number of protons} = 12$$

Again number of protons + number of neutrons = 24

$$\therefore 12 + \text{number of neutrons} = 24$$

$$\text{or number of neutrons} = 24 - 12 = 12$$

6. Let the % age of $^{10}_5B$ be x

Then % age of $^{11}_5B$ is $(100 - x)$

$$\text{Now, average mass of boron} = \frac{10 \times x}{100} + \frac{11(100 - x)}{100}$$

$$\text{or, } 100 \times 10.8 = 10x + 1100 - 11x$$

$$\text{or, } x = 1100 - 1080 = 20$$

$$\therefore \% \text{ age of } ^{10}_5B = 20\% \text{ and}$$

$$^{11}_5B = (100 - 20) = 80\%$$

7. The **charge number** is the charge present on the nucleus and is equal to the number of protons present in the nucleus. It is also known as **atomic number**.

For a neutral atom, charge number is also equal to number of electrons.

8. % age of abundance of $^{35}_{17}Cl = \frac{754 \times 100}{754 + 246} = 75.4\%$

$$\% \text{ age of abundance of } ^{37}_{17}Cl = \frac{246 \times 100}{754 + 246} = 24.6\%$$

$\therefore$ Relative atomic mass of natural chlorine

$$= \frac{35 \times 75.4}{100} + \frac{37 \times 24.6}{100} = 35.492$$

9. Weight of 10 protons + 10 neutrons < 20.20

This is due to the fact that atomic weight is an average isotopic weight.

1. (d) Orbital angular momentum = $\sqrt{l(l+1)} \frac{h}{2\pi}$; $l = 1$ for p -orbital.
2. (d)
3. (d)
4. (d) (a) $[\text{Ar}] 3d^3 4s^2$
 (b) $[\text{Ar}] 3d^{10} 4s^2 4p^3$
 (c) $[\text{Ne}] 3s^2 3p^3$
 (d) $[\text{Ar}] 3d^6 4s^2$ or $[\text{Ar}]$

↑	↑	↑	↑	↑	↓
---	---	---	---	---	---

↑	↓
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 (a), (b), (c) — 3 unpaired electrons (d) — 4 unpaired electrons.
5. (c) $7s$ orbital, $(n + l)$ is minimum with n minimum.
6. (d) $\text{Fe}^{2+} - [\text{Ar}] 3d^6$; number of d -electrons = 6
 $\text{Mg}(12) - 1s^2 2s^2 2p^6 3s^2$; number of s -electrons = 6
 $\text{Ne}(10) - 1s^2 2s^2 2p^6$; p -electrons = 6
 $\text{Cl}(17) - 1s^2 2s^2 2p^6 3s^2 3p^5$; p -electrons = 11

7. (c) Two electrons in the same orbital cannot have the same spins.
8. (b)
9. (c)
10. (a)
$$\bar{\nu} = \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$
$$= 1.097 \times 10^7 \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right]$$
$$\Rightarrow \lambda = 91.1 \times 10^{-9} \text{ m}$$
11. (a)
12. (c) In ${}^{90}_{38}\text{Sr}^{2+}$, Atomic number = no. of protons = 38
Since it has lost 2e⁻ so no. of electrons = 38 - 2 = 36
Atomic mass = no. of protons + no. of neutrons
90 = 38 + no. of neutrons
90 - 38 = no. of neutrons
52 = no. of neutrons
Hence, no. of neutrons = 52, no. of protons = 38.
13. (b, c)
(b) $l > n$ is not permissible; (c) $m > l$ is not permissible.
14. (a, c, d)
15. (a, b, c, d)
Angular momentum, $mvr = n \cdot \frac{h}{2\pi}$, n is an integer.
16. (a, d)
17. (a, b, c)
18. (b, c)
19. (a, b)
20. (a, b)
21. (a, c)
22. (b)
23. (c)
24. (a)
25. (b) Both assertion and reason are correct. Reason is **not** the correct explanation of assertion.
26. (a) Both assertion and reason are true and reason is the correct explanation of assertion.
Various types of electromagnetic radiations differ from one-another in wavelength (or frequency).
They on combination forms electromagnetic spectrum. The small portion (around 10^{15} Hz) is what is ordinarily called visible light. It is the only part which our eyes can see (or detect).
27. (a) Both assertion and reason are true and reason is the correct explanation of assertion.
Principal quantum number (n) can have positive integer values 1, 2, 3 etc. When $n = 2$, the total number orbitals present in it is $n^2 = 4$. Total number of orbitals present in a shell equals to n^2 .
28. (d) Assertion is false has reason is true.
Azimuthal quantum number can have the value $l = 0, \dots, n - 1$ while magnetic quantum number ml can have values $ml = -l, -(l - 1), \dots, 0, \dots, (l - 1), l$ (total $2l + 1$ value). It gives the information about the number of orbitals and its orientation.
29. (c) Assertion is true but reason is false.
Size and energy to a large extent is determined by the principal quantum number not azimuthal quantum number. For hydrogen and hydrogen like system (e.g., He^+ , Li^{2+}), it alone determines the energy and size of the orbital. The larger the value of n , the larger is the energy of the orbital. Therefore 5s has more energy than 4s.
30. (c) Assertion is true but reason is false.
 l identifies the subshell and $(2l + 1)$ orbitals are present corresponding to each type of subshell. As each orbital can have 2 electrons. Therefore total number of electrons in a subshell designated by $l = 2(2l + 1) = 4l + 2$.
31. (b) $A \rightarrow (p, q, r), (B) \rightarrow (s), (C) \rightarrow (q, r), (D) \rightarrow (p, s)$
32. 3
 $l = 3$ for f -orbital.
33. 6
 $n = 2, l = 1$ means $2p$ -orbital. Electrons that can be accommodated = 6 as p sub-shell has 3 orbital and each orbital contains 2 electrons.

Chapter

5

Chemical Bonding

INTRODUCTION

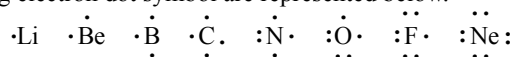
There are about 118 different kinds of elements which constitute all matters. Many of the elements are highly reactive. They interact with each other to form simple or complex substances. Only elements with the noble gas configuration have no tendency to react further since they have completely filled outermost electronic configuration of their orbitals or shells. An electronic structure of this type gives stability to an element, making it less reactive. Most elements do not have a noble gas configuration and consequently aim to achieve it. They attain this configuration by combining with each other either by gaining or losing electrons, a state of minimum energy.

The attractive force which holds together the constituent particles (atoms, ions or molecules) in a chemical species is known as chemical bond.

There is a definite way in which atoms of one element interact with other atoms. To understand how compounds are formed, we have to first know the types of bonds that can be formed between atoms. The tendency of elements to combine with one another is directly related to the valency or valence of the element. The valency is the ability or tendency of elements to combine with one another. Since the valency depends upon the electronic configurations of the atoms, the development of various theories of chemical bonding have closely been related to the understanding of the structure of atom.

LEWIS SYMBOLS OR ELECTRON DOT SYMBOLS

During formation of a molecule, the outershell electrons of the atom are involved. These are referred as valence electrons. The inner shell orbits are fully filled and well protected. Thus the inner shell electrons do not participate in bond formation. Therefore G. N. Lewis represented the atom by simple symbol and the outer shell electrons as dots surrounding the symbol of atom. These symbols are called Lewis symbols or electron dot symbols e.g. C_6 has electronic configuration $[He] 2s^2 2p^2$. Its electron dot symbol shows 4 valence electrons. Few atoms showing electron dot symbol are represented below.



The outermost shell electrons are farthest away from the nucleus and thus are not firmly bound to the nucleus. Therefore, these are easily attracted by the approaching atom.

Significance of Lewis Symbols : The number of dots represent the number of electrons present in the valence shell of the atom. These dots help us to find out the common valency of the element.

The common valency of the element is either equal to the number of dots in the Lewis symbol (if dots are equal to or less than 4) or it is 8 minus the number of dots (if dots are greater than 4). Therefore, Li, Be, B, C have common valencies 1, 2, 3 and 4 respectively as these have less than 4 electrons in their outermost shell (as represented by dots). Obviously, the elements N, O, F and Ne shall have common valencies as 3, 2, 1 and 0 respectively (i.e., 8 minus the number of dots in Lewis symbols).

ILLUSTRATION : 1

Write Lewis dot symbols for the following atoms and ions.

- (i) S and S^{2-} (ii) P and P^{3-} (iii) Al and Al^{3+} (iv) H and H^- (v) Na and Na^+

SOLUTION :

Lewis dot symbols are :

- (i) $:\ddot{S}\cdot$ and $:\ddot{S}:^{2-}$ (ii) $\cdot\ddot{P}\cdot$ and $:\ddot{P}:^{3-}$ (iii) $\cdot\ddot{Al}\cdot$ and $:\ddot{Al}:^{3+}$ (iv) $H\cdot$ and $H:^-$ (v) $\cdot Na$ and $:\ddot{Na}^+$

CHECK Point

✎ Write the Lewis dot symbols and predict the valencies you expect for the following elements :

Nitrogen, Fluorine and Neon

SOLUTION

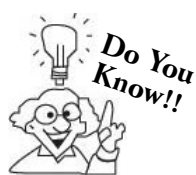
$:\ddot{N}\cdot$ (Valency = 5 or $8 - 5 = 3$) ; $:\ddot{F}\cdot$ (Valency = 7 or $8 - 7 = 1$) ; $:\ddot{Ne}:$ (Valency = 0)

CHEMICAL BONDING

Most of the substances are found in nature, are made up of clusters or groups of atoms. Such cluster in which atoms are held together and is electrically neutral is called a molecule. The molecules are made of two or more atoms joined together by some force acting between them. The force is termed as chemical bond. Thus, a chemical bond is defined as a force that acts between two or more atoms to hold them together as a stable molecule and to attain a state of minimum energy.

Types of Bonds

There are mainly three ways by which an atom may obtain a stable electronic configuration, namely by gaining, losing or sharing of electron(s) from one atom to other atom. The former two cases lead to electrovalent linkage, while the last one leads to a covalent linkage. Hence, mainly chemical bonds are of two types : **electrovalent** and **covalent**.



Nature of bond formed between two atoms depends upon electropositive and electronegative character of bonded atoms.

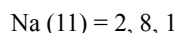
- (i) *Electropositive element + Electropositive element = Metallic bond*
 (ii) *Electropositive element + Electronegative element = Ionic bond*
 (iii) *Electronegative element + Electronegative element = Covalent bond*

IONIC BOND

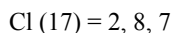
The atomic bonds which are formed by the transfer of electrons between the constituent atoms of a compound are known as ionic bonds (or electrovalent bonds). The compound thus formed is called an ionic or electrovalent compound. If you see the electronic configuration of Na^+ and Cl^- ions, they are completely filled. Therefore they easily separate or dissociate if dissolved in water. Generally ionic bonds are formed between metallic atoms having extra electron to spare and non-metallic atoms which are electron deficient. Magnesium Oxide (MgO), Potassium Chloride (KCl), Iron Oxide (FeO) are some of the examples of ionic compounds.

Formation of Sodium Chloride

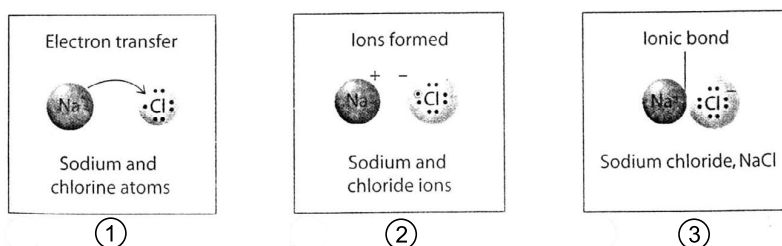
The electronic configuration of Na has 2 electrons in the first orbit (K shell), 8 electrons in the second orbit (L shell) and only one electron in the third (M shell). The last electron is loosely bound and Na atom willingly wants to give up its last electron, to obtain noble gas configuration (2, 8).



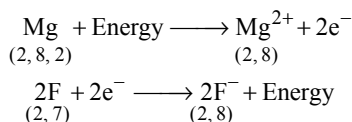
The electronic configuration of Cl has 2 electrons in the K shell, 8 electrons in the L shell and 7 electrons in the M shell. The Cl atom is short of one electron for completely filling the last orbit. Thus it picks up one electron from any one who is willing to donate **one**, to obtain noble gas configuration (2, 8, 8).



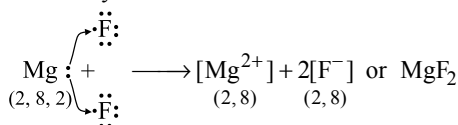
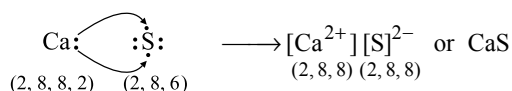
When a sodium and a chlorine atom come close, they complement each other's needs to loan-borrow electron and therefore easily bond together. The electron given up by a Na atom is easily accommodated in the last orbit of a Cl atom. The net reaction between Na and Cl is $\text{Na}^+ - \text{Cl}^-$, and the atoms are attracted to each other because of opposite electrical charges. Now you can see why Sodium (Na) and Potassium (K) do not form bonds.

**Formation of Magnesium Fluoride**

Magnesium (atomic number = 12, electronic configuration = 2, 8, 2) loses two electrons and is converted to magnesium cation (Mg^{2+}) and acquires the stable configuration of neon i.e., 2, 8. On the other hand fluorine (atomic number = 9, electronic configuration = 2, 7) on gaining one electron forms anion called fluoride ion.



The overall change may be represented as follows by Lewis structures

**Formation of Calcium Sulphide****ILLUSTRATION : 2**

Give the Lewis structure and empirical formulae for the ionic compounds formed between the following pairs of elements.

- (i) Ba, Cl (ii) Na, S (iii) Al, F (iv) Mg, N (v) Na, P

SOLUTION :

From the charges on the ions, the number of positive and negative ions in the compound can be found out. This gives the empirical formula (E. F.)

- (i) $\text{Ba} \cdot : \ddot{\text{Cl}} : \longrightarrow \text{Ba}^{2+}, \text{Cl}^{-} \text{ (ions)}$ Lewis formula : $[\text{Ba}]^{2+} 2[: \ddot{\text{Cl}} :]^{-}$, or BaCl_2
- (ii) $\text{Na} : \ddot{\text{S}} : \longrightarrow \text{Na}^{+}, \text{S}^{2-} \text{ (ions)}$ Lewis formula : $2[\text{Na}]^{+} [: \ddot{\text{S}} :]^{2-}$, or Na_2S
- (iii) $\text{Al} \cdot : \ddot{\text{F}} : \longrightarrow \text{Al}^{3+}, \text{F}^{-} \text{ (ions)}$ Lewis formula : $[\text{Al}]^{3+} 3[: \ddot{\text{F}} :]^{-}$, or AlF_3
- (iv) $\text{Mg} \cdot : \ddot{\text{N}} \cdot \longrightarrow \text{Mg}^{2+}, \text{N}^{3-} \text{ (ions)}$ Lewis formula : $3[\text{Mg}]^{2+} 2[: \ddot{\text{N}} :]^{3-}$, or Mg_3N_2
- (v) $\text{Na} \cdot : \ddot{\text{P}} : \longrightarrow \text{Na}^{+}, : \ddot{\text{P}} : ^{3-} \text{ (ions)}$ Lewis formula : $3[\text{Na}]^{+} [: \ddot{\text{P}} :]^{3-}$, or Na_3P

Properties of Ionic Compounds

- Ionic compounds are usually solid and crystalline in nature.
- Because of availability of ions, such compounds are good conductors of heat.
- Due to their crystalline nature, such compounds also have high melting points.
- They form electrolytes in molten or in solution state.
- They are non-conductors of electricity when solid, no free ions, all locked together in a crystal.

CHECK Point

- ☛ Why an ionic bond is formed between two elements having large difference in their electronegativity ?

SOLUTION

The more electronegative element will attract the shared pair of electrons to such a large extent than the other that it will account to transfer of electron resulting in the formation of ions.

Factors Affecting the Formation of Ionic Bond

There are three main factors

- (a) **Ionisation energy** : Ionisation energy (IE) is the amount of energy required to remove an electron from the outermost orbit of an isolated gaseous atom to form a gaseous positive ion (i.e., cation). Lesser the value of ionisation energy, greater is the tendency of the atom to form cation by loss of valence electrons.
- (b) **Electron affinity** : Electron affinity of an element is the amount of energy released when an electron is added to an isolated gaseous atom to form a negative ion (i.e., anion). Higher the value of electron affinity, greater will be the tendency of an atom to form anion.
- (c) **Lattice energy** : It is the amount of energy released when one mole of ionic solid is formed by combination of requisite number of cations and anions. Higher the value of lattice energy, greater will be the stability of ionic compound and thus greater will be the ease of its formation. Since the magnitude of lattice energy depends upon the charges of the ions and the distance between them, therefore, the value of lattice energy further depends upon the following two factors.
- (i) **Size of the ions** : Smaller the size of the ions, lesser is the internuclear distance and hence greater will be the interionic attraction. Therefore, the magnitude of lattice energy will be large.
For example, ionic size of K^{+} (133 pm) is larger than that of Na^{+} (95 pm). Consequently, the lattice energy of NaCl (769.8 kJ mole^{-1}) is greater than that of KCl (698.7 kJ mole^{-1}).
- (ii) **Charge on the ions** : Larger the magnitude of charge on the ions, greater will be the force of attraction between ions. Consequently, higher is the value of lattice energy.



- For dissolution of ionic solid in a solvent (water), hydration energy must be higher than the lattice energy.
- A number of ionic compounds are almost insoluble in water as their hydration energy is less than lattice energy.
e.g., BaSO_4 , AgCl , PbS , CaCO_3 , AgBr , AgI , CaF_2 , Ag_2CrO_4 , PbSO_4 , etc.

CHECK Point

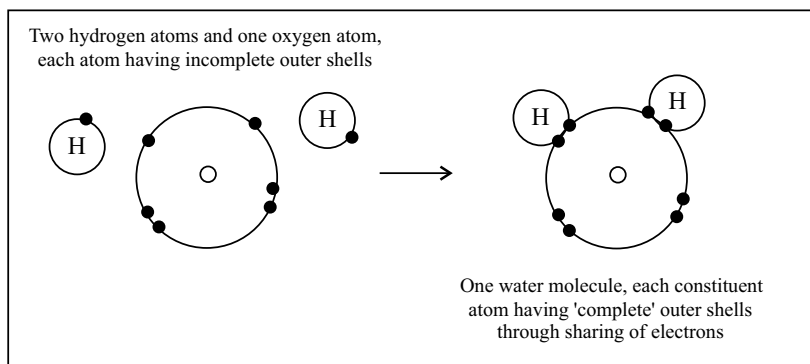
☛ Out of NaCl and MgO, which has higher lattice energy and why ?

SOLUTION

MgO has higher lattice energy because each ion carries two unit charge whereas in NaCl each ion carries one unit charge.

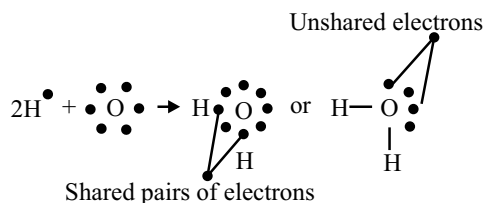
COVALENT BOND

There are many atoms where the electronic configuration of the last orbits or shells is not as simple as Na or Cl. For example, in an Oxygen (O) atom the electronic configuration has 2 electrons in the K shell and 6 electrons in the L shell. It would rather borrow two electrons from some other atom to complete its last shell than give up its six electrons. Hydrogen (H) atom has only one electron and needs to borrow only one extra electron to complete its first shell. Two atoms of H and one atom of O fulfill each other's needs by sharing electrons in their outermost orbits and form H_2O or a water molecule. In an H_2O molecule, the electrons are not totally given up, as in the case of $\text{Na}^+ - \text{Cl}^-$, but are shared by each of the neighbouring atoms.



Atomic bond formed by sharing of electrons is called a *covalent bond*. Compounds that are formed due to covalent bonding of atoms are called *covalent compounds*. Covalent bonds are generally formed between non-metals. Hydrogen gas (H_2), Nitrogen gas (N_2), Oxygen gas (O_2), Hydrochloric acid (HCl) are some of the examples of covalent bonded compounds. Most of the carbon bonds in organic materials like cells, sugar, etc., are examples of covalent bonds.

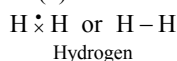
In terms of anion and cation electron-dot structure seen earlier, we can write the formation of covalent bonds as follows :

**Lewis Structure and Covalent Bond**

Electron dot structures, also known as **Lewis structures of covalent molecules**, are written in accordance with octet rule. Accordingly all atoms in a formula will have a total of eight electrons by sharing in the valence shell except the H-atom (hydrogen will have only two electrons).

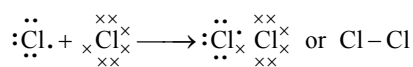
For writing Lewis structures for molecules, the atom which forms the largest number of bonds with other atoms is placed in the centre of the skeleton structure. Other atoms surround it to complete the octet. Thus structures in which valence electrons are represented by dots are called lewis structures. Whereas structures represented by lines (–) or dashes are known as **Copper structures**.

When each atom contributes one electron and hence two contributed electrons (*one pair*) are shared by each atom. The bond formed is known as **single bond** and is represented by a single line (–) between the two atoms.



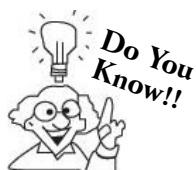
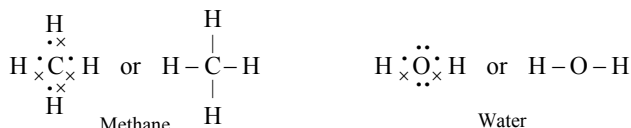
Each hydrogen atom possessing one electron in its shell on combination produces a shared pair which belongs simultaneously to both the hydrogen atoms i.e., each hydrogen atom has noble gas configuration of helium ($1s^2$).

Similarly, two chlorine atoms (each having seven electrons in the outermost shell) combine to form a molecule of chlorine.



Chlorine

Like-wise four covalent bonds are developed in the formation of CH_4 and two in the formation of water (H_2O).



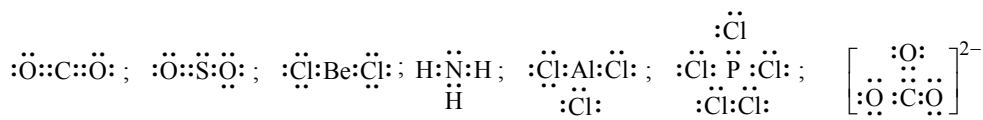
Covalent bonds, where larger than three pairs of electrons are shared, has not been observed.

CHECK Point

➤ Draw the Lewis structures of the following molecules and ions and tell in which case/cases the octet rule is violated.



SOLUTION



In BeCl_2 , AlCl_3 and PCl_5 octet rule is violated.

Knowledge ENHANCER

Conditions for writing the Lewis dot structures of molecules.

The main conditions for writing the electron dot (or Lewis) structures of covalent molecules are :

- (i) Each bond is formed as a result of sharing of an electron pair between the atoms.
- (ii) Each combining atom contributes one electron to the shared pair.
- (iii) Both the combining atoms attain the stable outer shell noble gas configurations as a result of sharing of the electrons. In other words, octets of both the atoms get completed.

Properties of Covalent Compounds

- The covalent compounds do not exist as ions but they exist as molecules.
- They exist at room temperature, as liquids or gases. However, a few compounds also exist in the solid state e.g. urea, sugar, etc.
- The melting and boiling points of covalent compounds are generally low.
- Covalent compounds are generally insoluble or less soluble in water and in other polar solvents.
- These are poor conductors of electricity in the fused or dissolved state.
- Since the covalent bond is localized in between the nuclei of atoms, it is directional in nature.

CHECK Point

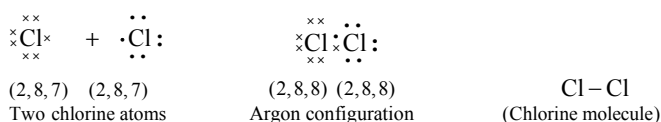
☛ Why NaCl gives a white precipitate with AgNO_3 solution but CCl_4 does not ?

SOLUTION

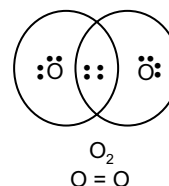
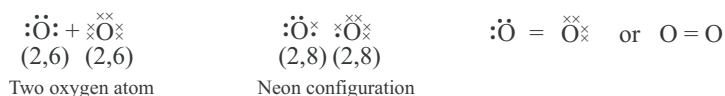
NaCl is an ionic compound and hence gives Cl^- ions in the solution which combine with Ag^+ ions given by AgNO_3 to form a white precipitate of AgCl but CCl_4 is a covalent compound and does not give Cl^- ions.

TYPES OF COVALENT BOND

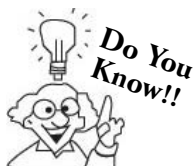
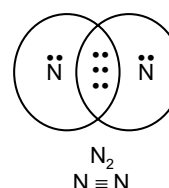
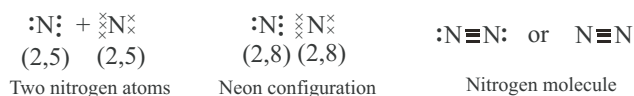
(A) Single bond : Chlorine molecule : Chlorine atom has seven valence electrons. Thus, each Cl atom requires one more electron to acquire the nearest noble gas configuration (Ar:2, 8, 8). This they do by mutual sharing of one pair of electrons as shown below.



(B) Double bond : Oxygen molecule : An oxygen atom has six electrons in its valence shell. As a result, it requires 2 more electrons to achieve the nearest noble gas configuration. When two oxygen atoms share two pairs of electrons this is achieved:



(C) Triple bond : Nitrogen molecule : Nitrogen atom has five electrons in its valence shell. It requires three more electrons to acquire a stable configuration of the nearest noble gas (neon). This is done by mutually sharing three pairs of electrons as shown below.

**Covalency :**

- (i) It is defined as the number of electrons contributed by an atom of the element for sharing with other atoms so as to achieve noble gas configuration.
- (ii) It can also be defined as the number of covalent bonds formed by the atom of the element with other atoms.
- (iii) The usual covalency of an element except hydrogen is equal to a number of the group to which an element belongs.

Lewis Dot Structures of Few Other Compounds

Molecules	Structural formula	Lewis dot formula
(i) Oxygen (O_2)	$O = O$	$:\ddot{O} :: \ddot{O}:$
(ii) Carbon dioxide (CO_2)	$O = C = O$	$\begin{array}{c} \times \times \\ \times O \times : C : \times \times \\ \times \times \end{array} \times O \times$
(iii) Hydrogen peroxide (H_2O_2)	$H - O - O - H$	$H \times \ddot{O} : \ddot{O} \times H$
(iv) Carbonate ion, CO_3^{2-}	$\begin{array}{c} O^- \\ \\ ^-O - C = O \end{array}$	$\left[\begin{array}{c} \times \times \\ \times O \times \\ \times \times \\ \times O \times \times C \times O \times \\ \times \times \end{array} \right]^{2-}$
(v) Nitrite ion, NO_2^-	$\begin{array}{c} N - O^- \\ \\ O \end{array}$	$\left[\begin{array}{c} \times \times \\ \times N \times O \times \\ \times \times \times \times \\ \times O \times \\ \times \times \end{array} \right]^-$
(vi) Peroxide ion, O_2^{2-}	$^-O - O^-$	$\left[\begin{array}{c} \times \times \times \times \\ \times O \times O \times \\ \times \times \times \times \end{array} \right]^{2-}$
(vii) Cyanide ion, CN^-	$^-C \equiv N$	$\left[\begin{array}{c} \times C : \times N \times \\ \times \times \end{array} \right]^-$
(viii) Hydronium ion, H_3O^+	$\begin{array}{c} H - O^+ - H \\ \\ H \end{array}$	$\left[\begin{array}{c} \times \times \\ H \times \ddot{O} \times H \\ \times \end{array} \right]^+$
(ix) Ammonium ion, NH_4^+	$\left[\begin{array}{c} H \\ \\ H - N - H \\ \\ H \end{array} \right]^+$	$\left[\begin{array}{c} H \\ \times \times \\ H \times N : H \\ \times \times \\ H \end{array} \right]^+$
(x) Sulphide ion, S^{2-}	S^{2-}	$\left[\begin{array}{c} \times \times \\ \times S \times \\ \times \times \end{array} \right]^{2-}$

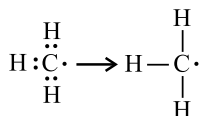
ILLUSTRATION : 3

In each of the following compounds, there is at least one multiple bond. Draw a Lewis structure for each molecule. Use lines to indicate electrons in bonds and dots to indicate nonbonding electrons.

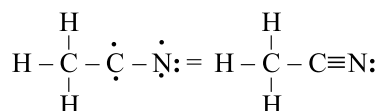
(a) H_3CCN (b) H_3CNO

SOLUTION :

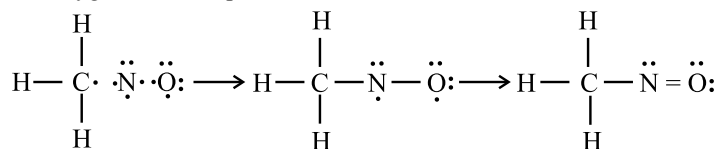
(a) In methyl group (CH_3) three covalent bonds are constructed by sharing three of carbon's four available electrons with the single electrons of the three hydrogens.



The other atoms, the carbon and the nitrogen, have four and five electrons, respectively in their valence shell. Formation of a triple bond between carbon and nitrogen completes the picture, and nitrogen is left with a non-bonding pair of electrons.

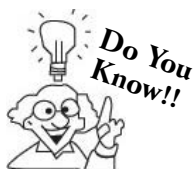


- (b) H_3CNO . Once again there is a methyl group. Nitrogen has five available electrons, and oxygen has six electrons. A carbon-nitrogen single bond and a nitrogen-oxygen double bond can be formed. Nitrogen is left with one pair of non-bonding electrons, and oxygen with two pairs.



DIFFERENCE BETWEEN COVALENT AND IONIC COMPOUNDS

Covalent compounds	Ionic electrovalent compounds
(i) Covalent compounds are formed by sharing of electrons between the two atoms. $\cdot \ddot{\text{X}} \cdot + \cdot \ddot{\text{Y}} \cdot \longrightarrow \cdot \ddot{\text{X}} : \ddot{\text{Y}} \cdot$	Ionic compounds are formed by complete transference of electrons from one atom to another. $\text{X} \cdot + \cdot \ddot{\text{Y}} : \longrightarrow \text{X}^+ \left[: \ddot{\text{Y}} : \right]^-$
(ii) In covalent compounds molecules are held together by weak vander Waal's forces of attraction.	In ionic compounds ions are held together with strong electrostatic forces of attraction
(iii) They have generally low melting and boiling points.	They have generally high melting and boiling points.
(iv) Covalent compounds are generally bad conductors of electricity.	Ionic compounds conduct electricity in dissolved state or in molten state.
(v) They are generally soluble in non-polar solvents such as benzene and insoluble in polar solvent such as water.	They are generally soluble in polar solvents such as water and insoluble in organic solvents.
(vi) Covalent bonds are rigid and directional.	Ionic bonds are non-directional and do not show isomerism.
(vii) These compounds undergo molecular reactions which are quite slow.	These compounds undergo ionic reactions which are almost instantaneous.
(viii) These compounds may be solids, liquids or gases	These compounds are generally solids.



Modern periodic table contains 18 vertical columns known as groups and 7 rows known as periods. All elements present in same group have same number of valence electrons. Elements of group 1 has one and of group 2 has two electrons in their valence shell. Similarly p-Block elements of group 13, 14, 15, 16, 17 and 18 have 3, 4, 5, 6, 7 and 8 electrons in valence shell respectively. Electrons present in valence shell participates only in formation of ionic and covalent bonds.

POLAR AND NON-POLAR COVALENT BONDS

Electronegativity is greatest for elements at the upper right of the periodic table and lowest for the elements at the lower left. Noble gases are not considered in this discussion as they rarely participate in bonding.

H 2.2																	He —				
Li 0.98	Be 1.57															B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne —
Na 0.93	Mg 1.31															Al 1.61	Si 1.9	P 2.19	S 2.58	Cl 3.16	Ar —
K 0.82	Ca 1.0	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	B 1.81	C 2.01	N 2.18	O 2.55	F 2.96	Ne —				
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe —				
Cs 0.79	Ba 0.89	La 1.10	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.8	Au 2.54	Hg 2.60	Tl 2.04	Pb 2.33	Bi 2.02	Po 2.0	At 2.2	Rn —				
Fr 0.7	Ra 0.9	Ac 1.1	Rf —	Db —	Sg —	Bh —	Hs —	Mt —	Uun —	Uuu —	Uub —										

Fig. The experimentally measured electronegativities of elements

When the two atoms in a covalent bond have the same electronegativity, no dipole is formed (as is the case with Cl_2) and the bond is classified as a **non-polar** bond. When the electronegativity of the atoms differ, a dipole may form (as with HF) and the bond is classified as a **polar bond**. A polar bond depends on the difference between the electronegativity values of the two atoms – the greater the difference, the more polar the bond.

As can be seen in figure, the greater the distance between two atoms in the periodic table, the greater the difference in their electronegativity, and hence the greater the polarity of the bond between them. So we can predict which bonds are more polar than others without reading the electronegativities. Bond polarity can be inferred by looking at the relative positions of the atoms on the basis of electronegativities of elements of the periodic table – the farther apart they are, especially when one is at the lower left and one is at the upper right, the greater the polarity of the bond between them.

The magnitude of bond polarity is sometimes indicated by the size of the crossed arrow or the δ^- and δ^+ symbols used to depict a dipole, as shown in figure.

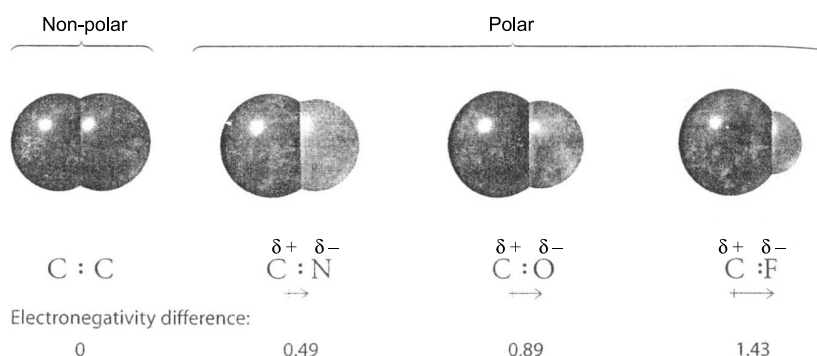
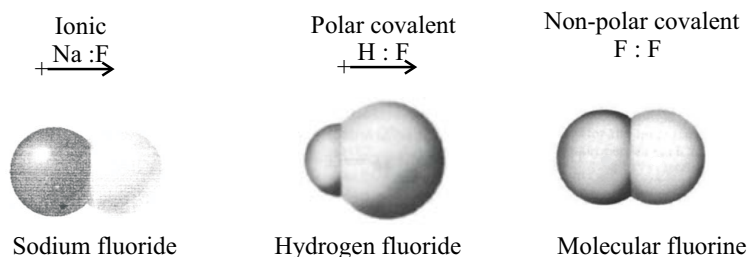


Fig. These bonds are in order of increasing polarity from left to right.



The electronegativity difference between atoms in an ionic bond can also be calculated. For example, the bond in NaCl has an electronegativity difference of 2.23, far greater than the difference of 1.43 for the C–F bond.

Atoms on opposite sides of the periodic table have great differences in electronegativity, and hence the bonds between them are highly polar – in other words, ionic. Non-metallic atoms of the same type have the same electronegativities, and so their bonds are non-polar covalent. The polar covalent bond with its uneven sharing of electrons and slightly charged atoms is between these two extremes.



Electronegativity difference between two atoms A and B is represented as $\chi_A - \chi_B$

Knowledge ENHANCER

Partial Ionic Character of Covalent Bond

The extent of partial ionic character is determined by the difference in electronegativities of the combining atoms. More is the difference in electronegativities, greater will be the ionic character.

- If the electronegativity difference between two atoms is 1.7, the bond is said to have 50% covalent character.
- If the electronegativity difference between the two atoms is more than 1.7, the partial ionic character of the bond is more than 50% and the bond is taken as ionic.
- If the electronegativity difference between two atoms is less than 1.7, the bond is predominantly covalent.

DIPOLE MOMENT

A measure of degree of polarity as discussed above in polar molecules, one end of molecule is negative and other end is positive separated by certain distance. Such polar molecules as a whole is electrically neutral, the negative and positive charges are equal in magnitude are said to possess an electric dipole. Therefore a new term was introduced known as dipole moment, to measure the degree of polarisation of covalent bond.

Dipole moment is defined as the product of magnitude of negative or positive charge on any one of the atom and the distance between the centre of two charges. It is usually represented by Greek letter ' μ '. Mathematically, it can be expressed as follows :

$$\text{Dipole moment } (\mu) = \text{Charge } (q) \times \text{distance between two atoms } (d)$$

$$\mu = q \times d.$$

ORBITAL OVERLAP CONCEPT OF COVALENT BOND

According to this concept a covalent bond is formed by the partial overlapping of two half filled atomic orbitals containing electrons in opposite spins. The term partial overlapping means partial merging of the electron cloud of two orbitals and the paired electrons become common. This can be illustrated by considering the formation of hydrogen molecule.

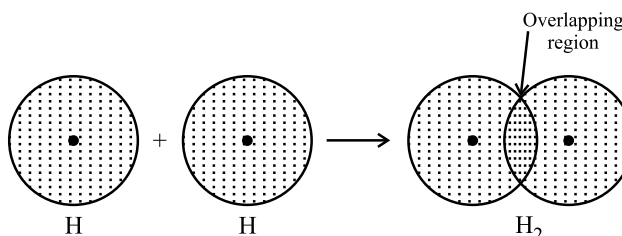


Fig. Overlapping of orbitals to form H_2 molecule.

A hydrogen atom has one electron in 1s-orbital. When two such atoms containing one electron each with opposite spins approach each other, then they merge to form a new orbital known as **molecular orbital**. These two electrons have greater probability of their presence in the region of overlap and thus get stabilised i.e., during overlapping energy is released. Thus, the requirement for overlapping of orbitals and its outcome may be summarised as follows.

- Overlapping occurs between half filled valence shell orbitals of the two atoms
- Resulting bond acquired a pair of electrons with opposite spins to get stability.
- Orbitals come closer to each other from the direction in which there is maximum overlapping
- Covalent bond has directional character.
- Extent of overlapping $\propto$ strength of chemical bond. Extent of overlapping depends on two factors :
 - Nature of orbitals** – p , d and f are directional orbitals $\rightarrow$ more overlapping.
 s -orbital is non directional $\rightarrow$ less overlapping
 - Nature of overlapping** –
 - Co-axial overlapping $\rightarrow$ extent of overlapping more.
 - Collateral overlapping $\rightarrow$ extent of overlapping less

In Co-axial overlapping extent of overlapping is $p-p > s-p > s-s$



(vi) As the value of n increases, bond strength decreases.

$$1-1 > 1-2 > 2-2 > 2-3 > 3-3$$

(vii) If n is same $2p-2p > 2s-2p > 2s-2s$

$$1s-2p > 2s-2p > 3s-3p$$

(viii) Electron which is already paired in valence shell can enter into bond formation, if they can be unpaired first and shifted to vacant orbitals of slightly higher energy of the same energy shell.

This point can explain the trivalency of boron, tetravalency of carbon, pentavalency of phosphorus etc.

(ix) Two types of bonds are formed on account of overlapping.

(A) Sigma (σ) bond (B) Pi (π) bond

CHECK Point

☛ Out of σ and π -bonds, which one is stronger and why ?

SOLUTION

σ bond is stronger. This is because σ -bond is formed by head-on overlapping of atomic orbitals and, therefore, the overlapping is large, π -bond is formed by sideways overlapping which is small.

Difference in σ and π bonds

Sigma (σ) bond	Pi (π) bond
1. It results from the end to end overlapping of two s orbitals or two p -orbitals or one s and one p -orbital.	It results from the sideways (lateral) overlapping of two p -orbitals.
2. Its bonded orbital consists of a single electron cloud symmetrical about internuclear axis.	Its bonded orbital consists of two electron clouds on above and the other below of the plane of participating atom
3. Stronger and less reactive	Less strong and more reactive
4. Bond energy 80 kcal	65 kcal
5. More stable	Less stable
6. Can be independently exist	Always exist along with a σ bond
7. The groups or atoms can undergo bond rotation about single sigma (σ) bonds	Due to resistance to rotation around the π bond the groups attached to it are not free to rotate.
8. The shape of the molecule is determined by the σ bonds present in the molecule	π bonding does not affect the shape of the molecule in which it occurs



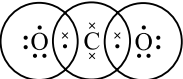
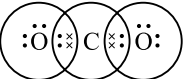
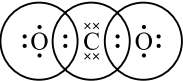
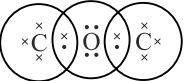
idea box

Go through the concept of ionic and covalent bonding thoroughly and divide them in two groups A and B and you also become a part of one group. Now group A will prepare 15 questions on ionic bonds and group B will also prepare 15 questions for covalent bonds. Students of group B have to answer the questions prepared by students of group A and vice-versa. The group which give more correct answer will win.

EXERCISE

DIRECTIONS (Qs.1 to 23) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- When two atoms combine to form a molecule
 - energy is released
 - energy is absorbed
 - energy is neither released nor absorbed
 - energy may either released or absorbed
- Which of the following compounds have highest melting point ?
 - BeCl_2
 - MgCl_2
 - CaCl_2
 - SrCl_2
- Which of the following is most covalent ?
 - AlF_3
 - AlCl_3
 - AlBr_3
 - AlI_3
- Element X is strongly electropositive and element Y is strongly electronegative. Both are univalent. The compound formed would be
 - X^+Y^-
 - $\text{X}-\text{X}^+$
 - $\text{X}-\text{Y}$
 - $\text{X} \rightarrow \text{Y}$
- The total number of electrons that take part in forming the bond in N_2 is
 - 2
 - 4
 - 6
 - 10
- Number of paired electrons in O_2 molecule is :
 - 7
 - 8
 - 16
 - 14
- Bonding in ferric chloride is
 - covalent
 - ionic
 - co-ordinate
 - None of these
- Consider the following statements:
 - A sigma (σ) bond is formed when two s-orbitals overlap
 - A pi (π) bond is formed when two p-orbitals axially overlap
 - A σ -bond is weaker than π -bond.Which of the above statements is/are correct ?
 - (i) and (ii)
 - (ii) and (iii)
 - (i) alone
 - (ii) alone
- Which of the following compounds has electrovalent linkage?
 - CH_3Cl
 - NaCl
 - CH_4
 - Cl_2
- In a double bond connecting two atoms there is a sharing of
 - 2 electrons
 - 4 electrons
 - 1 electrons
 - All electrons
- Outermost shells of two elements X and Y have two and six electrons respectively. If they combine, expected formula of compound will be
 - X Y
 - X_2Y
 - X_2Y_3
 - XY_2
- An atom of an element A has three electrons in its outermost shell and that of B has six electrons in the outermost shell. The formula of the compound between these two will be :
 - A_3B_4
 - A_2B_3
 - A_3B_2
 - A_2B
- In the formation of NaCl by combination of Na and Cl :
 - Sodium lose electrons
 - Chlorine gains electrons
 - Sodium loses and chlorine gains electrons
 - Sodium gains but chlorine loses electrons
- The bond between two identical non-metal atoms has a pair of electrons
 - unequally shared between the two
 - transferred fully from one atom to another
 - with identical spins
 - equally shared between them
- Hydrogen chloride molecule contains
 - polar covalent bond
 - metallic bond
 - co-ordinate bond
 - electrovalent bond
- The molecular formula of chloride of a metal M is MCl_3 the formula of its carbonate would be
 - MCO_3
 - $\text{M}_2(\text{CO}_3)_3$
 - M_2CO_3
 - $\text{M}(\text{CO}_3)_2$
- The rule which states that the atoms tend to form bonds until each atom has eight electrons in its outermost shell is called
 - Pauli's exclusion principle
 - Newland's rule
 - Octet rule
 - Dipole rule
- The electron dot structure is written considering the
 - electrons in the outermost orbit
 - electrons in the first orbit
 - protons present in the nucleus of an atom
 - the charge carried by an atom, positive or negative

19. In methane, a carbon atom shares
 (a) an electron with a hydrogen atom
 (b) two electrons with two hydrogen atoms
 (c) three electrons with three hydrogen atoms
 (d) four electrons with four hydrogen atoms
20. The most favourable conditions for the ionic bonding to occur are
 (a) high charge, large anions and small cation
 (b) high charge, small anion and large cation
 (c) low charge, small anion and large cation
 (d) low charge, large anion and small cation
21. What causes the atom of two different elements or same elements to go for bonding?
 (a) The opposite charges they carry
 (b) Their tendency to acquire a stable inert gas structure
 (c) The difference in their ionization energies
 (d) None of the above
22. Which of the following structures best describe the bond formation between carbon and oxygen to form a molecule of carbon dioxide?
 (a)  (b) 
 (c)  (d) 
23. The tendency of an atom of an element to attract the shared pair of electrons to form a covalent compound is called
 (a) Covalency (b) Electronegativity
 (c) Electron affinity (d) Electron attractive force
-
- DIRECTIONS (Qs. 24 to 37) :** This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.
24. Valency expresses
 (a) total electrons in valence shell of an atom
 (b) atomicity of an element
 (c) oxidation number of an element
 (d) combining capacity of an element
25. Ions are formed from neutral atoms by
 (a) loss of electrons
 (b) gain of electrons
 (c) sharing of electrons
 (d) simultaneous loss and gain of electrons
26. Which of the following compound(s) is/are covalent?
 (a) H_2 (b) CaO
 (c) $CHCl_3$ (d) BI_3
27. Which of the following statement(s) regarding covalent bond is/are true?
 (a) The electrons are shared between atoms
 (b) The bond is non-directional
 (c) The strength of the bond depends upon the extent of overlapping
 (d) The bond formed may or may not be polar
28. Which statement(s) is/are correct?
 (a) A sigma bond is weaker than a π -bond.
 (b) A sigma bond is stronger than a π -bond.
 (c) A double bond is stronger than a single bond.
 (d) A double bond is shorter than a single bond.
29. Which of the following statement(s) is/are correct for sigma and pi-bonds formed between two carbon atoms?
 (a) Sigma-bond determines the direction between carbon atoms but a pi-bond has no primary effect in this regard
 (b) Sigma-bond is stronger than a pi-bond
 (c) Bond energies of sigma- and pi-bonds are of the order of 264 kJ/mol and 347 kJ/mol, respectively
 (d) Free rotation of atoms about a sigma-bond is allowed but not in case of a pi-bond
30. Covalent compounds are soluble in
 (a) polar solvents (b) non-polar solvents
 (c) slightly polar solvents (d) all types of solvents
31. As compared to covalent compounds, electrovalent compounds generally have
 (a) low melting points
 (b) low boiling points
 (c) high boiling points
 (d) high melting points
32. The force(s) of attraction which holds two atoms together is/are called
 (a) Electrovalent bond (b) Covalent bond
 (c) Metallic bond (d) Coordinate bond
33. The factors which favour the formation of electrovalent bond is
 (a) high electron affinity
 (b) high lattice energy
 (c) low ionization potential
 (d) high ionization potential
34. Which of the following is/are an electrovalent (ionic) compound(s)?
 (a) Calcium chloride ($CaCl_2$)
 (b) Sodium sulphate (Na_2S)
 (c) Magnesium fluoride (MgF_2)
 (d) Carbon tetrachloride (CCl_4)
35. Which gas(es) has a triple bond in between its atoms?
 (a) Hydrogen (b) Nitrogen
 (c) Oxygen (d) Acetylene
36. Which of the statement(s) below is/are incorrect?
 (a) The covalent compound exist as ions.
 (b) The electrovalent compounds exist as molecules
 (c) The covalent compound exist as molecule not ions
 (d) The covalent compounds exist both as ions and molecules.
37. Which of the following exist as monoatomic?
 (a) Helium (b) Fluorine
 (c) Radon (d) Oxygen

Chemical Bonding

DIRECTIONS (Q. 38) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in Column I with entries in Column II.

38.	Column I	Column II
	(A) HCl	(p) Covalent compound with directional bond
	(B) CO ₂	(q) Ionic compound with non-directional bonds
	(C) NaCl	(r) Polar molecule
	(D) CCl ₄	(s) Non-polar molecule
	(A) (B) (C) (D)	(C) (D)
(a)	p, q, r	q, r
(b)	q	r
(c)	p, r	p, s
(d)	q	r

DIRECTIONS (Qs. 39 to 41) : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.

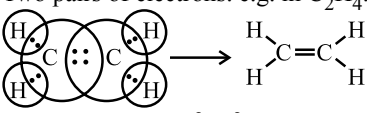
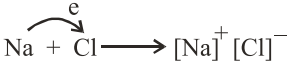
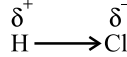
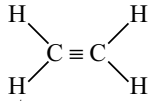
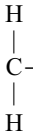
- (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
- (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
39. **Assertion :** Ionic compounds conduct electricity in solution as well as in molten form.
Reason : Ionic compounds are formed by sharing of electrons.
40. **Assertion :** The lesser the lattice enthalpy more stable is the ionic compound.
Reason : The lattice enthalpy is greater, for ions of higher charge and smaller radii.
41. **Assertion :** The atoms in a covalent molecule are said to share electrons, yet some covalent molecules are polar.
Reason : In a polar covalent molecule, the shared electrons spend more time than average near one of the atoms.

DIRECTIONS (Qs. 42 to 43) : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

42. Total number of σ -bonds present in the molecule of propyne is
43. What is the total number of pi bonds in Cyanobenzene ?

SOLUTIONS

Brief Explanations of Selected Questions

- (a) To attain stability energy is released.
- (d) As the size of cation increases polarising power decreases hence ionic character increases.
- (d) As the electronegativity difference decreases, covalent character increases.
- (a) $X^+ Y^-$
- (c) $N \equiv N : N :: N :$
- (d) O_2 - Oxygen ($Z = 8$) has following electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4$. i.e., 2 unpaired and 14 paired electrons.
- (b) Covalent bond on the basis of Fajan's rule.
- (c) Statement 1 is correct.
- (b) Out of given four compounds, only NaCl has electrovalent linkage & others have covalent linkages. Na has 1 e^- in its outermost shell thus tends to lose it and Cl has 7 e^- in its outermost shell thus tends to gain an e^- , hence the bond that is formed between Na and Cl is formed by transfer of electron i.e., electrovalent bond.
- (b) Two pairs of electrons. e.g. in C_2H_4 .

- (a) $X : + Y \xrightarrow{\times \times} X^{2+} Y^{2-}$
Hence the formula is XY
- (b) Clearly, valency of element A = 3
valency of element B = $8 - 6 = 2$
Hence, the compound
 $2A^{3+} + 3B^{2-} \longrightarrow A_2B_3$
- (c) In the formation of NaCl by combination of Na and Cl, sodium loses but chlorine gains electrons.

- (d) Electrons are equally shared in the bond formed between two identical non-metal atoms.
- (a) A gaseous HCl molecule has hydrogen and chlorine linked by a covalent bond. Here electronegativity of chlorine is greater than that of hydrogen. Due to this the shared pair of electron is more attracted towards chlorine. Thus, chlorine end of molecule has higher electron density and becomes slightly negative and the hydrogen becomes slightly positive. Hence the covalent bond in HCl has a polar character as shown below

- (b) $MCl_3 \rightarrow M^{3+} + 3Cl^-$ $2M^{3+} + 3CO_3^{2-} \rightarrow M_2(CO_3)_3$
- (c) According to octet rule all atoms in the formula of compound will have a total of eight electrons by sharing in the valence shell.
- (a) 
- (d) 
- (c) High charge and large anion results into polarisation of anion which favours covalent bond.
- (b) 22. (b) 23. (b)
- (a, d) It is the basic definition.
- (a, b) Cation and anion are formed by losing and gaining of electrons respectively.
- (a, c, d) All these compounds are covalent in nature.
- (a, c, d) Covalent bonds are directional in nature.
- (b, c, d) A σ -bond is stronger than a π -bond hence option (a) is not correct. Sigma (σ) bonds are formed by head on overlap of s-s, p-p or s-p orbitals hence σ bonds are strong bonds. Whereas pi (π)-bonds are formed by side ways overlap of p and d orbitals hence π bonds are weak bonds.
- (a, b, d) As sigma bond is stronger than the π (pi) bond, so it must be having higher bond energy than π (pi) bond.
- (b, c) Covalent compounds, being non-polar or less polar dissolve in non-polar or weakly polar solvents.
- (c, d) Ionic compounds generally have high melting and boiling points because of the strong electrostatic force of attraction between oppositely charged ions. Consequently, a considerable amount of energy is required to overcome strong attractive inter-ionic forces and to break down the crystal lattice.
- (a, b, c, d) A bond is a force which holds two atoms together.
- (a, b, c)
- (a, b, c) Only CCl_4 is covalent in nature as electronegativity difference between carbon and chlorine is not high.
- (b, d) $N \equiv N$ and  both has triple bond in between its atoms.
- (a, b, d) Covalent compound exist as molecules whereas in case of ionic compounds, electrons get transferred from one to other atom completely forming ions.
- (a, c) He and Xe exists as a monoatomic gas.
- (c) $A \rightarrow (p, r)$, $B \rightarrow (p, s)$, $C \rightarrow (q)$, $D \rightarrow (p, s)$
- (b) Both assertion and reason are true but reason is not the correct explanation of assertion. Ions become mobile and are able to move under the influence of an electric field and thus act as carrier of current.
- (d) Assertion is false but reason is true.
The greater the lattice enthalpy more stable is the ionic compound.
- (c) The shared pair of electrons remains attracted towards the more electronegative atom at all times.

- 6 $H-C \equiv C-H$. Thus, there are 6 σ -bonds present.

- 5 Cynobenzene has 5 pi bonds. 

Chapter 6

Chemistry in Everyday Life

INTRODUCTION

In our daily life we come across many chemical substances and chemical changes. Medicine for health care, dyes for imparting colour to clothes, and plastics material used for various applications are made up of polymeric material. It is difficult to imagine our life without these substances. Chemistry plays a key role in the development of mankind. Chemistry has helped in agriculture, food, medicine, transportation, computers and in our everyday life.

The various chemical changes which take place in living systems fall in the domain of biochemistry. Living systems are made up of a number of complex organic compounds called biomolecules. The various biomolecules are carbohydrates, proteins, enzymes, lipids, hormones etc. In this chapter we will discuss about these substances and their importance in our everyday life.

POLYMERS

Have you ever noticed the things around you at your home i.e. carpet, cloathings, plastic bottles, melamine dishes, bathroom curtain, your childhood toys and other materials made up of plastic and rubber and so many other things at home.

Have you ever tried to know about the material of which they are all made up of ?

They are all made up of a common material know as polymer.

Do you know, that DNA, the genetic blue print that defines people and other living things is a polymer. So are the proteins and starches in the food we eat, the wheels on skate board and in line skates and tires on our bikes and cars. In fact we are surrounded by polymers everyday, everywhere we go.

Now we shall discuss what are polymers ?

A polymer may be defined as a high molecular weight compound formed by the combination of a large number of one or more types of small molecules of low molecular weight. The small unit (or units) of which the polymer is made is known as *monomer* (or monomers). Many polymeric substances occur in nature such as cellulose, starch, rubber, proteins, and resins.

The work in the synthetic field of polymer is mainly due to W.H. Carothers. The synthetic polymers are manufactured generally from the small units by the process known as polymerisation. According to Carothers **polymerisation** may be defined as a chemical combination of a number of similar or different molecules to form a single large molecule. Hence now a **polymer** may be defined as a large molecule formed by the repeated combination (repeated units) of small and simple chemical compounds known as monomers. The molecular weight of the commercially important synthetic polymers is of the order of several thousands or higher.

The monomers can unite in two different manners to form large molecules, in other words we can say that the polymerisation can be achieved by two methods, viz., *condensation and addition polymerisation*. In the former the two monomers (same or different) link with each other by the elimination of a small molecule, such as water or methyl alcohol, as a by product to form the *dimer*, which again condenses with another molecule of monomer in the same way by the elimination of a small molecule to form *trimer*, which forms *tetramer*, and so on until a large molecule is formed.

On the other hand, addition polymerisation involves the self-addition of unsaturated molecules of one or two monomers without loss of any small molecule to form a single giant molecule, e.g., propylene polymerises to polypropylene.

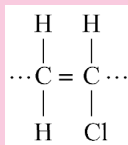
Some of the polymers are obtained only from one type of monomer, whereas the others are obtained from two different types of monomers, the former polymers are known as *homopolymers* whereas the latter are known as *copolymers*. Polypropylene, polyethylene, polyisoprene etc. are the examples of homopolymers; on the other hand, dacron, nylon, certain vinyl polymers, etc. constitute the examples of *copolymers*.

Knowledge ENHANCER

Some Common Examples of Addition and Condensation Polymers and their uses :

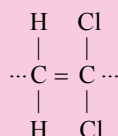
Addition Polymers	Repeating Unit	Common Uses/Applications
Polyethylene (PE)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \cdots \text{C} - \text{C} \cdots \\ \quad \\ \text{H} \quad \text{H} \end{array}$	Plastic bags, bottles
Polypropylene (PP)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \cdots \text{C} - \text{C} \cdots \\ \quad \\ \text{H} \quad \text{CH}_3 \end{array}$	Indoor-outdoor carpets
Polystyrene (PS)	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \cdots \text{C} = \text{C} \cdots \\ \quad \\ \text{H} \quad \text{C}_6\text{H}_5 \end{array}$	Plastic utensils, insulation

Polyvinyl chloride (PVC)



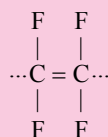
Shower curtains, tubing

Polyvinylidene chloride (Saran)



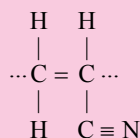
Plastic wrap

Polytetrafluoroethylene (Teflon)



Nonstick coating

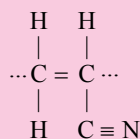
Polyacrylonitrile (Orion)



Yarn, paints

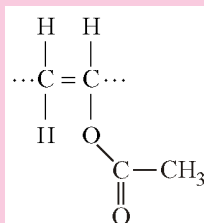
Condensation Polymers**Repeating Unit****Common Uses/ Applications**

Polymethyl methacrylate (Lucite, Plexiglas)



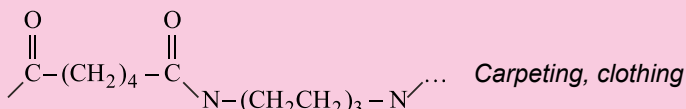
Windows, bowling balls

Polyvinyl acetate (PVA)



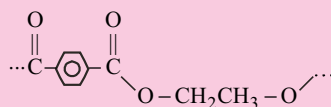
Adhesives, chewing gum

Nylon



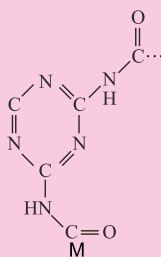
Carpeting, clothing

Polyethylene terephthalate



Clothing, plastic bottles

Melamine-formaldehyde resin (Melmac, Formica)



Dishes, countertops

CHECK Point

☛ In which classes, the polymers are classified on the basis of molecular forces ?

SOLUTION

On the basis of intermolecular forces of attraction operating between the different polymer chains, polymers are classified into the following classes :

1. Elastomers
2. Fibres
3. Thermoplastics
4. Thermosetting plastics.

Polymers may be Divided into two categories

- Natural polymers:** They are obtained from natural sources e.g. **polysaccharides** (starch, cellulose), **Proteins** (polymers of amino acids), gums, resins (cross linked polymers formed by compounds containing double or triple bonds slowly oxidised by atmospheric oxygen). **Natural rubber** (polymer of isoprene), **Nucleic acids** (polymers of nucleotides) silk and wool (polymers of amino acids).
- Synthetic polymers:** Polymers prepared by synthesis (man made), are known as synthetic polymers eg. polystyrene, nylon, PVC etc.

Rayon was originally called artificial silk but now a days it is a name given to artificial fibres derived from cellulose. Rayon can absorb over 90% of its own mass of water and it was not stick to wound.

Natural silk contains nitrogen while artificial silk may not have nitrogen.



idea box

Make a project report on importance of polymers in our daily life.

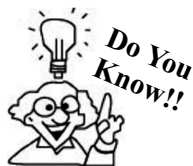
PLASTICS AND RESINS

There is no sharp line of demarcation between plastics and resins, and the terms are sometimes used interchangeably. Plastics are relatively tough substances of high molecular weights and soften on heating; on the other hand, resins are generally of low molecular weights and withstand high temperature. Moreover, resins are resistant to chemicals, whereas plastics are more readily attacked by chemicals. Due to their similarities they are studied together under two headings which are differentiated by the action of heat :

- Thermoplastics :** Those substances which soften on heating and can be remoulded again into desired shape and size.
- Thermosetting Resins or Thermosetting Plastics :** Those substances which get converted to an infusible material on heating and thus they are not subjected to heat-softening.

Differences between thermoplastic polymers and thermosetting polymers

	Thermoplastic polymers		Thermosetting polymers
1.	Soften and melt on heating.	1.	Do not soften on heating but burn if heated for a longer time.
2.	Less brittle.	2.	More brittle.
3.	Can be remoulded, recast and reshaped.	3.	Cannot be remoulded or reshaped.
4.	Insoluble in organic solvents.	4.	Insoluble in inorganic solvents because of three dimensional cross linking.
5.	Usually formed by addition polymerization.	5.	Formed by condensation polymerization.
6.	Usually have linear structures.	6.	Have three-dimensional cross-linked structures.
7.	Examples : Polyethene, PVC, nylon, teflon, etc.	7.	Examples : Bakelite, urea-formaldehyde resin, terylene.



The combination of polymeric substances with solid fillers is called filled or reinforced plastics. The filler acts as a reinforcing material while the polymer acts as binder, which links the filler particles. The filler or solid additives which modify the physical properties particularly the mechanical properties of basic polymer.

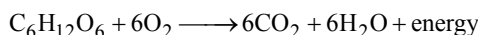
In many ways, our bodies can be thought of as chemical processing plants. Chemicals are taken in, processed through various types of reactions, and then distributed throughout the body to be used immediately or stored for later use. The chemicals used by the body can be divided into two broad categories : macronutrients, those substances that we need to eat regularly in fairly large quantities, and micronutrients, those substances that we need only in small amounts. Three major classes of macronutrients are essential to living organisms : carbohydrates, fats, and proteins.

CARBOHYDRATES

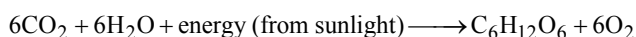
Carbohydrate is a fancy way of saying “sugar”. Scientists came up with the name because the compounds have many carbon atoms bonded to **hydroxide** groups.

Carbohydrates can be very small or very large molecules, but they are still sugars.

Carbohydrates are the main energy source for the human body. Chemically, carbohydrates are organic molecules in which carbon, hydrogen, and oxygen bond together in the ratio : $C_X(H_2O)_Y$, where X and Y are whole numbers. Animals (including humans) break down carbohydrates during the process of metabolism to release energy. For example, the chemical metabolism of the sugar glucose is shown below :



Animals obtain carbohydrates by eating foods like potatoes, rice, breads, and so on. These carbohydrates are manufactured by plants during the process of photosynthesis. Plants harvest energy from sunlight to run the reaction just described in reverse :



A potato, for example, is primarily a chemical storage system containing glucose molecules manufactured during photosynthesis. In a potato, however, those glucose molecules are bound together in a long chain. As it turns out, there are two types of carbohydrates, the simple sugars and those carbohydrates that are made of long chains of sugars - the complex carbohydrates.

Simple Sugars

All carbohydrates are made up of units of sugar (also called saccharide units). Carbohydrates that contain only one sugar unit (monosaccharides) or two sugar units (disaccharides) are referred to as simple sugars. Simple sugars are sweet in taste and are broken down quickly in the body to release energy. Two of the most common monosaccharides are glucose and fructose. Glucose is the primary form of sugar stored in the human body for energy. Fructose is the main sugar found in most fruits. Both glucose and fructose have the same chemical formula ($C_6H_{12}O_6$); however, they have different structures. Disaccharides have two sugar units bonded together. For example, common table sugar is sucrose, a disaccharide that consists of a glucose unit bonded to a fructose unit.

Sweetening power of common sugars :

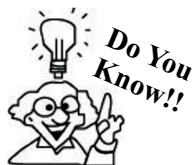
Fructose > Invert sugar > Sucrose > Glucose > Maltose > Lactose

Complex Carbohydrates

Complex carbohydrates are polymers of the simple sugars. In other words, the complex carbohydrates are long chains of simple sugar units bonded together. Therefore the complex carbohydrates can also be referred to as polysaccharides. The potato we discussed earlier actually contains the complex carbohydrate starch.

Starch is a polymer of the monosaccharide glucose. Starch is the principal polysaccharide used by plants to store glucose for later use as energy. Plants often store starch in seeds or other specialized organs; for example, common sources of starch include rice, beans, wheat, corn, potatoes, and so on. When human eat starch, an enzyme that occurs in saliva and in the intestines called amylase breaks the bonds between the repeating glucose units, thus allowing the sugar to be absorbed into the bloodstream. Once absorbed into the bloodstream, the human body distributes glucose to the areas where it is needed for energy or stores it as its own special polymer - glycogen. Glycogen, another polymer of glucose, is the polysaccharide used by animals to store energy. Excess glucose is bonded together to form glycogen molecules, which the animal stores in the liver and muscle tissue as an “instant” source of energy. Both starch and glycogen are polymers of glucose; however, starch is a long, straight chain of glucose units, whereas glycogen is a branched chain of glucose units.

Another important polysaccharide is cellulose. Cellulose is yet a third polymer of the monosaccharide glucose. Cellulose differs from starch and glycogen in terms of extra stability. Cellulose, also known as plant fiber, cannot be digested by human beings, therefore cellulose passes through the digestive tract without being absorbed into the body. Cellulose is a relatively stiff material, and in plants it is used as a structural molecule to add support to the leaves, stem, and other plant parts. Despite the fact that it cannot be used as an energy source in most animals, cellulose fiber is essential in the diet because it helps exercise the digestive track and keep it clean and healthy.



- *Glucose is called grape sugar or dextrose.*
- *Fructose is also called laevulose or fruit sugar.*
- *Lactose is also known as milk sugar.*
- *Starch is also known as amylum.*

AMINO ACIDS AND PROTEINS

Amino acids play central roles both as building blocks of proteins and as intermediate metabolism. The 20 amino acids that are found within proteins convey a vast array of chemical versatility. The precise amino acid content, and the sequence of those amino acids of a specific protein, is determined by the sequence of the bases in the gene that end protein. The chemical properties of the amino acids of proteins determine the biological activity of the protein. Proteins not only catalyze all (or most) of the reactions in living beings they also control virtually all cellular process. In addition, proteins contain within their an amino acid which sequences the necessary information to determine how that protein will fold in dimensional structure, and the stability of the resulting structure. The field of protein folding and stability been a critically important area of research for years, and remains today one of the great unsolved mystery however, being actively investigated, and progress is being made everyday.

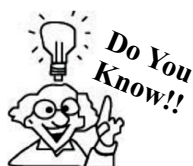
As we learn about amino acids, it is important to keep in mind that one of the more important reasons to understand amino acid structure and properties is to be able to understand protein structure and properties.

Essential Amino Acids

Humans can produce 10 of the 20 amino acids. The others must be supplied in the food. Failure to obtain of even 1 of the 10 essential amino acids, those that we cannot make, results in degradation of the body's proteins.

An essential or indispensable amino acid cannot be made by the body and must be supplied by food. These include *isoleucine*, *leucine*, *lysine*, *methionine*, *phenylalanine*, *threonine*, *tryptophan*, and *valine*. Another amino acid - histidine is considered semi-essential because the body does not always require dietary sources of it.

Non-essential amino acids are made by the body from the essential amino acids or normal breakdown of proteins. The nonessential amino acids are *arginine*, *alanine*, *asparagine*, *aspartic acid*, *cysteine*, *glutamine*, *glutamic acid*, *glycine*, *proline*, *serine*, and *tyrosine*.



Leucine, isoleucine, and valine are called branched-chain amino acid (BCAAs). Human beings cannot survive unless these amino acids are present in the diet. The combination of these three amino acids makes up approximately one-third of skeletal muscle in the human body.

The classification of amino acids as essential or nonessential does not reflect its importance, because all 20 amino acids are necessary for health.

In addition to their role in protein and enzymes synthesis, amino acids are extremely crucial for good health. Amino acids contribute significantly to the health of the nervous system, muscular structure, hormone production, vital organs and cellular structure. Some affects low levels of the essential amino acids result in hormonal imbalances, irritability, low concentration, and depression.

CHECK Point

☛ Is a diet consisting mainly of rice an adequate diet ? Why or Why not ?

SOLUTION

A diet consisting mainly of rice is not an adequate diet because it is deficient in lysine and threonine which are essential amino acids required for growth and maintenance of health and hence their deficiency has to be supplemented by other protein rich diets like pulses, etc.

Structure of Amino Acids

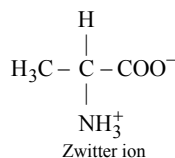
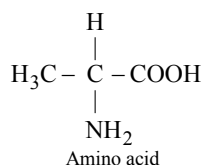
The amino acid is an organic acid in which one or more hydrogen atoms are replaced by NH_2 group. It is one of the organic compounds, which represent the end product of the protein breakdown. All proteins are made up of amino acids. An amino acid contains a central carbon atom to which a amino group and a carboxylic group (COOH) are attached.

Properties of Amino Acids

Besides these two groups, the other two valencies of carbon are satisfied by an alkyl group and hydrogen atom. Amino acids are amphoteric in reaction and form salts with both acids and bases. The amino acids in the body are derivatives of saturated fatty acids. Amino acids are colourless crystalline substance, soluble in water, easily diffusible and (except glycine) optically active. When the amino and carboxyl groups of amino acids combine acid residues. A peptide thus consists of 2 or more amino acid residues linked by peptide bonds.

The amino acid has the tendency to transfer the H^+ ion from $-\text{COOH}$ group to $-\text{NH}_2$ group. As a result of this, the molecule carries a negative charge on one end and positive charge on the other end. It is this charge separation which makes the molecule dipolar and the dipolar ions thus formed, are called zwitter ions.

Examples



CHECK Point

- ☛ Why an electrolysis in acidic solution amino acids migrate towards cathode while in alkaline solution these migrate towards anode ?

SOLUTION

In acidic solution, an amino acid exists as a positive ion, $\text{R} - (\text{NH}_3^+) - \text{COOH}$. Therefore, when an electric field is applied, the amino acid migrates towards the cathode.

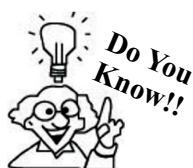
In alkaline solution, an amino acid exists as a negative ion, $\text{R} - \text{CH}(\text{NH}_2) - \text{COO}^-$. Therefore, when an electric field is applied, it migrates towards the anode.

PROTEINS

You have already learned about amino acids. Proteins are made of amino acids. Even though a protein can be very complex, it is basically a long chain of amino acids, all twisted around like a knot. **Proteins** (also known as **polypeptide**) are *organic compounds* made of *amino acids* arranged in a linear chain and folded into a globular form. The amino acids in a *polymer* are joined together by the *peptide bonds* between the *carboxyl* and *amino* groups of adjacent amino acid *residues*.

Like other biological macromolecules such as polysaccharides and nucleic acids, proteins are essential parts of organisms and participate in virtually every process within cells. Many proteins are enzymes that catalyze biochemical reactions and are vital to metabolism. Proteins also have structural or mechanical functions, such as actin and myosin in muscle and the proteins in the cytoskeleton, which form a system of scaffolding that maintains cell shape. Other proteins are important in cell signaling immune responses, cell adhesion, and the cell cycle. Proteins are also necessary in animals diets, since animals cannot synthesize all the amino acids they need and must obtain essential amino acids from food. Through the process of digestion, animals break down ingested protein into free amino acids that are then used in metabolism

- Insulin is a globular protein and myosin is a fibrous protein.
- Globular proteins on heating or on treatment with mineral acids undergo coagulation to give fibrous protein that is insoluble in water called denaturation of protein.
- Haemoglobin is a globular protein found in red blood cells.



The animals can live for a long time without fat or carbohydrate but not without protein. Protein mainly supply new tissues, repair working parts and make up the loss in the vital process. Only plants can build up proteins from inorganic materials like nitrates, ammonium sulphate, carbon dioxide and water.

CHECK Point

- ☛ Where does the water present in the egg go after boiling the egg ?

SOLUTION

When the egg is boiled, the proteins undergo denaturation and the water present in the egg gets absorbed/adsorbed in the denaturated proteins probably through H-bonding.

PEPTIDES

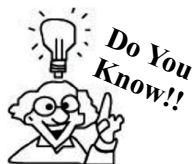
Peptides are short polymers formed from the linking in a defined order, of an amino acid residue and the next is called an amide bond or a peptide bond.

Peptides play a crucial role in fundamental physiological and biochemically functions of life. For decades now, peptide research is a continuously growing field of science.

Peptide is a molecule formed by joining two or more amino acids. When the number of amino acids is less than about 50, these molecules formed by peptides while larger sequences are referred to as proteins.

The amino acids are coupled by a peptide bond, a special linkage in which the nitrogen atom of one amino acid binds to the carboxyl carbon atom of another.

Peptides (proteins) are present in every living cell and possess a variety of biochemical activities. They appear as enzymes, hormones, antibiotics, receptors, etc.



In fact no sharp distinction can be made between proteins and high molecular weight polypeptides. Generally, the peptides having 10 to about 50 amino acid residues are known as polypeptides while the peptides having more than 50 amino acid residues are known as proteins.

CHECK Point

- ☛ The sucrose that we eat in daily life is converted into glucose and fructose. Name the enzyme which facilitates this chemical reaction.

SOLUTION

Invertase or sucrase.

VITAMINS

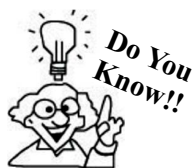
A vitamin is an organic compound required as a nutrient in tiny amounts by an organism. The term vitamin was derived from “vitamin”, a combination word from vital and amine, because it was suggested that the organic micronutrient food factors which prevented beriberi and perhaps other similar dietary-deficiency diseases, might be chemical amines. This proved incorrect for the micronutrient class, and the word was shortened. Today, a chemical compound is called a vitamin when it cannot be synthesized in sufficient quantities by an organism, and must be obtained from the diet.

Vitamins are classified by their biological and chemical activity, not their structure. Thus, each “vitamin” refers to a number of *vitamer* compounds that all show the biological activity associated with a particular vitamin. Such a set of chemicals are grouped under an alphabetized vitamin “generic descriptor” title, such as “vitamin A”, which includes the compounds retinal, retinol, and four known carotenoids. Vitamers by definition are convertible to the active form of the vitamin in the body, and are sometimes inter-convertible to one another, as well.

Vitamins have diverse biochemical functions. Some have hormone-like functions as regulators of mineral metabolism (e.g. vitamin D), or regulators of cell and tissue growth and differentiation (e.g. some forms of vitamin A). Other functions as antioxidants (e.g. vitamin E and sometimes vitamin C).

The largest number of vitamins (e.g., B complex vitamins) function as precursors for enzyme cofactors, that help enzymes in their work as catalysts in metabolism. Although assisting enzyme-substrate reactions are vitamins’ best-known function, the other vitamin functions are equally important.

Until the 1900s, vitamins were obtained solely through food intake, and changes in diet can alter the types and amounts of vitamins ingested. Vitamins have been produced as commodity chemicals and made widely available as inexpensive pills for several decades, allowing supplementation of the dietary intake.



- *Provitamins are the biologically inactive compounds which have almost similar structure as vitamins and can be converted easily into active vitamins e.g., β -carotene is a provitamin of vitamin A.*
- *Of all the vitamins, vitamin B_{12} does not occur in plants but occur only in animals and microorganism. In fact it is exclusively synthesized by the microorganism and is conserved in liver. Vitamin B_{12} has been found in rain water where its presence is attributed to microorganism sucked up by the winds.*

Vitamins are classified as either water soluble or fat soluble. In humans there are 13 vitamins 4 fat-soluble (A, D, E and K) and 9 water-soluble (8 B vitamins and vitamin C). Water-soluble vitamins dissolve easily in water, and in general, are readily excreted from the body, to the degree that urinary output is a strong predictor of vitamin consumption. Because they are not readily stored, constant daily intake is important. Many types of water-soluble vitamins are synthesized by bacteria. Fat-soluble vitamins are absorbed through the intestinal tract with the help of lipids (fats). Because they are more likely to accumulate in the body, they are more likely to lead to hypervitaminosis than are water-soluble vitamins. Fat-soluble vitamin regulation is of particular significance in cystic fibrosis.

CHECK Point

- ☛ Explain why vitamin C cannot be stored in the body ?

SOLUTION

Vitamin C is water soluble, therefore, it is readily excreted in urine and hence cannot be stored in the body.

Knowledge ENHANCER

The discovery of vitamins and their sources

Year of discovery	Vitamin	Food source
1913	Vitamin A (<i>Retinol</i>)	Cod liver oil, milk, liver butter, egg yolk, carrot, spinach, shark liver.
1910	Vitamin B_1 (<i>Thiamine</i>)	Rice bran, yeast, egg, in all cells as its pyrophosphate ester, certain vegetables.
1920	Vitamin C (<i>Ascorbic acid</i>)	Lemons, Fresh fruits, oranges, grapes, green vegetables and potatoes.
1920	Vitamin D (<i>Calciferol</i>)	Cod liver oil, egg, milk.
1920	Vitamin B_2 (<i>Riboflavin</i>)	Yeast, green vegetables, milk, meat, liver, kidney.
1922	Vitamin E (<i>Tocopherol</i>)	Wheat germ oil, seed germ oil, cotton seed oil, green leaves, cosmetics and liver.
1926	Vitamin B_{12} (<i>Cyanocobalamin</i>)	Liver, meat, egg and rain water.
1929	Vitamin K (<i>Phylloquinone</i>)	Alfafa, green plants, fish, egg yolk and liver.
1931	Vitamin B_5 (<i>Pantothenic acid</i>)	Liver
1931	Vitamin B_7 (<i>Biotin</i>)	Liver
1934	Vitamin B_6 (<i>Pyridoxine</i>)	Rice bran, meat, egg yolk and whole cereals.
1936	Vitamin B_3 (<i>Niacin</i>)	Liver
1941	Vitamin B_9 (<i>Folic acid</i>)	Liver

List of Vitamins

Each vitamin is typically used in multiple reactions and, therefore, most have multiple functions.

Vitamin generic descriptor name	Solubility	Deficiency disease	Overdose disease
Vitamin A	Fat	Nightblindness and Keratomalacia	Hypervitaminosis
Vitamin B ₁ large	Water	Beriberi, Wernicke-Korsakoff syndrome	Drowsiness of music relaxation with doses
Vitamin B ₂	Water	Ariboflavinosis	
Vitamin B ₃ other	Water	Pellagra	Liver damage (doses > 2g/day) and problems
Vitamin B ₅	Water	Paresthesia	Diarrhoea; possibly nausea and heartburn
Vitamin B ₆	Water	Anemia peripheral neuropathy nerve damage (dose > 100 mg/day)	Impairment of proprioception,
Vitamin B ₇	Water	Dematitis, enteritis	
Vitamin B ₉	Water	Deficiency during pregnancy is deficiency, other effects	May mask symptoms of vitamin B ₁₂ associated with birth defects, such as neural tube defects
Vitamin B ₁₂	Water	Megaloblastic anemia	No known toxicity
Vitamin C	Water	Scurvy	Vitamin C megadosage
Vitamin D	Fat	Rickets and Osteomalacia	Hypervitaminosis D
Vitamin E	Fat	Deficiency is very rare; mild hemolytic anemia in newborn infants	Increased congestive heart failure seen in one large randomized study
Vitamin K	Fat	Bleeding diathesis	Increases coagulation in patients taking warfarin.

CHECK Point

- ☛ Fresh tomatoes are a better source of vitamin C than those which have been stored for some time. Explain.

SOLUTION

On prolonged exposure to air, vitamin C present in tomatoes is destroyed due to aerial oxidation.

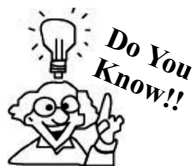


idea box

Make a list of choice of your food items which you take regularly and those which you never or very rare eat. Now try to investigate which component are part of your diet which are not. Also draw a conclusion with the help of your mother wheather you are taking a balance diet or not.

OIL AND FATS

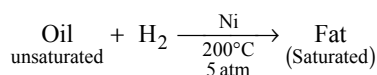
Fats consist of a wide group of compounds that the generally soluble in organic solvents and largely insoluble in water. Chemically, fats are generally triesters of glycerol and fatty acids. Fats may be either solid or liquid at room temperature, depending on their structure and composition. Although the words “oils”, “fats”, and “lipids” are all used to refer to fats, “oils” is usually used to refer to fats that are liquids at normal room temperature, while “fats” is usually used to refer to fats that are solids at normal room temperature. “Lipids” is used to refer to both liquid and solid fats, along with other related substances. The word “oil” is used for any substance that does not mix with water and has a greasy feel, such as petroleum (or crude oil) and heating oil, regardless of its chemical structure.



- *One gram of metabolised fat or oil yield 9 kJ while the corresponding values for carbohydrate and protein are 4 kJ and 5.5 kJ respectively.*
- *The number of grams of iodine which will add to 100 gms of fat or oil is known as iodine number of that oil.*
- *Saponification is a process in which oil is treated with alkali (NaOH or KOH) to prepare sodium or potassium salts of higher fatty acids*

Hydrogenation of Oils

Unsaturated vegetable fats and oils can be transformed through partial or complete hydrogenation into fats and oils of higher melting point. The hydrogenation process involves “sparging” the oil at high temperature and pressure with hydrogen in the presence of a catalyst, typically a powdered nickel compound. As each double-bond is broken, two hydrogen atoms each form single bonds with the two carbon atoms. The elimination of double-bonds by adding hydrogen atoms is called saturation; as the degree of saturation increases, the oil progresses towards being fully hydrogenated. An oil may be hydrogenated to increase resistance to rancidity (oxidation) or to change its physical characteristics. As the degree of saturation increases, the oil’s viscosity and melting point increase.



Drying and Non-Drying Oils

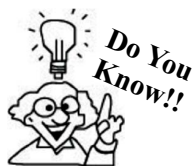
Drying oils absorb oxygen from the air and are thereby converted into plastic, elastic, resin-like substances. Hence, when exposed in a thin layer, as in painting, they form a tough, elastic, waterproof film which adheres tightly to the painted surface and protects it from the weather. The two chief drying oils are linseed oil and tung (chinawood) oil, which find wide use in the manufacture of paints, varnishes, artificial rubber, linoleums, and other coverings.

The non-drying oils find a wide variety of industrial uses : they enter into soaps and cleansers, cosmetics, lubricants, leather dressings, and candles.

Between the drying oils and the non-drying oils is a group of oils which, while they possess the property of absorbing oxygen, do not do so sufficiently to qualify them as drying oils. They are changed more or less when exposed to the atmosphere, but not as completely as linseed tung and certain other drying oils. They are often termed semi-drying oils. Soy bean oil and corn oil are examples of semi-drying oils. When exposed to the atmosphere in a thin layer they thicken but do not form a hard, dry film.

HORMONES

A hormone is a chemical released by a cell in one part of the body, that sends out messages that affect cells in other parts of the organism. Only a small amount of hormone is required to alter cell metabolism. It is essentially a chemical messenger that transports a signal from one cell to another. All multicellular organisms produce hormones; plant hormones are also called phytohormones. Hormones in animals are often transported in the blood.



Hormones are the substances produced by ductless or endocrine glands and poured directly into the blood stream.

Effects of Hormones

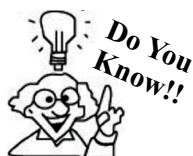
Hormones have the following effects on the body :

1. Stimulation or inhibition of growth
2. Mood swings
3. Activation or inhibition of the immune system
4. Regulation of metabolism
5. Preparation of the body for mating, fighting, and other activity
6. Preparation of the body for a new phase of life, such as puberty, parenting, and menopause
7. Control of the reproductive cycle

A hormone may also regulate the production and release of other hormones. Hormone signals control the internal environment of the body through homeostasis.

Important Hormones and their Functions

Name	Organ of secretion	Functions
1. Sex hormones		
(a) Androgens	Testes	Control the development and normal functioning of male sex organs
(b) Estrogens	Ovary	Control the development and normal functioning of female sex organs
(c) Gestorens	Corpus luteum	Control the development and maintenance of pregnancy
2. Adrenal cortical hormones	Adrenal cortex	Control the metabolism of fats, carbohydrates and proteins; Control the balance of water and minerals in the body
3. Oxytocin	Posterior pituitary gland	Control the contraction of the uterus after child birth and releases milk from the mammary glands
Vasopressin		Controls the reabsorption of water in the kidney
Angiotensin-II	Blood plasma of persons	Contracts the blood vessels
Insulin	Pancreas	Controls the metabolism of glucose
Adrenaline	Adrenal body	Alarming hormone. Increases pulse rate and blood pressure, releases glucose from glycogen and fatty acids from fats
Thyroxine	Thyroid gland	Controls the metabolism of carbohydrates, lipids and proteins



Cortisone is responsible for the metabolism of water, mineral salts, fats, proteins and carbohydrates.

CHECK Point

- ☛ Name a polypeptide hormone which maintains glucose level in the blood.

SOLUTION

Insulin.

Comparison of Vitamins and Hormones

The two types of compounds resemble in the sense that both are required in very small amounts and deficiency of a particular compound results in specific disease. However, they differ from each other in the following points.

- Hormones are produced within the body while vitamins are supplied from outside in the form of diet (except vitamin D).
- Hormones are never stored in the body while vitamins may remain stored in the body.

CHEMOTHERAPY

CHEMICALS IN MEDICINE

Chemical substances administered to a human body or to an animal, either for treatment of diseases or to reduce suffering from pain, are called medicines or drugs.

Most of the diseases in human beings are caused by the entry of the micro-organisms in human body. The treatment of disease by chemical compounds, which destroy the micro-organism without affecting the tissues of the human body, is known as **chemotherapy**. The chemical compounds used for curing are termed as chemotherapeutic agents. The essential feature of the drug is that its action should be localised at the site where it is required.

Chemicals used as therapeutic agents are broadly classified according to the purpose for which these are used e.g., analgesics relieve pain, antipyretics reduce body temperature and antibiotics destroy bacteria and other micro-organisms. The various types of medicinal compounds are discussed below.

Antiseptics

The chemicals which prevent or check the sepsis of wounds. Examples are : *dettol* (chloroxylenol + terpeneol), *Bithional* (added to soaps), *Salol*, *Acriflavin*, *Savlon*, *Gention violet*, *Mercurio Chrome*, *Salicylic acid*, *picric acid*, *resorcinol*, *phenol*, *iodoform*, *boric acid*, *iodine*, *methylene blue*, *potassium permanganate*.

Disinfectants

The chemical substances which completely destroy the micro organisms or stop their growth but are harmful to human tissues are called disinfectants. Examples are : *1.0% phenol*, *SO₂* etc. They are used to disinfect floors, clothes, utensils etc.

Antipyretics

They lower the body temperature i.e. fever reducing. Examples are : *aspirin*, *para-cetamol*, *phenacetin*, *analgin*.

Analgesics

They are pain relieving. Examples are : *Aspirin* and *analgin*, both are antipyretic and analgesics. *Novalgin* is most widely used analgesic.

Certain narcotics like *Codeine*, *morphene*, *pethidine hydro-chloride*, *methadone* and *heroin* etc. are also used as analgesics.

Tranquilizers

Also called psycho-therapeutic drugs, reduce anxiety, induce sleep, and cure mental diseases. Examples are : *Barbituric acid* and its derivatives, *seconal*, *luminal*, *Methyldopa* and *Hydralazine*, *Equanil*.

CHECK Point

1. Sleeping pills are recommended by doctors to the patients suffering from sleeplessness but it is not advisable to take the doses without consultation with the doctor. Why ?
2. Pick out the odd one from among the following compounds on the basis of their medicinal properties mentioning the reason.
Penicillin, *streptomycin*, *aspirin*, *chloroamphenicol*.

SOLUTION

1. Most of drugs taken in doses higher than recommended may produce harmful effects and, act as poison and cause even death. Therefore, a doctor must always be consulted before taking the drug.
2. *Aspirin*. It is an antipyretic whereas other three are antibiotic.

Sedatives and Hypnotics

They are central nervous system depressants reduce restlessness, emotional tension and induce sleep. Examples are : *Phenobarbital*, *Glute thimide*, *Valium* etc.

Antianxiety Agents

Examples are : *Meproamate* and *Diazepam*.

Tranquilizers, Sedatives & Hypnotics and Antianxiety agents are central nervous system stimulants.

Anaesthetics

The chemical substances which produce insensibility to the vital functions of all types of cell especially of nervous system temporarily are called anaesthetics. Examples are, **General anaesthetics** - *Chloroform*, *Fluothane*, **Local anaesthetics** - *Cocaine*, α -*Eucaine*, β -*Eucaine*.

Narcotics

The chemical substances which act as depressant and analgesic. Examples : *Heroin*, *Opium* and *Pethedine*.

Antibiotics

The chemical substances produced by micro-organisms that inhibit the growth of bacteria or even destroy them are called antibiotics. Examples : *Penicillin* is the first antibiotic discovered by Alexander Fleming. It is effective against pneumonia, bronchitis and sore throat etc.

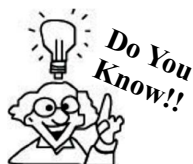
Antimalarials

These are the drugs which cure malaria. Examples are : **Plasmoquin** (*Plasmochin*), **Atebrin** (Mepacrine), **Chloroquine**.

Antacids

Antacids are the drugs which neutralize excess acid in the gastric Juices and give relief from acid indigestion. They remove the excess acid and raise the pH to appropriate level in stomach. There are mainly weak bases.

Examples– $\text{Mg}(\text{OH})_2$, KHCO_3



- *Serotonin is a compound of particular interest because it is important in maintaining stable mental processes.*
- *Antibiotics which are effective against several different types of harmful microorganism are called broad spectrum antibiotics e.g. Tetracyclins, chloromycetin.*
- *Sedatives are given to patients who are mentally agitated and violent. Sedatives give the feeling of calmness, relaxation and drowsiness in the body. Their high dose induces sleep.*

CHECK Point

☛ Name macromolecules which are chosen as drug targets.

SOLUTION

Proteins, carbohydrates, lipids and nucleic acids are chosen as drug targets.

EXERCISE

DIRECTIONS (Qs. 1 to 31) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Which is not true about polymers?
 - Polymers do not carry any charge
 - Polymers have high viscosity
 - Polymers scatter light
 - Polymers have low molecular weight
- An example of biopolymer is
 - Teflon
 - Rubber
 - Nylon-66
 - DNA
- Polymer obtained by condensation polymerization is
 - Polythene
 - Teflon
 - Phenol-formaldehyde
 - Nitrile rubber
- A condensation polymer among the following is
 - Dacron
 - PVC
 - Polystyrene
 - Teflon
- $F_2C = CF_2$ is a monomer of
 - Teflon
 - Glyptal
 - Nylon-6
 - Buna-S
- P.V.C. is formed by polymerisation of
 - Vinyl chloride
 - Ethene
 - Propene
 - 1-Chloropropene
- The sugar that is characteristic of milk is
 - maltose
 - ribose
 - lactose
 - galactose
- Deficiency of vitamin A results in
 - Loss in appetite
 - Skin diseases
 - Sterility
 - Retarded growth
- Polythene is a resin obtained by polymerisation of
 - Butadiene
 - Ethylene
 - Isoprene
 - Propylene
- Night-blindness is caused due to the deficiency of
 - Vitamin D
 - Vitamin A
 - Vitamin C
 - Vitamin B
- Substances which bring body temperature down are known as
 - Antipyretics
 - Analgesics
 - Antibiotics
 - None of these
- Fats and oils are :
 - Polyhydric alcohols
 - Solid and liquid hydrocarbon
 - Lipids
 - Weak carboxylic acids
- What is fat ?
 - Acid
 - Alcohol
 - Esters
 - Cellulose
- Which of the following gives maximum energy in metabolic process ?
 - Carbohydrates
 - Fats
 - Proteins
 - Vitamins
- Alkaline hydrolysis of esters is known as
 - esterification
 - saponification
 - dehydration
 - alkalination
- Deficiency of vitamin E causes
 - Beriberi
 - Scurvy
 - Hemolytic anemia
 - None of these
- Which among the following is the simplest sugar?
 - Glucose
 - Starch
 - Cellulose
 - None of these
- Amino acids are the building blocks of
 - fats
 - proteins
 - vitamins
 - carbohydrates
- The function of enzymes in the living system is to
 - transport oxygen
 - provide energy
 - provide immunity
 - catalyse biochemical reactions
- Scurvy is caused due to the deficiency of vitamin
 - B_1
 - C
 - K
 - A

21. Which carbohydrate is essential constituent of plant cell?

- (a) starch (b) glucose
(c) cellulose (d) sucrose

22. The linkage present in proteins and peptides is

- (a) $\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{NH}- \end{array}$ (b) $\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{O} \end{array}$
(c) $\begin{array}{c} \text{O} \quad \text{O} \\ || \quad || \\ -\text{C}-\text{O}-\text{C}- \end{array}$ (d) $-\text{NH}-$

23. Commonly used antiseptic 'Dettol' is a mixture of

- (a) o-chlorophenoxylenol + terpineol
(b) o-cresol + terpineol
(c) phenol + terpineol
(d) chloroxylenol + terpineol

24. Which one of the following can possibly be used as analgesic without causing addiction and mood modification?

- (a) Diazepam
(b) Morphine
(c) Paracetamol
(d) Tetrahydrocannabinol

25. The number of essential amino acids in man is

- (a) 8 (b) 10
(c) 18 (d) 20

26. Cellulose is a polymer of

- (a) Glucose (b) Fructose
(c) Ribose (d) Sucrose

27. Which of the following is not an amino acid?

- (a) Glycine (b) Adenine
(c) Histidine (d) Benzidine

28. Amino acids are the building blocks of

- (a) Carbohydrates (b) Vitamins
(c) Fats (d) Proteins

29. What is the monomer of a polypeptide?

- (a) Amino acid (b) Glucose
(c) Nucleoside (d) Nucleotide

30. Peptides on hydrolysis give

- (a) Amines (b) Amino acids
(c) Ammonia (d) Alcohols

31. Insulin, a hormone, chemically is a

- (a) Fat (b) Steroid
(c) Protein (d) Carbohydrate

DIRECTIONS (Qs. 32 to 44) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

32. Which of the following is/are a macromolecule(s) ?

- (a) DNA (b) Starch
(c) Palmitate (d) Proteins

33. Which one of the following is/are vitamin(s)?

- (a) Calciferol (b) Ascorbic acid
(c) Oxalic acid (d) Cyanocobalamine

34. The general formula for most of carbohydrates is $\text{C}_n(\text{H}_2\text{O})_m$; in this formula the value of n and m may be

- (a) $n > m$ (b) $n < m$
(c) $n = m$ (d) None of these

35. Which of the following statement(s) is/are true ?

- (a) All amino acids, including glycine, are ampholyte.
(b) An amino acid may be neutral, acidic or basic.
(c) The basic functional group in amino acids is $-\text{COO}^-$.
(d) Amino acids exist in zwitter ionic form in aqueous solution

36. On the basis of mode of formation, polymers can be classified as

- (a) addition polymers only
(b) condensation polymers only
(c) copolymers
(d) synthetic polymers

37. Which of the following is/are example of addition(s) polymer ?

- (a) Polystyrene (b) Nylon
(c) PVC (d) Polypropylene

38. Which of the following statement(s) about vitamin B_{12} is/are correct ?

- (a) It has a cobalt atom
(b) It is not present in significant amount in plants
(c) It is also present in rain water
(d) It is needed for human body in very small amounts

39. Glucose can be classified as

- (a) hexose (b) carbohydrate
(c) aldose (d) oligosaccharide

40. Which of the following is/are polysaccharide(s) ?

- (a) Lactose (b) Starch
(c) Cellulose (d) Glycogen

Chemistry in Everyday Life

41. Fats and oils are formed from :
- Glycerol and long chain unsaturated acids
 - Glycerol and long chain saturated acids
 - Ethylene glycol and long chain saturated and unsaturated acids
 - Ethylene glycol and long chain saturated and unsaturated acids
42. The human body produce(s)
- Vitamins
 - Hormones
 - Enzymes
 - DNA
43. Which of the following is water-insoluble?
- Vitamin E
 - Vitamin K
 - Vitamin A
 - Vitamin B
44. Which one of the following is/are protein(s)?
- Wool
 - Nail
 - Hair
 - DNA

DIRECTIONS (Qs. 45 to 48) : Study the given paragraph(s) and answer the following questions.

Passage-1

Amino acids contain an -NH_2 (basic) as well as a -COOH (acidic) group. They exist as zwitter ions

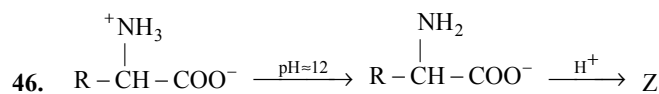


which explain their several characteristic properties, like decomposition on heating, solubility in water, large dipole moment.

Thus in solution, amino acids may exist as dipolar (neutral pH), cation (in strongly acidic solution), or anion (in strongly basic solution).

Amino acids undergo usual reactions of the -COOH group as well as -NH_2 group.

45. Amino acid are
- basic as a typical amine and acidic as a carboxylic acid
 - less basic than a typical amine and less acidic than a -COOH
 - more basic than a typical amine and more acidic than a -COOH
 - nothing is certain



Compound Z is

- $\begin{array}{c} \text{NH}_2 \\ | \\ \text{R}-\text{CH}-\text{COO}^- \end{array}$
 - $\begin{array}{c} ^+\text{NH}_3 \\ | \\ \text{R}-\text{CH}-\text{COO}^- \end{array}$
 - $\begin{array}{c} ^+\text{NH}_3 \\ | \\ \text{R}-\text{CH}-\text{COOH} \end{array}$
 - $\begin{array}{c} \text{NH}_2 \\ | \\ \text{R}-\text{CH}-\text{COOH} \end{array}$
47. How many dipeptides are possible from two molecules of a typical α -amino acid ?
- 1
 - 2
 - 3
 - 4
48. α -Amino acids are
- acidic due to -COOH group and basic due to -NH_2 group
 - acidic due to -NH_3^+ group and basic due to -COO^- group.
 - neither acidic nor basic.
 - None of the above is true

DIRECTIONS (Qs. 49 to 53) : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - If **Assertion** is **correct** but **Reason** is **incorrect**.
 - If **Assertion** is **incorrect** but **Reason** is **correct**.
49. **Assertion :** Hydrogenation is the process of converting an oil into a fat, called vegetable ghee.
Reason : Hydrogenation as carried out in presence of a catalyst usually finely divided nickel.
50. **Assertion :** Proteins on hydrolysis produce amino acids.
Reason : Amino acids contain -NH_2 and -COOH groups
51. **Assertion :** Valine is an essential amino acid.
Reason : The lack of essential amino acids in the diet causes Kwashiorkor.
52. **Assertion :** The drugs which act on the central nervous system and help in reducing anxiety are called antibiotics.
Reason : Pencillin is an antibiotic.

53. Assertion : Antiseptics are applied to living tissues.

Reason : Iodine is a powerful antiseptic

DIRECTION (Q. 54) : Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.

54.	Column-I (Carbohydrate)	Column-II (Constituent unit)
	A. Starch	(p) Fructose
	B. Sucrose	(q) Galactose
	C. Cellulose	(r) β -D-Glucose
	D. Lactose	(s) α -D-Glucose

	A	B	C	D
(a)	p,q	q,r	p,r	r
(b)	q	r	p	s
(c)	p,r	p,s	q	p,s
(d)	s	p, s	r	q, r

DIRECTIONS (Qs. 55 and 56) : Following are integer based questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive).

- 55.** How many of the following are essential amino acids?
Phenylalanine, alanine, glutamine, valine and lysine.
- 56.** How many of the following are antibiotics?
Plasmoquin, Pencillin, Streptomycin, aspirin and salol.

SOLUTIONS

Brief Explanations of Selected Questions

1. (d)
2. (d) Deoxyribonucleic acid is a polymer found in living being.
3. (c) All other three polymers are obtained by addition polymerisation
4. (a) 5. (a)
6. (a) P.V.C. (Poly vinyl chloride) is formed by polymerisation of vinyl chloride, $\text{CH}_2=\text{CHCl}$.
7. (c) It is found in the milk of all animals and imparts sweetness to milk (hence named milk sugar).
8. (d) In childhood, lack of vitamin A retards growth and hence like other vitamins, it is also said to be a growth promoting factor. In mild deficiency it leads to night blindness. Its prolonged deficiency leads to xerophthalmia.
9. (b) Polyethylene (Polyethene). It is addition polymer of ethylene.
10. (b) Night-blindness is caused due to the deficiency of vitamin A.
11. (a) Antipyretics are those compounds which reduce the body temperature in fever. Examples are aspirin (acetyl salicylic acid), paracetamol, phenacetin, novalgin and analgin.
12. (c) Oil and fats are also called as lipids.
13. (c) Chemically, fats and oils are triesters of glycerol with higher fatty acids.
14. (b) Oils and fats are food materials which provide energy. Actually, among food materials, fats and oils provide highest amount of energy.
15. (b) Alkaline hydrolysis of esters is known as saponification.

$$\text{R}-\text{COOR}' + \text{NaOH} \longrightarrow \text{R}'\text{OH} + \text{RCOONa}$$
16. (c)
17. (a) Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is the simplest molecule which is monosaccharide while others are polysaccharides which on hydrolysis give monosaccharides.
18. (b) α -Amino acid is the building block unit of protein which is formed by polymerisation of amino acid through peptide linkage.

$$\left[\begin{array}{c} -\text{NH}-\text{C}- \\ \parallel \\ \text{O} \end{array} \right]$$
19. (d) The function of enzymes in the living system is to catalyse biochemical reactions which occur in living systems. e.g. invertase, pepsin, amylase.
20. (b) Scurvy (bleeding of gums) is caused due to deficiency of vitamin 'C' (ascorbic acid).
21. (c) Cellulose is a polymer of carbohydrate which is essential constituent of plant cell.
22. (a) Proteins and peptides are linked by peptide linkages

$$\begin{array}{c} \text{O} \\ \parallel \\ (-\text{C}-\text{NH}) \end{array}$$
23. (d) The mixture of chloroxylenol and terepeneol are dettol which is used as antiseptic.
24. (c) We know that N-acetyl-para-aminophenol (or paracetamol) is an antipyretic which can also be used as an analgesic to relieve pains.
25. (b) There are 20 amino acid in man out of which 10 amino acids are essential amino acids. These essential amino acids are supplied to our bodies by food which we take because they cannot be synthesized in the body. These are (1) valine (2) leucine (3) Isoleucine (4) Phenyl alanine (5) Threonine (6) Methionine (7) Lysine (8) Tryptophone (9) Arginine (10) Histidine.
26. (a) We know that cellulose ($\text{C}_6\text{H}_{12}\text{O}_6$)_n is the chief constituent of cell walls of plants. It is the most abundant organic substance found in nature. It is a polymer of glucose with 3500 repeat units in a chain.
27. (d) 28. (d) 29. (a) 30. (b) 31. (c)
32. (a, b, d) Palmitate is just a salt or an ester of palmitic acid (a fatty acid of the formula, $\text{C}_{15}\text{H}_{31}\text{COOH}$)
33. (a, b, d) Vitamin C is chemically ascorbic acid. Calciferol is vitamin D and cyanocobalamine is vitamin B₁₂.
34. (a, c) The value of n in $\text{C}_n(\text{H}_2\text{O})_m$ may be either same as m as in monosaccharides or more than m as in oligosaccharides, but never less than m .
35. (a, c, d) Every amino acid has at least one basic and one acidic group, so it acts as an amphoteric compound (ampholyte). An amino acid may have one $-\text{NH}_2$ and one $-\text{COOH}$ group (neutral amino acids), or two $-\text{NH}_2$ and one $-\text{COOH}$ group (basic amino acids), or one $-\text{NH}_2$ and two $-\text{COOH}$ groups (acidic amino acids). In amino acids, $-\text{NH}_3^+$ acts as an acid while $-\text{COO}^-$ acts as a basic group.
36. (a, b)
37. (a, c, d) Nylon is a condensation polymer.

38. (a, b, d) It is found in liver, egg, milk, meat, and fish. Minute amounts are probably present in all animal cells. Peculiarly, unlike other vitamins, B₁₂ is not found in significant amounts in green plants.
39. (a, b, c) Glucose is aldohexose. Glucose is a monosaccharide, i.e. it can not be hydrolysed further to simple sugars. Oligosaccharides on hydrolysis give 2-10 molecules of monosaccharides.
40. (b, c, d) Lactose is a disaccharide.
41. (a, b) Fats and oils are formed from glycerol. These are long chain saturated and unsaturated acids. Fats which are solids at ordinary temperatures contain a large proportion of saturated acids. The acids present in glycerides are almost exclusively straight chain acids and always contain an even number of carbon atoms. The chief saturated fatty acids are lauric (C₁₂), [myristic (C₁₄)], palmitic (C₁₆) and stearic (C₁₈). The main unsaturated acids are oleic, linoelic and linolenic acids.
42. (a, b, c) The human body does not produce DNA.
43. (a, b, c) Vitamin B is water soluble whereas all other are water insoluble.
44. (a, b, c)
45. (b) The acidic character of an amino acid molecule is due to -NH_3^+ group, which is less acidic than a -COOH group, while the basic character of an amino acid is due to -COO^- group which is less basic than -NH_2 group.
46. (b) -NH_2 is a better base than -COO^- , resonance in -COO^- decreases basic character of the -COO^- group.
47. (a) Two molecules of an α -amino acid will form only one dipeptide, recall that four different dipeptides are formed when two α -amino acids are different.
48. (b) Amino acids exist as Zwitterions in which acidic character is due to -NH_3^+ and basic due to -COO^- group.
- $$\begin{array}{ccc} \text{R} & & \text{R} \\ | & & | \\ \text{H}_3\text{N}^+\text{CHCOOH} & \xleftarrow{\text{acid}} & \text{H}_3\text{N}^+\text{CHCOO}^- \\ & & | \\ & & \text{R} \\ & & \downarrow \text{base} \\ & & \text{H}_2\text{NCHCOO}^- \end{array}$$
49. (b) Hydrogenation or hardening of oil is a process in which various unsaturated radicals of fatty glycerides are converted into more highly or completely saturated glycerides by the addition of hydrogen in the presence of a catalyst, usually finely divided nickel.
50. (b) 51. (b) 52. (d) 53. (b)
54. (d) A - (s); B - (p, s); C - (r); D - (q, r)
55. 3 Phenylalanine, valine and lysine are essential amino acids.
56. 2 Penicillin and streptomycin are antibiotics.

